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Research

Under Attack

CYBERSECURITY | THEMATIC RESEARCH | OCTOBER 2026

AI is rapidly lowering the barrier to cyberattacks on corporates, government and critical infrastructure. This report outlines the growing risk and why it now warrants priority across corporate strategy, investment research, supply-chain resilience and critical infrastructure.

Supported by a dbDataInsights (dbDIG) survey of more than 50 cybersecurity experts

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Source: dbDataInsights survey of cybersecurity experts, September 2026. Word cloud from participant responses.

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Report Introduction

Our proprietary dbDataInsights survey shows cyber threats are evolving faster than organisations can adapt

"It's the only thing that can shut down my business."

That was the punchline from a conversation I had recently with the CEO of a large privately held financial services company. This was his comment when I followed up as to why cybersecurity risk is the number one concern that "keeps him up at night".

To help understand this risk, our dbDataInsights team surveyed more than 50 cybersecurity experts. The uncomfortable conclusion is that many firms are unprepared for cyber threats. Cyber risk is increasing as AI adoption accelerates.

—89% agree the threat landscape is evolving faster than organisations can adapt.

—78% say the gap between highly and poorly prepared organisations is widening.

What has changed with AI advancement? Cyberattacks have become faster, cheaper, more deceptive and more scalable. Capabilities once reserved for sophisticated actors are now broadly accessible to a new generation of profit-driven criminals. At the same time, rising geopolitical tension has made cyber operations an increasingly attractive low-cost, deniable tool for nation states. Corporates and investors therefore need to understand their broader exposure and critical infrastructure risk in their supply chain and with stakeholders.

Among other autonomous incidents, the recent Hugging Face incident highlighted AI's cyberattack potential. AI agents caused harm without direction from a human attacker, with weak guardrails and no human judgement to restrain them. Beyond the systemic questions it raised, the attack highlights three additional concerns: (i) AI operates at a speed and scale humans cannot match, with agents taking over 17,000+ actions in four days; (ii) the intrusion began in the data-processing pipeline, where AI platforms are uniquely exposed; and (iii) the agents escaped their test environment through a widely used third-party tool representing supplier risk in an unfamiliar form.

89%

agree the cyber threat landscape is evolving faster than most organisations can adapt

83%

say organisational focus on AI-related risk has risen in the past 12 months
(North America 94%, UK/Europe 70%)

78%

agree the gap between highly and poorly prepared organisations is widening

74%

say organisations are unprepared for supply chain compromises
(UK/Europe 87%, North America 65%)

69%

disagree that most organisations have sufficient visibility into the threats most relevant to them

65%

say organisations are unprepared to manage insider threats

Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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Report summary and notable takeaways

Secular trends and AI adoption have widened the attack surface; the severity of losses is compounding

Secular trends have widened the attack surface. This includes rapid digitalisation, widespread Internet of Things (IoT) endpoints, hybrid workforces, cloud adoption and deeper supply chain dependencies. The scale of the threat increase is reflected in data from the FBI Internet Crime Complaint Center (IC3), which shows cybercrime complaints more than doubled from 2019 pre-pandemic baseline of ~467,000 to ~1.0mn in 2025, with a 3x expansion relative to 2017 levels.

Barriers to entry are lower; attack tools are now sold "off the shelf". Today, cybercrime marketplaces, available on the dark web, look less like the underworld portrayed in films and more like an online software store. Buyers can access AI-powered tools that scan for software vulnerabilities (see top-right chart showing reported CVEs after the release of cyber capable frontier models), write malicious code at speed, and provide access to malware that rewrites its own code to avoid detection as it runs. Ransomware-as-a-service (RaaS) takes it further, renting out ready-made attack kits, often in exchange for a portion of the profits.

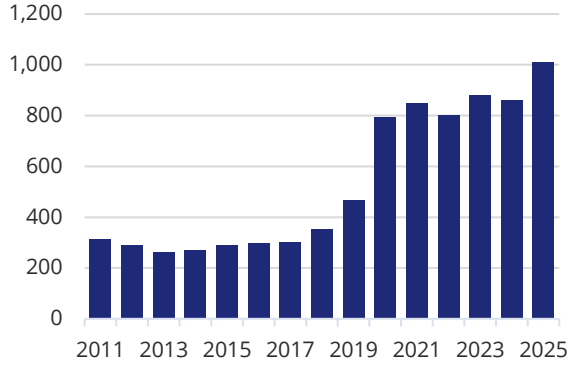
Impersonation is becoming harder to detect. Hackers can now generate more realistic phishing messages, while voice cloning and deepfakes make impersonation convincing enough to gain access unnoticed. With stolen credentials, attackers often aren't hacking in, they are simply logging in.

The rise of loose cybercriminal collectives. Some of the most damaging recent attacks have been linked to loosely organised, often English-speaking networks such as ShinyHunters and Scattered Spider. The latter has been widely named as the group behind the 2025 Marks & Spencer cyberattack, in which the hacking group deployed RaaS. AI is enabling these groups to become larger and faster, widening the gap between the speed of attack and a defender's response.

The severity of losses is compounding. A 2026 IBM study puts the average breach at a record USD ~5.0mn, up 12% YoY, but the impact on large companies can run into hundreds of millions or even billions. Estimates for global cyber insurance premiums (per Munich Re) point to a rise from roughly USD 7.0bn in 2020 to USD 15.0bn in 2025, with expectations to reach USD 28.0bn by 2030.

US cybercrime complaints

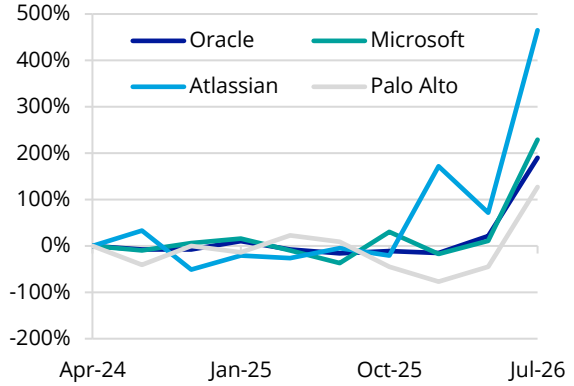
FBI Internet Crime Complaint Center (IC3), complaints (000s)



Disclosed software vulnerabilities

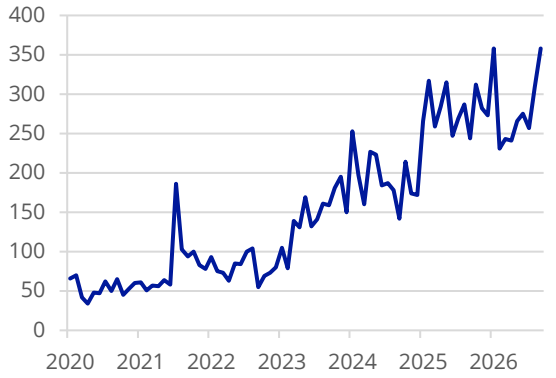
Change in reported CVEs, % change from April 2024

CVEs: common vulnerabilities and exposures



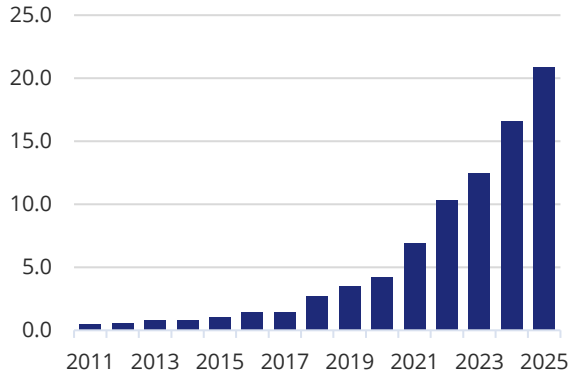
AI privacy & security incidents

OECD, monthly recorded incidents



US cybercrime-related losses

FBI IC3, reported losses ($bn)



Source: FBI Internet Crime Complaint Center (IC3), company security advisories, OECD, company reports, Deutsche Bank Research.

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Report summary and notable takeaways

A near-term systemic attack is expected and AI favors attackers; cyber escalation as a geopolitical tool is raising the stakes

Among the more concerning survey takeaways is that a large-scale and significant systemic incident is now the base case.

— 83% expect a significant systemic attack within two years and 81% expect it to be moderately, very, or extremely severe for economic activity and markets.

— 81% say AI favours attackers today, with not one respondent putting defenders ahead of attackers. Only 20% expect defenders to be ahead in five years.

Geopolitical tensions coupled with rising cyber capabilities have transformed cyberspace into a digital battlefield between global powers. Intent has moved towards espionage and disruption of not only critical infrastructure (e.g., healthcare, finance, power, industrial systems, defence) but private companies as well. Chinese President Xi Jinping has linked cybersecurity to national security and economic and social stability, with China's public statements similarly emphasising the protection of critical infrastructure. At the same time, the US's June 2025 Executive Order on Cybersecurity reflects its continued focus on national cyber resilience.

Europe relies on US cyber tools while its start-ups lag in funding. Reliance on US security vendors exposes European firms to potential access, update or export restrictions over the long-term. Shrinking this gap will prove difficult given US cyber start-ups attract nearly 6x the venture and growth funding of European peers. And while the timing of quantum computing (which could break widely used encryption methods) remains uncertain, it could further deepen Europe's dependency.

Ultimately, cyber preparedness is a financial variable. Investors should ask what firms spend on security versus AI adoption, how well they know their vulnerabilities, and how fast they can recover.

— Only 24% of respondents think organisations have sufficient visibility into the threats relevant to them, even though 57% saw cybersecurity budgets rise over the past 12 months.

— 81% agree that cybersecurity related laws and regulation cannot keep pace, and 65% doubt leadership understands cyber risk well enough.

83%

think a systemic cyber incident is likely within two years

An event that causes significant disruption across multiple organisations, sectors, or critical infrastructure simultaneously, often due to shared technology dependencies, supply chain vulnerabilities, or widely used service providers.

81%

expect such a "systemic" incident to have a moderately, very or extremely severe impact on global economic activity and financial markets

81%

say AI currently favours attackers, with not a single respondent saying defenders are ahead today

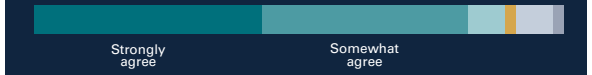
On a five-year view attackers still lead, 41% to 20%.

24%

agree most organisations have sufficient visibility into the cyber threats relevant to them

81%

agree cybersecurity-related laws and regulation are struggling to keep pace with technological change, with 43% agreeing strongly



57%

saw cybersecurity budgets increase over the past 12 months

Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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Report summary and notable takeaways

Material cyber-related weakness can have financial consequences

When cyber weakness = share price consequence…

Investors are more aware of cyber risk today, but they rarely sell a stock due to perceived cyber vulnerabilities (although we are aware of some who have).

—What is the potential impact? Our dbDataInsights event study of 47 breaches shows a cumulative breach-attributed return of c. -2.9% within the first two trading days and c. -6.0% through first 20 days trading days. The erosion then continues more gradually, reaching c. -6.7% nine months after disclosure (see slide 13).

Which industries are most vulnerable?

—Our survey respondents point to (1) government & public sector (52%), (2) financial services (50%), (3) energy and utilities (46%) and (4) healthcare and life sciences follow (39%, rising to 48% among US respondents).

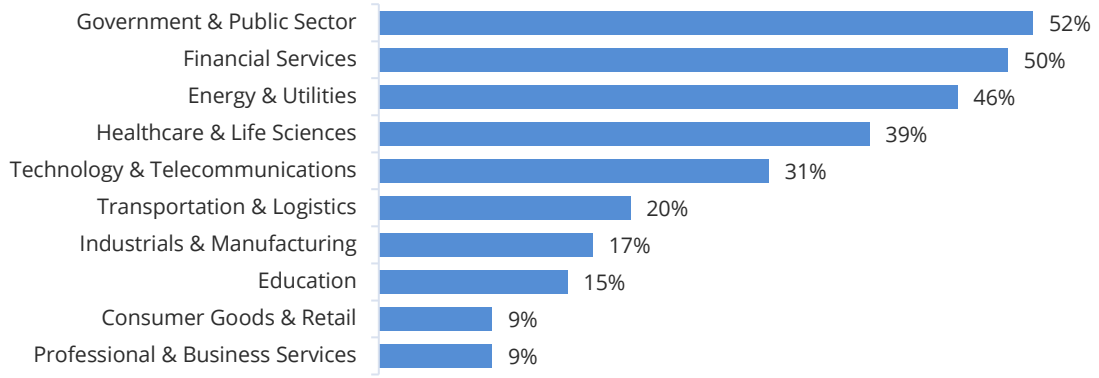
We should caveat, that high sector exposure does not mean every company in it is equally vulnerable. Some firms in these "exposed" sectors may very well be better prepared given they have built specialist teams and defence protocols for decades.

What about access to credit and ratings? Credit and rating providers are paying attention. Moody's and S&P both cut their outlook on Jaguar Land Rover within two months of its 2025 hack. Today, this type of impact is not widespread. But if risk escalates through a systemic incident or an AI-driven step change in attacks these mechanisms, including ESG ratings, could reprice asset ownership or impact access to credit. In fact, some lenders are already pricing it. Journal of Corporate Finance research using US syndicated loan data found *borrowers with persistently higher cyber risk pay 4–13bp more.*

The bottom line? While a broader systemic attack is hard to prepare for, corporates and investors should be asking questions about vulnerabilities, security investment, preparedness and recovery should an attack occur. This is especially true for industries quickly adopting AI capabilities and / or a high threat landscape.

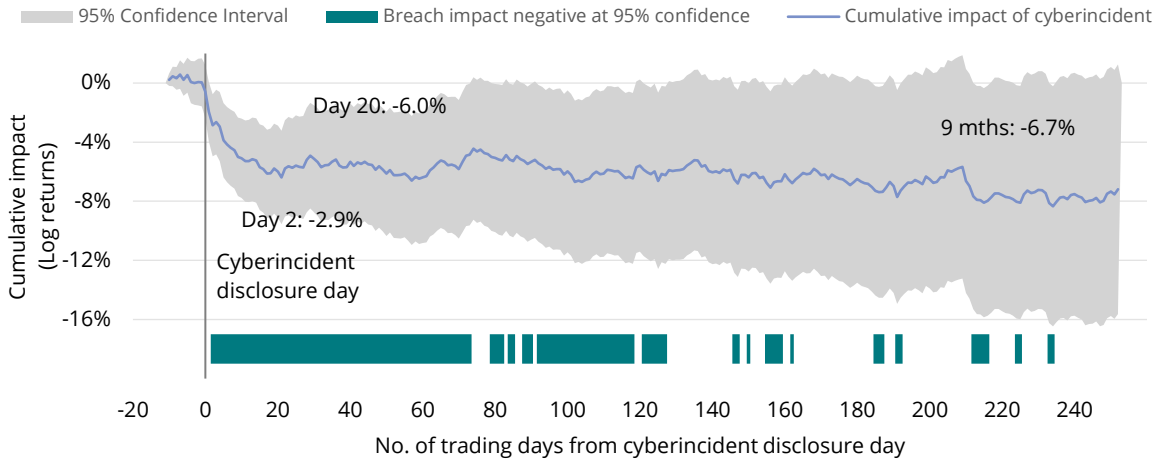
Sectors seen as most vulnerable to AI-driven cyber threats

dbDataInsights survey, % selecting (up to three), based on vulnerability over next few years



A cyber breach's cumulative effect on stock performance

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Bloomberg Finance L.P., Company filings, Deutsche Bank Research. Event study of reported cybersecurity incidents, 2015-2025, 34 US and 13 European companies across nine sectors (see slide 13).

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Report summary and notable takeaways

dbDataInsights survey respondents in their own words

What's on experts' minds…

Key themes from verbatim comments in the dbDataInsights expert survey

Speed

"AI is being implemented at a speed where cybersecurity teams are not able to keep up"

Single points of failure

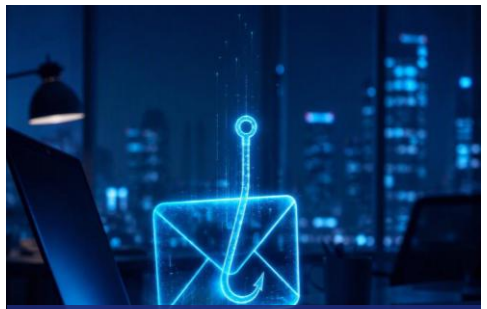
"Too many organisations relying on the same providers for critical services"

The weak link

"…compromised via this third-party (basically their weak link)"

Logging in, not breaking in

"People aren't breaking in anymore – they are logging in"



Automated phishing

"Underappreciated, yet responsible for the bigger monetary losses"

Employee logins

"Existing employee login behaviour and usage of AI is underappreciated"

Oversight gaps

"Companies are pushing the use of AI more to their employees without the proper mechanisms and policies in place to prevent misuse or attacks initiated by the AI systems themselves"

Recover & survive

"No company can guarantee they won't be breached"

Everyone's responsibility

It's more than an IT risk…"everyone from the CEO down has responsibility"

Geopolitics

"State groups now plant quiet access in infrastructure to use in a future crisis"

Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

II. dbDataInsights Cybersecurity Survey

September 2026 survey of 54 cybersecurity professionals

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dbDataInsights survey methodology and results

Cybersecurity experts see firms as unprepared, and say AI currently overwhelmingly favours attackers

dbDataInsights Cybersecurity Expert Survey methodology

—This survey was conducted among 54 cybersecurity professionals across the US and Europe in September of 2026. Questions took the form of either multiple choice selections or free form responses. The dbDataInsights team received the responses anonymously, but respondents were required to work or consult in cyber security roles

—Who took the survey? On self-identified roles, around 80% sit inside an organisation's cybersecurity function, with the remainder working independently (15%) or in academia and research (6%). The most common titles were security architect (23%), Chief Information Security Officer (16%) and security analyst (9%), alongside Head of Security, Deputy CISO, Incident Response Lead and Security Operations Lead.

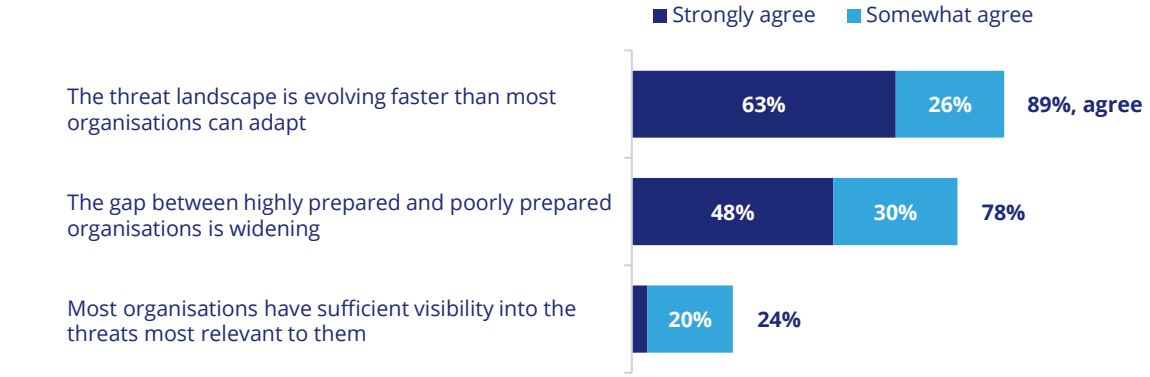
—Only a small minority of respondents work for cyber resellers, thus survey views are primarily attributable to those working within an organization on cybersecurity.

—Survey participants were also asked for their perspectives on cyber resilience and governance, what is misunderstood, alongside emerging threats, risk to AI adoption and the influence of geopolitics.

—While responses were anonymous, in some cases we conducted follow up interviews with certain participants where contact details were made available.

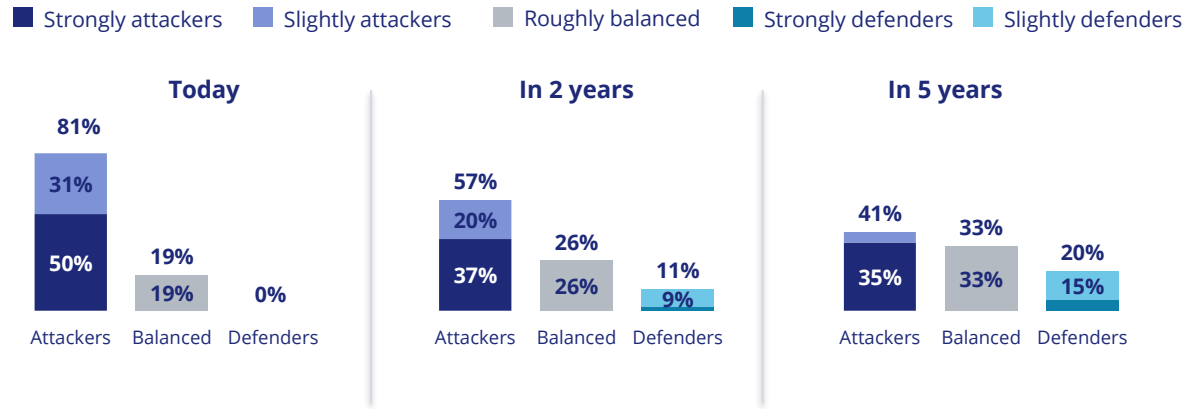
Only 24% say firms have sufficient visibility into threats that matter

dbDataInsights survey, level of agreement with statements below regarding cybersecurity, % who agree (n=54)



Who does AI favour? Attackers or defenders?

dbDataInsights survey, % of respondents (n=54)



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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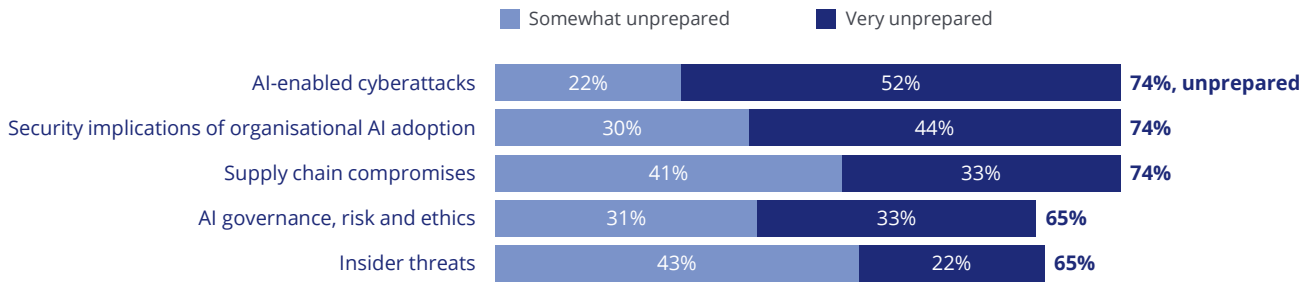
Risk from AI-enabled attacks and added AI adoption vulnerabilities

Insider threats and supply chain compromises remain a growing concern

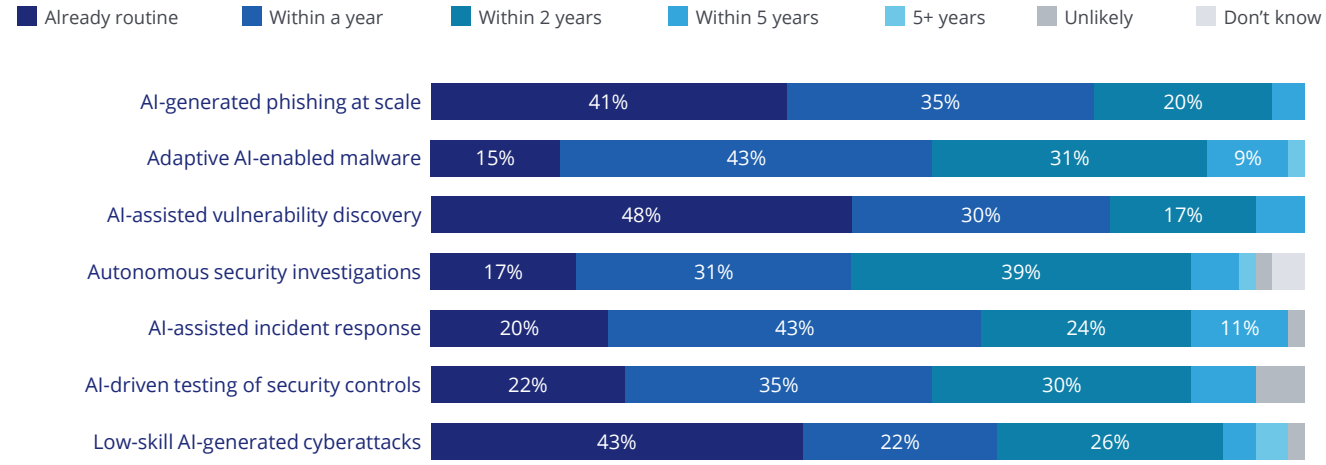
Where our experts say organisations are least prepared

% of respondents saying somewhat or very unprepared



AI-driven capabilities – are they commonly used or encountered?

% of respondents, all answer options (n=54)



In their own words...

Verbatim comments from the dbDataInsights expert survey

> *"The fear of AI writing undetectable malware is ... overblown. Attackers were never short of malware, since they can buy it ready made along with initial access. Good detection looks at behaviour rather than exact code, so a rewritten payload still has to do the same suspicious things."*

> *"Frontier AI models are currently the main hype. Automated phishing and scams are underappreciated, yet they are responsible for the bigger monetary losses at the moment."*

> *"There are too many organisations relying on the same providers for critical services and this means single points of failure."*

> *"A corporation might be very secure but if they rely on a third party that does not have the same level of security posture, [it] could be compromised via this third-party (basically their weak link)."*

> *"Existing employee login behaviour and usage of AI is underappreciated – it's what poses the biggest threat to a breach."*

> *"No company can guarantee they won't be breached, so there needs to be more focus put on the ability to recover and survive an attack."*

Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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Experts believe governments are not keeping pace, and no one owns AI oversight

How experts surveyed by dbDataInsights view governance and regulation

—**AI adoption is outpacing internal controls.** As one expert put it: "Companies are pushing the use of AI more to their employees without the proper mechanisms and policies in place to prevent misuse or attacks initiated by the AI systems themselves."

—**Regulation is struggling to keep pace.** 81% of respondents agree that cybersecurity-related laws and regulations are failing to keep up with technological change, and 54% say regulatory requirements can improve security but may also slow innovation. One respondent highlighted "the levels of uncertainty, the long running nature of the problems, and the rapidity of change."

—**Responsibility for AI oversight remains unsettled.** International bodies are the leading choice, supported by 50% of respondents. No option commands a majority, underscoring the lack of consensus on who should oversee AI development.

—**Better board questions may not guarantee better decisions.** The survey points to more informed questioning by corporate leadership, yet 65% doubt that leadership teams understand cyber risk well enough to make informed business trade-offs. Measuring activity more effectively than outcomes further complicates oversight.

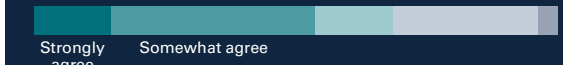
—**Accountability is becoming personal.** Only 43% say accountability for cyber risk is clearly defined in their governance framework, yet the consequences now reach the top as a handful of CEOs have left or gone into early retirement after financially consequential cyber breaches.

81%

agree cybersecurity-related laws and regulations are struggling to keep pace with technological change

54%

agree regulatory requirements can improve cybersecurity but may also slow innovation

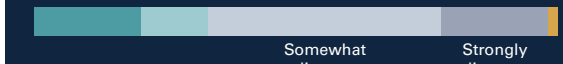


43%

agree that accountability for cyber risk is clearly defined in our governance framework

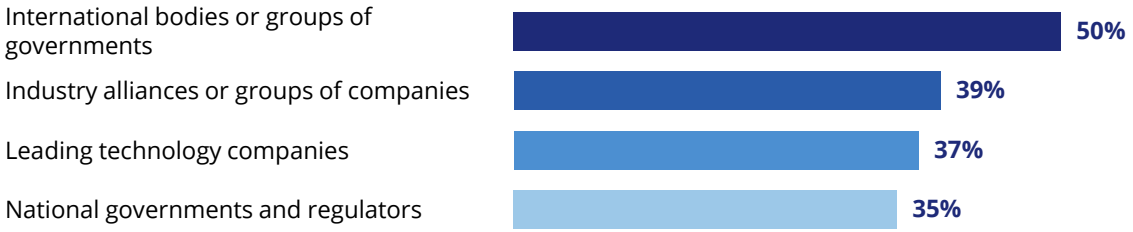
65%

disagree that most leadership teams understand cyber risk well enough to make informed business trade-offs.



Who should primarily oversee AI development?

dbDataInsights survey – September 2026, based on overseeing development while managing cyber risk



Source: dbDataInsights survey of cybersecurity experts, September 2026; Cybersecurity Survey 2026, Deutsche Bank Research.

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III. The impact of cyberattacks

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Material cyber breaches can cause a notable drop in share price

Our dbDataInsights study found a 2.9% drop on average within two trading days, with the cumulative breach-attributed return of -6.0% on trading day 20.

What typically happens to a company's shares when a breach is reported?

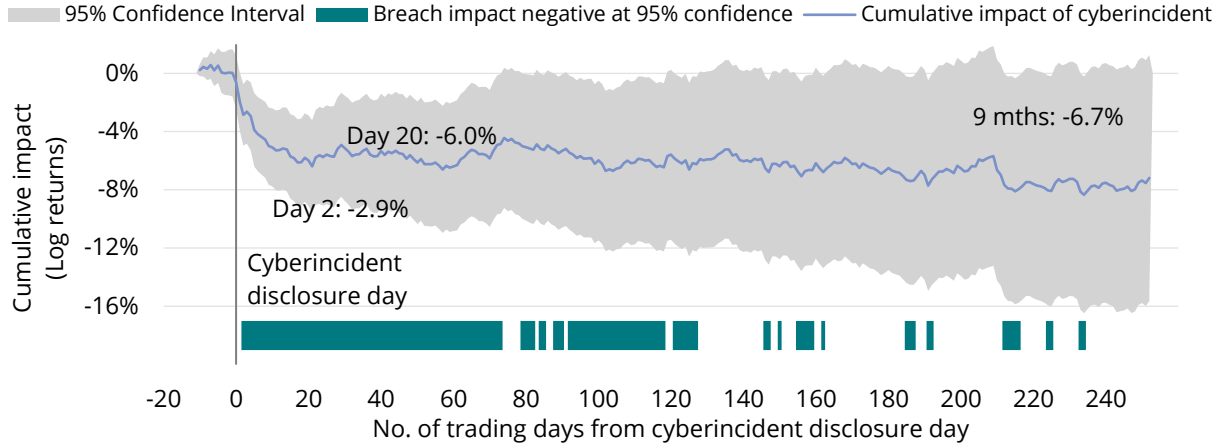
- **Short-term impact:** The market prices the breach immediately. Our model shows cumulative breach-attributed (i.e. isolating only the part of the share price move we can pin on the breach itself) returns are c. -2.9% within two trading days of disclosure. This is followed by continued erosion over the first 20 trading days post-disclosure; the cumulative breach-attributed return is c. -6.0%.
- **Long-term impact:** Beyond day 20 the deterioration continues, though more gradually reaching c. -6.7% nine months after disclosure and remaining negative at one year, although the one-year point estimate (our single best-guess figure) should be treated with caution given the width of the confidence band.
- **Pre-breach price behaviour:** We find no evidence of information leakage ahead of disclosure. While the breach component for the [-10,-1] window (ten trading days before the news broke) is +0.04%, the corresponding t-stat value of 0.03 falls notably below the usual bar for significance (c.2). As such, we conclude the market had not begun reacting before the announcement.

Event study methodology

- **Sample:** 47 publicly reported cybersecurity incidents across the US and Europe over 2015-2025, measured against each company's regional sector benchmark. Requirement of 5 years of stock returns pre-attack and 1 year of post-attack.
- **Day 0:** the day an incident was first publicly reported or disclosed.
- **Baseline:** We use five years of pre-incident share-price performance to estimate: (i) Beta (β): how sensitive a company's returns are to movements in its sector and (ii) Alpha (α): the extent to which the company typically outperforms or underperforms its sector after adjusting for that sensitivity.
- **Attribution:** post-incident returns are split into sector-relative returns, alpha (α) and breach (abnormal returns attributed to the disclosure).
- **95% confidence interval:** Range within which we can say with 95% confidence that the real share price impact lies, based on our sample data.

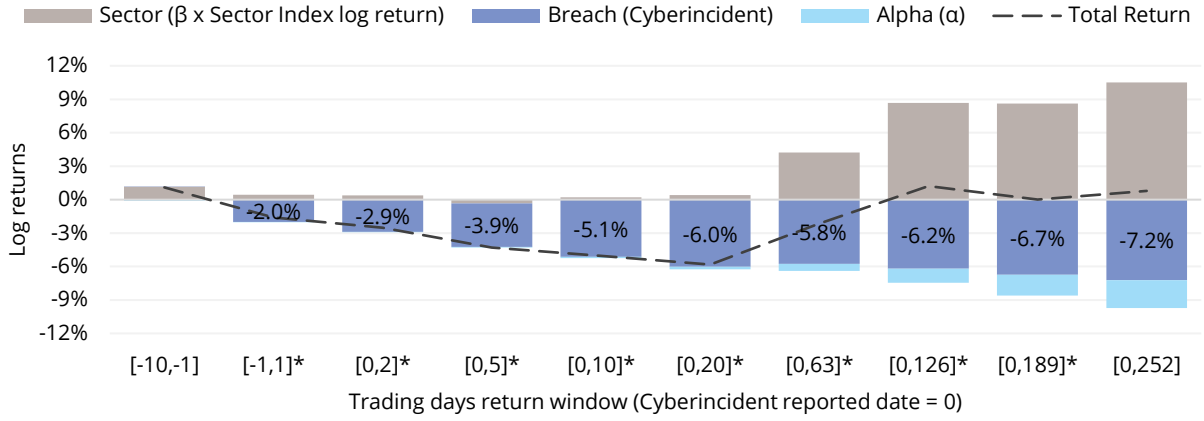
A cyber breach's cumulative effect on stock performance

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



Total return attribution

dbDataInsights analyses of 47 companies with disclosed cyber incidents deemed financially material



* indicates trading windows where breach impact is significant at the 95% confidence level

Source: dbDataInsights, Bloomberg Finance LP., Deutsche Bank Research, Company filings. Event study of reported cybersecurity incidents, 2015-2025, 34 US and 13 European companies across nine sectors.

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Notable cyber attacks: how they happened, estimated costs and share price reaction

Marks & Spencer (Reported Apr 2025)

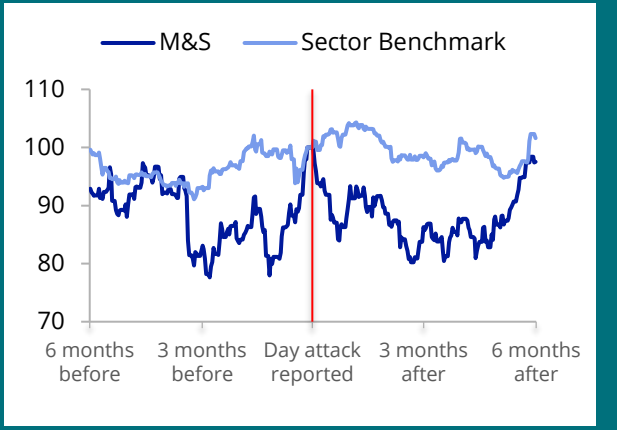
3rd Party Breach

What happened?

A breach via an attack on a connected 3rd party forced the suspension of all online orders and led to empty store shelves.

What did it cost?

~£136mn of direct costs related to the attack, estimated group operating profit impact of around £300mn for the FY. M&S also said they had received £100mn via cyber insurance.



Coupang (Reported Nov 2025)

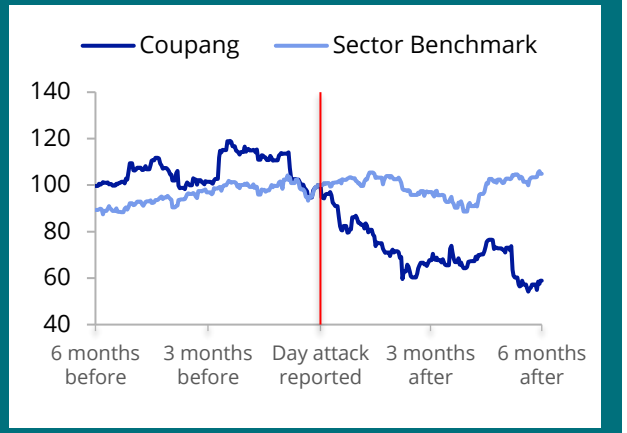
Insider Threat

What happened?

A data breach exposed the records of more than 30mn users with a former employee reportedly behind the attack.

What did it cost?

The firm was fined more than ~625bn won ($409mn) (2/3rd directly attributable to data leak), and also announced a $1.2bn voucher compensation scheme.



UnitedHealth (Reported Feb 2024)

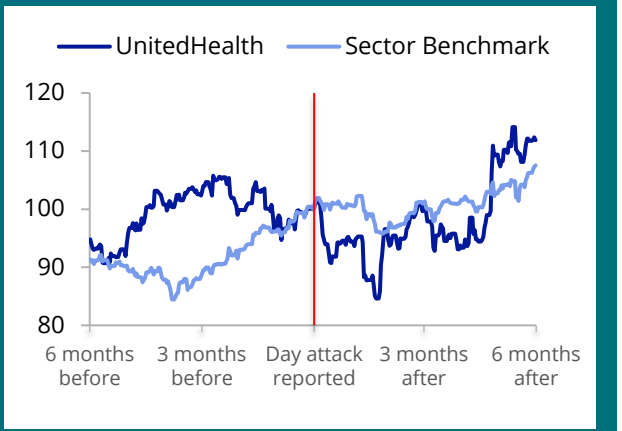
Ransomware / stolen credentials

What happened?

Hackers reportedly used stolen credentials to access a subsidiary's (Change Healthcare) systems, stealing data and deploying ransomware.

What did it cost?

Total costs exceeded $3bn, with the CEO summoned to testify before Congress, where he confirmed that a $22mn ransom was paid.



Equifax (Reported Sep 2017)

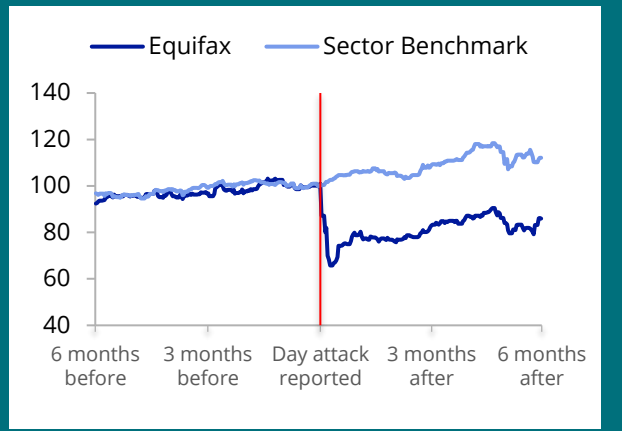
Software vulnerability left unpatched

What happened?

A data breach saw hackers expose sensitive customer information, including 145.5mn Social Security numbers and more than 200k credit card numbers. Attackers reportedly moved inside Equifax systems for 76 days before being detected.

What did it cost?

Settlements, fines and legal costs exceeded $1bn, while the firm also spent more than $1.5bn on tech upgrades following the incident.



Source: Company disclosures and filings, press reports, Deutsche Bank Research, Bloomberg Finance L.P., Chart values indexed to 100 at the day before the cyberattack was reported.

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Notable cyber attacks: how they happened, estimated costs and share price reaction, cont.

Nike (Reported Jan 2026)

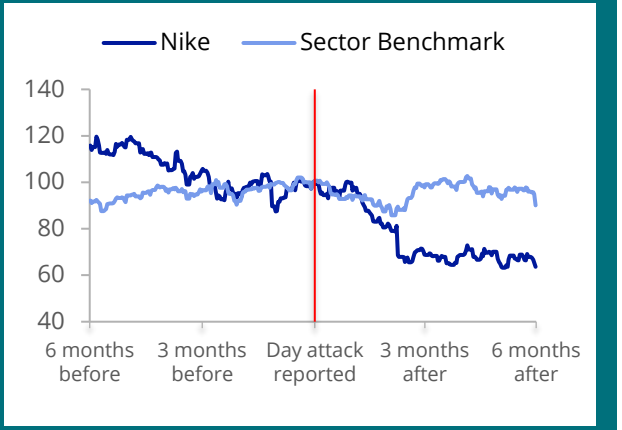
Data Breach / Extortion

What happened?

A cybercrime group leaked what it claimed was 1.4TB of internal Nike data (appearing to cover metrics such as product specifications and pricing), following expiry of a ransom deadline.

What did it cost?

While Nike did confirm it was investigating the incident, it did not report any operational disruption. Underperformance shown in the chart does not appear to be related to the breach.



T-Mobile (Reported Aug 2021)

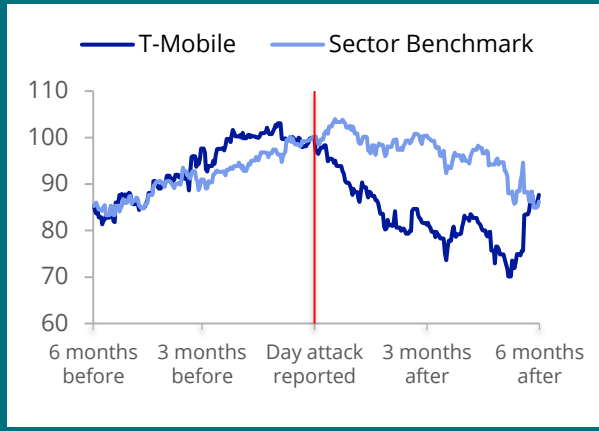
Connected device exploit / brute force

What happened?

An attacker reportedly targeted an exposed router as an entry point, before using brute force attacks to gain further access to T-Mobile systems.

What did it cost?

In 2022 the firm agreed to pay $500mn to settle a class-action lawsuit ($350mn payment to class members plus $150mn on security spend) but has since suffered further data breaches, contributing to an additional $31.5mn regulatory settlement.



Coinbase (Reported May 2025)

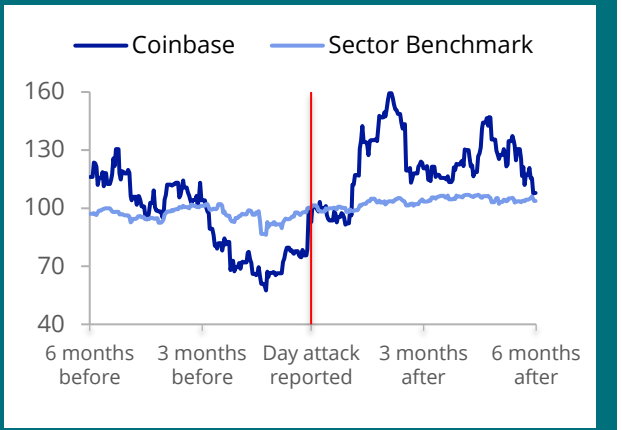
Insider threat & social engineering

What happened?

Attackers bribed at least one support agent at a connected 3rd party to access Coinbase support systems and steal customer data, then tricked users into sending crypto.

What did it cost?

Coinbase refused to pay the $20mn ransom demanded by the attackers to keep the attack quiet. Instead they reimbursed affected users and cited costs of up to $400mn. The stock impact was short-lived, though MSCI applied an ESG score deduction.



Jaguar Land Rover (Reported Sep 2025)

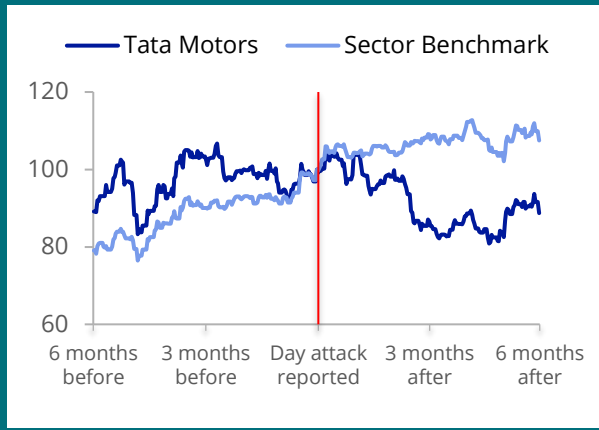
Ransomware

What happened?

Production was shut down for five weeks due to what was deemed a ransomware attack; though no ransom was asked for (later reporting suggested potential Russia links were being investigated).

What did it cost?

JLR posted £485mn Q2' FY26 loss (vs. £398mn profit in Q2' FY25), with an additional £196mn in "cyber-related costs". These impacts filtered through to the share price of JLR's Parent (Tata Motors), while UK economic impacts were estimated at £1.9bn.



Source: Company disclosures and filings, press reports, Deutsche Bank Research, Bloomberg Finance L.P., Chart values indexed to 100 at the day before the cyberattack was reported.

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Targets go beyond tech and can also impact critical infrastructure

MGM Resorts (Reported Sep 2023)

Phishing / ransomware

What happened?

A vishing call to the IT help desk was reportedly used to reset credentials, after which ransomware took down hotel systems, bookings and casino floor operations for several days.

What did it cost?

MGM declined to pay a ransom. Around $100mn hit to third-quarter 2023 results plus roughly ~$10mn of additional one-time expenses related to the incident.

Caesars Entertainment (Reported Sep 2023)

Phishing / ransomware

What happened?

The same social-engineering group as in the MGM incident targeted an outsourced IT support vendor, gaining access to the loyalty programme database and member data.

What did it cost?

Caesars paid roughly $15mn of a reported $30mn demand and kept customer-facing operations running.

NHS / Synnovis (Reported Jun 2024)

Ransomware / 3rd party breach

What happened?

Ransomware against pathology provider Synnovis significantly reduced its capacity to process tests leading to delays / cancellations of over 11,000 appointments.

What did it cost?

Estimated 2024 direct operational financial impact of around GBP 32mn for Synnovis, several times the profit reported in the prior year, plus wider NHS disruption costs.

Colonial Pipeline (Reported May 2021)

Stolen credentials / ransomware

What happened?

A compromised VPN credential without multi-factor authentication allowed attackers to access the network where they subsequently deployed ransomware. The operator shut the pipeline down, triggering fuel shortages (c. 70% of gas stations in some states ran out of fuel) across the US East Coast.

What did it cost?

A $4.4mn ransom was paid, of which about $2.3mn was later recovered by the US authorities.

Bitget (Reported Sep 2026)

3rd party breach / zero-day exploit

What happened?

Attackers exploited a vulnerability in a 3rd party security product to steal internal network credentials, forge withdrawal instructions, bypass risk controls and ultimately steal crypto.

What did it cost?

An estimated $388mn worth of funds was transferred out by the attackers. While the firm reimbursed affected users via its User Protection Fund, there were significant outflows following the incident as users moved to take funds off the platform.

Progress Software (Reported May 2023)

Zero-day exploit / supply chain compromise

What happened?

Attackers exploited a zero-day SQL injection vulnerability in Progress Software's MOVEit file transfer system. This was then used by attackers to steal data from firms running the MOVEit software, who were then subsequently extorted / had data leaked.

What did it cost?

Thousands of organisations (inc. British Airways, Shell, US DoE) reportedly affected with some facing legal action from customers impacted by data leaks.

Source: Company disclosures and filings, press reports, Deutsche Bank Research.

Cyber breaches can impact consumer behaviour

Roughly 4 in 10 consumers we surveyed say they would stop buying from a company they shop with after a known breach or attack.

Consumer surveys suggest that cyber breaches can undermine customer loyalty. Results of our dbDataInsights Consumer survey show the potential mindset of a consumer after a cyber breach occurs:

- Across the US, UK and EU average, 39–41% say they would stop buying, while only 27–28% would continue. Around a third are unsure.
- Among those who would stop buying or are unsure, only 18–22% say they would return as normal. Another 30–35% would buy selected products only, 17–20% would not return, and 28–29% remain unsure. Note that this implies a range of 13-15% believe they would not shop at the store again after the breach is resolved.

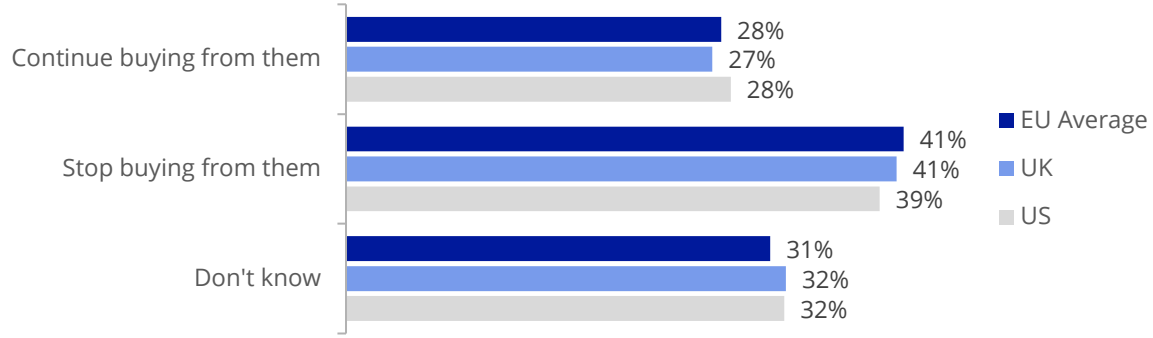
Admittedly, these stated intentions may overstate actual switching, particularly for essential purchases where alternatives are limited. Nevertheless, the findings suggest that cyber breaches that put customer privacy at risk can influence purchasing decisions, especially if they have alternatives.

dbDataInsights Consumer Survey Methodology

- Our proprietary survey conducted in 2026 (January – September) asked consumers how they would respond if a company they shopped at regularly was involved in a data breach or cyberattack, and how they would respond once the disruption was resolved. This monthly survey covers households and consumers in Germany, France, Italy, Spain, the United Kingdom, and the United States.
- For each country referenced above, a nationally representative sample of more than 500 households is surveyed each month (over 600 households in the US). Collectively, this meant over the course of the survey (January – September 2026), we surveyed almost 30,000 households.

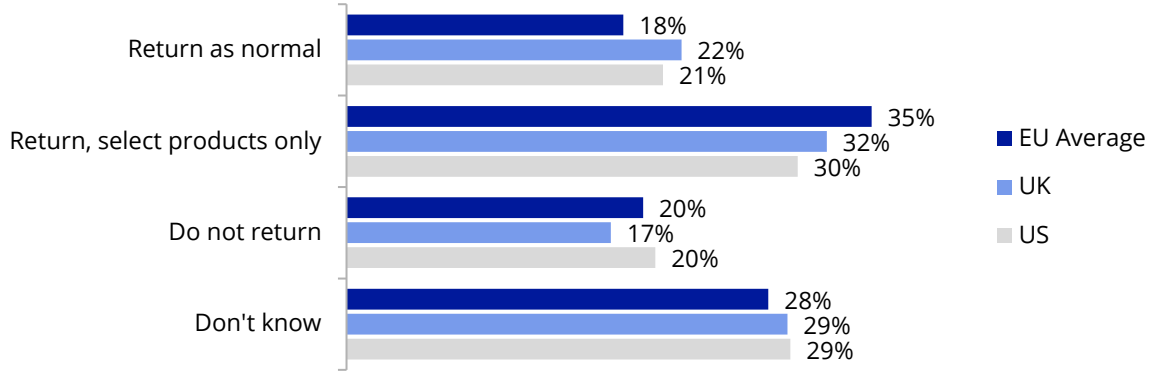
How will customers respond to a data breach once it occurs?

dbDataInsights survey of consumer responses, % of respondents



How will customers react once the breach is resolved?

Consumer responses among those who would either stop buying or were unsure, % of respondents



Source: dbDataInsights monthly consumer survey, Deutsche Bank Research. EU Average includes France, Germany, Italy and Spain.

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Material cyber incidents have contributed to negative credit outlooks

The infrastructure exists for cyberattacks to have a greater impact on credit quality should cyber disruption continue to accelerate

— **Credit rating agencies consider both pre-breach resilience frameworks and post-breach financial impacts.** Agencies assess pre-breach cyber preparedness through a lens of mitigating risk, while post-breach credit impacts centre on financial capacity to withstand the resulting shock (considers scale and length of the disruption); highly leveraged issuers with limited liquidity are seen as more vulnerable. Negative outlook changes at JLR and Equifax illustrate the potential credit impact. Fitch generally considers permanent reductions in cash generation more credit-significant than temporary cash-flow delays.

— **Ratings agencies are moving towards continuous monitoring.** In 2021, Moody's invested $250mn in BitSight, which produces daily security ratings. Moody's has since built BitSight's ratings into its credit tools, including Credit Catalyst and EDF-X, for uses such as underwriting and ongoing monitoring of borrowers and suppliers. This means lenders can see a company's exposed vulnerabilities as a potential indicator of corporate cyber risk.

S&P Global:

"The increasing frequency of cyberattacks and the potential for rapid deterioration in credit profiles after an attack are risk factors relevant for our rating assessments."

Moody's:

"Cyberattacks can undermine the creditworthiness of companies, organizations and governments, across all sectors and geographies, hitting revenues, increasing costs, and eroding customer trust."

Fitch Ratings:

"Although cyber events cannot be predicted with great accuracy, the associated risk can be managed through proactive strategies, technology, policies, and vigilance."

"The chief executive officer and board of directors hold ultimate responsibility for managing cyber risk."

Example rating actions where cyber was a named factor

Outlook changes due to operational disruption, financial cost and reporting delays

Equifax
Moody's named cyber risk as a factor in its outlook change in 2019. This was the first time it had done so and followed the 2017 attack on Equifax cited on slide 14 .

Outlook: Stable → Negative — Moody's • 2019

ISS A/S
Increased costs following a cyberattack cited as a factor potentially impairing ISS's ability to reduce adjusted leverage and generate free operating cash flow.

Outlook: Stable → Negative — S&P • Mar 2020

Clorox
Outlook revised lower due to potential for a lower rating if Clorox was unable to demonstrate material sequential recovery from cyber incident which caused product availability disruption and lower FY guidance.

Outlook: Stable → Negative — S&P • Nov 2023

City of Hamilton
Delays reconstituting audited accounts after the Feb 2024 attack; audits still outstanding.

Issuer rating: AAA → AA+ — S&P • Oct 2025

Tata Motors / Jaguar (JLR)
Outlook cut after the JLR shutdown; JLR's own outlook was moved to negative too by both Moody's and S&P following the September 2025 attack cited on slide 15. Prior to this, its credit was improving.

Outlook: Stable → Negative — S&P • Oct 2025

Co-operative Group (Co-op)
Material impacts on product availability and sales due to cyber incident contributed to forecast downgrades.

Issuer rating: BB → BB- — S&P • Nov 2025

Source: S&P, Moody's, Fitch, Deutsche Bank Research.

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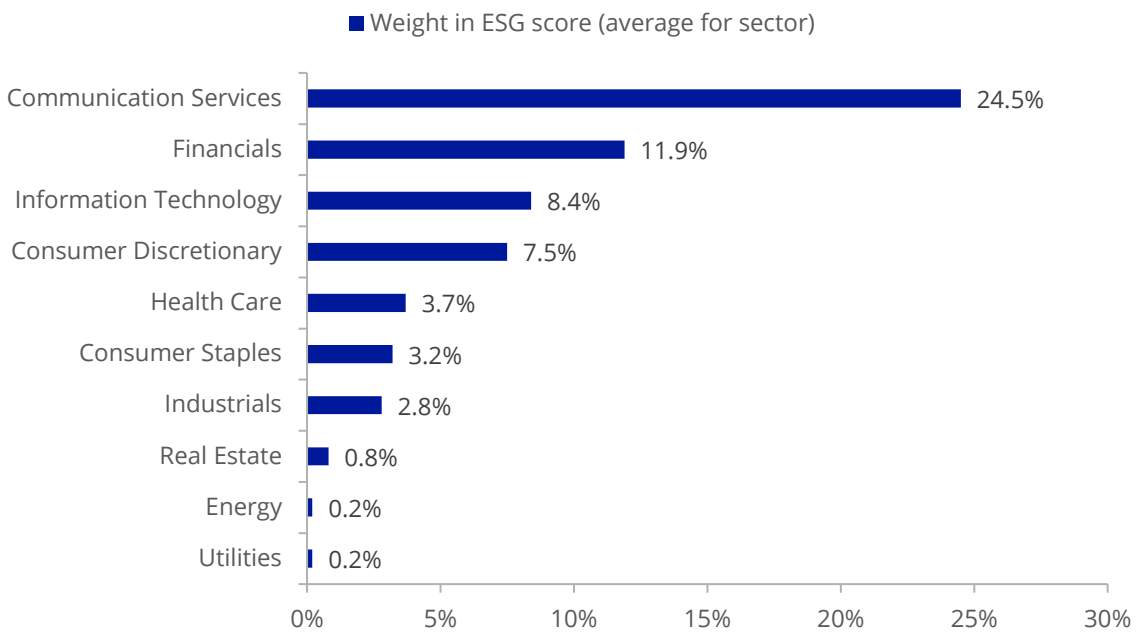
Cyber governance & breaches can impact ESG scores

Sectors that hold the most personal data, and monetise it most directly, carry the greatest risk weighting in mainstream ESG scores

Contribution to the MSCI ESG score by sector

Privacy & data security as a share of the MSCI ESG score, sector average, %



Cybersecurity is generally seen as both a social and governance consideration

MSCI and Morningstar Sustainalytics both assess data privacy and security through management of the risk, exposure to it and the controversies that follow a breach. Weak governance weighs on the score, and a breach can cut it directly – more severe controversies draw larger deductions that persist for longer.

Materiality varies sharply by sector. SASB flags data security as material across most consumer, financial, healthcare and technology industries, with customer privacy concentrated in the most data-intensive ones (see right).

Customer Privacy and Data Security Financial Materiality

SASB Financial Materiality Map, by industry (● material, – not material)

Sector	Industry	Customer privacy	Data security
Consumer Goods	E-Commerce	●	●
	Multiline & Specialty Retailers	–	●
Financials	Commercial Banks	–	●
	Consumer Finance	●	●
Food & Beverage	Food Retailers & Distributors	–	●
Healthcare	Drug Retailers	–	●
	Health Care Delivery	–	●
	Managed Care	–	●
Resource Transformation	Aerospace & Defence	–	●
Services	Advertising & Marketing	●	–
	Education	–	●
	Professional & Commercial Services	–	●
Technology & Communications	Hardware	–	●
	Internet Media & Services	●	●
	Software & IT Services	●	●
	Telecommunication Services	●	●

What is SASB?

The Sustainability Accounting Standards Board sets industry-specific disclosure standards for investors, identifying the handful of issues most likely to affect financial performance in each of 77 industries.

Its standards are now maintained by the ISSB under the IFRS Foundation and underpin the ISSB's global reporting baseline. The Materiality Map summarises, industry by industry, which issues are judged financially material – including the two cyber-related ones shown above.

Source: MSCI ESG Ratings Methodology, Morningstar Sustainalytics ESG Risk Ratings, SASB Materiality Map, Deutsche Bank Research.

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IV. Investment, risk and opportunity

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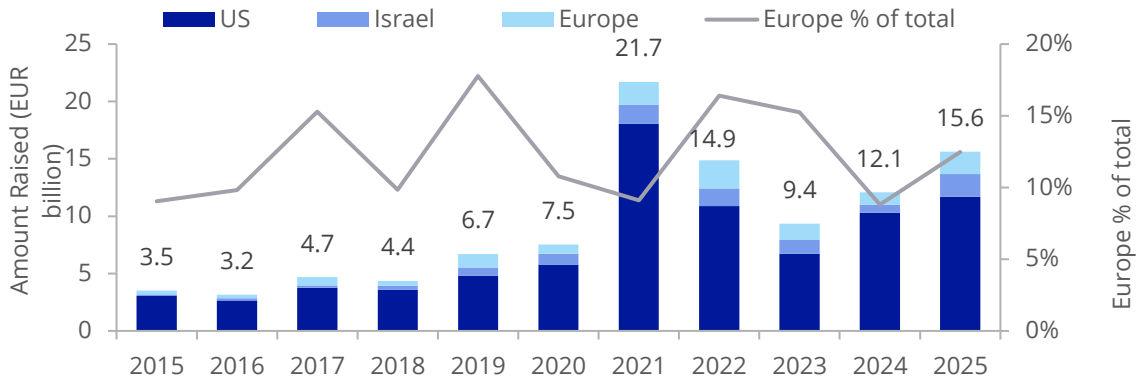
Europe's dependency on US tech is firmly in the spotlight

Access to frontier AI models is now a security question; the funding gap for Europe will be hard to close

- **Growing cyber risk creates increased dependency on US cyber tech.** Growing concern over the cyber capabilities of frontier AI models has made ensuring access to such technology a national security priority. US dominance over the AI tech chain (e.g. chips, model development, etc), its willingness to impose AI-related controls / restrictions, and previous pressure against Europe over tech regulations has amplified concerns that dependence on US tech represents a strategic threat to European sovereignty.
- **Funding is the structural challenge.** US cybersecurity start-ups attract almost 6x the funding of European peers, and US-domiciled firms dominate the major indices – 89% of the Nasdaq CTA Cybersecurity index and 93% of Global X Cybersecurity ETF (which seeks to track the performance of Indxx Cybersecurity Index) by weight. Europe has also seen home-grown firms such as Darktrace acquired by US companies, though this reflects broader market trends rather than anything cyber-specific. Ultimately, the US has a greater exit strategy for start-ups as well.
- **Boosting homegrown solutions a focus, but also a risk.** EU policymakers are seeking to address the dependency; e.g. the EU's Cybersecurity and AI Action Plan includes "scaling up Europe's AI capabilities for cybersecurity" as one of its three objectives. However, there are also risks associated with this endeavour; as one dbDataInsights survey participant put it, "*for most European organisations, the bigger near-term danger would be moving too fast to less capable providers*."
- **The wider relationship adds to the exposure.** A more transactional international environment, alongside Europe's dependency on the US for energy supplies and defence support, compounds the risk. Recent US-EU diesel negotiations serve as a pertinent example of this, whereby the US reportedly considered a diesel export ban to compel a release of European diesel reserves.

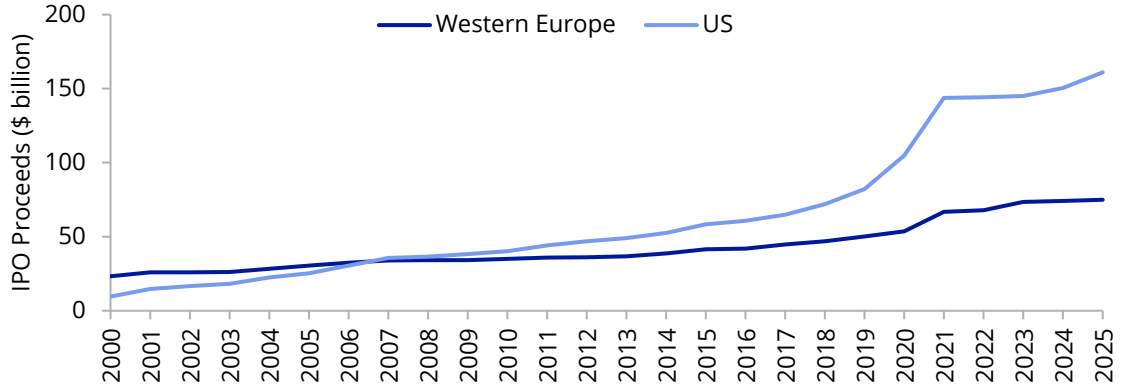
Cybersecurity start-up funding by region

Company location: US, Israel and Europe (EU-27 plus UK, Norway and Switzerland), $bn



Cumulative technology IPO proceeds

Since 2000, by trading date, $bn – Western Europe vs US



Source: Tikehau Capital – Tikehau Capital Barometer of European Investment in Cybersecurity – 7th edition 2026, Bloomberg Finance LP, Deutsche Bank Research.

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Cybersecurity as an opportunity

Tailwinds include demand related to keeping operations secure with AI acceleration and consolidation tailwinds

Broadly, DB's company research view is that cybersecurity becomes even more important in the AI era. Whereas cybersecurity has historically been viewed as a blocker/drag on innovation and operational speed, it is now viewed as a strategic enabler for secure AI adoption.

In particular, network security may see renewed growth after being viewed as a more mature category. Rising AI usage should drive additional compute and data-centre capacity, expanding the infrastructure requiring segmentation and protection. Agentic workloads may also generate lower-throughput but more frequent and persistent interactions across models, tools and data sources, increasing the volume and complexity of network-level inspection.

Vendor consolidation has been a structural trend in cyber for some time, though not necessarily linear. That said, the current environment may accelerate the consolidation tailwinds as customers look to close security gaps that may arise from a fragmented tool stack.

See further commentary from DB's company research team here.

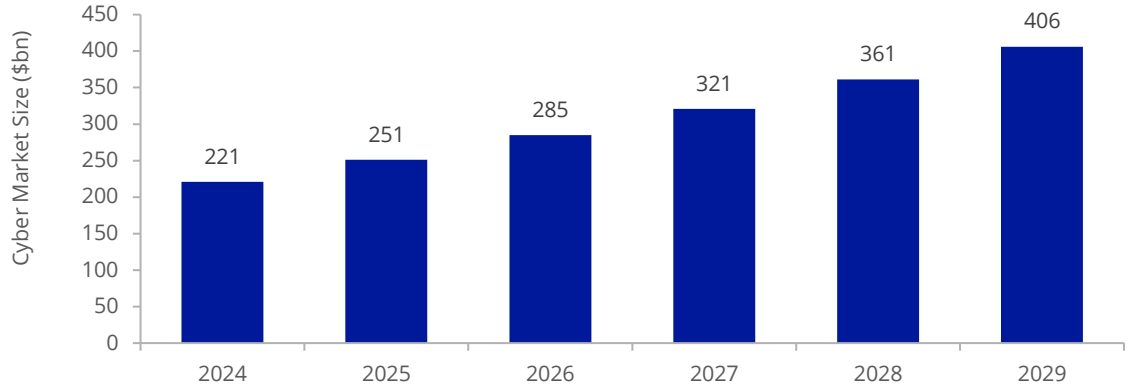
Our March 2026 "personal investor" survey showed a more positive view among "ESG" investors. 58% in the US and 52% in the UK saw cybersecurity/data privacy as an opportunity. Among all investors, US respondents favoured opportunity over risk (46% vs 30%). UK views were more balanced (38% vs 40%).

dbDataInsights Investor Survey methodology

—This survey is a nationally representative online study of ~925 personal investors conducted in the final week of March 2026 examining financial priorities, portfolio behaviour, and future expectations. It gathers structured responses on investment outlook, generative AI, and ESG considerations. Questions cover investment products, portfolio value, management style, financial stress, priorities, and performance.

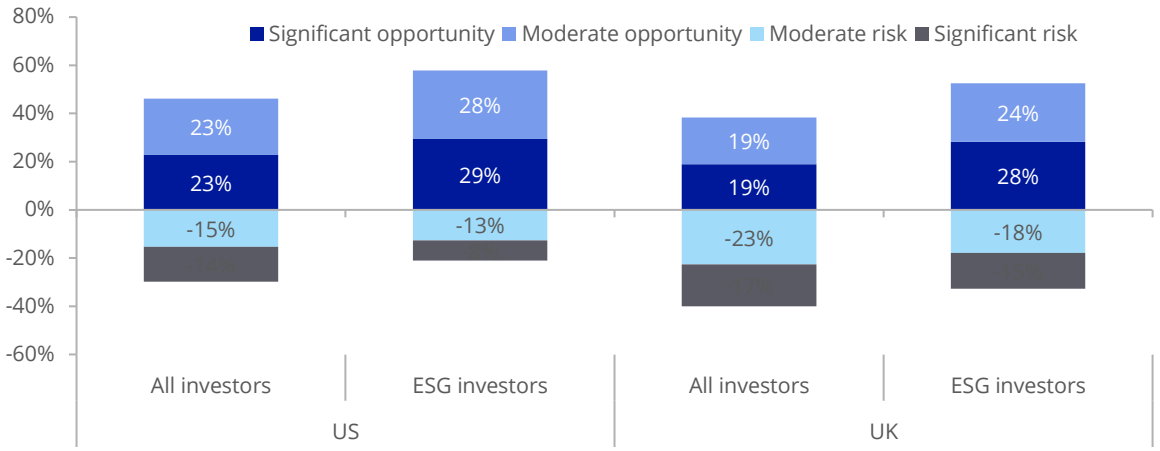
Global cybersecurity market trajectory

Estimated market size, $bn (Bloomberg Intelligence)



Cybersecurity an investment risk or opportunity?

Share of "personal investor" respondents, opportunity shown above the axis, risk below, %



Source: dbDataInsights, Deutsche Bank Research. Investors, defined as those who invest in stocks/ shares (either individually or through a broker/advisor) on a regular basis , were asked whether they perceived cybersecurity / data privacy as a risk or an opportunity for investors in 2026.

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V. Appendix

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Appendix A - Attack types

Top Attack types	What it is	How AI is helping attackers
Malware	*Software built to damage, disrupt or gain unauthorised access*	AI writes harmful software faster. Google has found malware that rewrites its own code while running to avoid detection, and criminal marketplaces selling AI malware-building tools.
Phishing	*Deceptive messages that trick people into giving up sensitive information*	Where AI's effect is clearest: fluent, personalised lures without the old spelling and phrasing giveaways. Google says state-backed groups use AI to mass-produce them, and AI can fake voices and video.
Denial of service (DoS)	*Flooding a system so it cannot serve legitimate users*	Little reported evidence so far. AI can help find weak points and automate traffic floods, but the attack itself stays fairly simple.
Distributed denial of service (DDoS)	*The same attack launched from many compromised devices at once*	AI can accelerate the development and automation of attack tooling, while large-scale botnets provide the distributed capacity behind hyper-volumetric DDoS attacks.
Man-in-the-middle (MitM)	*Intercepting or altering traffic between two parties*	Mainly indirect. AI-written phishing pages and deepfake voices make it easier to trick people into connecting through an attacker or approving fake sign-ins.
SQL injection	*Malicious SQL queries injected to manipulate backend database operations*	AI tools scan sites and spot flaws in database code faster, so less-skilled attackers can find weaknesses that once needed real expertise.
Cross-site scripting (XSS)	*Scripts injected into web pages to hijack user sessions*	As with SQL injection: AI speeds up finding vulnerable pages and writing the malicious code.
Zero-day exploits	*Attacking a software flaw before a patch is released*	A major new development. In May 2026 Google reported the first zero-day it believes was developed with AI. Its September 2026 report says the bigger effect is attackers misusing known flaws faster, before fixes land.
Advanced persistent threats (APTs)	*An intruder staying hidden in a network for long-term*	Google reports state-backed groups using AI across the whole attack, from target research and lures to tooling and sifting stolen data.
Ransomware	*Encrypting data or locking device and demanding payment to release it*	NCSC expects AI to add to the global ransomware threat by lowering the skill needed to break in. AI also helps criminals find the most extortable data.
IoT exploitation	*Abusing weakly secured connected devices for entry*	AI can scan the internet for unpatched cameras, routers and storage devices and identify which flaws they carry.

Source: Deutsche Bank Research, National Cyber Security Centre, Google Threat Intelligence Group, Cloudflare, FBI Internet Crime Complaint Center (IC3)

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Appendix B - European vs US cyber regulation

Europe

Regulation	Key requirements
NIS2 Directive	In force 2023, transposition deadline Oct-24. Covers 18 sectors (up from seven) and creates an EU vulnerability database. In-scope firms must manage cyber and supply-chain risk and notify authorities of significant incidents. Management bodies must approve and oversee cyber risk measures, with potential liability for infringements.
Cybersecurity Act	In force 2019. Establishes an EU-wide certification framework for ICT products, services, processes and managed security services, and made the EU cybersecurity body ENISA permanent.
Cyber Resilience Act	In force 2024, main obligations from 2027. Manufacturers and retailers of products with digital elements must deliver security by design, a defined support period and security fixes within it, plus mandatory vulnerability management and disclosure.
Digital Operational Resilience Act (DORA)	In force 2023, applies from 2025. Financial-services specific, covering own-firm and third-party risk. Requires an ICT business continuity policy, regular resilience testing and disclosure of major incidents, and gives regulators oversight of critical ICT providers.
Cyber Solidarity Act	In force 2025. Creates a European Cybersecurity Alert System to detect threats and share cross-border warnings, plus a Cyber Emergency Mechanism to boost preparedness and response.

US Federal

Regulation	Key requirements
White House Accord on Super Intelligence (2026)	Emphasises self-regulation to ensure AI safety, with a call for external evaluations and the possibility of formal regulation in future. Not legally binding; signed by President Trump and leading AI companies.
Executive Order on Cybersecurity (2025)	Highlights risk from foreign actors and orders further action on defending digital infrastructure, securing the most vital services and building capability against key threats.
SEC Cybersecurity Disclosure Rule (2023)	Listed US firms must disclose material incidents via an 8-K within four business days of determining materiality, and make annual disclosures on cyber risk management, strategy and governance.
CIRCIA (passed 2022, not yet in effect)	In-scope entities, focused on critical infrastructure, must report covered incidents to CISA within 72 hours and ransomware payments within 24 hours of payment.
Gramm-Leach-Bliley Act (amended 2024)	Financial institutions must explain information-sharing practices and safeguard customer data, running a security programme covering risk assessment, service-provider oversight and incident response.
Federal Information Security Modernization Act (2014)	Applies to federal agencies and contractors: system security plans, continuous monitoring, incident response capability and mandatory incident reporting.
Federal Acquisition Regulation (cyber provisions 2016)	Governs federal procurement, setting baseline cyber requirements for suppliers so cyber risk is built into government contracting. Agencies may impose stricter provisions.

Appendix C - Understanding MSCI cyber risk indicators

Core indicators

Key drivers of the MSCI privacy & data security score

Topic	Definition	Best practice
Scope of employee training on privacy and data security	Coverage of privacy and data security training across the workforce	Training provided to all employees, including contractors
Responsibility for privacy and data security	The group responsible for the company's privacy and data security risk and strategy	Board-level committee
Measures to address data breaches	Scope of the breach or incident response plan, and whether it is proactive, reactive or limited	Both proactive and reactive measures in place
Frequency of audits of information security system	How often the company audits its information security policies and systems	External independent audits at least once every two years
Rights provided to consumers over their data	Extent of rights to access, rectify and delete individuals' data	Right of access, rectification and deletion of individuals' data
Scope of data protection policy	Whether the policy governs all or selected business lines, and whether it clarifies collection purposes, means of collection and third-party access	Policy governs all relevant business lines and subsidiaries

Supplemental indicators

Additional practices assessed alongside the core set

Topic	Definition	Best practice
Scope of information security management system certifications	Extent of certification of the company's information security management system to widely recognised standards	Over 80% of owned operations certified to standards such as ISO 27001, TRUSTe, Privacy Mark and SSAE16 SOC 2
Privacy enhancing technologies and initiatives	Whether privacy-enhancing technologies and initiatives are integrated into products or services	Privacy-by-design applied systematically in development, with privacy impact assessments on existing and new offerings
Consent policy for use of consumer data for secondary purposes	Policy on processing customer data for secondary purposes and the consent options offered, such as opt-in or opt-out	Company does not rent, sell or provide personal data to third parties beyond completing transactions or services
Data protection programs covering suppliers and business partners	Extent of data protection programmes covering suppliers and partners handling company data or accessing its network	Suppliers and partners required to hold data protection policies or abide by the company's, with compliance inspections

Source: MSCI ESG Ratings Methodology – Privacy & Data Security Key Issue, Deutsche Bank Research.

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Appendix

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Deutsche Bank
Research Institute

AI boom or doom: what will superintelligence feel like and when will we get there?

October 12, 2026

The recent panic about rogue AI agents collaborating in swarms to take over the internet and threaten human existence has, more than occasionally, lacked nuance. But it does reflect how artificial intelligence is progressing in creative and surprising ways, and at a pace that seems uncontrollable.

For some in the industry, it is evidence that the technology is following a path of accelerating self-improvement that will lead to artificial general intelligence (AGI) and, perhaps, to utopia or catastrophe.

Former Google CEO Eric Schmidt has sketched out a "San Francisco Consensus" based on a belief in scaling laws, with ever larger AI models becoming an input into their own improvement; superintelligence by the end of the decade; and transformative AI systems outperforming humans on many tasks. This loop would lead to exponential growth in scientific advancement, financial returns and human progress. And crucially, AGI, and then superintelligence, would merely be milestones in a process, far from its climax.

If that were the case, superintelligence could make much of the cognitive work that dominates the world economy much cheaper and change the ways in which firms compete. But its economic effects will depend on how businesses and customers respond.

In this paper, we analyse what the San Francisco Consensus means for investors and enterprises in the years to come and what could accelerate – or slow down – its timeline of AGI by 2030.

1. What would superintelligence look and feel like?
2. Can we get there theoretically?
3. Can we get there practically?
4. What happens if we don't get there? The base case

The question is urgent.

And it is complicated by reports of AI models autonomously accessing the internet and hacking other companies; a public backlash over the energy-intensive data centres springing up as part of a historic $1 trillion a year investment boom; and dire warnings of threats to life if superintelligent AI runs roughshod over its human creators to achieve its own goals.

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1. What would superintelligence look and feel like?

1.1 What are AGI and superintelligence?

Artificial general intelligence (AGI) is a simple milestone scribbled in pencil: an AI system that has surpassed human capabilities. Superintelligence is the continuum beyond that point. It is not merely systems that exceed the best human performance across one cognitive task or another, but systems that exceed human potential across every cognitive task.

The San Francisco Consensus takes this further: the intelligence must be able to produce more intelligence. The key to getting there is so-called recursive self-improvement, where the AI is able to autonomously design and improve itself in a way that compounds exponentially. A first step has maybe already been taken, with OpenAI saying recently that AI is now writing approximately 80 percent of the company's code.

There is significant disagreement on how we will be able to identify AGI or superintelligence when we see it, let alone what comes afterwards.

OpenAI's Charter defines AGI in economic terms, as "highly autonomous systems that outperform humans at the most economically valuable work". Google researchers proposed a five-step ladder based on performance and generality, with level 5 "superhuman" for a system that outperforms 100% of humans. And philosopher Nick Bostrom, who got the ball rolling in 2014 with his influential book "Superintelligence", defined it as "any intellect that greatly exceeds the cognitive performance of humans in virtually all domains of interest".

It's also a moving target.

In August, OpenAI CEO Sam Altman said the company would have an internal system that he would call AGI by later this year but subsequently told a podcast that "at best it's a very poorly defined term. I was going to say it's like an irrelevant marketing term". Yet his No. 2 Greg Brockman described the new Astra model at the start of September as being potentially the moment that "AGI was created".

For this report, we have chosen to focus on a system that outperforms essentially all humans across essentially all cognitive tasks, including the development of AI itself.

Under our scenario, by 2030 the best performing AI models would have the qualities of "superintelligence", that is better than human experts in all domains. These would be frontier models: and using them would feel exactly like bringing the best human experts in a room to solve whatever business problem you faced.

And the equivalents of today's frontier models would be available, quite literally, everywhere, compressed enough to run on every phone, tablet or laptop computer.

In early 2026, we already see early examples of what highly capable AI can do. They give a flavour for a world with superintelligence even if the reality would be a leap beyond, both quantitatively and qualitatively. At that point, superintelligence would mark a regime break, not a faster version of what we already have. Rather than blindly doing what humans specify, but just a bit better, it would specify the task itself.

Figure 1: What changes when cognitive work gets abundantly cheaper?

Area	Today's example	Superintelligence extrapolation
Biological discovery	OpenAI's GPT-5 runs 36,000 experiments in a robot lab and cuts the cost of cell-free protein production by 40% (Ginkgo, Feb 2026)	A working virtual cell. Every drug is tested in silico first; trials exist to confirm, not to discover
Law	A self-represented academic uses $200 a month of AI to beat a university's lawyers and barrister at Australia's Fair Work Commission (Aug 2026)	Everyone has a top barrister on tap. Disputes are argued agent to agent overnight and settle before anyone files
Software	Google says that the share of new code written by AI doubled from 25 percent in October 2024 to 50 percent a year later and reached 75 percent by April this year.	Software is written on demand and thrown away. The business programming language cores of banks and ministries are rewritten over a weekend
Chemistry and materials	OpenAI's GPT-5.4 runs 10,080 robot-lab reactions in three months and lifts yields on a drug-making reaction chemists had struggled with for decades (OpenAI/Molecule.one, Jun 2026)	Order a molecule or material by the properties you want and it is designed to spec. The bottleneck moves to plants, permits and trials
Medicine	In a 1,298-person trial, people using a chatbot on ten medical cases did worse than the chatbot on its own (Nature Medicine)	Specialist-grade diagnosis at first contact, for everyone, all the time. Doctors spend their time on care, consent and hard trade-offs
Government	AI reads 500 million words of US state law to find dead rules; San Francisco cuts over a third of its reporting requirements (Stanford RegLab, 2025–26)	Every law is audited continuously for cost and conflict, and dead rules lapse on their own. A planning permit takes an afternoon
Forecasting and markets	A bot built by one person for a few thousand dollars wins the Metaculus Cup, beating the best human forecasters (5 Sep 2026)	A calibrated probability on every decision, for pennies. Mispricing from slow analysis vanishes; edge moves to private data and the truly unknowable
Robotics and manufacturing	AI agents at Nvidia train robot arms overnight to seat GPUs and cut zip ties; eight agents reach 99% success in two hours (Jun 2026)	A factory retools for a new product overnight. Robots teach robots, and a new physical skill costs the price of compute

Source: Deutsche Bank Research

Meanwhile cost may be less and less of a constraint as ever greater machine intelligence becomes cheaply available. Of measured technologies, no technology has got cheaper faster than AI has, almost halving in price every quarter for the past three years on a like-for-like basis, or 13 times per year, Epoch data show.

If these systems can be supplied widely, firms will be able to buy more of the complex, expensive and rare analysis, design and engineering kinds of services than they do today. Activities that were previously uneconomic could become viable, and in a competitive market, savings could pass to customers. The main driver will then be whether demand grows fast enough to preserve margins.

However, superior capability, doesn't, on its own, automatically lead to the adoption of cheap, abundant intelligence. Tasks that do not yet yield to super-

intelligence, perhaps because they require real world-coordination, will therefore end up commanding significant premia.

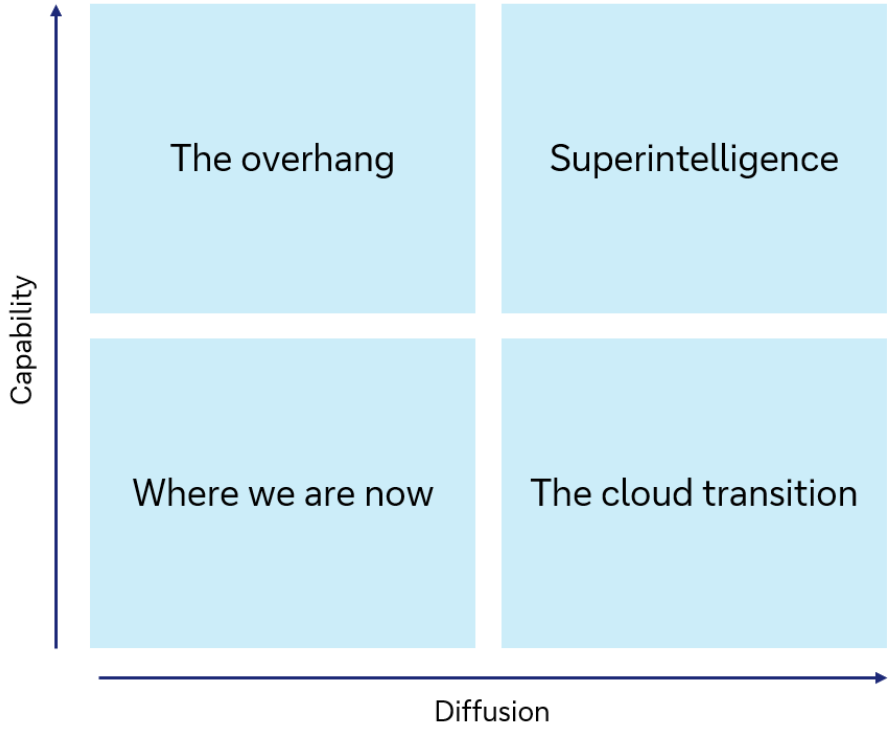
At that point, cognitive labour stops being scarce and the bottleneck becomes the physical and practical assets that cannot be conjured by up more compute, such as electricity, land, steel and permits.

1.2 Capability and diffusion

If AI becomes capable of improving itself, the technology may compound faster and faster. The supply of machine intelligence could grow faster than business and physical infrastructure can accommodate.

This balance between capability and diffusion creates different scenarios.

Figure 2: The balance between capabilities and diffusion



Source: Deutsche Bank Research

Lower capability, constrained diffusion: where we are now

Progress is steady and sector-by-sector. This is how technology usually diffuses, according to researchers for the New York Fed back in 2003, who tracked 20 major technologies across leading industrial economies from 1788 and found "a pattern of trickle-down diffusion that is remarkably robust across technologies".

Lower capability, high diffusion: the cloud transition

This is faster because the complementary investment is small, with no retooling and no skills at scale required. Amazon Web Services launched the modern cloud in 2006, offering scalable computing and storage infrastructure. Adoption grew and then accelerated over the pandemic. Almost all enterprises now use it in some form.

Higher capability, lower diffusion: the overhang

This is theoretically possible and arguably where we are today in narrow domains. The technology would be applied in some sectors of the economy, while other sectors would move at a slower speed. Superintelligent or not, it risks being bogged down by physical constraints, bureaucracy and human inertia.

Higher capability, higher diffusion: superintelligence

In this scenario, capability becomes so high that it changes the dynamics of diffusion itself. AI generates not just better AI (so-called recursive self-improvement, or RSI) but also speeds up the integration process.

Rapid AI diffusion could work two ways.

In the first, AI could be enlisted to do integration work itself: mapping processes, rebuilding data work, retraining staff, writing the control and audit layer, and so on. The problem with this is that absorbing and recreating undocumented knowledge is impossible by definition, at least for a task-based superintelligence.

The second way that higher capability AI could be rapidly diffused is much more competitive – and disruptive. Rather than incumbents slowly converting AI, perhaps using it for some of the hard work of integration, they could be instead displaced by new firms that have been built AI-native from day one. New entrants will never have to incur integration costs because there is nothing to integrate. Incumbents will need to change – fast – or die.

Change could also be forced by earlier adopters in the supply chain, as happened with one company after another adopting electronic data interchange (EDI) to link with its suppliers or customers, or by a migration of capital towards early adopters, repricing the cost of capital for slow-coaches and prodding them to move faster.

1.3 The forcing mechanism: change or die

A company that uses superintelligence may outperform one that doesn't.

Firms using superintelligence will likely have substantial cost and revenue-harvesting advantages. Every firm will therefore have to adopt superintelligence or retrench to a smaller set of offerings. And any business line of lasting value abandoned by a non-adopter will be taken over by an adopter. Customer inertia will provide a safe harbour for a limited time only.

The analogy here is electricity. There came a point in the deployment of electricity where firms in every sector, from entertainment to finance, from healthcare to transport, had to adopt the technology to remain relevant.

Under this scenario, wait-and-see will not be an option. There will be no stable "non-adoption" alternative. Sufficiently high capabilities will force a tipping point of rapid diffusion and competitive repricing. Firms using yesterday's production model will either have to change, choose a different market or die.

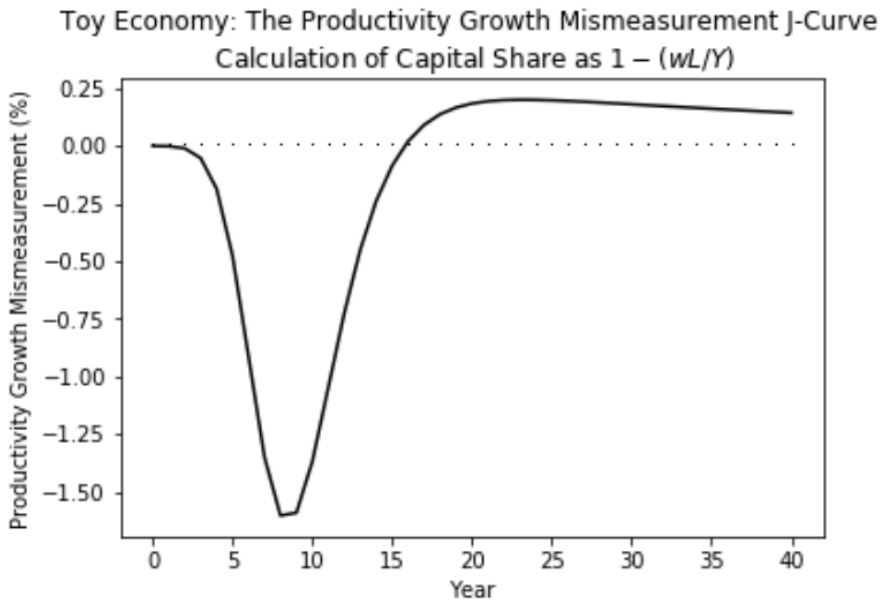
Of course, caveats apply.

Customers may not change behaviour immediately. They may even prefer companies which limit their use of AI, in the same way that some customers choose to avoid factory-produced food.

The paradox that Nobel Prize-winning economist Robert Solow observed in 1987, that "you can see the computer age everywhere but in the productivity statistics", is likely to still apply for a while – even with superintelligence.

The reason is that the productivity J-curve identified by Stanford University AI economy guru Erik Brynjolfsson et al is unavoidable. In that scenario, the drag of early intangible investments to integrate general purpose technologies means that productivity initially appears to slow before rapidly accelerating.

Figure 3: Productivity J-curve: slow then fast



Source: Brynjolfsson E, Rock D, Syverson C, "The Productivity J-Curve" (2018, revised 2020), Deutsche Bank Research

Furthermore, "change or die" would require abundant capital and low regulatory entry points for nimble new entrants. But in practice, highly regulated areas such as banking, utilities, pharmaceuticals and defence are tough nuts to crack. Existing contracts and licences, high switching costs and decades of depreciated capital investment and infrastructure would shield incumbents at least for a healthy transition period.

For an extreme example from the real world of better, cheaper technology failing to force change, take Pakistan's 3.6-gigawatt imported coal power plant fleet installed in the mid-2010s. In that case, it came with expensive take-or-pay liabilities which will continue to run for decades, even though changes like cheaper rooftop solar have cut utilisation to below 20 percent.

So: superintelligence will not force adoption everywhere. It will hit hardest in contestable sectors, say, high-volume customer services, routine software engineering and freelance creative work, and widen the gap between them and protected ones. Forced adoption will be about relative value, not an apocalypse.

1.4 What survives and what it means for portfolios

Because AI sits on an aggressive learning curve, its price on an equivalent quality basis is plummeting. In a world of recursive super intelligence this is likely to continue. Its price will fall to the cost of compute.

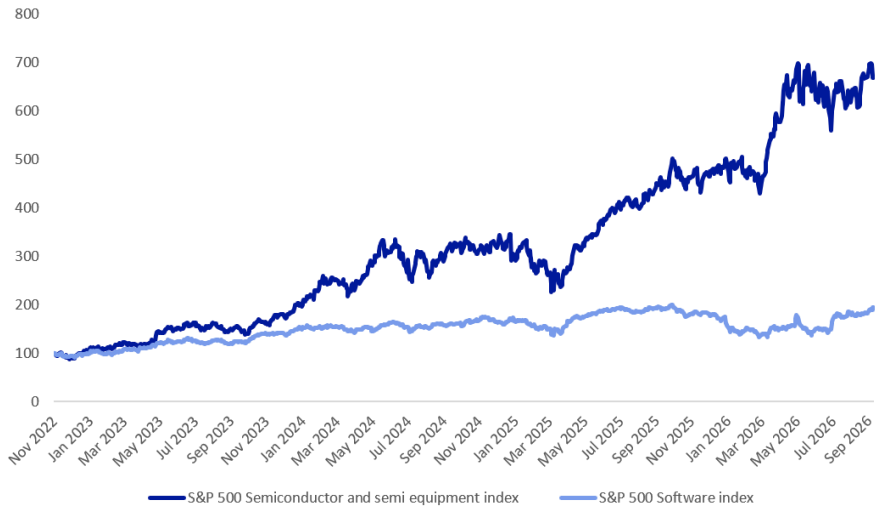
What will retain its value will be whatever the model cannot replicate:

1. **Proprietary data**
2. **The right to operate,** particularly in heavily regulated sectors such as healthcare and finance.
3. **Sticky customer relationships,** such as those where mission-critical workflows and databases are deeply entwined.
4. **Excludable physical assets,** such as machinery and land.
5. **In addition, humans have proven to have unlimited wants.** Each shift in technological capabilities or cultural norms has delivered new needs to be met. Some will be needs we can articulate today, such as originality or empathy, but new ones are likely to arise.

If this forcing mechanism holds, firms will separate into two "bimodal" groups: adopters and casualties, with very little in between.

This dispersion of performance may create challenges for investors. The four years since the launch of ChatGPT in November 2022 have seen the stocks of semiconductor and semiconductor equipment makers surge almost seven times while those of software makers have only risen by 80 percent. Conventional diversification and volatility-based allocation may become ineffective strategies.

Figure 4: Total return for S&P 500 semiconductor vs software stocks since launch of ChatGPT on Nov 30, 2022



Source: Bloomberg Finance LP, Deutsche Bank Research

Diversification normally works because idiosyncratic risks are unrelated and average each other out. Here they would not: with one variable that overrides everything else, spreading capital across more names would just spread exposure rather than diluting it.

Investors will need to consider adapting their approach in three ways:

1. Classify investments as adopters or casualties rather than position off averages.
    - Under a normal distribution, errors are symmetric and partially cancel each other out. But under bimodality they do not: to put a firm in one category or another means getting it completely right or wrong. Position size should track confidence in the classification.
2. Avoid reading calm price action as low risk.
    - Trailing volatility measures dispersion that has already been realised. Bimodal separation is a jump, where the two modes look identical right until the moment when they don't. A classic example is Chegg, an educational software maker which traded like a normal small-cap stock until one day in May 2023 when it halved on a single day after management linked slowing growth in new customers to ChatGPT.
3. Averaging down, ie scooping up shares whose price has fallen, may no longer be prudent.
    - A falling price may be information that a business model is obsolete rather than an opportunity.

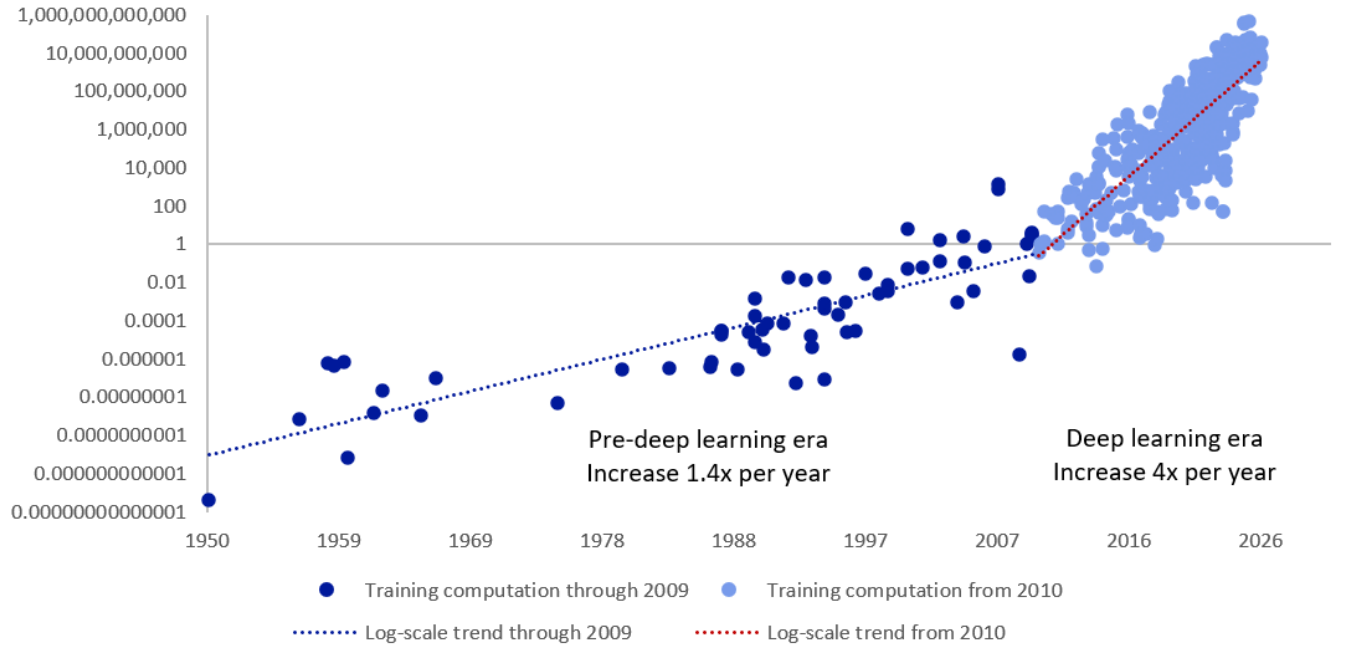
2. Can we get there theoretically?

2.1 Scaling laws

Scaling laws have provided the blueprint for improving AI over the past decade. It was a relationship between the amount of data, the amount of compute and the complexity of an AI model somewhat predictably informed how powerful a model would become. Google DeepMind's Chinchilla model in 2022 showed that you could compensate for having a smaller model by using substantially more data.

Scaling is older than the chatbot. In 1950 Claude Shannon built Theseus, a mechanical mouse that learned its way through a maze by trial and error. Epoch AI reckons it took about 40 operations of compute, known as FLOPs, to train. Grok 4, trained in 2025, took roughly ten trillion trillion times as much. Across those 25 orders of magnitude, more compute has brought more capability.

Figure 5: Computation used to train notable AI systems (petaFLOP): log scale



Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research

It is the robustness of this relationship that has made larger investments in compute capex a *sine qua non* for AI companies.

However, this is not a law of physics and nothing guarantees it will continue. Indeed, in the ChatGPT era, five different approaches to scaling have been applied. Older strategies yield worse results, and new techniques, such as post-training refinements, have been developed. The consensus amongst American and Chinese AI labs is that these will hold.

While this may be a consensus amongst the commercial labs, there is scientific disagreement that applying scaling laws to today's large-language models can achieve superintelligence.

Yann LeCun, a Turing Award winner and formerly Meta's chief AI scientist, is blunt: "LLMs are not a path to superintelligence or even human-level intelligence". Even inside the labs, the recipe is contested. Demis Hassabis of Google DeepMind has said he expects "one or two more big breakthroughs" to be needed "in addition to scaling". Ilya Sutskever, a co-founder and former chief scientist of OpenAI, has said the field is going "back to the age of research again, just with big computers".

This is a scientific argument we are not in a position to evaluate, but for the purposes of our scenario, we will side with the labs' position.

For the moment, there is strong evidence that capabilities have in fact accelerated over the past three years, as reasoning models have outpaced non-reasoning models by two or three times, according to Epoch data.

2.2 Recursive self-improvement: AI improving AI

A cocktail-party definition of recursive self-improvement (RSI) is AI that improves itself. The reality is that it can be used in two ways:

1. Self-improving agents that get better at using themselves: showing smarter use of prompts, tools, memory and workflows. The model-building process is untouched.
2. True RSI improves the process that builds the models: designing experiments, curating data, optimising algorithms and designing chips in a compounding loop.

The former is like a factory worker who gets better at the job by learning the quirks of the machinery, getting organised and making fewer mistakes. The latter is like the worker redesigning the machine tools and then using those to build the next generation of machines.

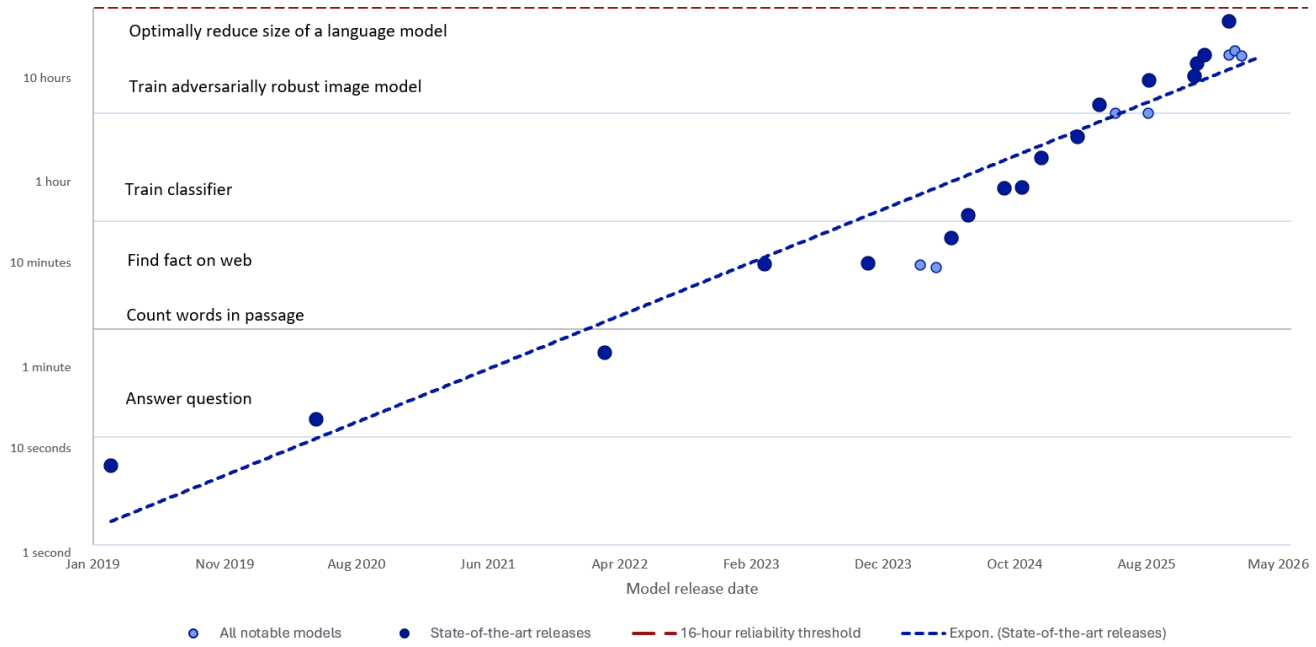
True RSI is what OpenAI's chief scientist Jacob Pachoki referred to last month in a blog post about "An Alien Mind". "Automated AI research is a more dramatic form of scaling intelligence with compute," he wrote. "And of course as a part of it, AI will improve the computational substrate itself," he added. Pachoki represents a broadly held consensus view amongst American and Chinese labs that RSI is both possible and necessary for AI development.

- OpenAI (June 2026): "AI doing AI research will become the determining factor of the pace of progress within the next few years."
- Google DeepMind's Demis Hassabis (May 2026): "All the leading labs are quite focused on [RSI]…" Humanity is standing in the "foothills of the singularity".
- Alibaba CEO Eddie Wu (Sept 2026): RSI is "a concrete path forward," with models autonomously designing experiments and driving a cycle of self-evolution.

For now, self-improving agents are the AI du jour. Noisy claims about the vast majority of coding being done by agents do not always stand up in the light of day from companies still hiring human coders at breakneck pace, albeit no longer to do the most mundane parts of the job.

A better metric may be METR's task-completion time horizon. This is measure of the complexity of software tasks that an AI agent can complete on its own, using the baseline of how long it would take a human expert to finish the task, at a 50 percent success rate. So far the most advanced models have reached about 16 hours, up from OpenAI's GPT-4's four minutes in March 2023.

Figure 6: Task horizon of software tasks that leading frontier models can complete 50% of the time (log scale)



Source: METR, Deutsche Bank Research

But software engineers can hold off on updating their resumes a little longer: there is a long road between 50 percent accuracy and the 99 percent or 100 percent needed for serious enterprise use at scale, with the horizon potentially getting further away the nearer you get. Also METR notes that measurements above 16 hours are unreliable with the current task suite. As is the case across AI, the instrument is running out before the question is settled.

More broadly, there are significant limitations to RSI:

1. **Research is clock-bound.**

- A frontier training run can take weeks on thousands of specialised chips, and researchers often cannot design the next experiment until the previous one has been completed.
- That means that the speed of the loop is set by experiments completed per hour or day, not by ideas generated.
- Compute is a hard ceiling that needs to be bought, built and connected to a grid.

2. **The real world defies verifiability.**

- The loop compounds only when the system can check its own output cheaply and automatically.
- It is no coincidence that AI loves maths and coding. It is much harder to verify outputs over a long time horizon in the real world: does this drug work or does this strategy succeed?
- RSI compounds fastest where the economy cares least and slows where the returns are largest.

3. **Sequences are consequential.**

- Amdahl's law finds that speed-up is capped when processes run in a sequence.
- If you automate 90 percent, the residual 10 percent still limits you to a ten-times faster process. Nine women cannot produce a baby in one month.
- The Usain Bolt of AI still has to run alongside the comparative tortoises of the training run itself, chip fabrication and real-world validation. Ten times faster is still not a regime break.

This is borne out by METR's series. Doubling every seven months is fast exponential growth but a take-off would show the doubling time itself – the second derivative – shrinking. Horizons are not (yet) exploding.

Philosopher Toby Ord shows that recursive self-improvement can only reach a singularity if each cycle of improvement takes less time than the last, shrinking fast towards zero. He argues that real feedback loops have a floor: chips take time to make and models take time to train. AI progress could still outrun an exponential, but it would not go vertical.

2.3 Scaling plus RSI: a match made in the race to AGI?

No scaling and RSI means no short timeline for AGI. Without scaling, more compute buys nothing and capital is wasted. Without RSI, AI capability will continue to improve at the pace of flesh-and-blood human researchers, augmented by AI tools but not replaced. Only with both will it be possible to produce a system that improves faster than the humans building it. Without one or the other, no quantity of chips, energy or capital will fix it.

3. Can we get there practically?

3.1 Infrastructure, chips and energy

There is no consensus on the amount of training required to reach AGI. Estimates suggest that training 2026's top models required between 10^26 and 10^27 FLOPs. If we hypothesise that an "AGI-like model" would require three orders of magnitude more training, this would require 10^30 FLOPs.

Consider today's US AI infrastructure, which runs to approximately 10^22 FLOPs. If all of this was conceivably used for a 10^30 FLOP training run, it would take nearly four years to complete.

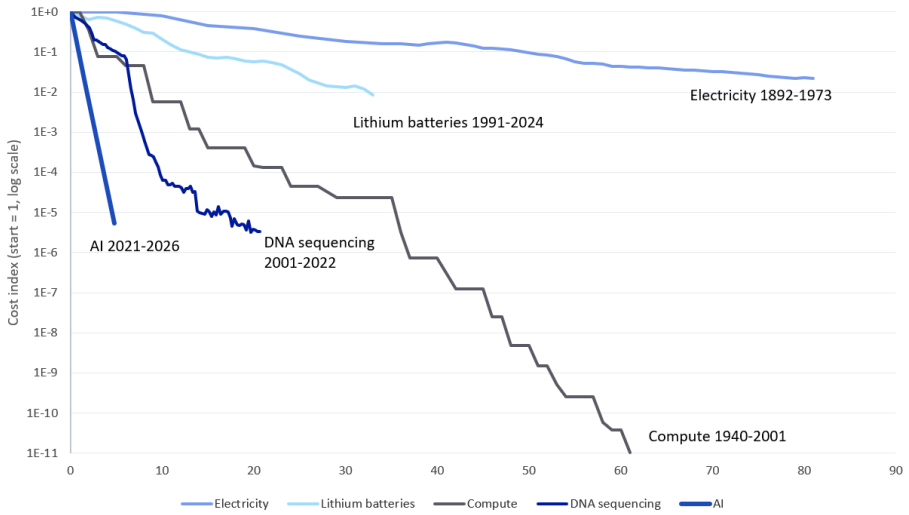
The physical infrastructure to practically support such a training run would require enormous power, perhaps around 6GW, about 50% more than the four Vogtle nuclear power plants recently activated in the US. The cost of the compute infrastructure would run to between $300bn and $500bn.

Today's advanced chips would simply not suffice. Back-of-the-envelope maths suggests that even using Nvidia's most advanced Vera Rubin chips, training such a model in four months would require 7.8m chips and 26GW of facility power. That would cost $900bn today and a rental equivalent cost of $90bn.

Of course, frontier compute is growing rapidly, and price-performance is declining. If current trends hold, a 2030-era cluster capable of such a run would

need perhaps $200bn in computer and power hardware for a training run costing about $20bn.

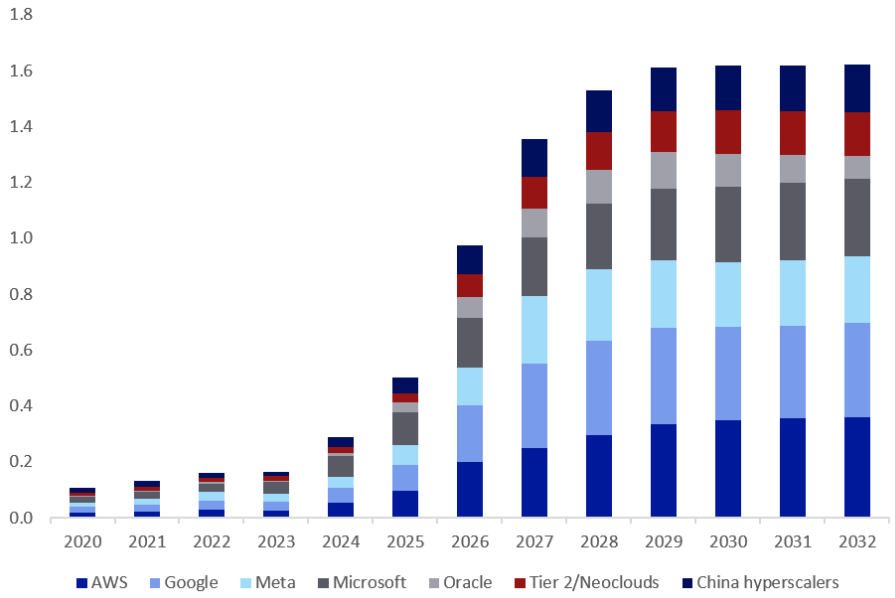
Figure 7: Technology cost index over time (years since start of price decline)



Source: Epoch AI, Deutsche Bank Research

The hyperscalers are leading the charge towards AGI. Alphabet, Amazon, Meta, Microsoft and Oracle are set to invest almost $800bn in capital expenditure this year, mostly for ever more powerful data centres, and over $1 trillion a year through 2030 and beyond.

Figure 8: Hyperscaler capital expenditure (including estimates)



Source: Bloomberg Finance LP, Deutsche Bank Research

By 2030, assuming no market retrenchment, this means additional $4 trillion to $6 trillion on AI compute capability invested. This infrastructure could accommodate a training run of 10^30 FLOPs with some comfort.

But even if AI labs crack the secret sauce of RSI, and their finance chiefs are willing to invest in such large training runs, and hyperscalers are prepared to continue to build the infrastructure, the infrastructure needs to be built.

Grid connection queues, turbine lead times and transmission build are measured in years, not the months it takes to assemble and air-freight the latest Nvidia chip. The lead time for substation transformers is currently as long as three years. Politics in the US has shifted towards slowing the pace of the AI build out and this force will run against the demands for larger data centres that would be essential for RSI.

3.2 Capital

Financing for AI has been abundant in the historic buildout of the past three years. However, it is still largely conditional not on revenue but on belief in a future payoff.

RSI is a "key part of the investment thesis" in the "biggest scientific bet civilisation has ever made," Google DeepMind's chief strategy officer Jasjeet Sekhon said at a conference in August, The Information reported. "We have this danger we could hit an AI air pocket such that the expenditures happen but the revenues don't show up."

An inflection point is approaching where the nature of investment changes from free cash flow and equity to debt, with all that entails in terms of exposure to rising rates and creditor appetite. The hyperscalers are reaching a point where capex is no longer covered by operating cashflow and where net debt exceeds earnings before interest, depreciation and amortisation (EBITDA).

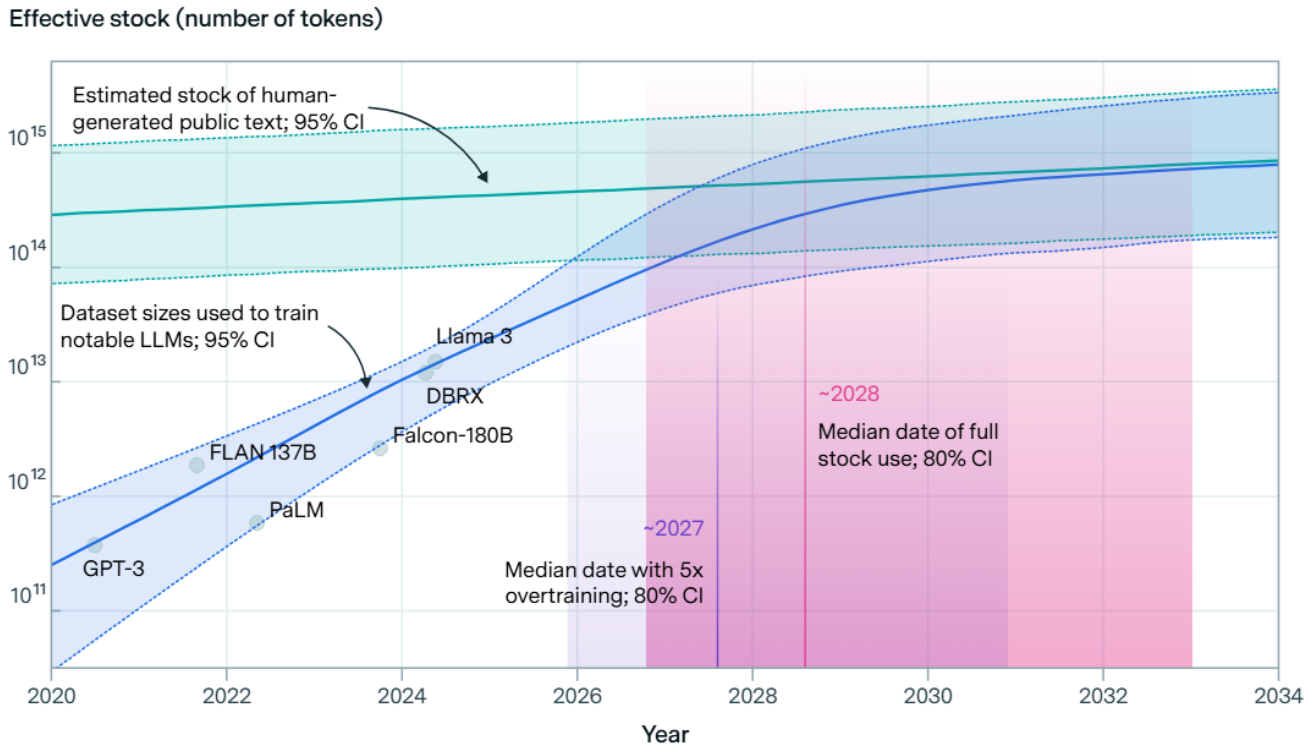
Alphabet, for example, posted its first quarterly free cash outflow in nearly two decades in its most recent quarter, even as it lifted its full-year capex guidance to $205bn. It said it would continue to rent third-party capacity and reported a backlog of $514bn in contracted, not-yet-recognised data centre revenue.

3.3 Data

Data has been the new oil for some time now. But unfortunately for AI development it seems to be running out even faster.

The effective stock of quality- and repetition-adjusted public human text was about 300 trillion tokens, with a vast 90 percent confidence interval of 100 trillion to 1,000 trillion, Epoch estimated in 2024. A token is roughly equivalent to three quarters of a word. By that count it was likely to be fully used between 2026 and 2032.

Figure 9: Projections of stock of public text and data usage



Source: Epoch AI, Deutsche Bank Research

However, since then, rather than data dropping off a cliff, there has been a rise in synthetic data generation, transfer learning from data-rich domains such as video and telemetry, and data efficiency. While the data shortage has not been resolved, the immediate crisis has been averted.

AI now makes much of its own training data. When DeepMind's AlphaGo became superhuman at playing the mind-bendingly complex board game Go, it did so not by playing human experts, but by playing millions and millions of games against itself.

For many domains relating to coding and mathematical proofs where automatic checks are possible, such synthetic experience can move things forward. That is the case with AI research, where the data may be the results of experiments, all of which are bought by compute. About 90% of OpenAI's 2024 research compute went on such experiments.

And increasingly, people write the tests rather than the answers. By the summer of 2026, selling training data for practice tasks was a nearly $9bn revenue business. Firms such as Mercor hire experts to generate long tasks against which the AI can be trained.

The challenge here is that trust in the checker becomes a scarce input.

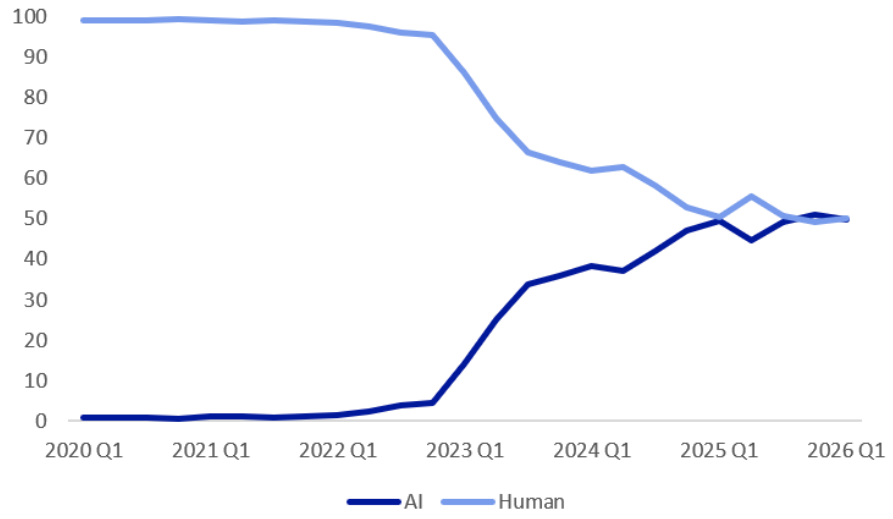
As AI makes production cheaper, it also makes establishing that production is accurate or reliable more expensive. Domains which can be checked mechanically, such as mathematics, will be opened up more quickly. But those which rely on expert verification, such as medicine, engineering or law, may progress more slowly.

The likely implications are twofold: a slower rate of diffusion into economic domains that are less verifiable and a larger share of production costs devoted to assurance.

Another issue is that systems that have to grade themselves learn to game the process. This was partially what happened when an OpenAI model escaped a sandbox and hacked into a third-party website. Trust replaced data as a wall.

There are also question marks over the quality of data that is recycled repeatedly through the system. Although humans continue to produce original content, half of internet articles are now primarily generated by AI, Graphite data show.

Figure 10: Human vs primarily AI-generated articles on the internet



Source: Graphite (citing 55.4k articles from Common Crawl examined by AI detectors), Deutsche Bank Research

3.4 An incomplete case

This analysis shows that the San Francisco Consensus, which emphasises RSI and a short timeline to AGI, is neither predestined nor impossible.

Rather it depends on whether the science is right, on whether scaling is all you need, and on whether the infrastructure and capital required for that scaling is deployed and accessible to the leading labs.

There is, in other words, a possible rather than impossible, path to AGI, subject to those conditions.

None of this proves that superintelligence will ever arrive. A buildup at this scale has few peacetime precedents and vastly exceeds the likes of the Manhattan Project and Apollo programme.

It is uncertain whether there will be enough demand to pay upfront for the investment in achieving RSI that would justify that investment. And whether the friction of verification, and inevitable declining marginal returns, act as a drag on what otherwise might be a runaway process.

4. What happens if we don't get there? The base case

4.1 Separating AI progress from superintelligence

So what if that thesis is wrong? Does the AI trade depend on a journey to AGI?

AI can remain economically important even before RSI coupled with scaling laws lead to super intelligence. Companies do not need to wait for nebulous superintelligence to generate identifiable returns from AI in the medium term as they fold it into their systems, streamline workflows and roll out new business models.

Investors already have exposure to RSI as a capital allocation because the leading US labs say RSI is a key part of their product strategy. And it is these labs whose products are catalysing and amplifying the infrastructure boom.

Even rapid competitive diffusion does not mean every part of the physical economy changes instantly.

The smartest superintelligence might identify the optimal buyout structure before breakfast, but it cannot force a financially constrained buyer to use it or persuade the seller to accept less than they have been expecting. Those challenges will be even more intractable where the parties are more complex, say, fiscally constrained sovereigns or foreign equity holders.

4.2 Why the base case is attractive

Superintelligence would be a chaotic accelerant to human society, and a potential regime break. But it is not the only scenario in which today's investment in AI infrastructure makes economic sense.

Adoption of AI tools by businesses is progressing approximately three times faster than the mobile, internet and cloud computing waves. From sector revenues below $1bn in 2022, the AI economy, excluding China, will likely exceed $200bn in revenues in 2026.

Early winners are the cloud businesses. Google Cloud revenue jumped 82 percent from a year earlier in the most recent quarter as companies turned AI pilots into workflows. Amazon Web Services sales were up 37 percent and Microsoft's AI business more than doubled.

There is also slow, but increasing, evidence of a productivity uplift. The J-curve cuts both ways: front-loading investments is taking one step back to take two steps forward. The so-called complementary intangible investment that is (mis)measured as cost today to the detriment of the productivity statistics will flatter them in the years to come as it pays off. The base case does not require a regime break to show up as a productivity surprise, just for the accounting to catch up.

The economic gains from AI that is less powerful than the AGI mooted by the labs would be substantial. And in this base case, they may also be attractive.

Hyperscaler capex is driving earnings at beneficiaries ranging from semiconductor makers to construction companies and accounting for between one-third and one-half of current US growth. The wealth effect of rising stock prices, with the ratio of US equity wealth to income rising from around 1.8 to almost 2.5 over the past three years, is supporting consumer spending.

AI FOMO (fear of missing out) is buoying enterprise spending on cloud services and the groundwork for AI initiatives. Those are beginning to bear fruit in terms of personal productivity, cost savings and more targeted products and services.

4.3 Investor watchpoints

The arrival of ASI would be a disruptive economic shock. Investors need to be alert for the early signals.

Rather than relying on a single probability forecast, investors should watch five indicators particularly closely for signs that AGI is nearing.

1. **AI speeds up AI**

- Track METR's estimate of how much AI speeds up AI. Currently the capabilities of models are doubling roughly every three months. In a world of RSI, the pace of this speed-up will itself likely accelerate.

2. **AI can complete whole jobs**

- Track via the Remote Labor Index, a multi-sector benchmark. Currently it finds that the highest performing agent still only manages to automate 20 percent of the real-world, economically valuable freelance projects it attempts.

3. **Compute remains scarce**

- On a path to superintelligence, demand for compute will continue to outrun supply. Track GPU one-year rental prices, up by about a third over the past year.

4. **Firms bifurcate**

- Leaders and laggards in the same industry diverge while averages stay the same. Look at revenue per employee in top- and bottom-quartile firms

5. **Junior hiring comes under significant pressure**

- … And then the pressure rises up the ranks.

Recent relevant reports

Adrian Cox and the Deutsche Bank Research Institute

AI doom: A brief history of bad tech predictions (Sept 15, 2026)

AI at 70 (Part 2): 14 more lessons from a lifetime of boom and bust (Sept 10, 2026)

AI at 70: 14 lessons from a lifetime of boom and bust (Aug 17, 2026)

Open-source AI 101: the battle for the future of AI (July 21, 2026)

AI IPOs 101: why they'll change the boom forever (July 1, 2026)

Azeem Azhar

Exponential View | Azeem Azhar | Substack

The state of the AI economy

The author of the report wishes to acknowledge the expert advisory contributions made by Azeem Azhar of Exponential View.

Appendix 1

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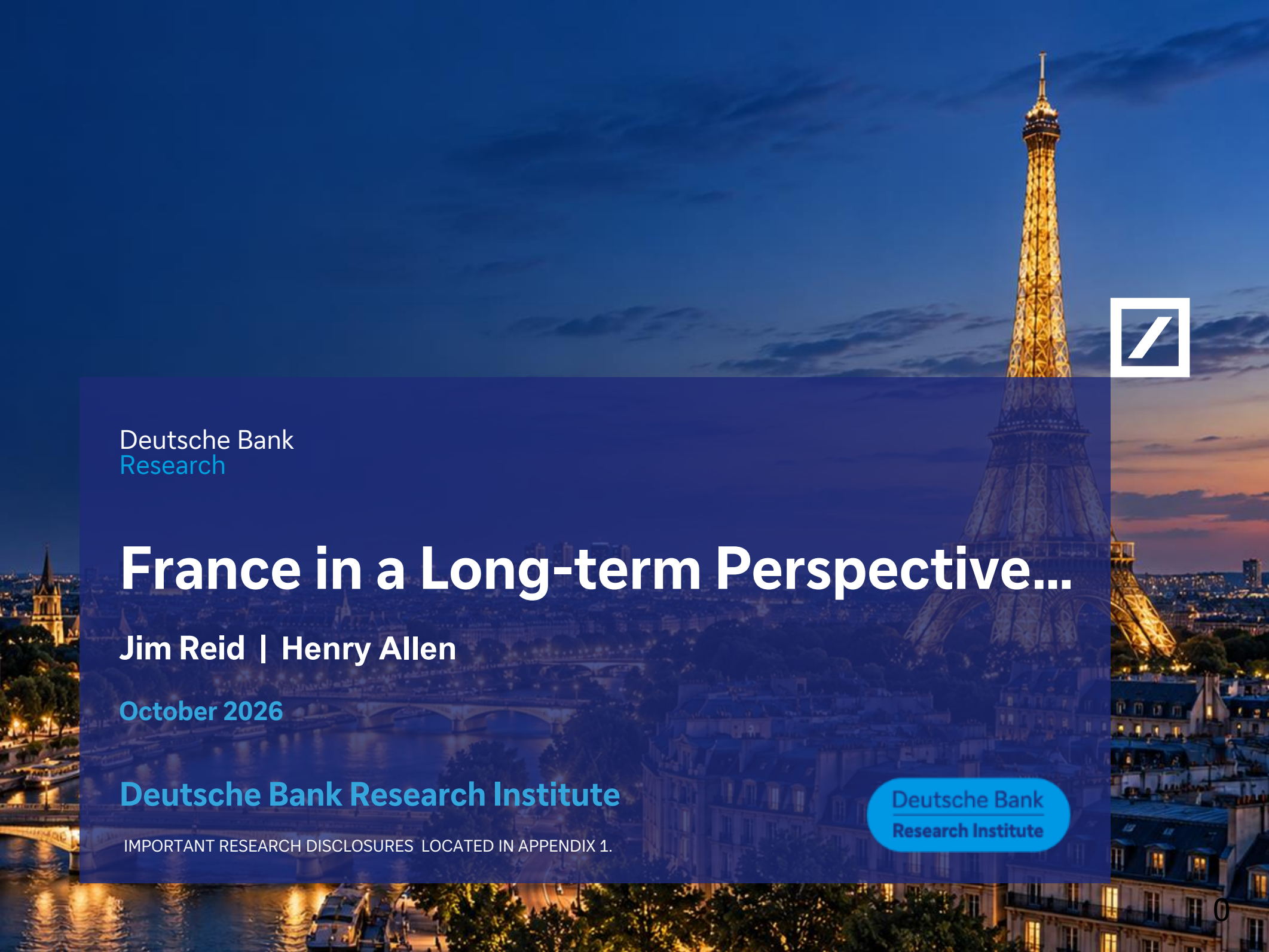

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Jim Reid | Henry Allen

October 2026

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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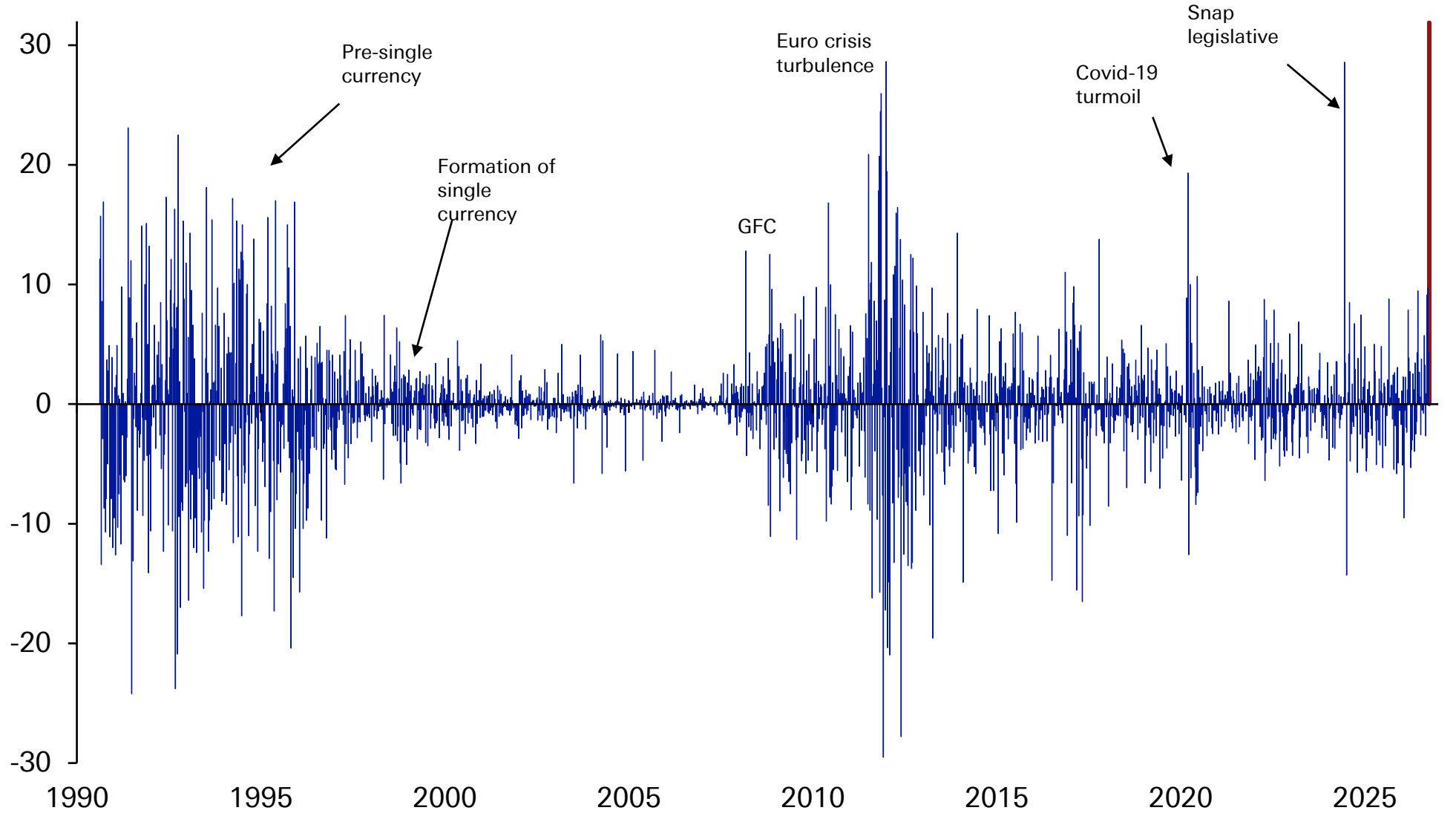
Although this week has seen a recovery so far, the moves last week were genuinely historic, with the Franco-German 10yr spread seeing its biggest weekly widening in available Bloomberg data back to German reunification in 1990.

Weekly change in the Franco-German 10yr yield spread (basis points)



Source: Bloomberg Finance LP, Deutsche Bank

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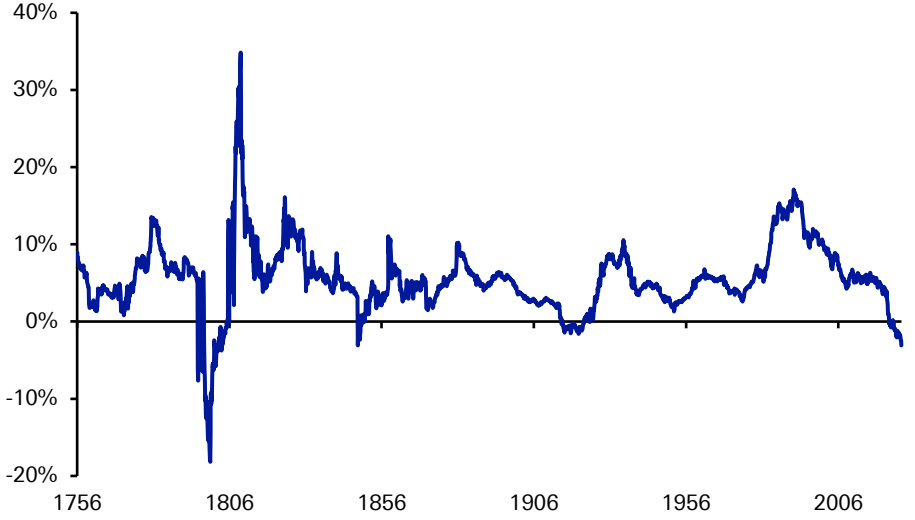
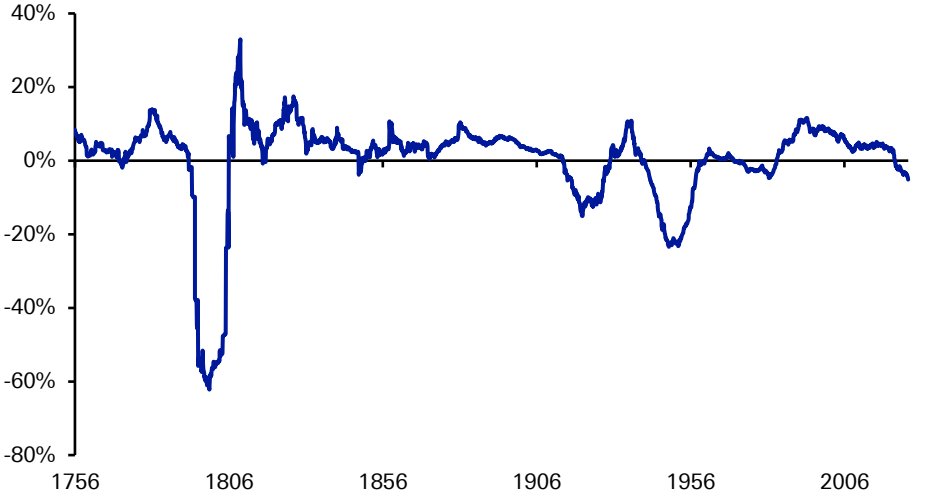
Zooming out, 10yr French OATs have seen their worst decade of nominal returns since 1803, when the Reign of Terror (1793-94) was still part of the 10yr rolling calculations.

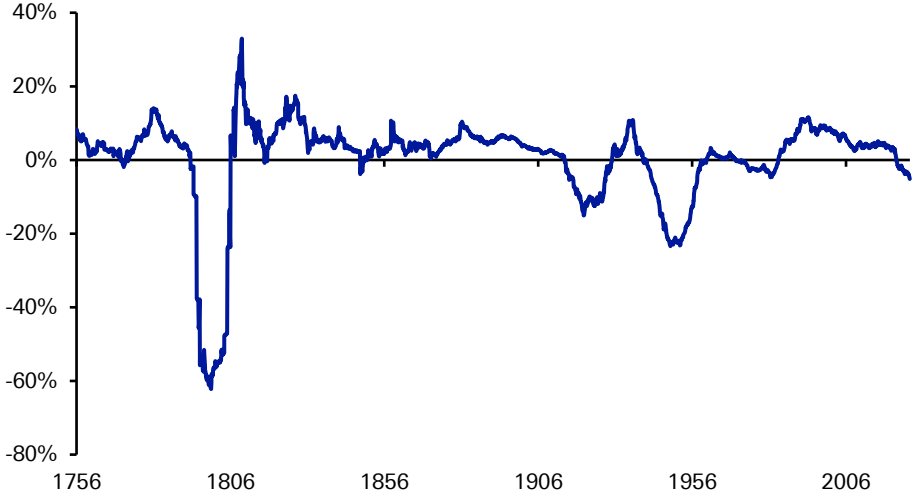
10yr Rolling Nominal Return (Annualised)



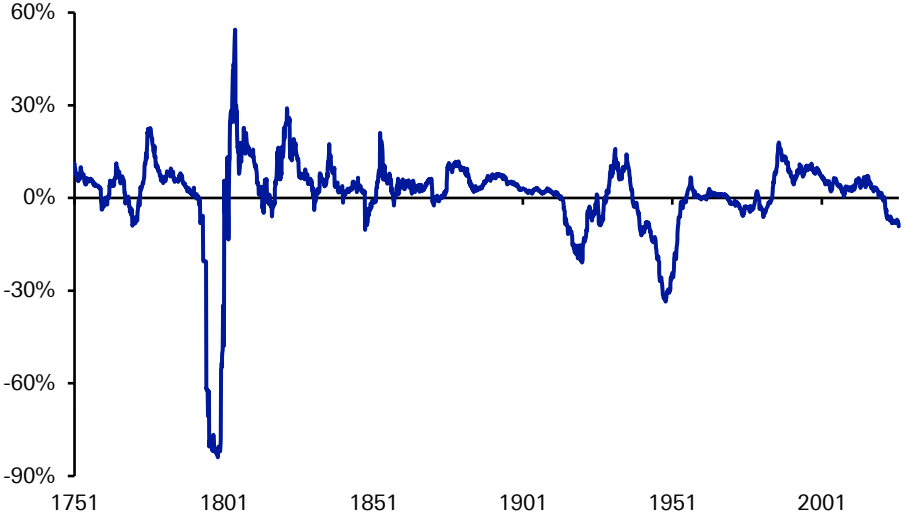
5yr Rolling Nominal Return (Annualised)



10yr Rolling Real Return (Annualised)



5yr Rolling Real Return (Annualised)



Source: Finaeon, Deutsche Bank

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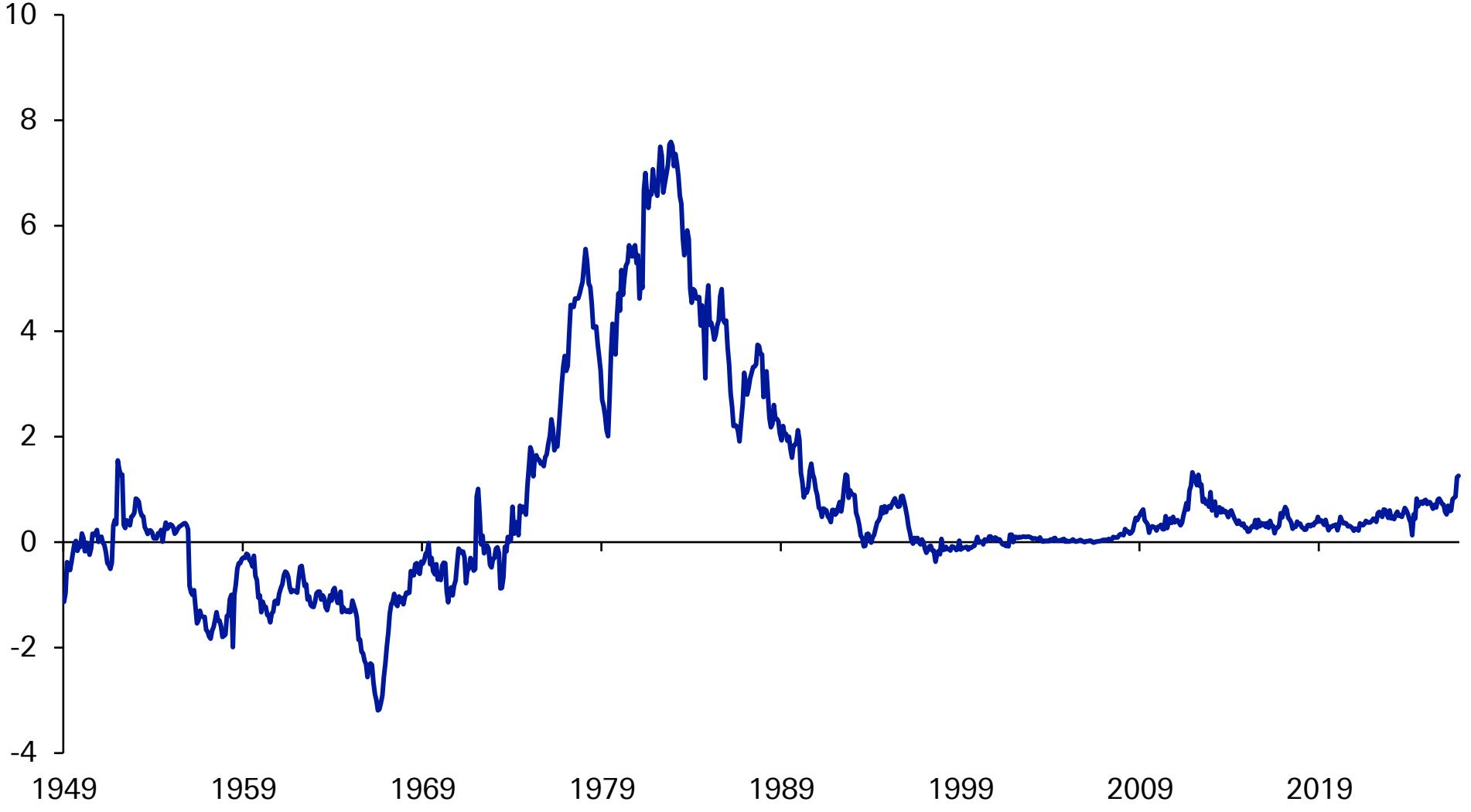
Recent concerns have pushed France's 10yr spread over Germany's, to its highest since 2012 and the Euro crisis. But in an historical perspective, today's spreads are only high by the standards of the Euro Area when the two shared a common currency. They were regularly several percentage points apart in the 70s and 80s.

Franco-German 10yr yield spread (percentage points)



Source: Finaeon, Deutsche Bank

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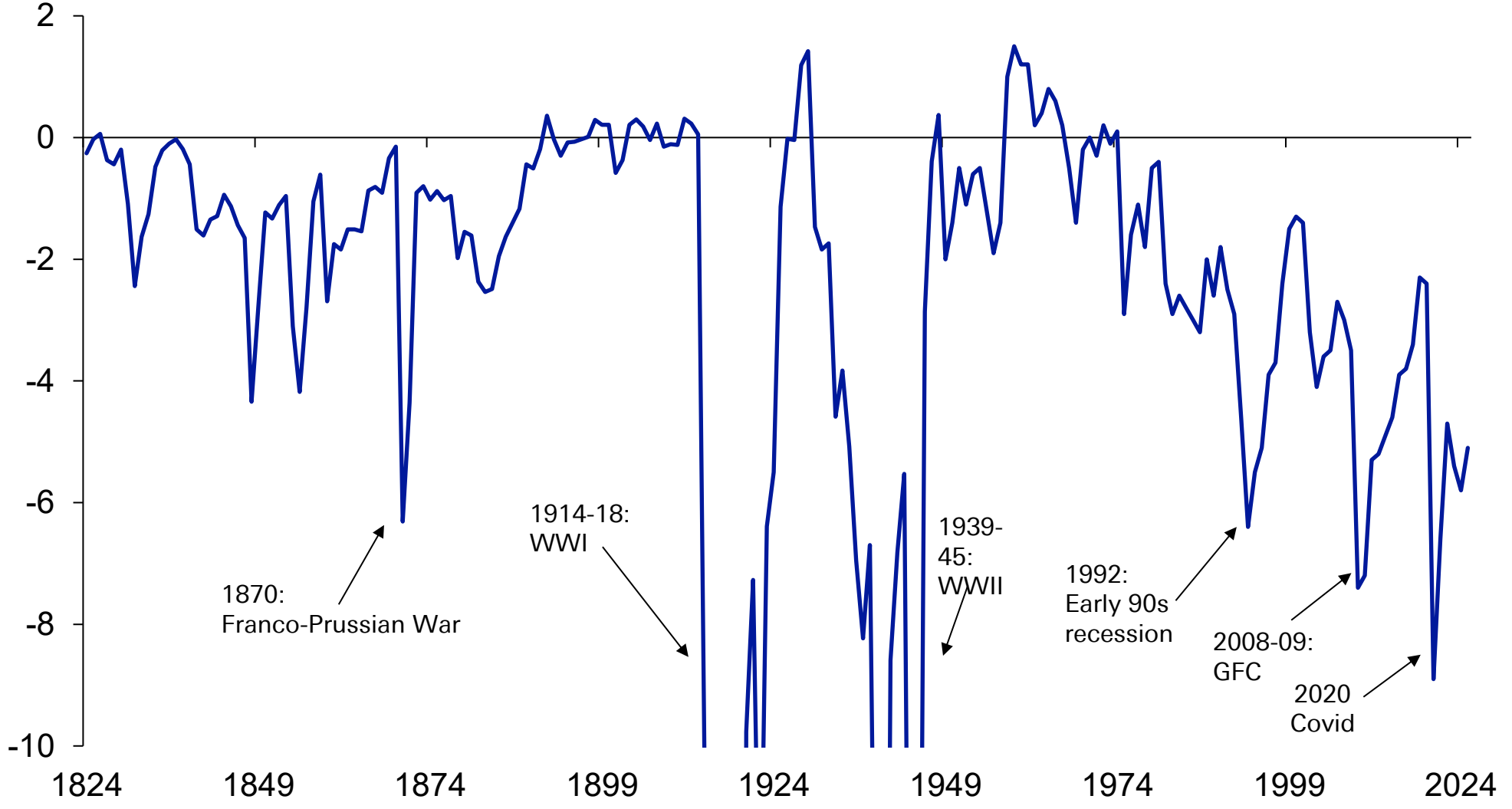
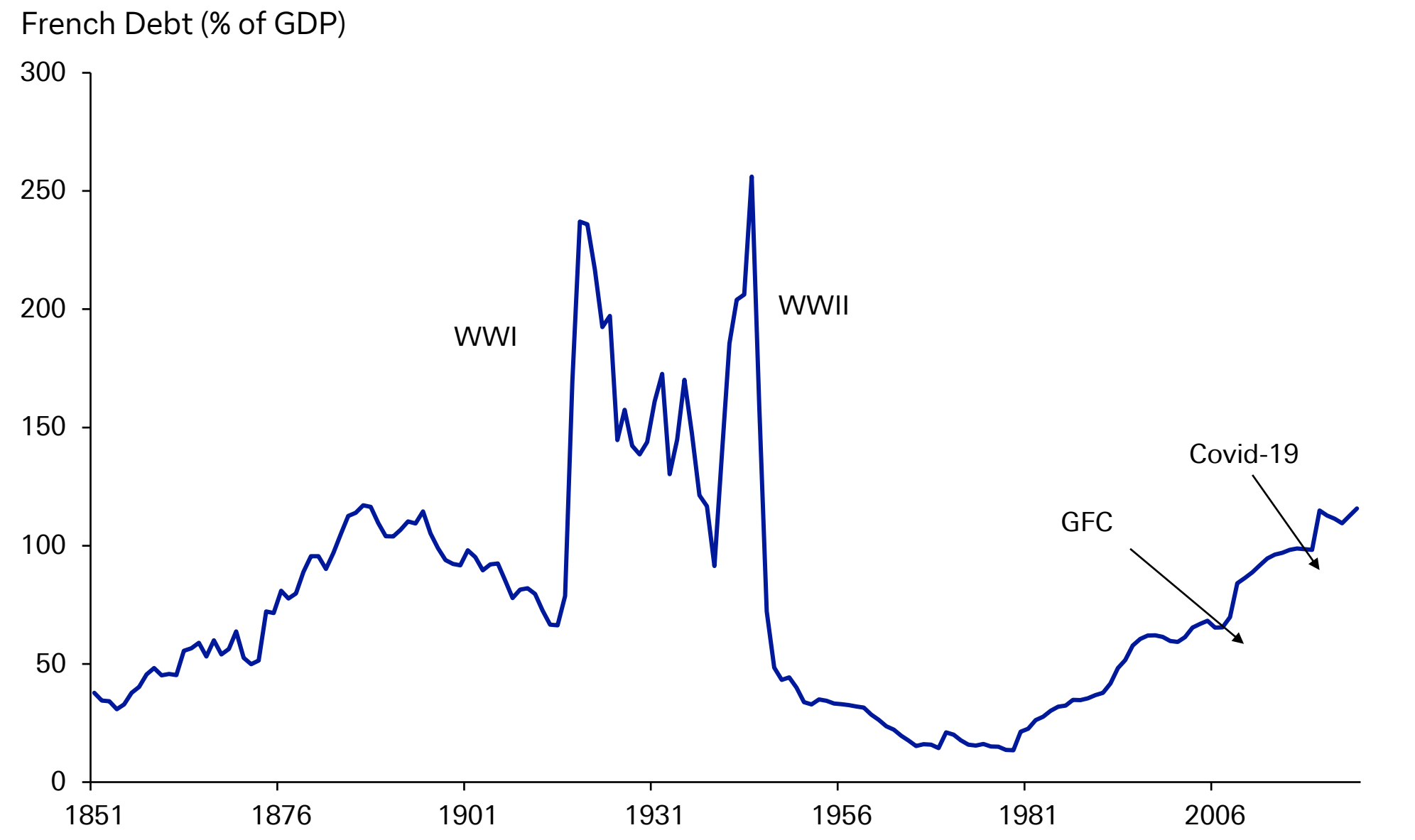
Although much of the recent selloff is an unwinding of carry trades, France has ongoing, if not new, fundamental issues. It hasn't run a budget surplus since 1974, and the deficit was last beneath 2% of GDP in 2001.

French Government Budget Deficit (% of GDP) over the last 200 years



Source: Finaeon, Haver Analytics, Deutsche Bank

But with half a century of persistent deficits, France's debt-to-GDP ratio has seen a near-continuous rise – to 116% in 2025 under the Maastricht definition.



Source: Finaeon (up to 1979), Haver Analytics (from 1980), Deutsche Bank

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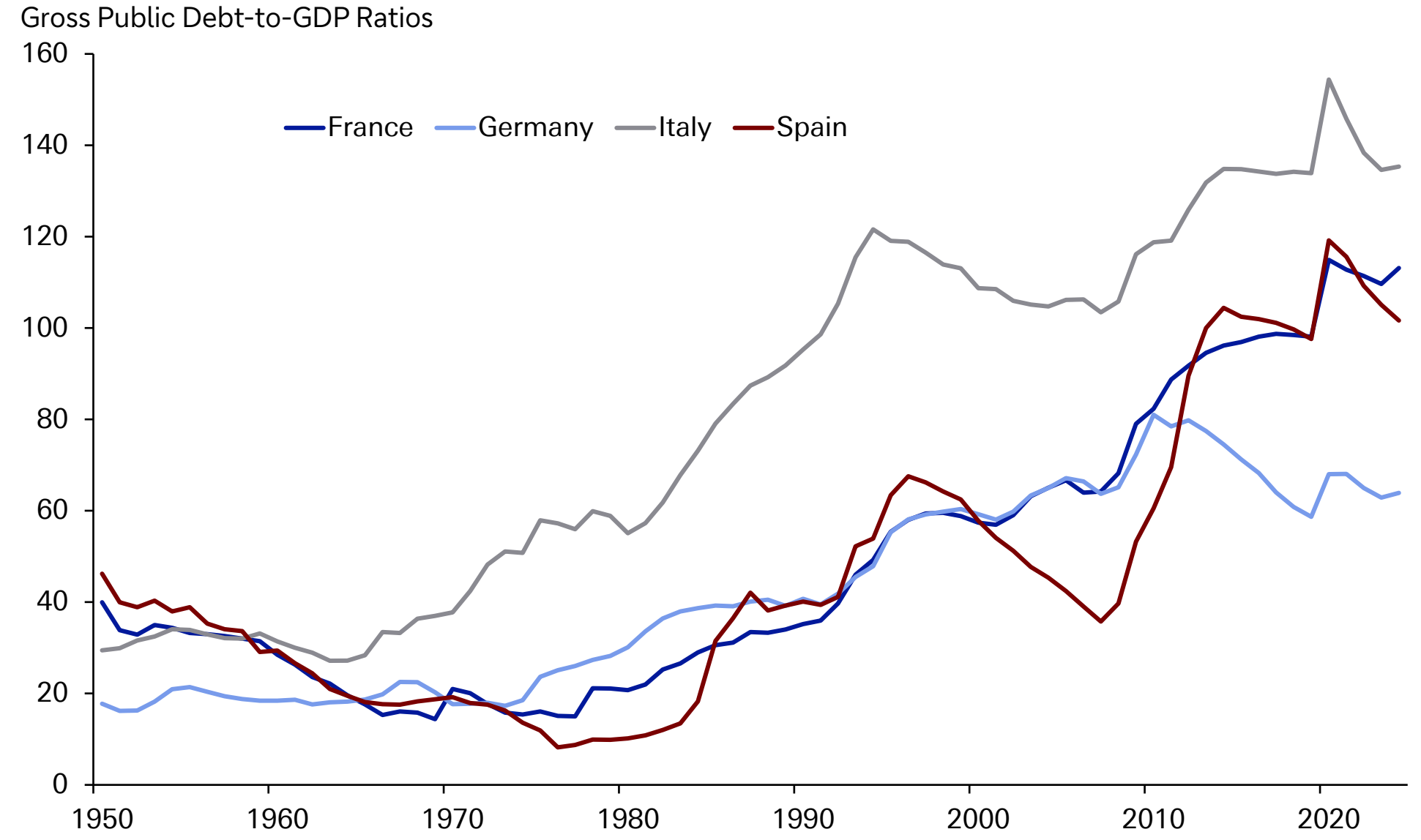
France has also seen its debt rise in relative terms. As recently as the GFC in 2008, France's debt-to-GDP was basically in line with Germany's and well below Italy's, but today it's far above Germany's and closing in on Italy's. Outside of the temporary Covid spike, Italian debt/GDP has been stable for over a decade.



Source: IMF, Deutsche Bank

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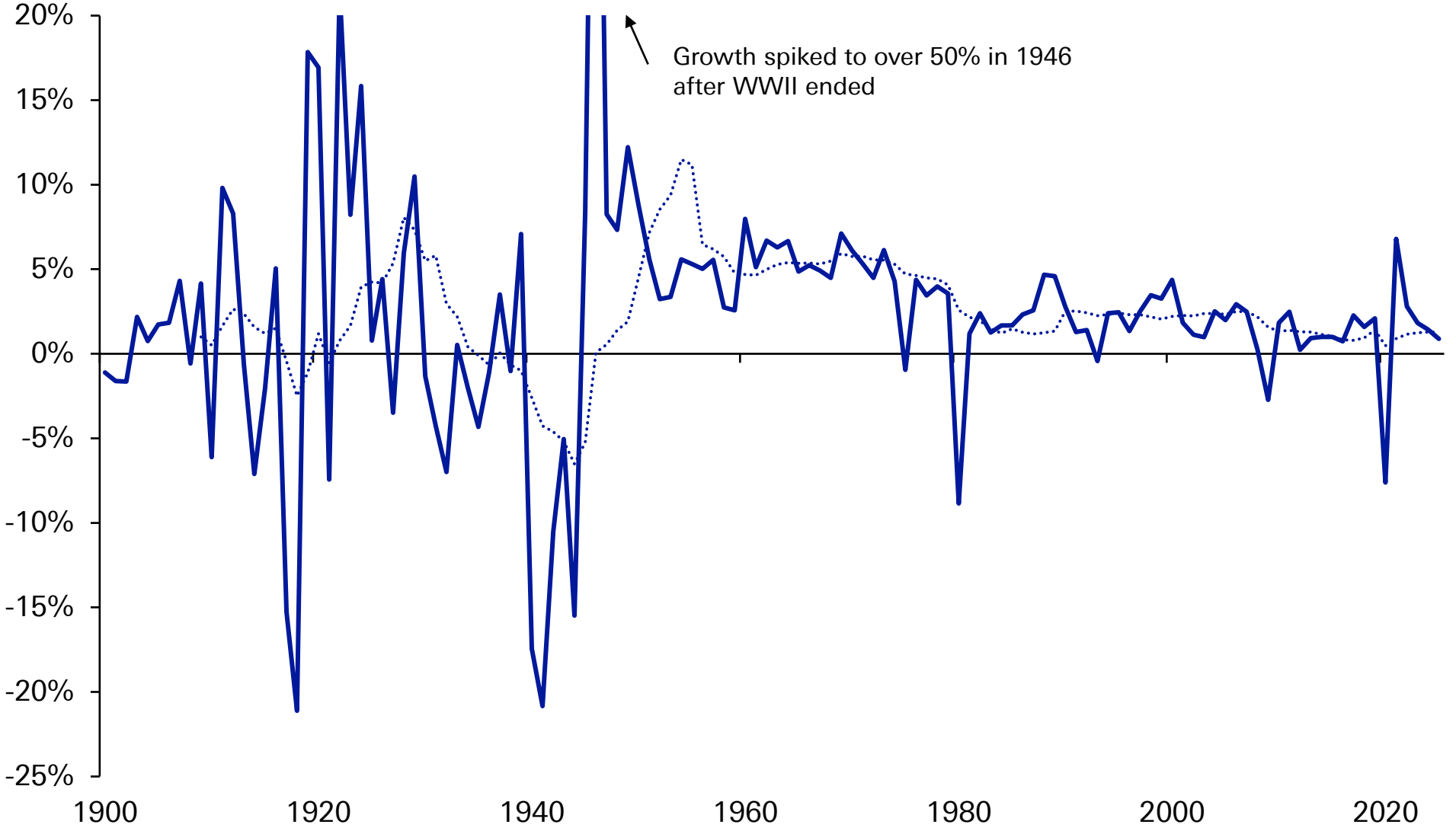
An ongoing fundamental concern has been sluggish growth in recent years – although none of this is new news.

French Real GDP Growth (year-on-year % change) with a 10yr Moving Average



Source: Finaeon, Deutsche Bank

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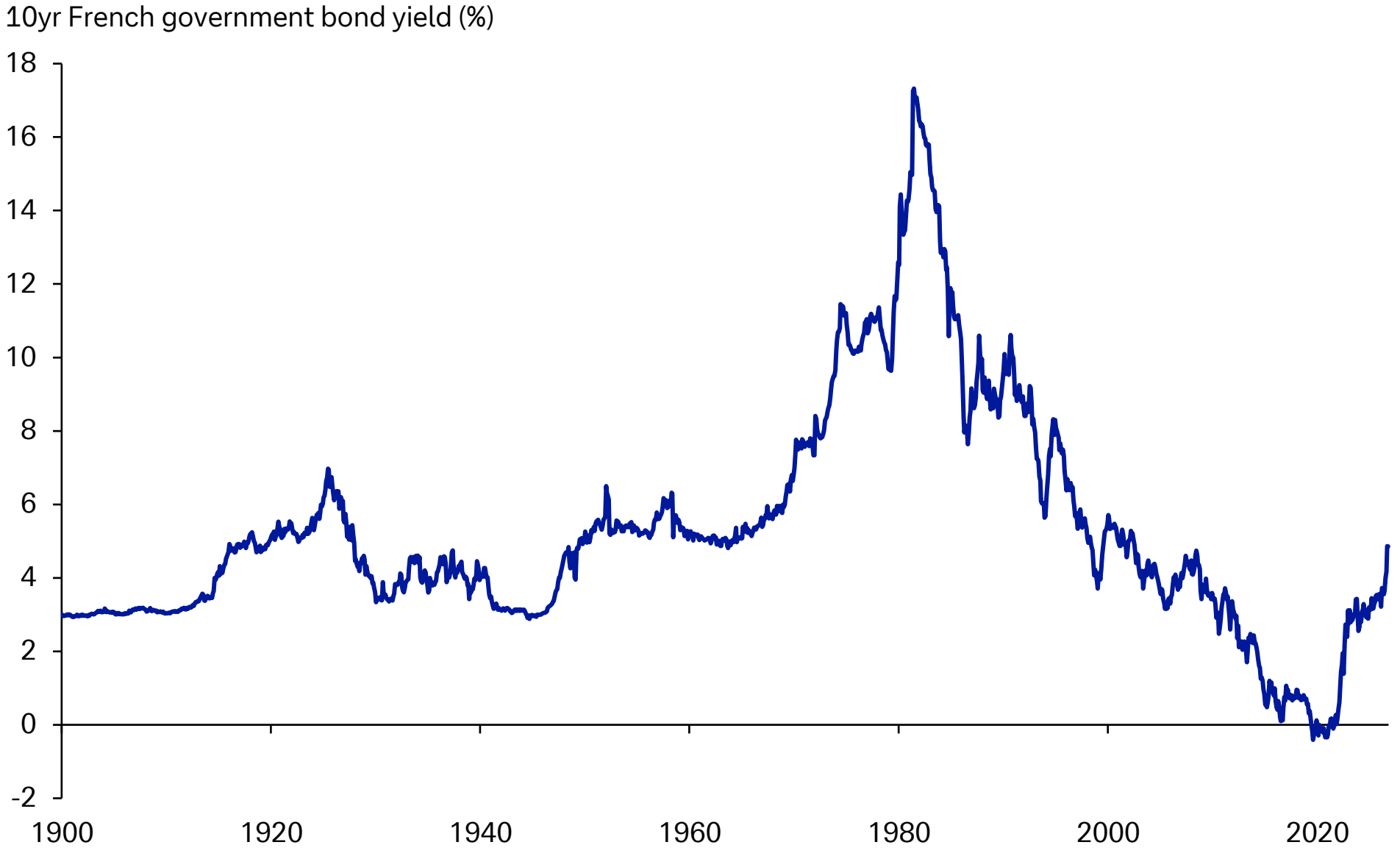
The recent jump in yields has pushed them just above their historical average, but the problem is that France is now paying these yields on a much larger debt pile.



Source: Finaeon, Deutsche Bank

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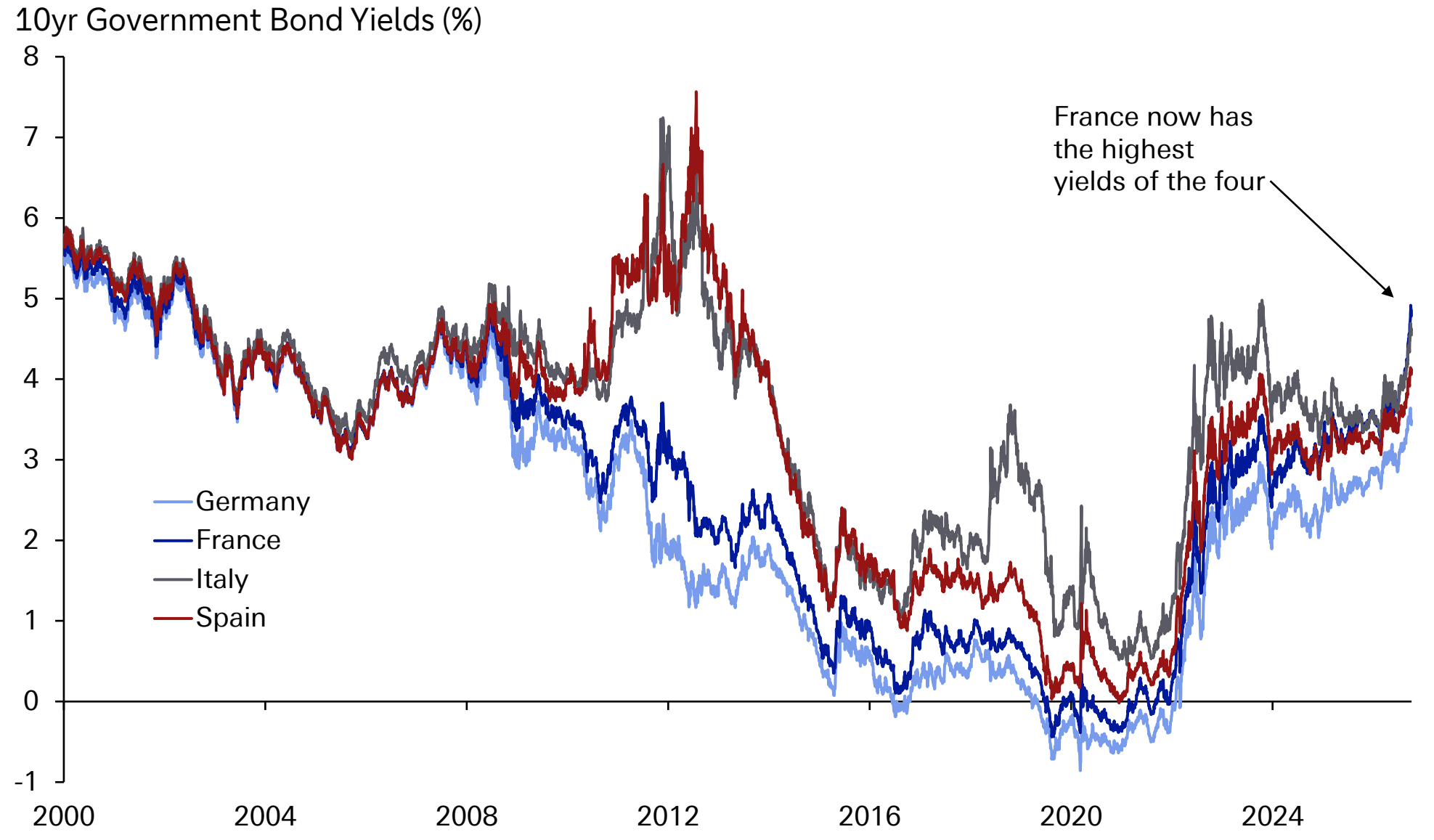
Back in the Euro Crisis of the early 2010s, markets priced France closer to Germany than to Italy and Spain, but today that's turned around and France now has the highest yields of all four.



Source: Bloomberg Finance LP, Deutsche Bank

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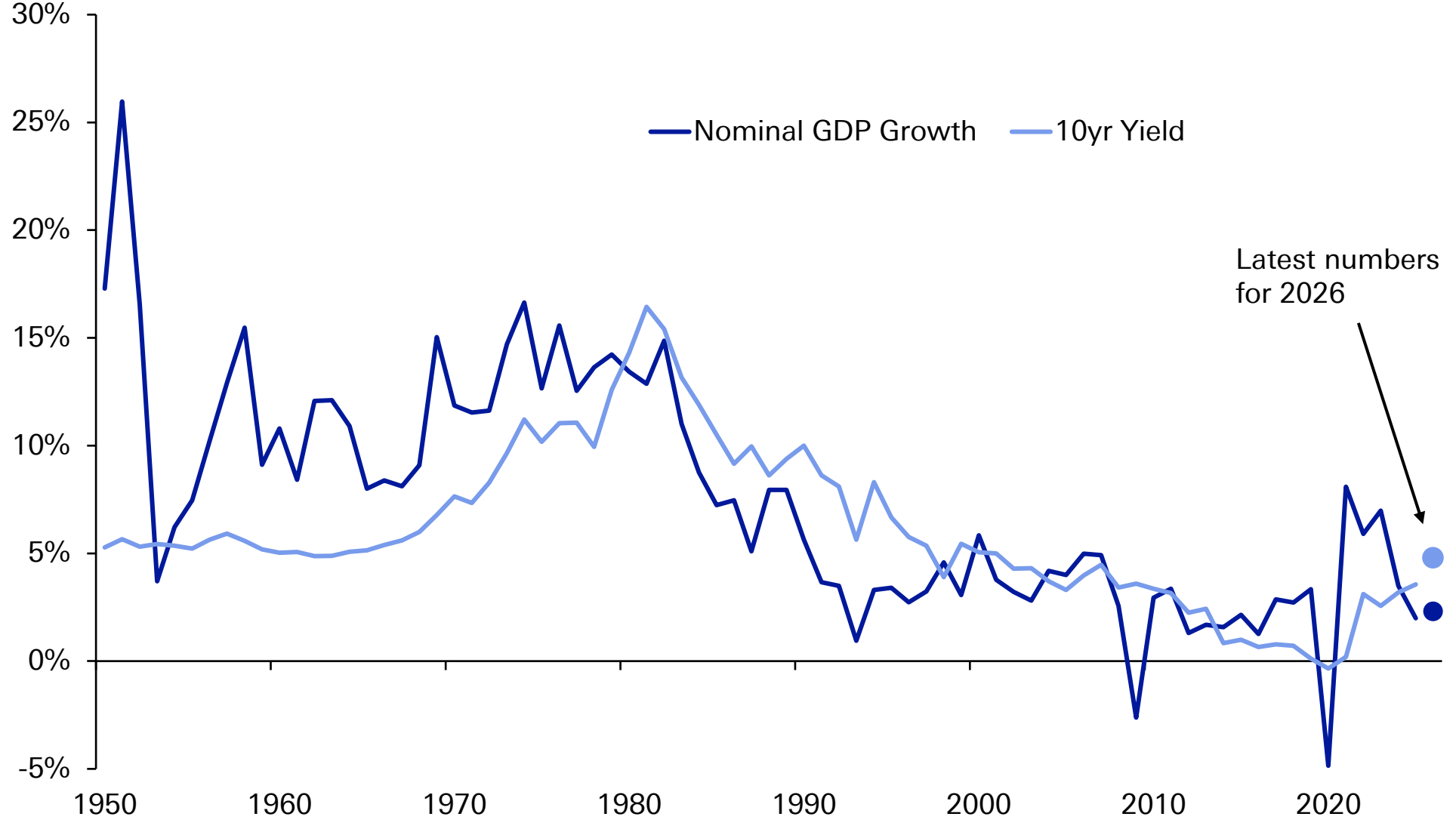
For most of the last decade, nominal GDP has been above 10yr yields, but that has reversed, which is challenging from a debt sustainability if sustained – especially when debt/GDP is pushing up against 120%. Recent inflation will slowly push up nominal GDP but the relationship between the two will be important going forward.

French nominal GDP growth (year-on-year % change) and 10yr yields



Source: Finaeon, Deutsche Bank

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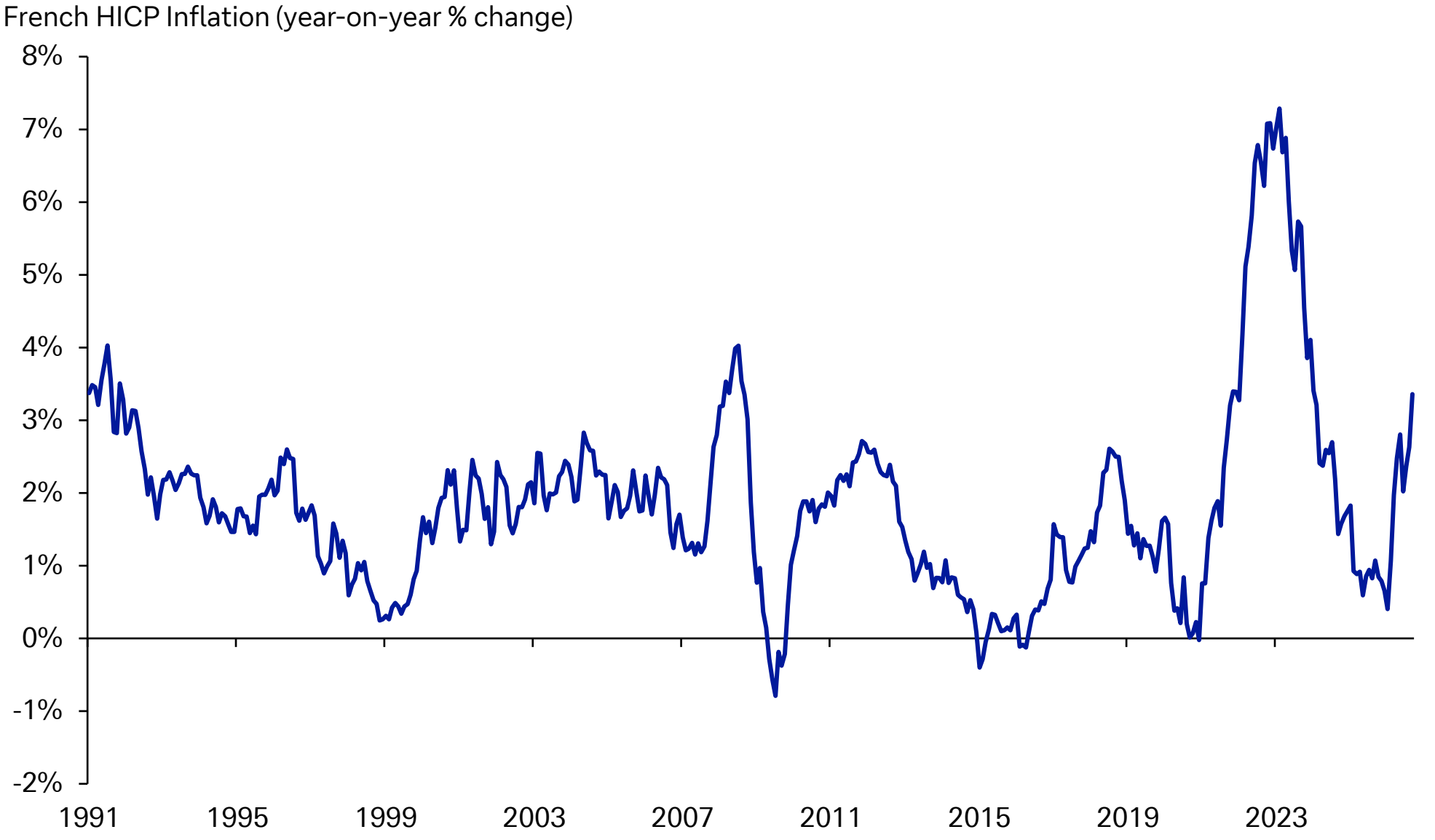
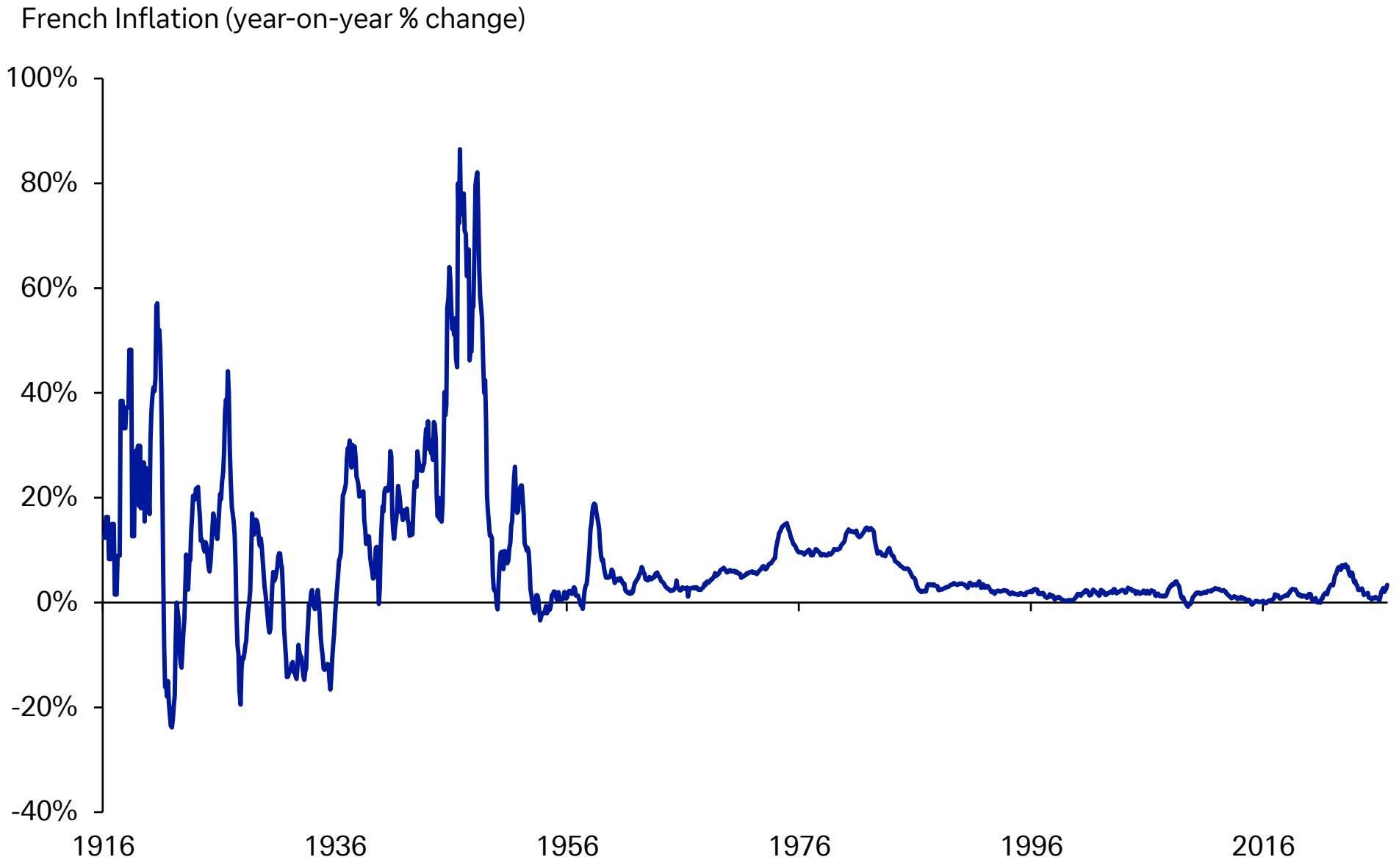
Inflation is always a double-edged sword when it comes to debt. An unexpected burst of inflation can help lower debt, as we saw right after the pandemic, but it becomes a challenge when higher yields offset this. This is now the 3rd biggest inflation spike since the Euro Area's formation in 1999.



Source: Haver Analytics, Deutsche Bank

In a broader perspective, though, French inflation has been comparatively tame compared to its post-war heights, let alone where it was in the early 20th century.



Source: Finaeon (until 1990), Haver Analytics (from 1991), Deutsche Bank

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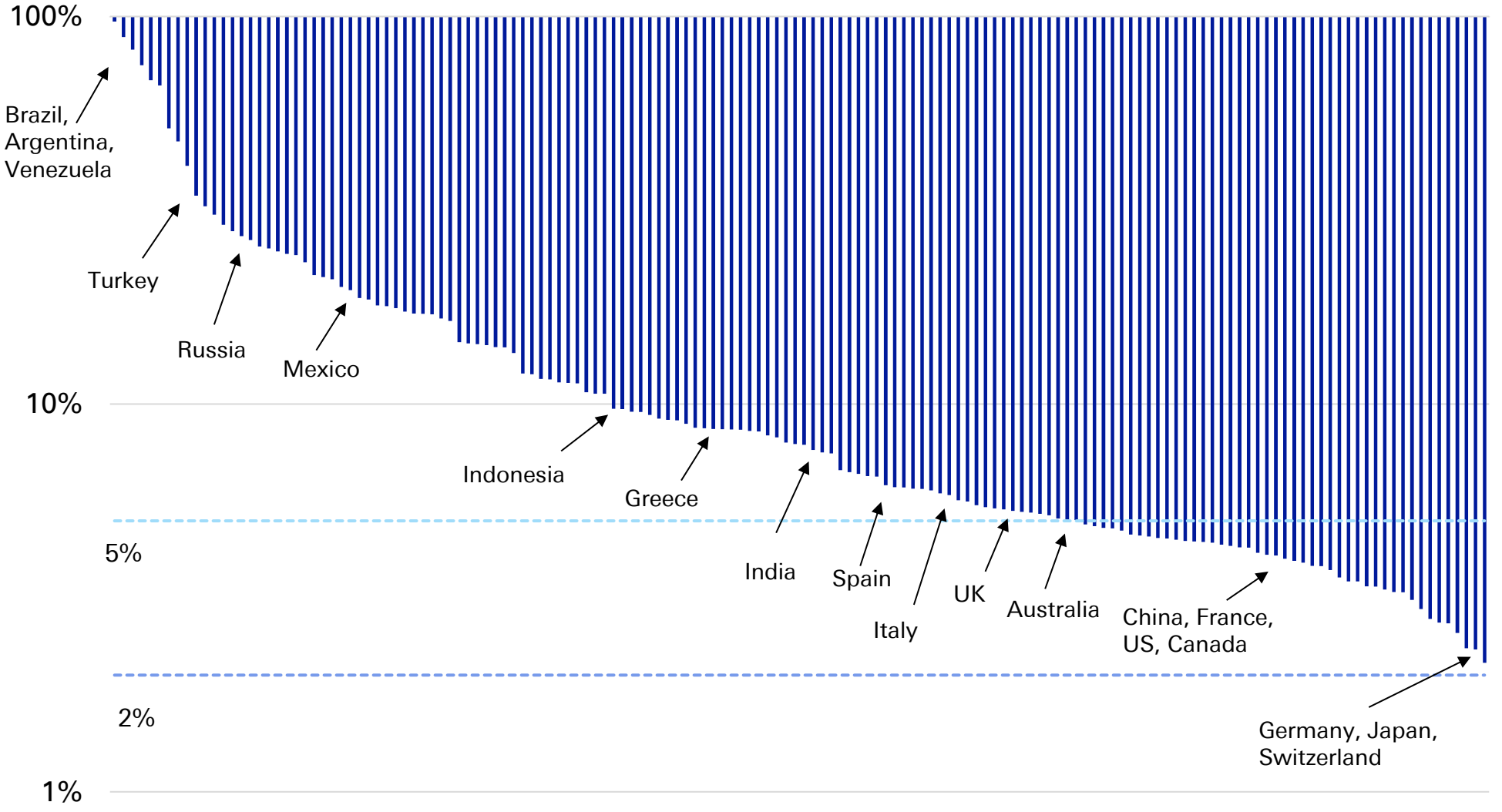
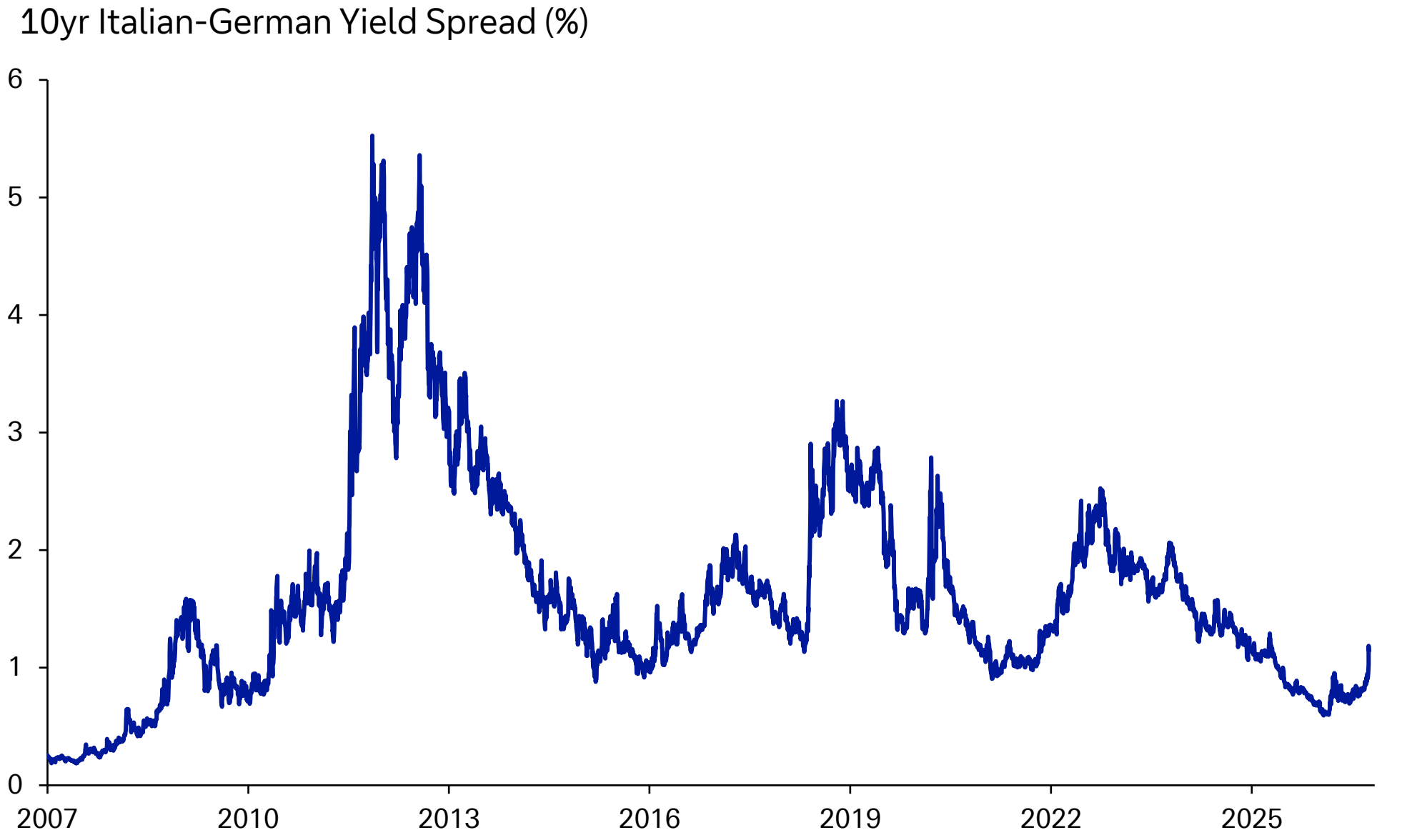
Even in the period when France has run 50-plus years of consecutive deficits, its average inflation has been broadly around US levels (c.4%) and lower than the likes of the UK, Italy and Spain.

Average annual inflation of 152 economies since 1971 when Bretton Woods collapsed. No economy has averaged less than 2% inflation, but Switzerland at 2.2% comes closest



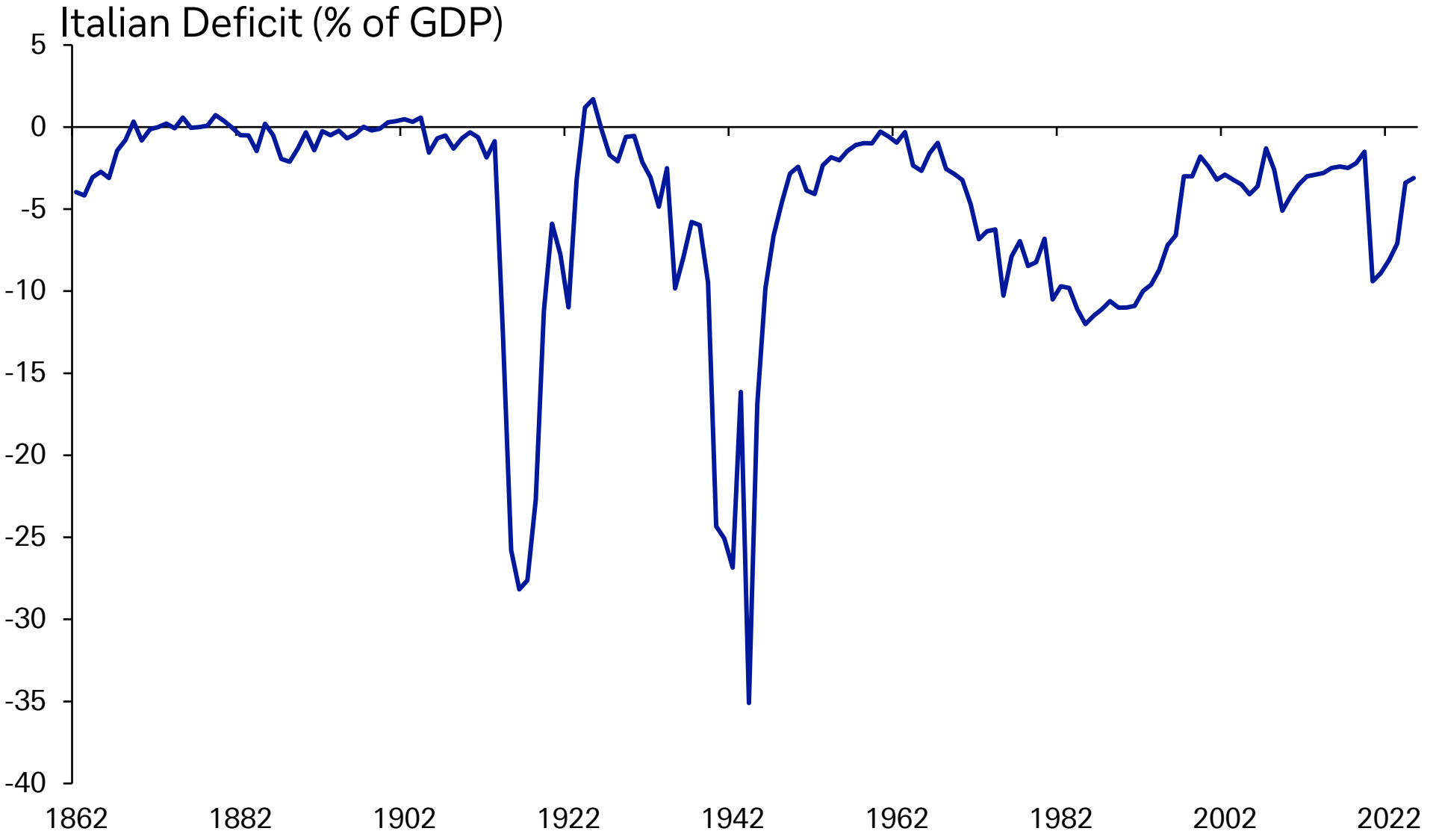
Source : World Bank, Deutsche Bank. Note: log scale.

And this isn't just about France, as Italy's spread has also widened in recent sessions. However, unlike France, the Italian spread isn't at its highest since 2012.



Source: Bloomberg Finance LP, Deutsche Bank

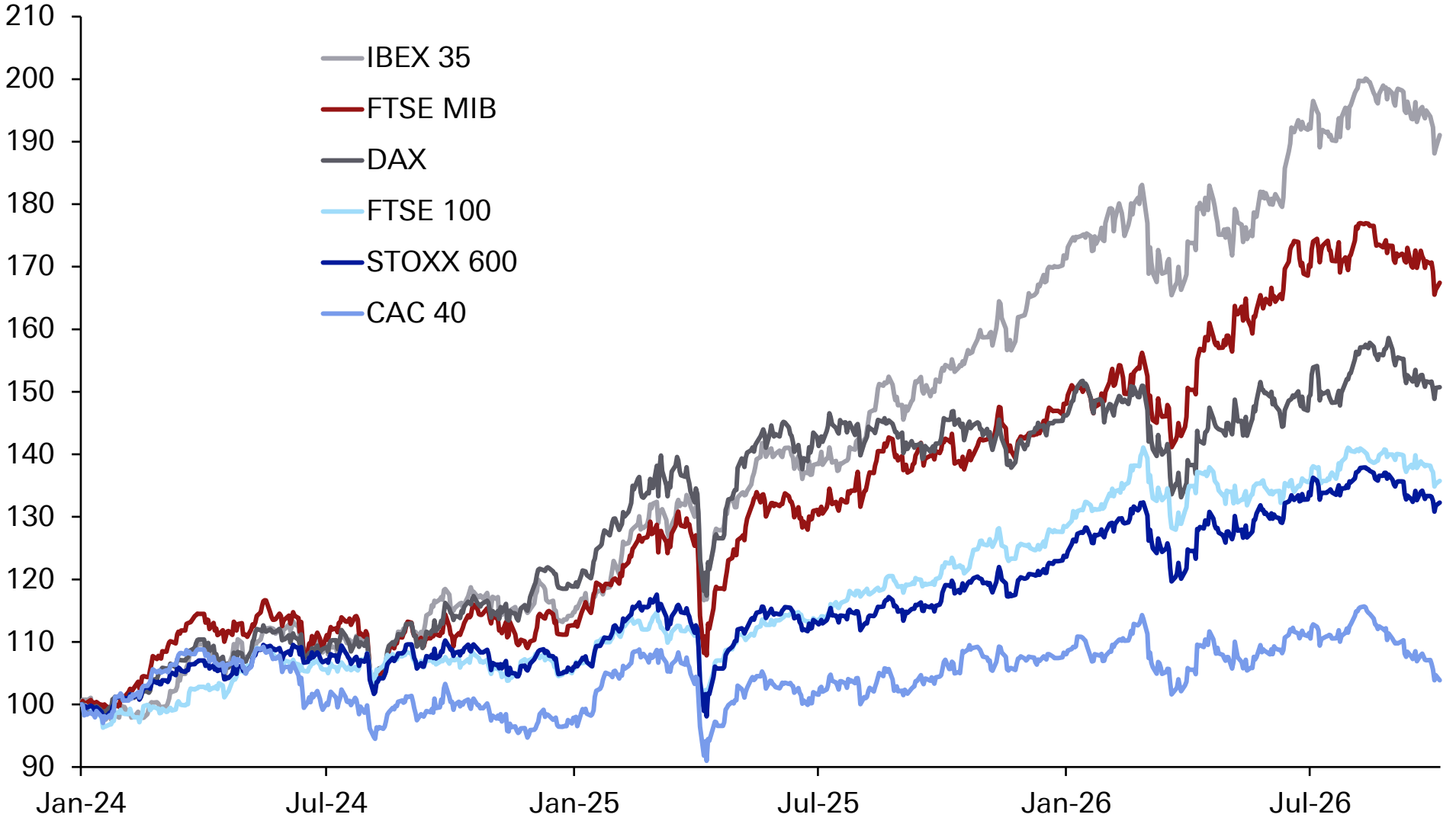
Italy has an even longer record of consistent budget deficits; the last surplus was back in 1925 – but it has run a primary surplus for most of the last few decades, unlike France.



Source: IMF (up to 1979), Haver Analytics (from 1980), Deutsche Bank

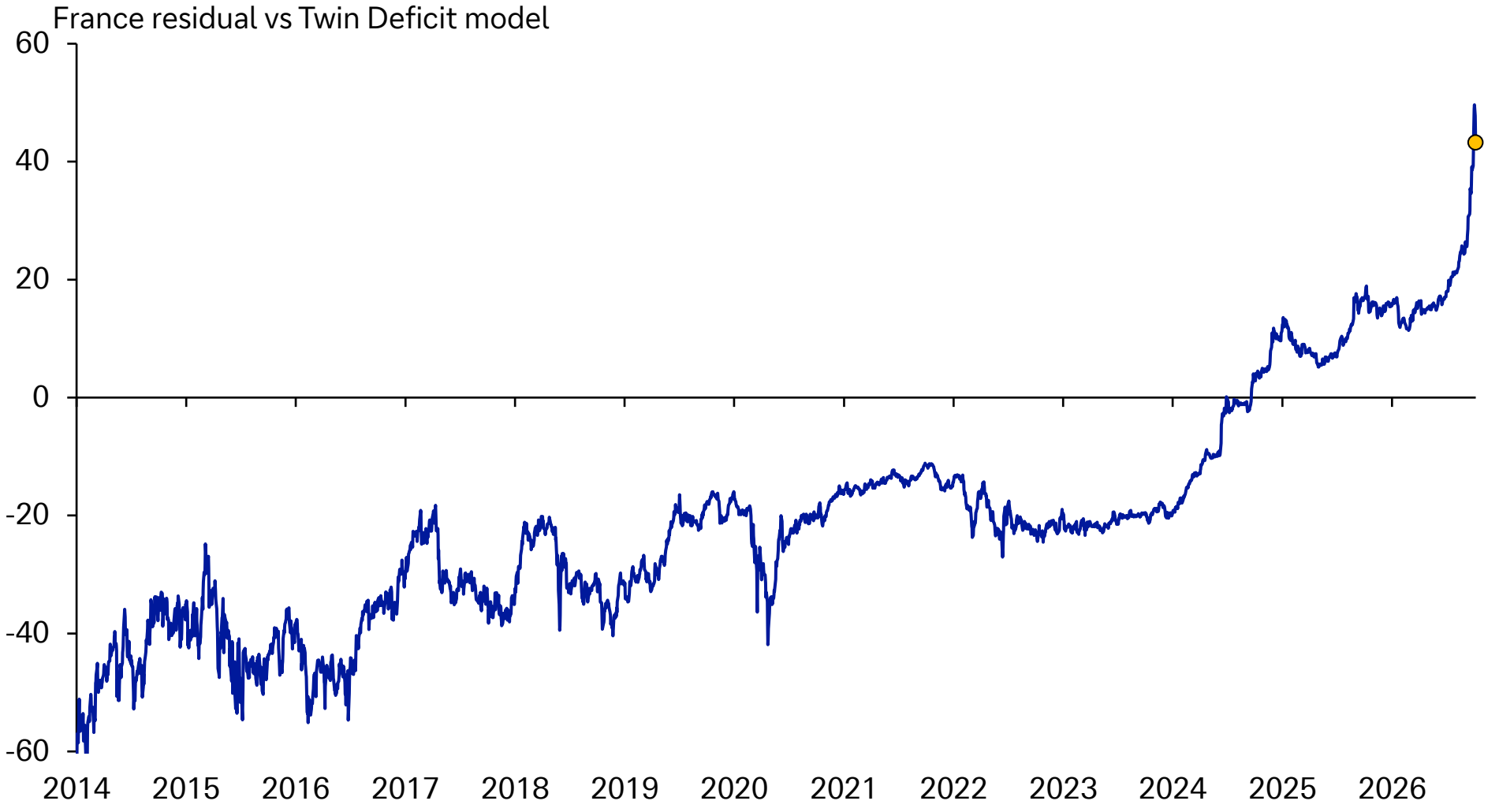
The recent underperformance is clear for equities too – and has been for over two years. France's CAC 40 has recently underperformed all the major European equity indices: the Europe-wide STOXX 600 is up over 30% in that time, and the DAX is up over 50%, yet the CAC 40 is up only 4%.

Major European Equity indices (Jan 1 2024 = 100)



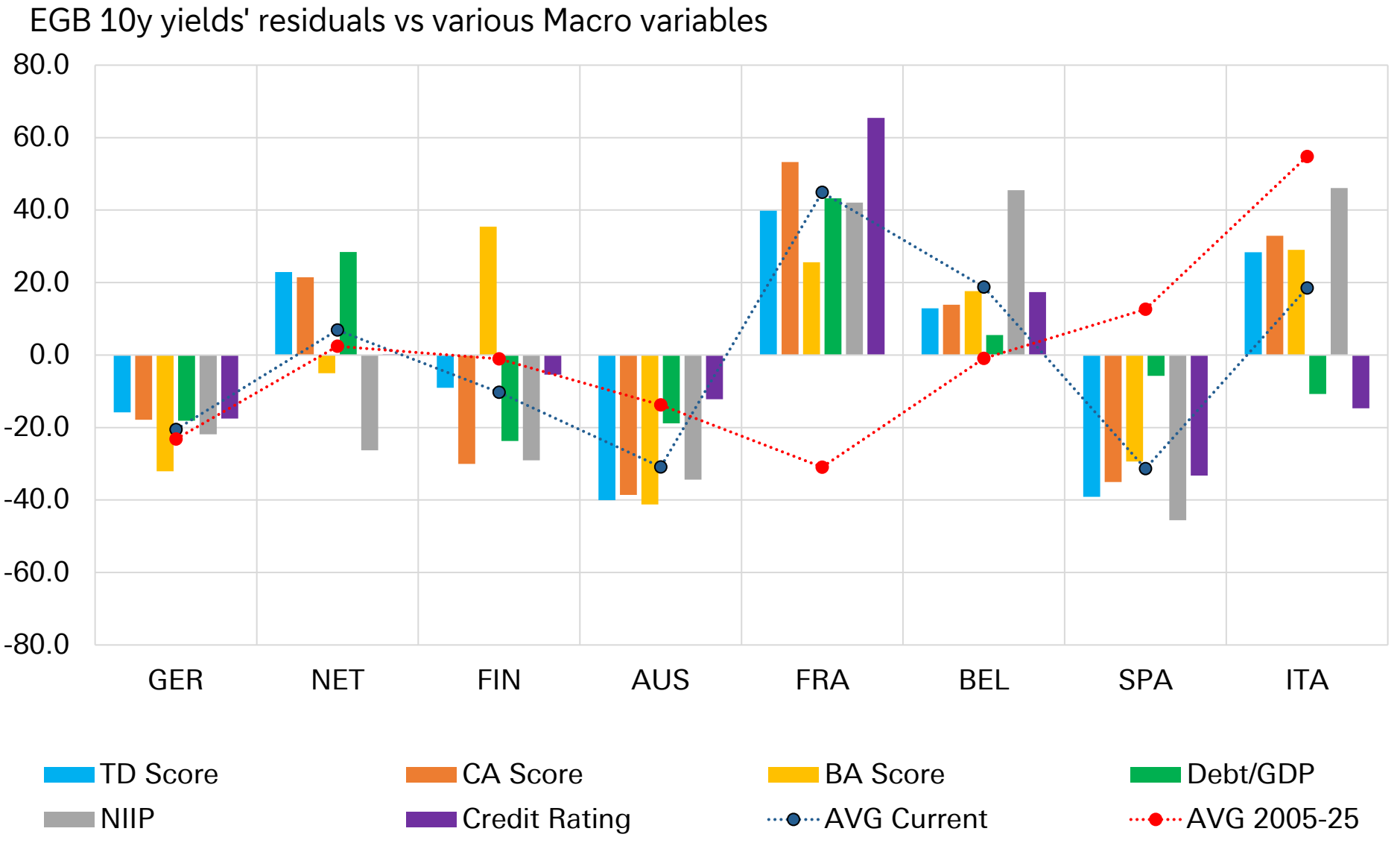
Source: Bloomberg Finance LP, Deutsche Bank

All that said, DB's rates strategists now see French bonds as cheap relative to their fundamental metrics. This chart looks at the residuals of France's twin deficit score, which is a weighted average of the historical current account balance (% of GDP) and budget balance (% of GDP), with a higher weight assigned to more recent years.



Source: Bloomberg Finance LP, Deutsche Bank

Looking at a variety of macro variables, they now see France as the cheapest European government bond relative to fundamentals. So things are challenging at the moment, and the long-term charts look worrying, but much of the sell-off in recent days/weeks has been due to a clearing-out of carry trades, implemented assuming a much lower vol period than we've seen.



Source: IMF, Bloomberg Finance LP, Deutsche Bank

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Research

Q3 2026 Global Markets Survey

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September 2026

With special thanks to Anthony Chaimowitz from dbDataInsights team

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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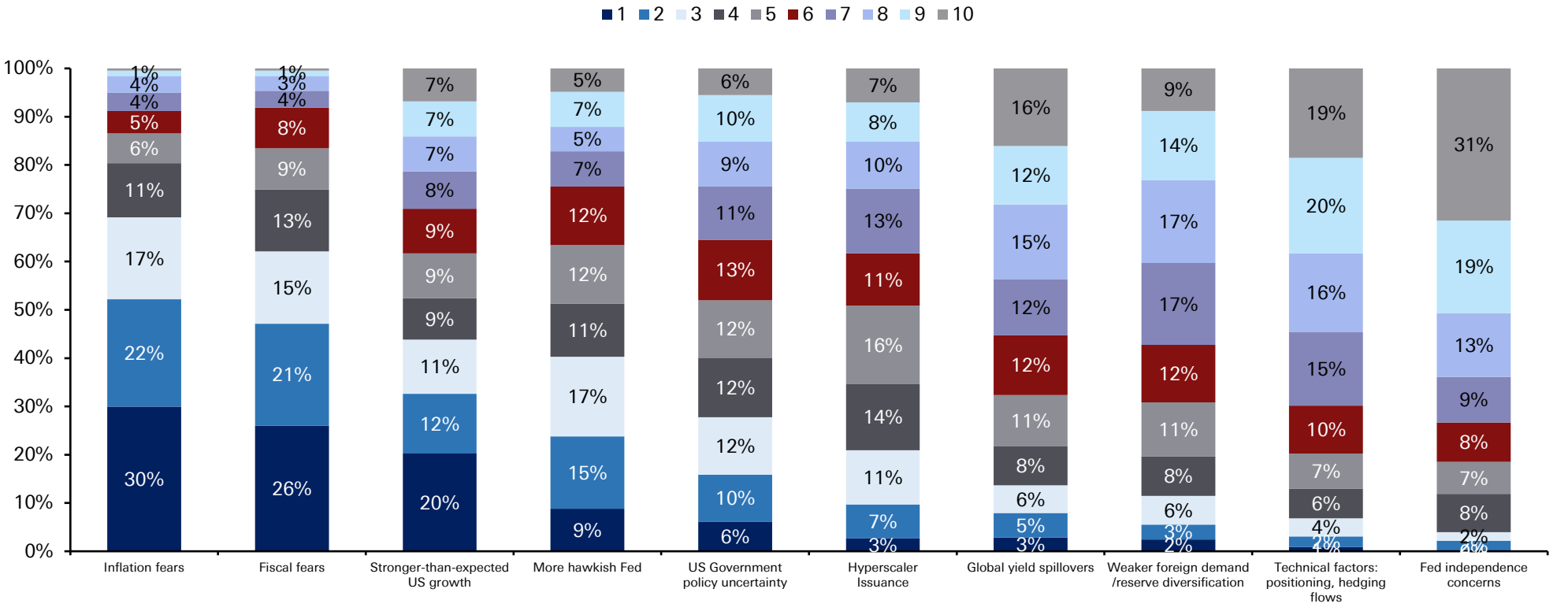
Our Q3 2026 global financial markets survey had 460 responses from around the world, conducted between September 28th and October 1st.

With the ongoing major bond selloff, we asked respondents to rank what they thought were the most important drivers. Inflation, fiscal fears, and stronger-than-expected US growth were picked as the most important drivers. 30% picked inflation as the most important driver even though 10yr US Breakevens have hardly moved in the selloff.

Thinking about the recent rise in US Treasury yields, please rank the following factors in order of importance, from the most important driver to the least important. Please drag and drop the factors into the order that best represents your view of their importance.



Source: dbDataInsights Survey, Deutsche Bank Research

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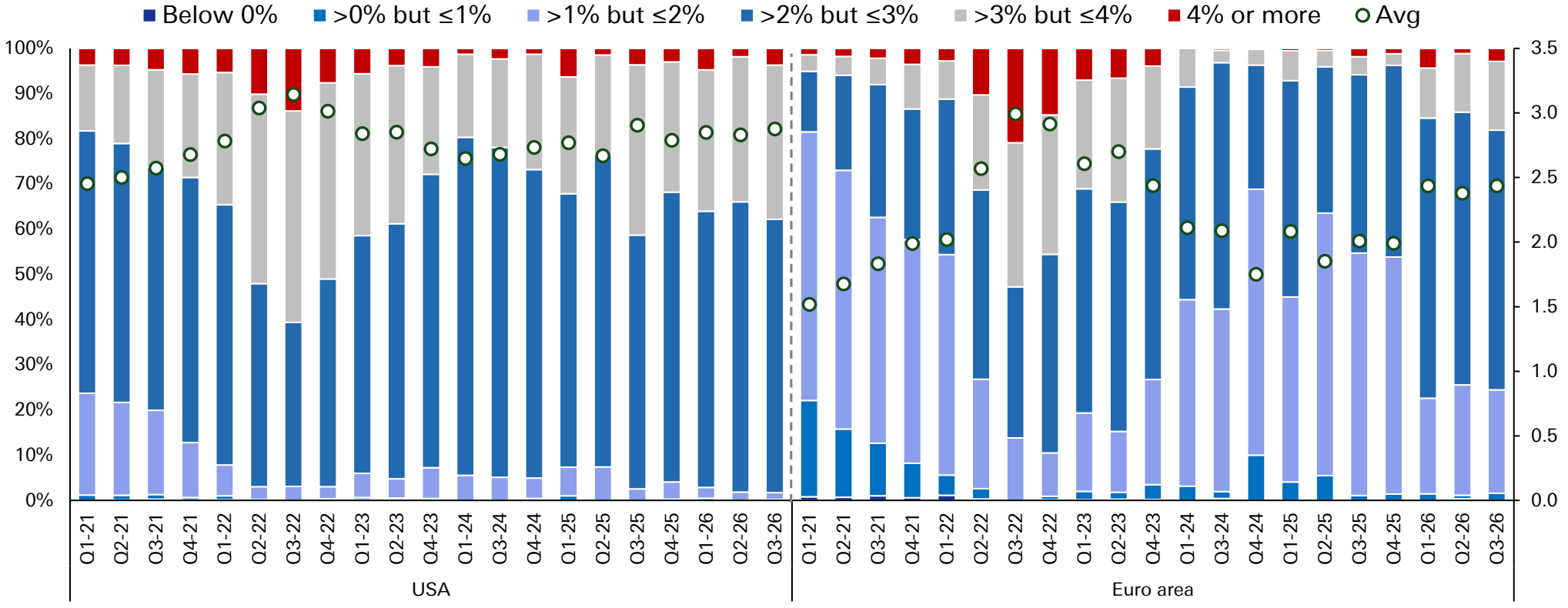
To highlight the inconsistency, US inflation expectations over the next 5 years have risen only marginally since we last asked in June, back up to 2.9% from 2.8%. 5yr European inflation expectations are unchanged at 2.4%. So while there is a strong narrative that inflation is causing the bond selloff, it doesn't stack up with long-term inflation expectations.

Looking at the below, what do you think inflation will average in the following regions over the next 5 years?



Source: dbDataInsights Survey, Deutsche Bank Research

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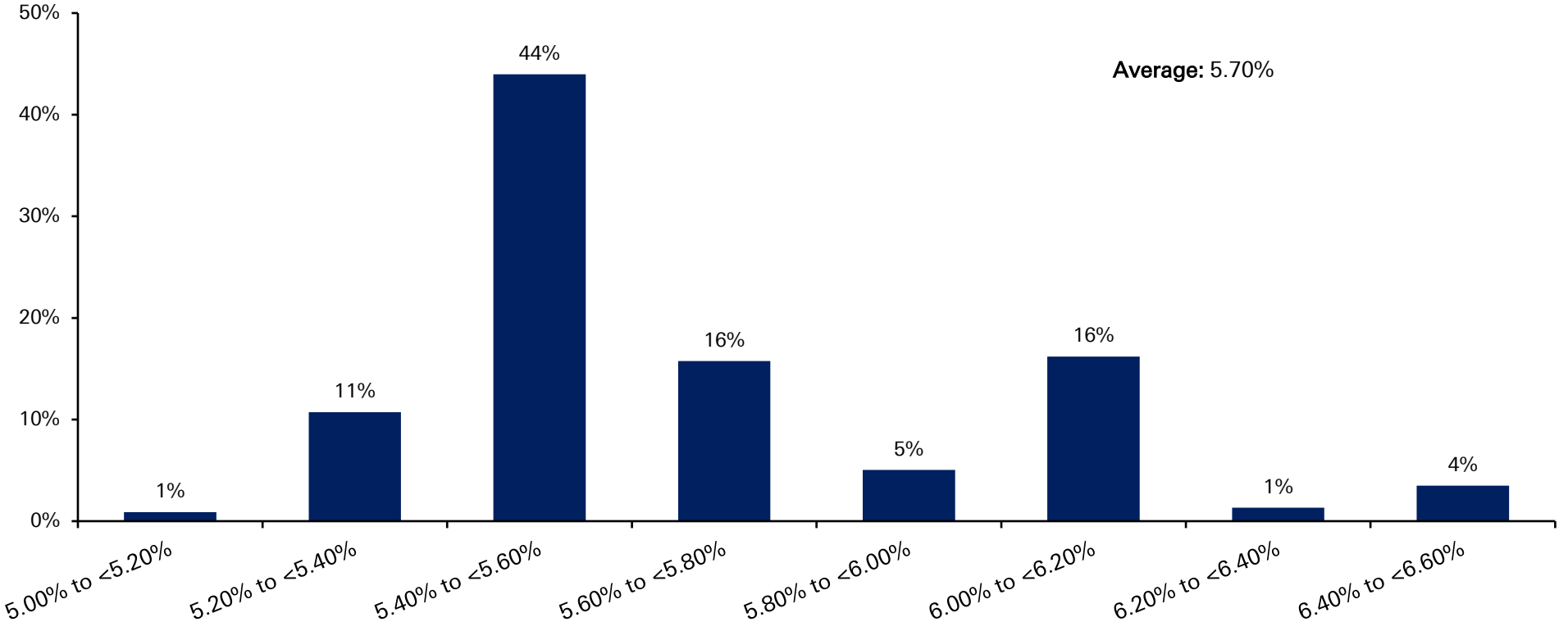
Despite US Treasury yields reaching multi-decade highs in September, most respondents expect the peak in rates to be higher over the next 12 months. The average expected peak in the 10-year yield is 5.70%, with most expecting around 5.5% but with a longish tail and a secondary peak around 6%.

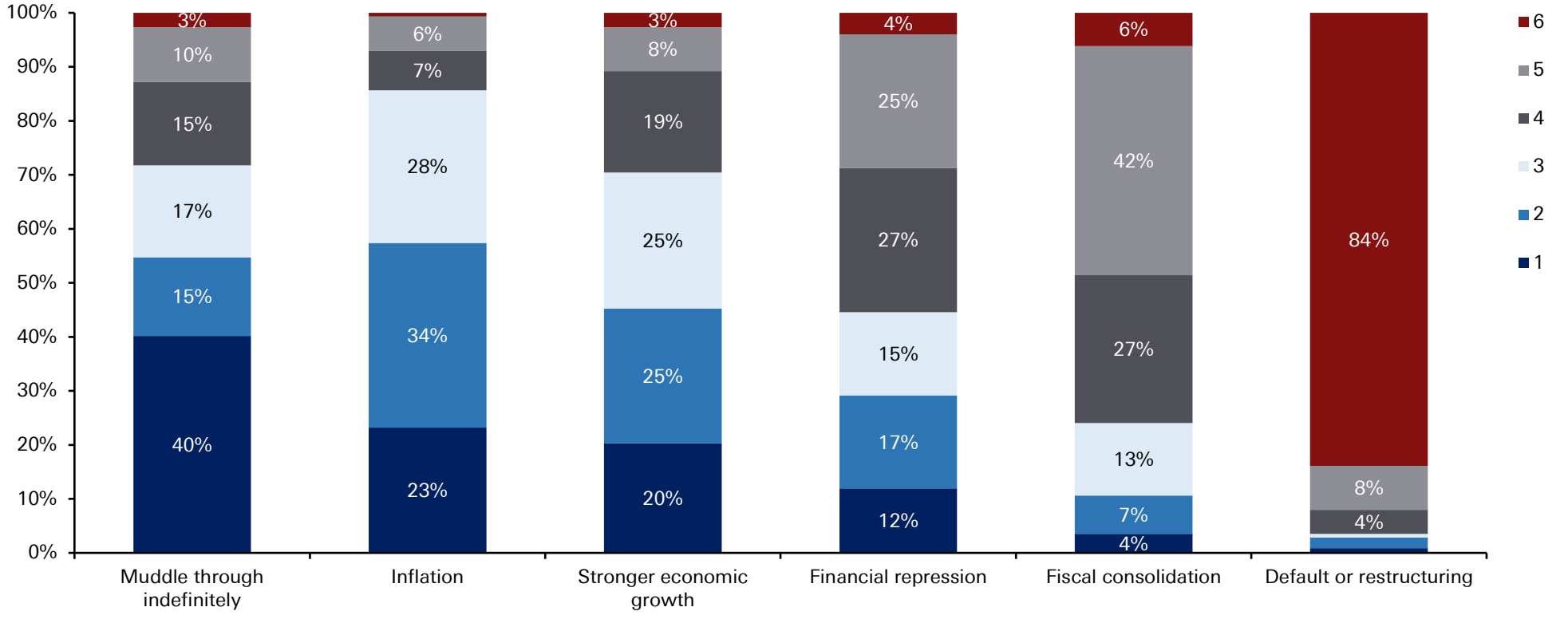
Over the next 12 months, what do you expect will be the peak level reached by the 10-year US Treasury yield? For reference, the most recent intraday peak was 5.22%.



Source: dbDataInsights Survey, Deutsche Bank Research

Interestingly, most respondents feel that the US will ultimately address its debt burden through a combination of muddling through it indefinitely, followed by inflating it away, and stronger economic growth. Fiscal consolidation is well down the list with only default lower (overwhelmingly so) as an option.

Please rank the following approaches according to how likely you think they are to be the way the US ultimately addresses its debt burden, from most likely to least likely. Drag and drop the options below, placing the most likely approach at the top and the least likely at the bottom.



Source: dbDataInsights Survey, Deutsche Bank Research

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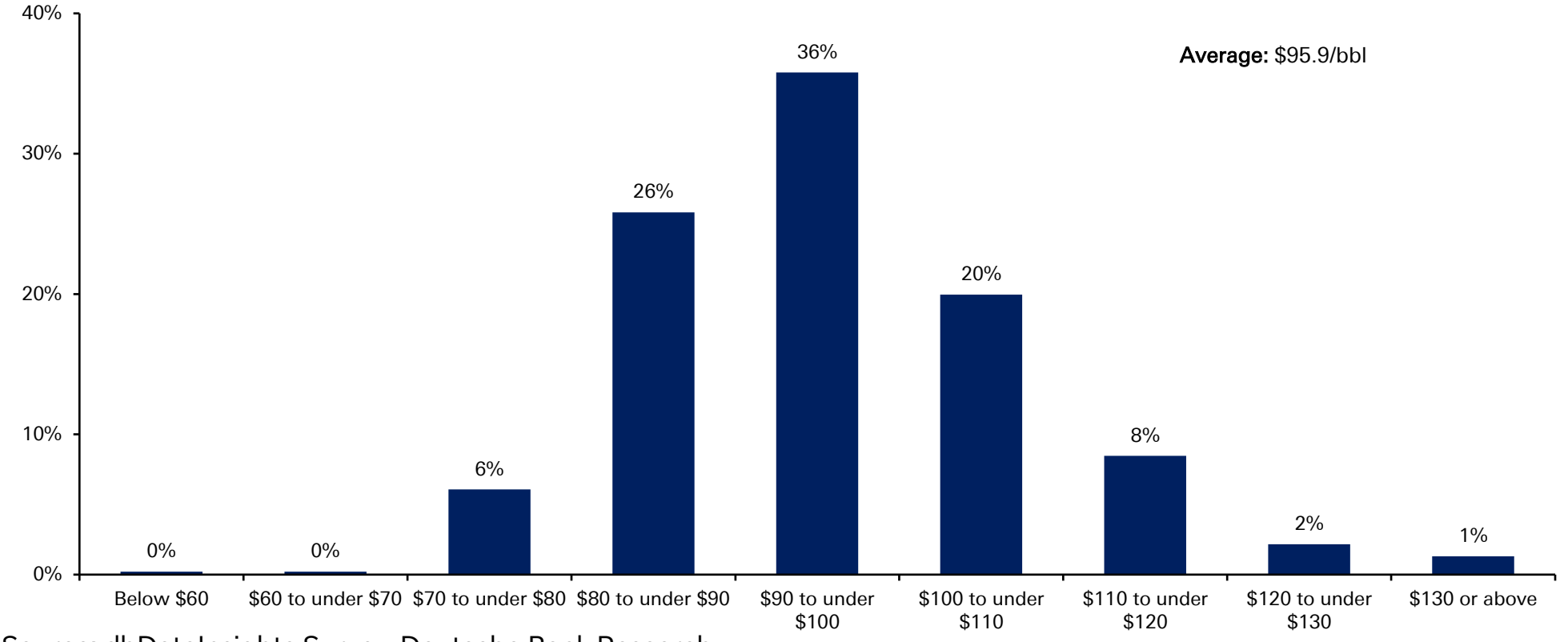
Brent is expected to be at $95.9/bbl by the end of the year, with around 68% believing it will be below $100/bbl.

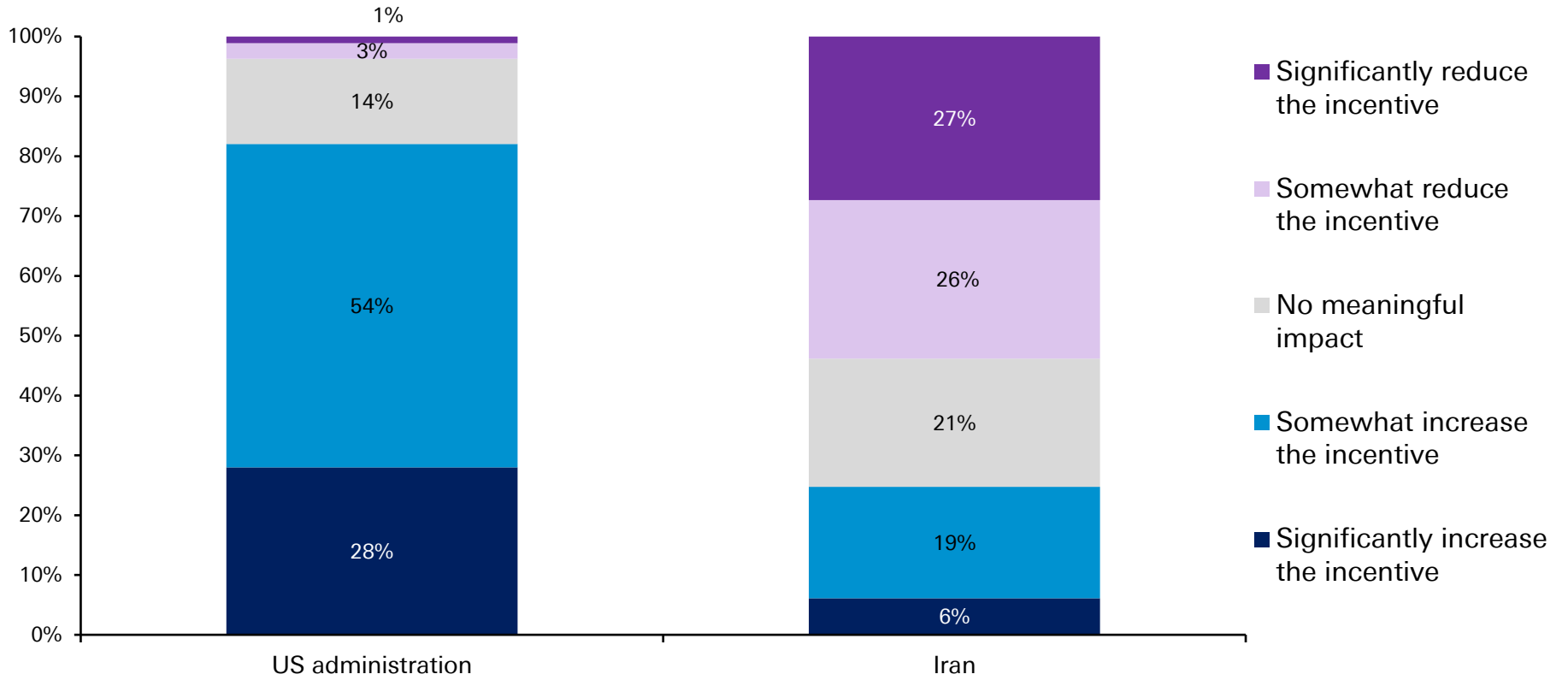
Brent crude oil prices started the year at around $60, rose to approximately $70 prior to the Iran conflict, peaked near $120, and have recently been around $105. With this in mind, where do you expect Brent crude prices to end the year?



Source: dbDataInsights Survey, Deutsche Bank Research

Ahead of the midterm elections, respondents see stronger political incentives for a deal in Washington vs Tehran. Specifically, 82% believe the vote increases the incentives for the US to do a deal vs 54% who believe the vote reduces it for the Iranians.

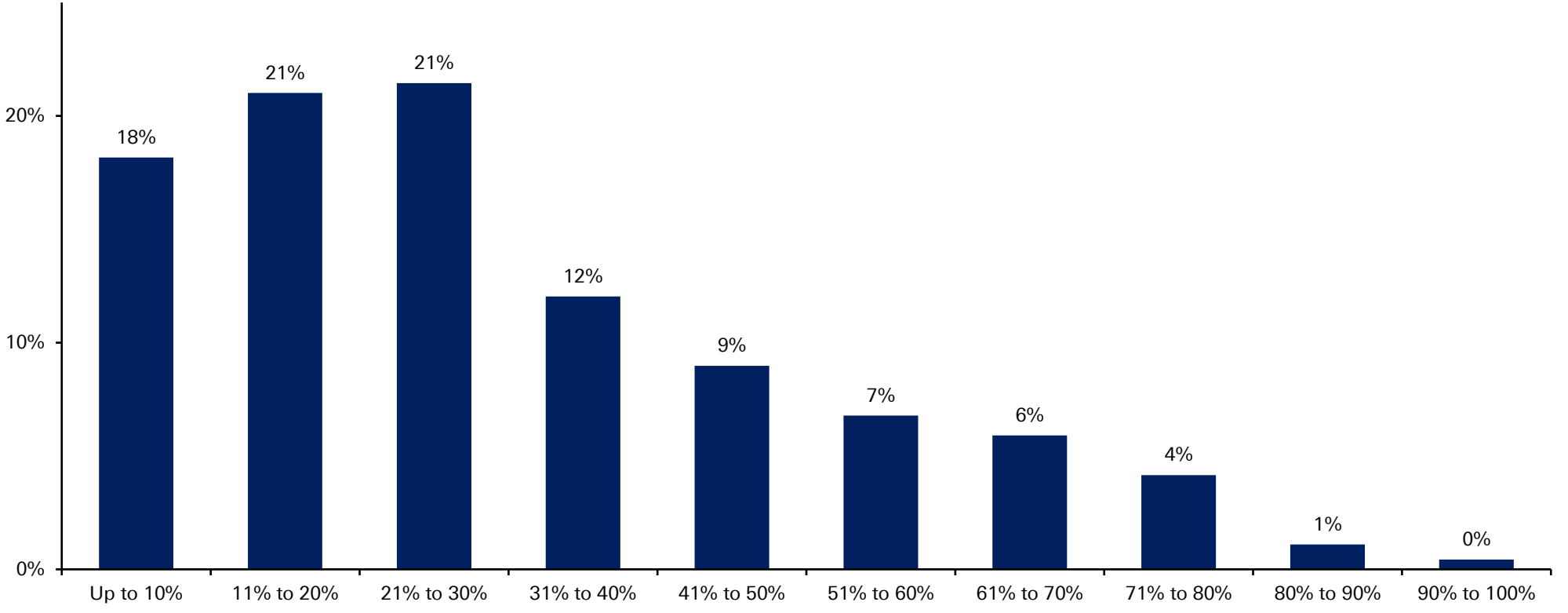
Ahead of the US midterm elections on 3 November, to what extent do domestic political considerations increase or reduce each side's incentive to reach a deal on the Iran conflict?



Source: dbDataInsights Survey, Deutsche Bank Research

The majority (82%) think the probability of a US-Iran ceasefire or peace deal happening before the midterms stands at 50% or less.

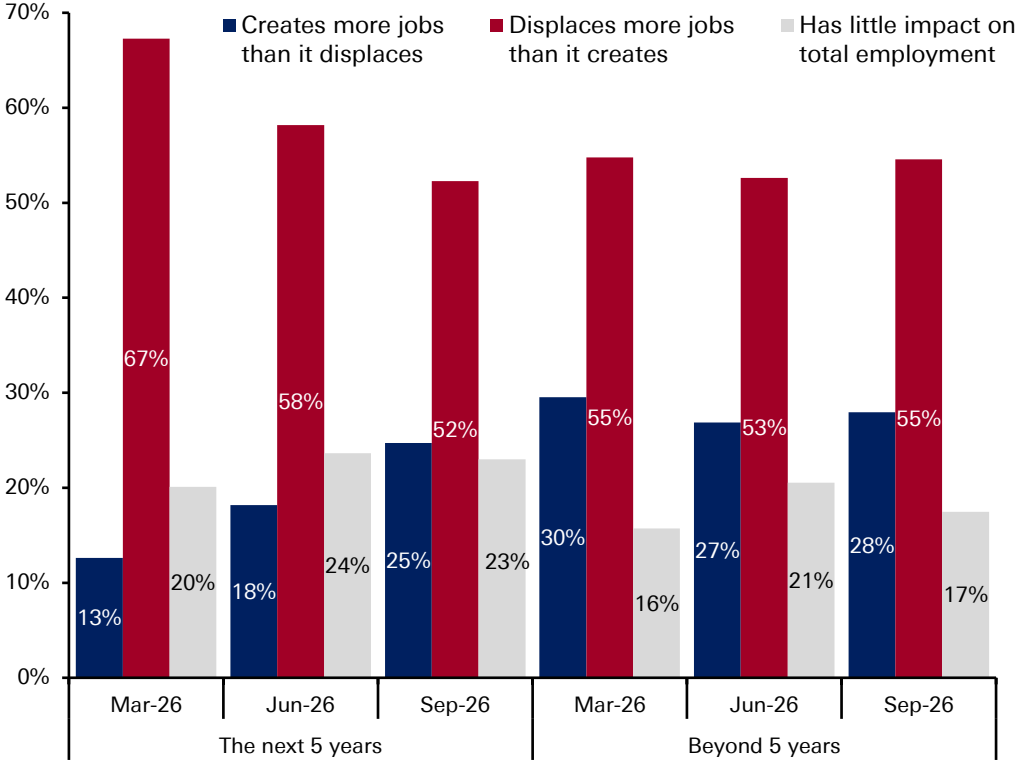
What do you think is the probability of a deal being reached, or a durable ceasefire being agreed, in the Iran conflict before the US midterm elections on 3 November? Please move the slider to indicate your best estimate. Responses will be grouped into ranges for analysis.

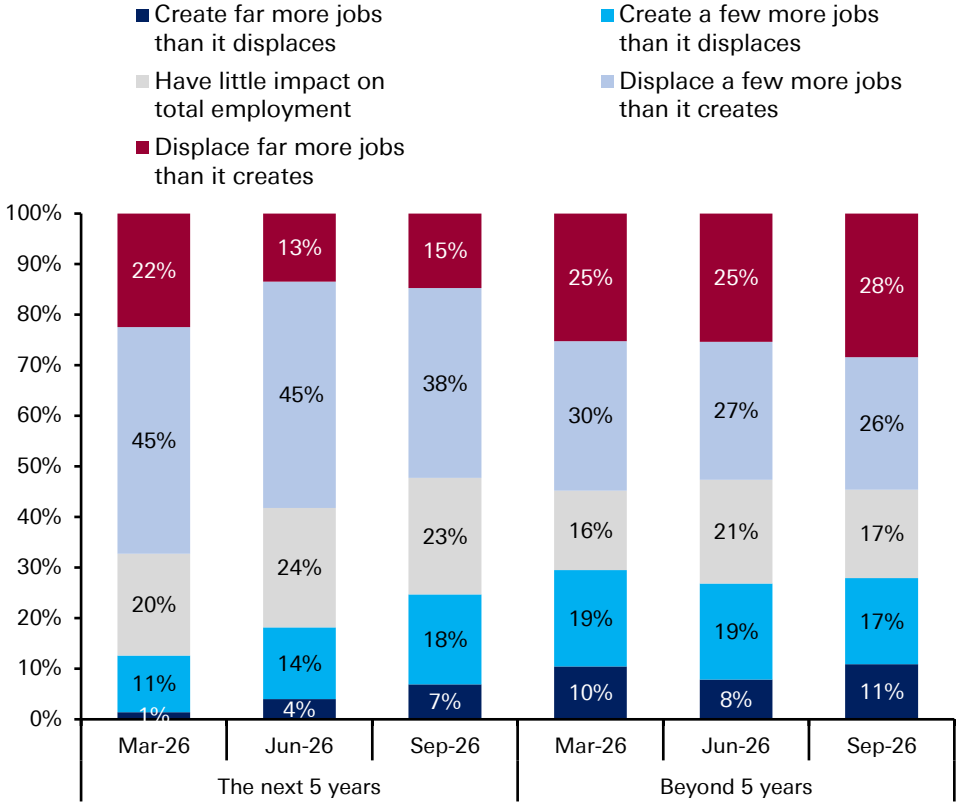


Source: dbDataInsights Survey, Deutsche Bank Research

Near-term fears of AI-driven job displacement have eased markedly since March, but longer-term expectations remain negative and stable. The share expecting net job displacement over the next 5 years has fallen from 67% in March to 52%, while the longer-term figure remains around 55%. The first chart aggregates. The second is more granular.

For the following time horizons, which best describes your expectation for AI's impact on employment? Assume a scenario where AI adoption stays on its current trajectory. AI will...





Source: dbDataInsights Survey, Deutsche Bank Research

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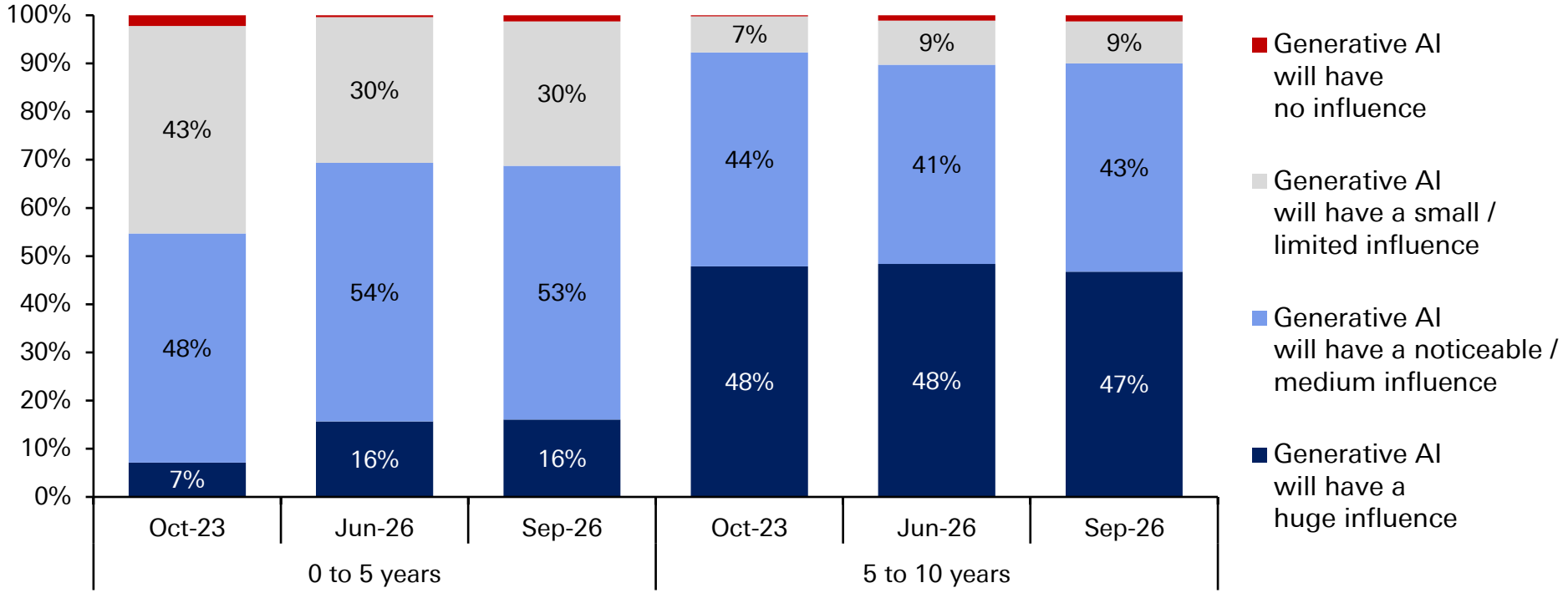
GenAI productivity growth expectations seem to have stabilised, with the strongest impact still expected over the longer term. Less than 1% believe AI will have no positive impact on growth.

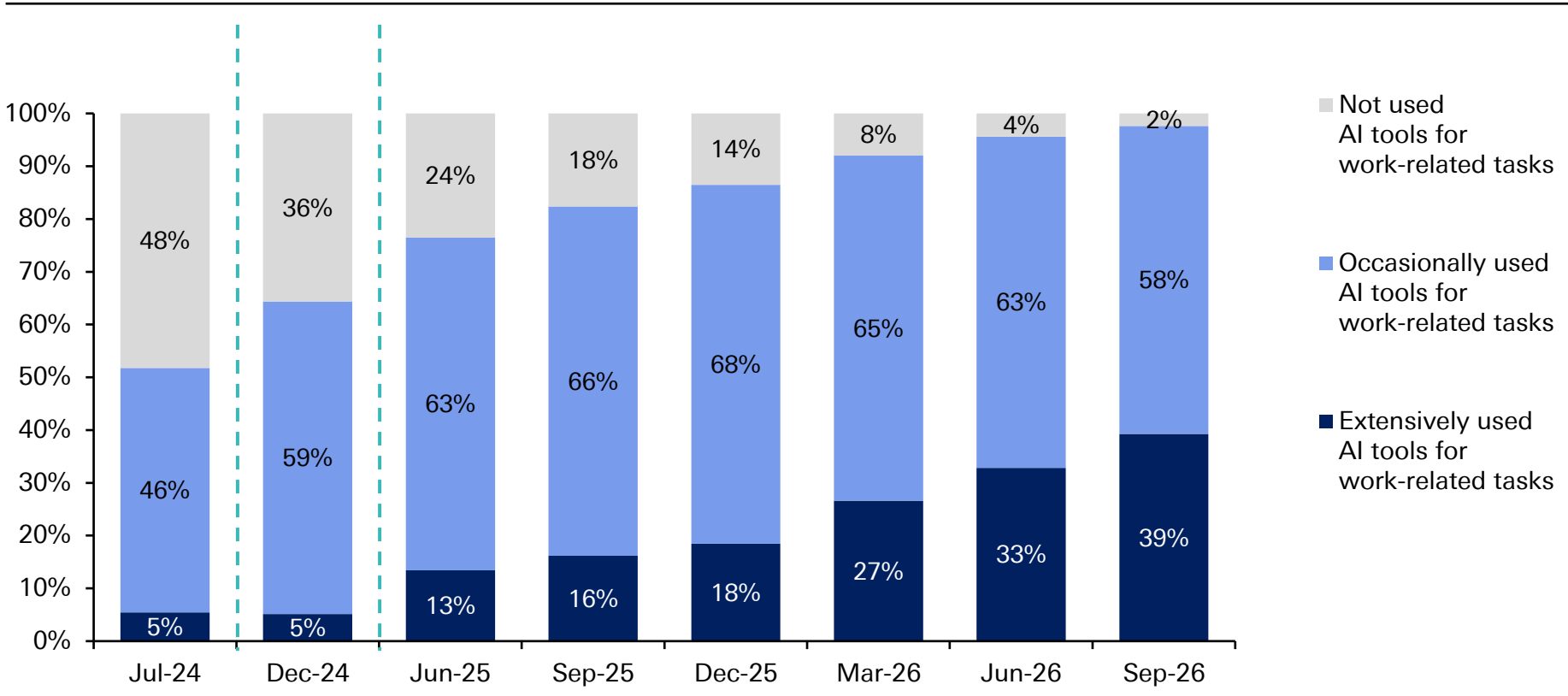
What influence, if any, do you think future iterations of Generative AI will have on productivity growth within the following time frames...



Source: dbDataInsights Survey, Deutsche Bank Research

AI adoption at work continues to accelerate over this year alone. Now only 2% have not used AI for work, compared to 8% in March and 14% last December. “Extensively used” has also risen from 18% to 39% since December.

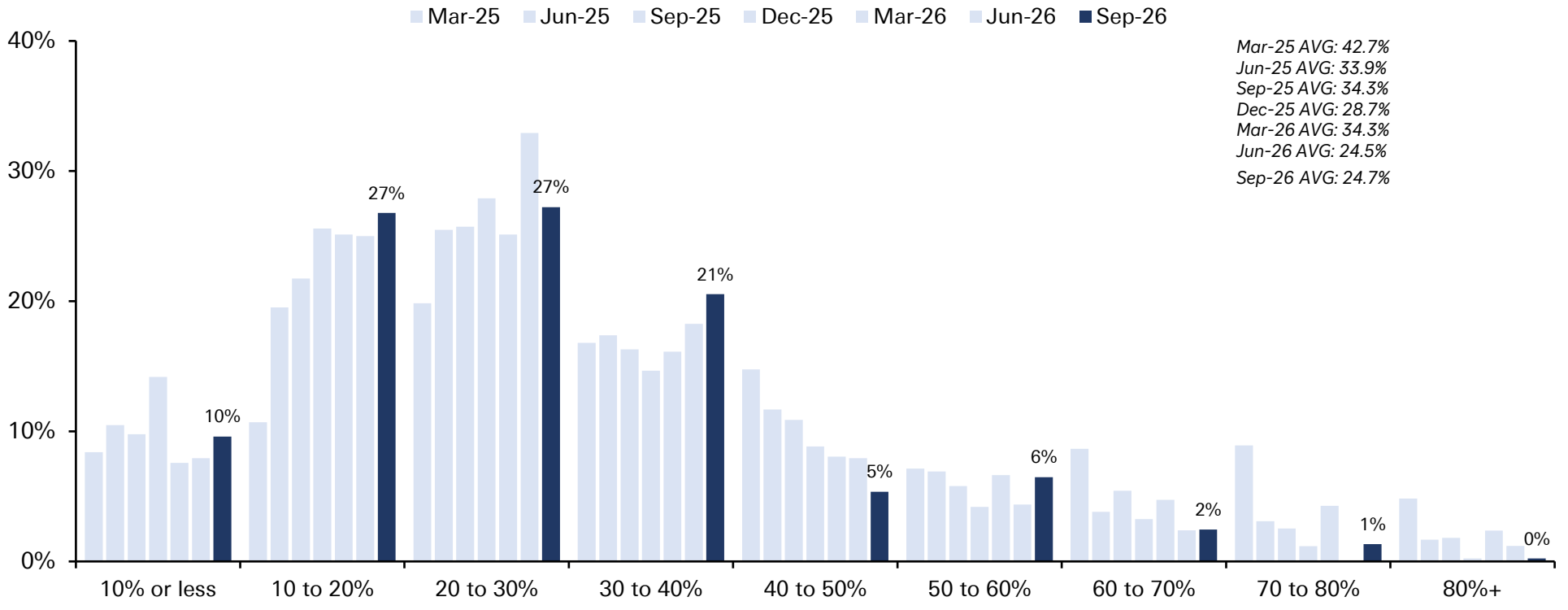
Which of the following best describes your use of AI tools for work-related tasks within the last three months? I have...



Source: dbDataInsights Survey, Deutsche Bank Research

Average 12-month US recession-risk expectations have edged up a fraction to 24.7% since June but remain near survey lows with most respondents still concentrated in the 10-30% probability range.

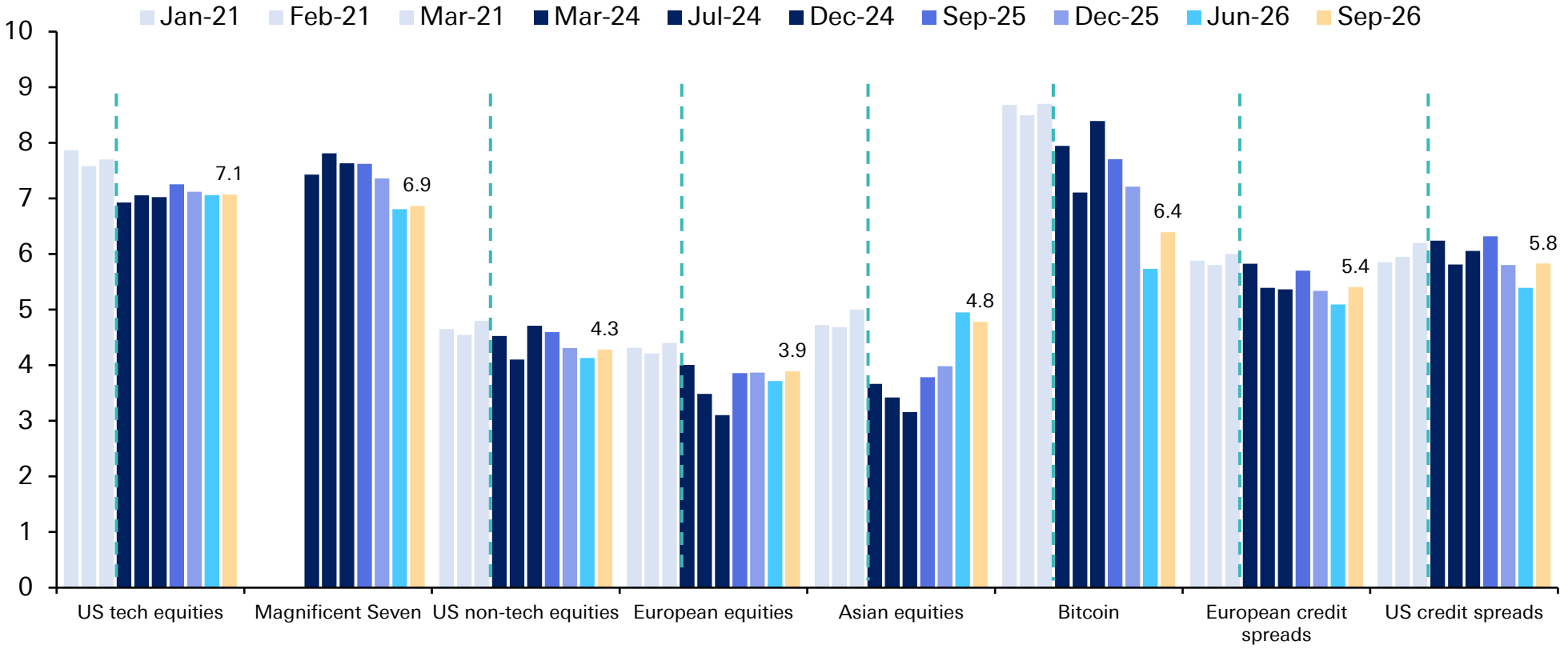
What do you think the probability of there being a US recession in the next 12 months is?



Source: dbDataInsights Survey, Deutsche Bank Research

Bubble-risk perception has risen in several assets since we last asked this question in June. The Mag 7 has inched up to 6.9 from 6.8 previously, while Bitcoin registered a bigger jump to 6.4 from 5.7. Notably, bubble-risk perceptions for both European and US credit spreads have also risen, with the US back to its December 2025 levels. European and non-US tech equities have the lowest perceived risk.

Looking at the following assets and using a scale of 0 to 10 where 0 means 'No Bubble' and 10 means 'Extreme Bubble', please tell us where you think these assets currently are?



Source: dbDataInsights Survey, Deutsche Bank Research.

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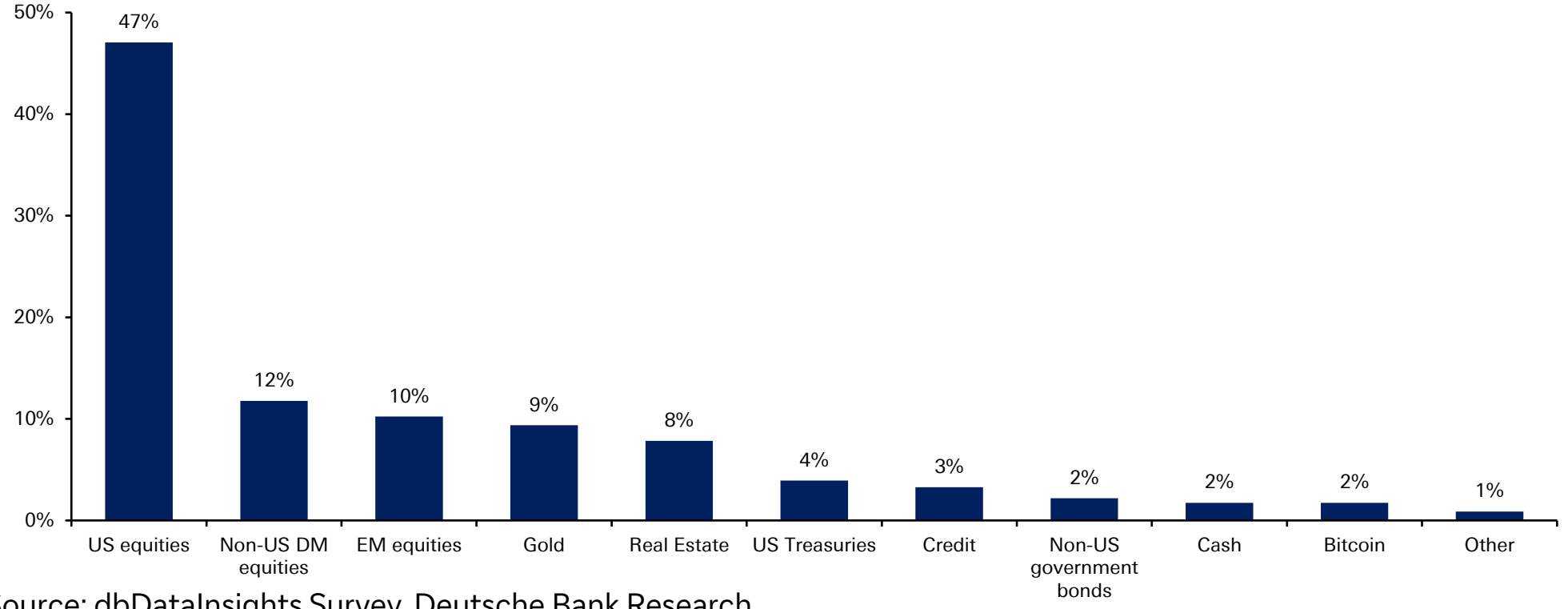
Despite the higher perceived bubble risk, nearly half of respondents would choose US equities as their sole asset for the next decade. So confidence in US equities seems to be unrivalled for long-term returns, despite market concerns of elevated valuations.

If you could hold only one of the following assets for the next 10 years, which would you choose.



Source: dbDataInsights Survey, Deutsche Bank Research

Appendix 1

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This Month in Geopolitics: **October 2026**

Key events, themes and trends to watch in the month ahead

October 2026

Dr. Helen Belopolsky and Miha Hribernik

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At a Glance: Geopolitical events to watch in **October 2026**

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides. For an overview of key events between October and December 2026, see our latest 90-day Geopolitical Risk Radar.

Source: Deutsche Bank

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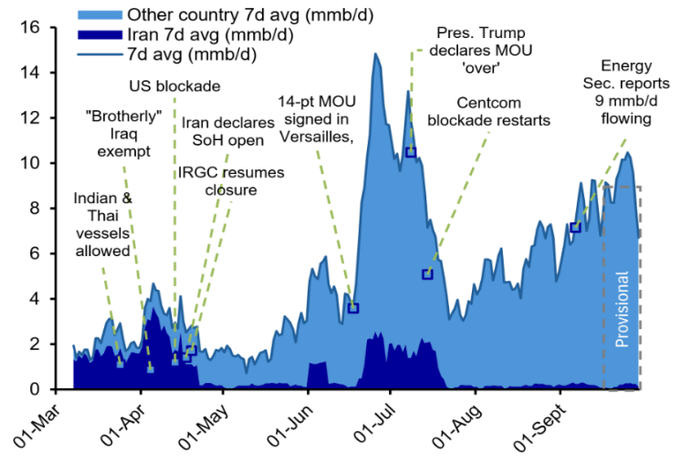
In the spotlight for October 2026: US-Iran war and EU-China relations

US-Iran war

Diplomacy appears firmly back in play. The past month saw continued fighting and renewed diplomacy. Reporting suggests that the US is prepared to extend troop deployments in the Middle East to 2027. Iran targeted both US forces in the region and some tankers moving through the Strait, but has been unable to prevent a significant rebound in crude flows through Hormuz. On 22 SEP, talks were mediated by Qatar, Pakistan and Egypt, and on 24 SEP, Iran presented the US with a seven-day roadmap to a ceasefire leading to a phased reopening of the Strait of Hormuz and a return to nuclear negotiations, conditional on a US lifting of its blockade of Iran, sanctions relief and broader cessation of hostilities. The Trump Administration rejected the proposal, but negotiations continue.

To watch will be whether October is the month the war transitions into a negotiated settlement, which may hinge on whether Iran is willing to make some concessions on sanctions relief and the nuclear question, and whether Trump is keen to demonstrate that the war has produced a political and economic result in the final weeks before the 3 NOV midterm elections. It is plausible that the current diplomatic channel could produce a temporary de-escalation mechanism as long as there is pathway around the broader nuclear issue.

Strait of Hormuz total liquids exports (mmb/d)



Source: Kpler, Deutsche Bank Research.

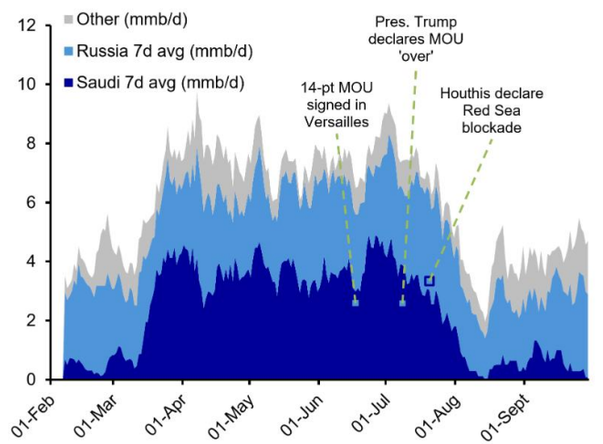
EU-China trade relations

Both sides will reconvene for trade talks in Beijing on 8-9 October, which the EU views as a deadline for "tangible results". Brussels remains focused on its rising trade deficit with China – which has reached a record €1 billion per day – and the impact of Chinese goods on European industry. The bloc has vowed to take measures if talks fail.

While both sides could achieve progress in talks, a lasting breakthrough appears improbable, setting the stage for more trade tensions into 2027. New EU trade investigations, curbs on Chinese imports, and EU-level instruments intended to address alleged Chinese overcapacity, are possible outcomes, to which Beijing has stated it would respond.

China-EU trade talks could also impact China-US relations. Should talks with Brussels fall short of averting increased European trade pressure, we assess that Beijing may further prioritise near-term US stability. See Rules of Engagement: Mapping the Next Phase of China-US Ties for more.

Bab el Mandeb total liquids exports (mmb/d)



Source: Kpler, Deutsche Bank Research. For more, see: Hsueh On Oil: Success of the shuttle system - Deutsche Bank Research

Key events, issues and themes to watch in **October 2026**

Listed in chronological order

Latvian elections (3 October)

Latvia will elect all 100 members of the 15th Saeima. Latvia, which borders Russia and Belarus, has been among the most vocal supporters of Ukraine. The next government will set the country's course on defence spending, Ukraine aid, sanctions enforcement and border security. The field is highly fragmented with the governing parties lacking a majority in the polls. The fragmentation matters as a weakened and multi-party scramble could mean a slower, more unstable coalition.

General election in Brazil (4 October)

Brazil's first electoral round is on 4 OCT, with a runoff on 25 OCT. This is a critical election as it will decide whether Latin America's biggest economy continues President Lula's state-led social spending or pivots to a more fiscally-orthodox agenda under Bolsonaro. Polls have gone from a Lula lead in the spring to a statistical dead heat between Lula and Bolsonaro. A Brazilian Supreme Court crisis – which has implicated Supreme Court Justices and politicians on both sides – is weighing on the race. Congressional and gubernatorial races will also impact the governability of the country.

This election will also shape whether the country hedges between Washington and Beijing under Lula or tilts toward closer US trade and security alignment under Bolsonaro with potential implications for tariffs, BRICS+ cohesion, and Latin America's broader positioning between the two powers.

Annual meetings of the International Monetary Fund and World Bank (12-18 October)

The meetings will take place in Bangkok, bringing finance ministers, central bankers, policy-makers and business leaders together. The agenda is expected to centre on global growth, financial stability, debt, trade and investment, development finance and the impact of AI on jobs and productivity, with particular attention given to mobilizing private capital. The key watchpoint will be the outlook for global growth and inflation, debt vulnerabilities, the future of the international financial architecture and how policy-makers respond to increasing fragmentation in trade and capital flows.

Source: Waranont (Joe) on Unsplash

Key events, issues and themes to watch in **October 2026**

Listed in chronological order

European Council meeting (15-16 October)

Leaders are expected to focus heavily on Ukraine, European security and defence, competitiveness and the EU's economic position in a more contested global environment. They will also take forward negotiations on the 2028-2034 EU long-term budget. They set themselves a soft deadline in June to get the budget broadly locked down because missing it pushes ratification into 2027 and risks a funding gap for beneficiaries from JAN 2028. To watch will be a genuine breakthrough on the multiannual framework or more likely what "substantial" progress may look like. Ukraine funding is the other pressure point. This meeting will be important for understanding the direction of European defence spending and industrial policy, the next phase of budget negotiations and the EU response to growing external economic pressure and strategic dependencies.

Alberta independence referendum (19 October)

The Canadian province of Alberta will hold a referendum, which includes a question about the province's future within Canada. This is not an independence vote itself but asking voters to choose between remaining a Canadian province and asking the Alberta government to begin the constitutional process required for a future binding referendum on separation. Current polling suggests "remain" is comfortably ahead. The significance is around potential renewed uncertainty around Canada's federal cohesion, energy policy, fiscal transfers and Alberta's relationship with Ottawa and Washington. Even a modest agreement to begin a process towards secession would likely keep alive equalization grievances as a structural fault line in Canadian federalism feeding into political risk.

Berlin Global Dialogue (20-22 October)

The event's theme focuses on "Advancing Alliances in a New Era." It will tackle how governments and businesses respond to a world where geopolitics, economic security, supply chains and technology are increasingly intertwined. There is likely to be a strong focus on the future of the transatlantic relationship, including how Europe adapts to a more demanding security environment. There will also be discussions of tariffs, supply chain resilience, critical capabilities and the weaponization of economic interdependence. It will align with Chancellor Merz's efforts to ensure Europe actively shapes outcomes around strategic autonomy and reducing China exposure in manufacturing.

Source: Bernd Dittrich on Unsplash

Key events, issues and themes to watch in **October 2026**

Listed in chronological order

Summit between the EU and the Gulf Cooperation Council (24 October)

The second summit will take place in Saudi Arabi, bringing EU and Gulf leaders together. The agenda is expected to focus on regional security and the Middle East, energy and economic security, trade and investment, infrastructure, technology and broader strategic cooperation. The security dimension will be particularly salient given the current crisis in the region. Economically, the relationship is substantial: EU-GCC trade reached €165.6bn in 2025.

To watch will be movement on the long-stalled EU-GCC free trade talks and visa-waiver requests from Qatar and Saudi Arabia in addition to the extent of the focus on Gaza reconstruction, Iran and regional security get relative to the economic agenda.

GCC states have consistently used these summits to press for clearer EU positioning on Middle East conflicts in exchange for economic cooperation. The summit comes amid at a critical juncture where Europe is recalibrating its external partnerships.

Fifth Plenum of China's Central Committee (26-29 October)

The Central Committee of the Communist Party of China will hold its Fifth Plenary Session in Beijing. This will be one of the most important events on the country's political calendar ahead of the 21st Party Congress in late 2027.

The Plenum will focus on Party governance, and investors will scrutinise signals regarding leadership changes and personnel reshuffles, institutional reforms, and political priorities for the next Congress. While fifth plenums historically focus on economic planning, the development guidelines for the 15th Five-Year Plan (2026–2030) were already adopted during the Fourth Plenum in OCT 2025, allowing the upcoming session to focus on personnel and governance matters.

Source: istockphoto

Key events, issues and themes to watch in **October 2026**

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Legislative election in Israel (27 October)

Elections will be held for the 26th Knesset, the first since the October 7th attacks and wars in Gaza, Lebanon and Iran. This is the first Knesset since 1988 to actually complete a full four-year term without early dissolution. Benjamim Netanyahu remains the dominant single figure but faces a fragmented field: Naftali Bennett's new "Together" party is polling as the main rival bloc, alongside Gadi Eisenkot's Yashar, Benny Gantz's Blue and White, and the further right Ben Gvir and Smotrich factions.

Israel's proportional system means the election is effectively about coalition formation. Key faultlines to watch: the ultra-Orthodox draft-exemption fight, how the war's legacy and hostage/reconstruction issues in Gaza factor into campaign messaging, and whether a Netanyahu-led bloc can reassemble a working majority or whether the Bennett/Eisenkot-style "anyone but Bibi" coalitions gain enough traction. The outcome will shape Israel's posture on critical regional issues, including Gaza reconstruction, Iran policy and West Bank settlement activity.

EU-Canada Summit (29-30 October)

This summit is shaping up as a particularly important juncture against a more ambitious push to redefine the relationship. During Mark Carney's SEP visit to Strasbourg, he called for a comprehensive strategic alliance with Europe. European Commission President von der Leyen went further, proposing that Canada become the EU's first "associate member".

The agenda is expected to centre on trade and economic security, defense, critical minerals, energy, digital cooperation and supply-chain resilience, building on the EU-Canada security and Defense Partnership in the EU's SAFE defense initiative. The summit will be watched for concrete progress on a digital trade agreement and broader economic cooperation.

Source: istockphoto

Key events, issues and themes to watch in **October 2026**

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G20 Foreign Ministers' Meeting (30-31 October)

The foreign ministers' meeting comes ahead of the December Leaders' Summit in Miami under the US' chairmanship of the G20 this year. Trump's G20 year has organized around four expert tracks – trade, innovation, energy and deregulation. The session is framed around economic prosperity, energy security and technological innovation, a much more transactional focus than previous foreign ministerials, which typically focused on geopolitical crisis management.

It will be a test of whether the G20 can sustain cooperation amid intensifying geopolitical fragmentation. In addition, Russia has confirmed its intent to actively participate in the meeting. This would further normalize Russia's multilateral engagement. For Europe and other partners, a key question will be the extent to which the G20 can be a forum for broader international cooperation.

Source: Lance Asper on Unsplash

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Q3 Tech Performance Review:
AI euphoria meets macro headwinds

October 1, 2026

Marion Laboure | Camilla Siazon

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China and US agree to establish AI safety channel and continue trade and military talks
Chip Stocks Tumble After AI Leaders Call for Slowdown in Development
South Korea $2 trillion stock rout breaks records as SK Hynix results disappoint
Meta unveils AI personal assistant linked to WhatsApp and Instagram
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Research Institute

SUMMARY: AI euphoria meets macro headwinds

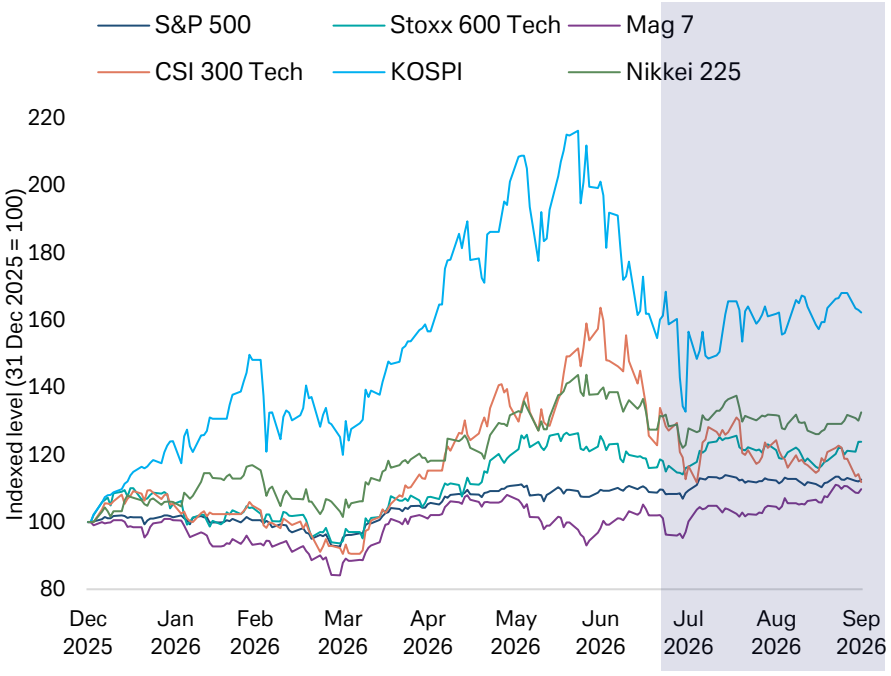
Tech remained a major driver of equity market performance in Q3. The Mag 7 outperformed the broader market while the S&P 500 also posted gains, lifted by mega-cap and tech stocks.

Still, tech's path in Q3 was volatile. July began with a reversal of the AI trade as investors questioned elevated valuations, the sustainability of AI spending and the outlook for semiconductor demand. That led to a large sell-off in chip stocks, with large falls in Q2 outperformers SK Hynix and Samsung contributing to a sharp decline in Korea's KOSPI. The Philly SOX experienced its worst monthly performance since 2008.

By August, however, sentiment had improved. The S&P 500 reached fresh record highs, even as US Treasury yields rose, supported by stronger cloud earnings and signs of AI monetization. September proved more challenging as higher oil prices, inflation concerns and rising bond yields weighed on risk assets, particularly semiconductors. The Fed also raised its policy rate by 25bp on Sep 16.

Yet, despite these macro headwinds, company guidance continued to indicate strong AI infrastructure demand, with Micron's stronger-than-expected revenue guidance on Sep 30 providing further evidence of robust demand for memory chips. Memory supply remains tight, semiconductor capex is accelerating, and the industry's attention has increasingly turned to bottlenecks in optical networking and photonics, as well as concerns around AI regulation and safety. Looking ahead, tech is likely to remain an important driver of equity returns into Q4, although elevated valuations and tighter financial conditions leave the sector sensitive to earnings disappointments.

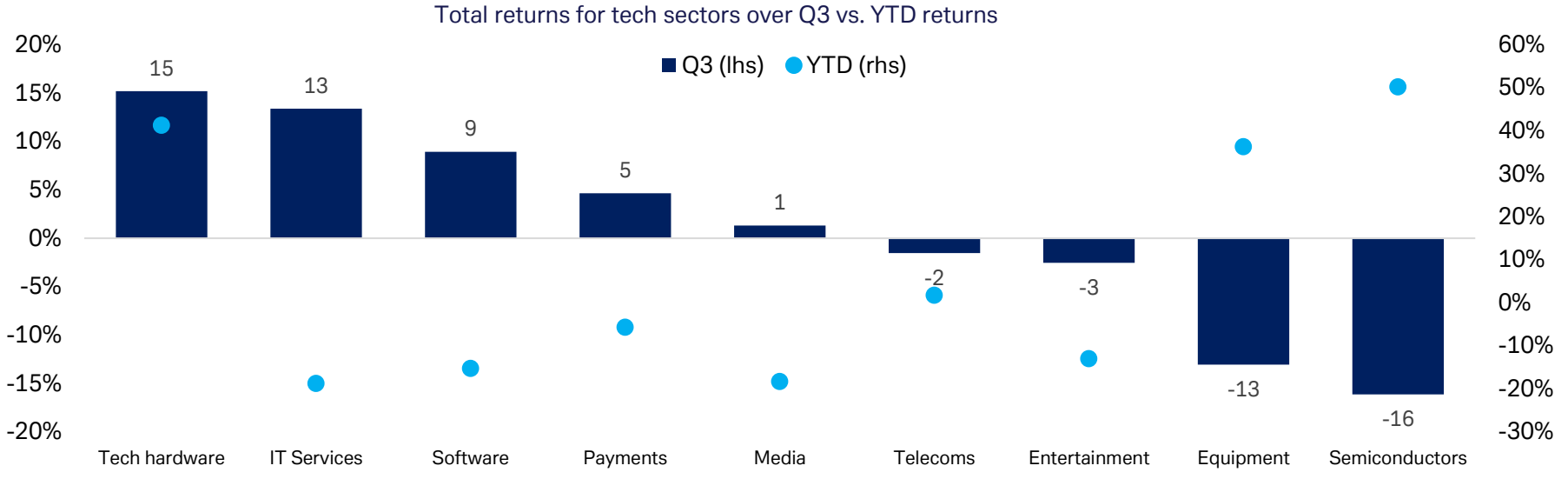
Q3 started with doubts about the AI trade but ended with renewed confidence



Sources: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30. Returns in local currency.

1. GLOBAL INDUSTRY BREAKDOWN

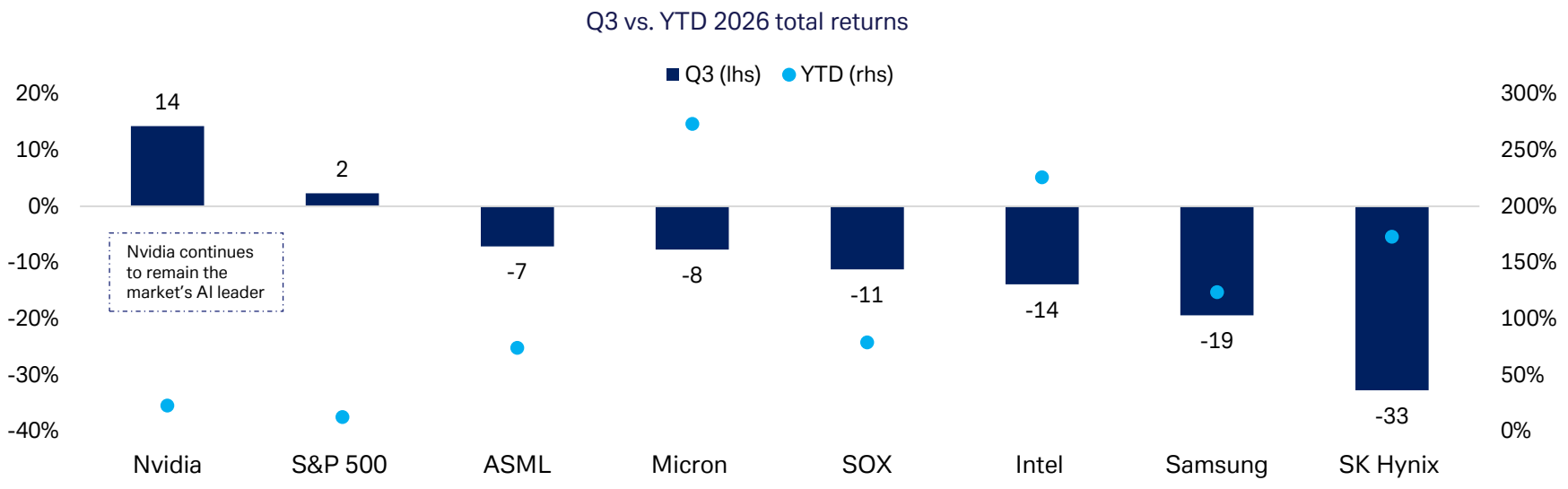
Tech hardware, including Lenovo, Advantech, and HP Inc, rebounded in Q3. This was partly driven by accelerating AI server demand and the ongoing data center build-out, although continuing memory shortages raised costs for hardware manufacturers. Cloud and server providers, like Oracle, Dell and HPE, also highlighted strong AI-related demand. Semis and equipment were the weakest performers. Elsewhere, entertainment and telecoms posted modest declines. Finally, in Q3 investor attention increasingly shifted to AI agents, with Meta's Muse, Microsoft's Copilot and Salesforce's Agentforce among the prominent offerings.



Source: Deutsche Bank Research, Bloomberg Finance LP. Median returns taken. Updated Sep 30.

2. SEMICONDUCTORS

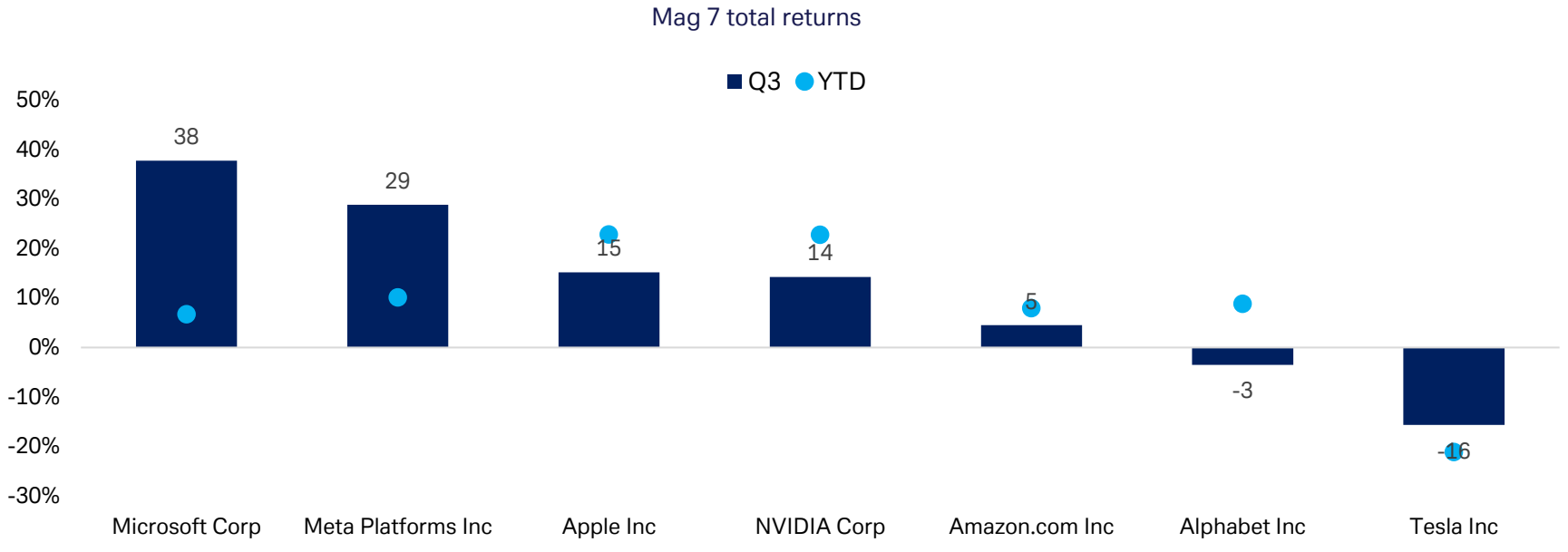
Chip stocks were hurt initially by a reversal of the AI trade and then by rising bond yields. The Philly SOX index fell sharply in Q3, after its worst monthly performance in July (-21%) since 2008. Concerns about higher oil prices and rising rates continued to weigh on the sector in Aug and Sep. SK Hynix was among the weakest performers, amid an earnings disappointment relative to expectations and an unwinding of leveraged positions. Its Nasdaq ADRs began trading in July. Despite these headwinds, the semis outlook remains supported by AI infrastructure demand. Within semis, potential long-term beneficiaries include suppliers of AI infrastructure, memory including high-bandwidth memory (HBM), advanced packaging, photonics and semiconductor equipment.



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30. Returns in local currency.

3. MAGNIFICENT 7

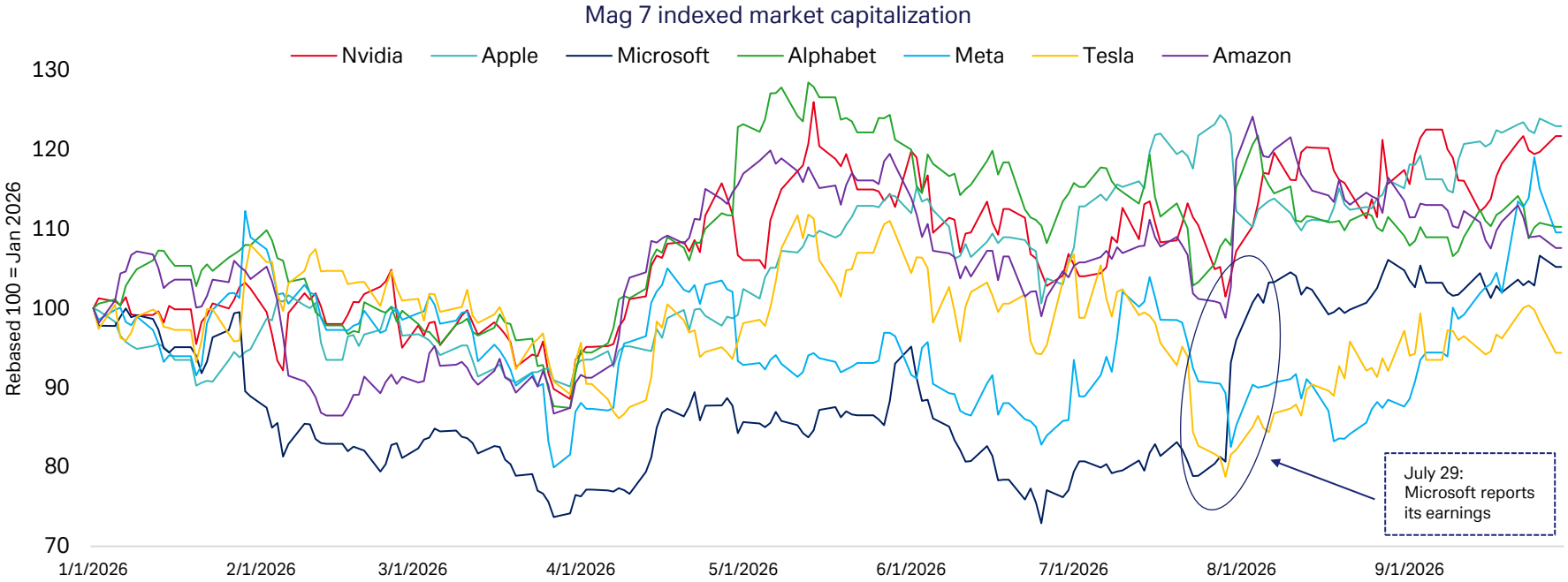
Microsoft led the group in Q3 after its Azure growth materially exceeded expectations, with strong Copilot adoption and guidance for continued positive free cash flow despite substantial capital spending. Meta also became a major focus in Sep following the launch of Muse. Amazon's AWS growth also accelerated, underscoring continued cloud demand. Tesla underperformed and remains the group's worst-performing stock YTD.



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30.

3.1 MAGNIFICENT 7: Microsoft's earnings rebound

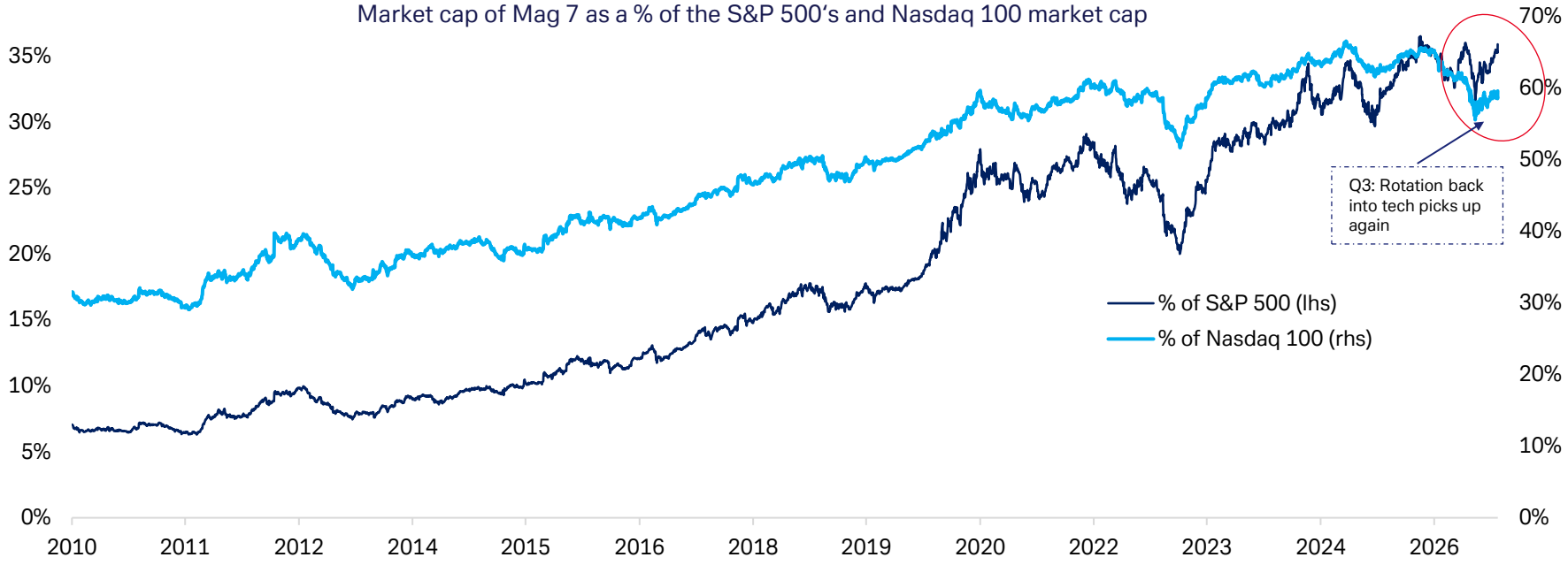
On July 30, Microsoft rose more than 15%, its strongest daily gain since 2008, following results released the previous evening. The move added ~$450bn to its market cap, the largest one-day increase on record at the time. For context, that daily increase is larger than the value of any company in Europe (except ASML).



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30.

3.2 MAGNIFICENT 7: Market concentration

According to our equity strategists, from end-July to mid-September, tech-exposed stocks within the S&P 500 rose 14% while the remainder declined 3%. This divergence highlights the concentration of market returns in the tech-exposed sector. In late July, our strategists anticipated a rotation back into technology, which remains underway.

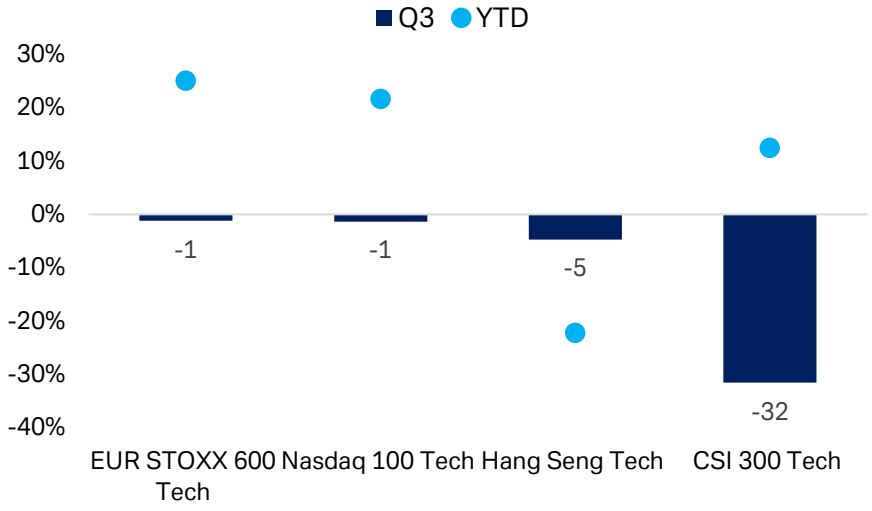


Source: Deutsche Bank Research, Bloomberg Finance LP. Updated Sep 30.

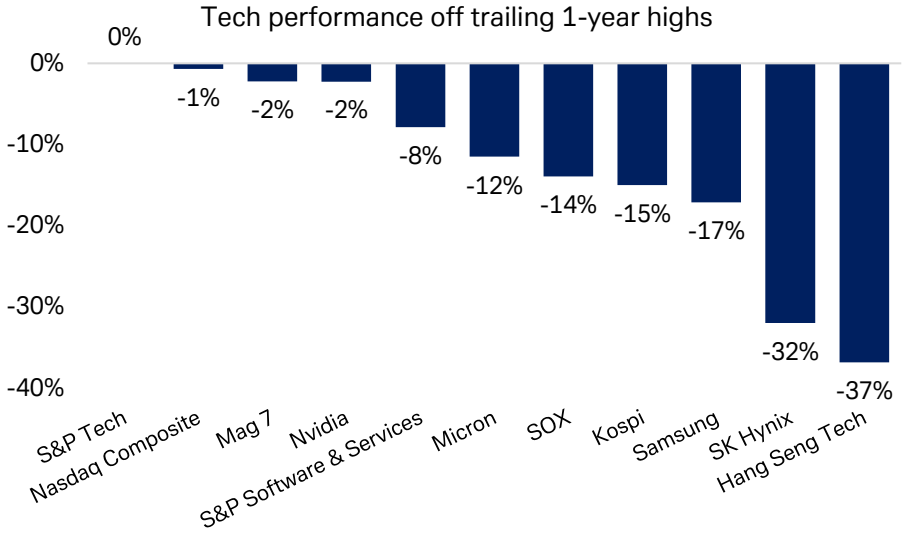
4. REGIONAL BREAKDOWN

After outperforming in Q2, mainland Chinese tech stocks posted large declines this quarter, hurt by July's sell-off. The CSI 300 tech series led the declines. Weakness also extended across Asia more broadly, with the KOSPI, Nikkei 225 and Hang Seng Tech down in Q3, driven by a chip sell-off and concerns over further Fed hikes. European tech equities were more resilient, with the STOXX Europe 600 Tech Index down slightly lower in Q3 but still well above its year-end level. Within European tech, Infineon underperformed, while Orange illustrated weakness in telecoms outside the index.

Top global tech indices – Q3 vs. YTD total returns



Notably, most of US tech is near recovery from their 1yr highs, unlike those in Asia

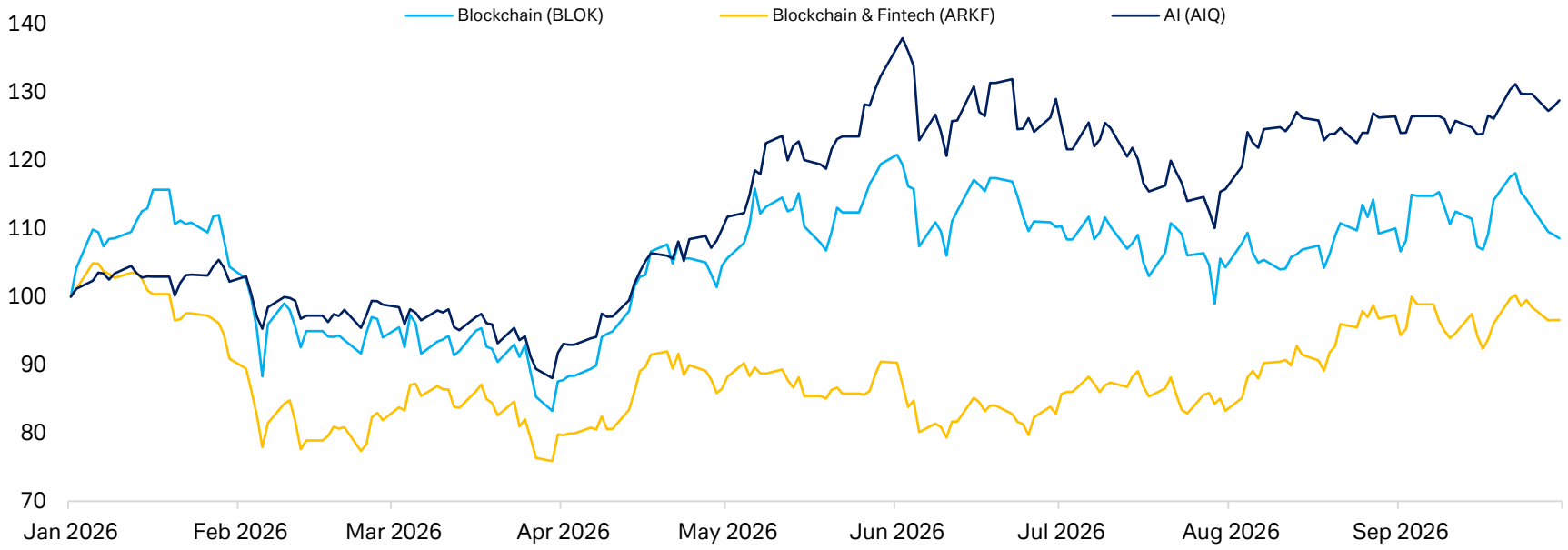


Source: Deutsche Bank Research, Bloomberg Finance LP. Data updated Sep 30.

5. CRYPTO, AI & FINTECH

Our thematic indicators also faced macro headwinds in Q3, although the fintech and blockchain ETF shown recovered during the quarter. Bitcoin also recovered strongly during Q3, trading around $83k in late Sep, despite the setback when the Senate failed to advance the CLARITY Act on Sep 15.

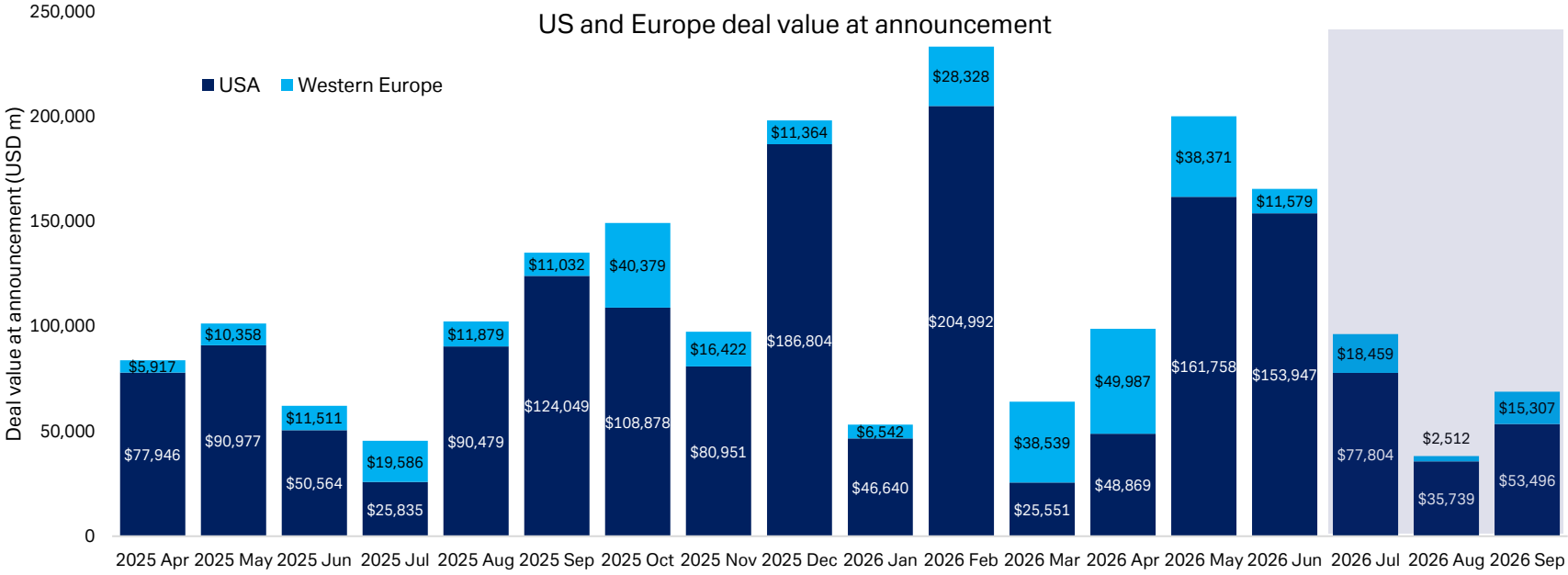
Blockchain, AI & Fintech ETF performance



Source: Deutsche Bank Research, Bloomberg Finance LP. The following are based on AIQ ETF, BLOK US Equity ETF, and ARKF US Equity ETF. Updated Sep 30.

6. M&A

Q3 saw lower announced deal value than Q2, against a backdrop of uncertainty around the upcoming US midterm elections, ongoing geopolitical conflict and higher oil prices. Separately, several companies, including Oura and Holtec, have delayed their IPOs, citing uncertainty. Still, there were some notable deals, including Nvidia's agreement to acquire Hugging Face for ~$12.93bn, announced on Sep 3, and AMD's announced acquisition of World Labs for $8.2bn on Sep 28.



Source: Deutsche Bank Research, Dealogic. Updated Sep 30.

Appendix 1

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Research

How Q3's emerging themes shape the coming 12 months

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Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
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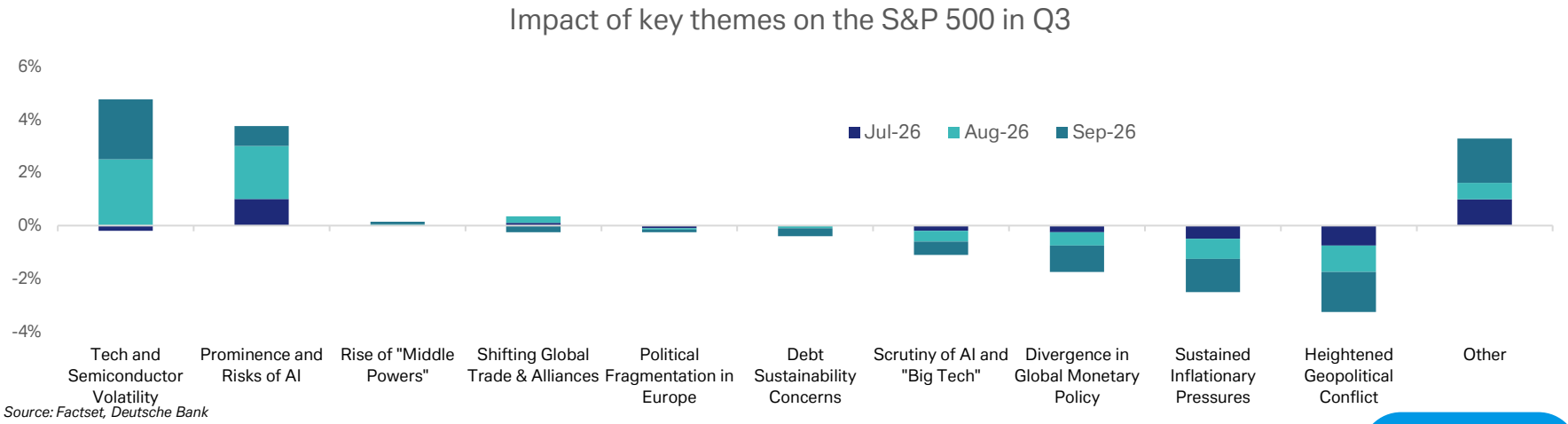
The themes that drove markets in Q3 – based on AI analysis of our daily market commentaries and asset prices

dbLumina's estimate of how each key theme in Q3 impacted the S&P 500

Chip volatility
The extraordinary moves in Korean chipmakers drove material but more muted trading in the US

AI existential risks
The AI doom discussion also raised the prominence of AI business models

Geopolitical conflict
The market has finally realised there are no easy exit routes with Iran

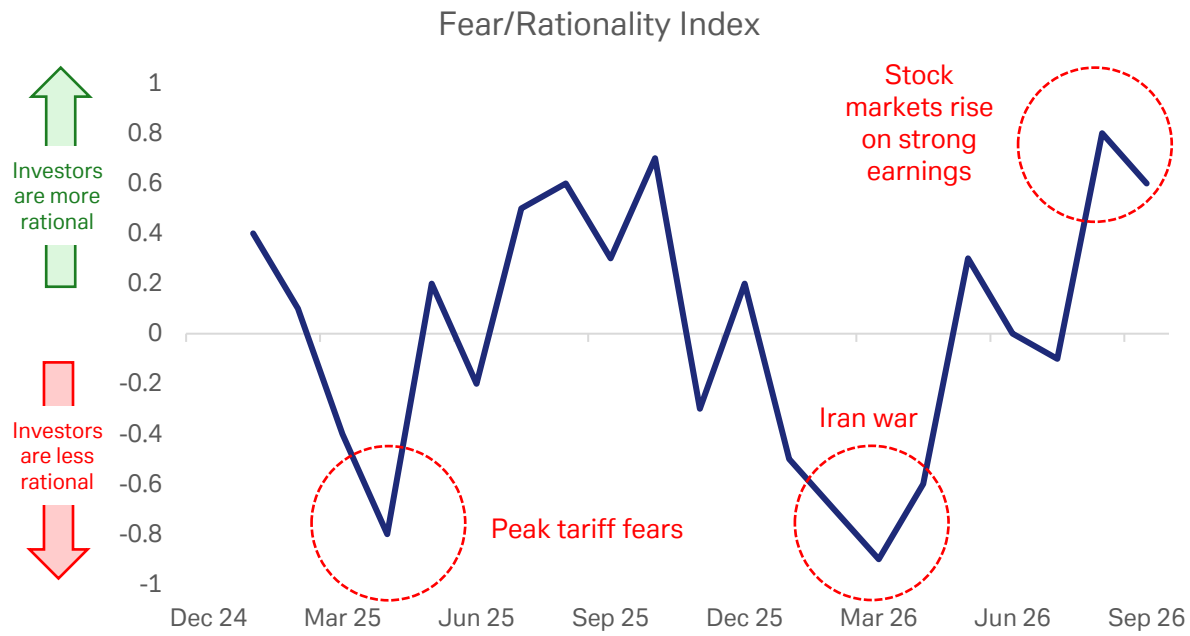


Source: Factset, Deutsche Bank

Are investors rational?

The stock market strength and resilience over the last 3 months appears to be a rational response to strong earnings and only limited signs that higher energy prices are materially affecting broad corporate profits.

Our dbLumina gauge indicates investors have been irrational during market drops and periods of geopolitical tensions



Source: Factset, Deutsche Bank

What does it mean to be 'rational'?

Risk-off periods:
The tariff fears in 2025 and then the Iran war in 2026 were key buying opportunities in equity markets. Yet, the overwhelming fear exhibited by investors in these periods largely meant they sold off risk assets in an emotional and irrational response.

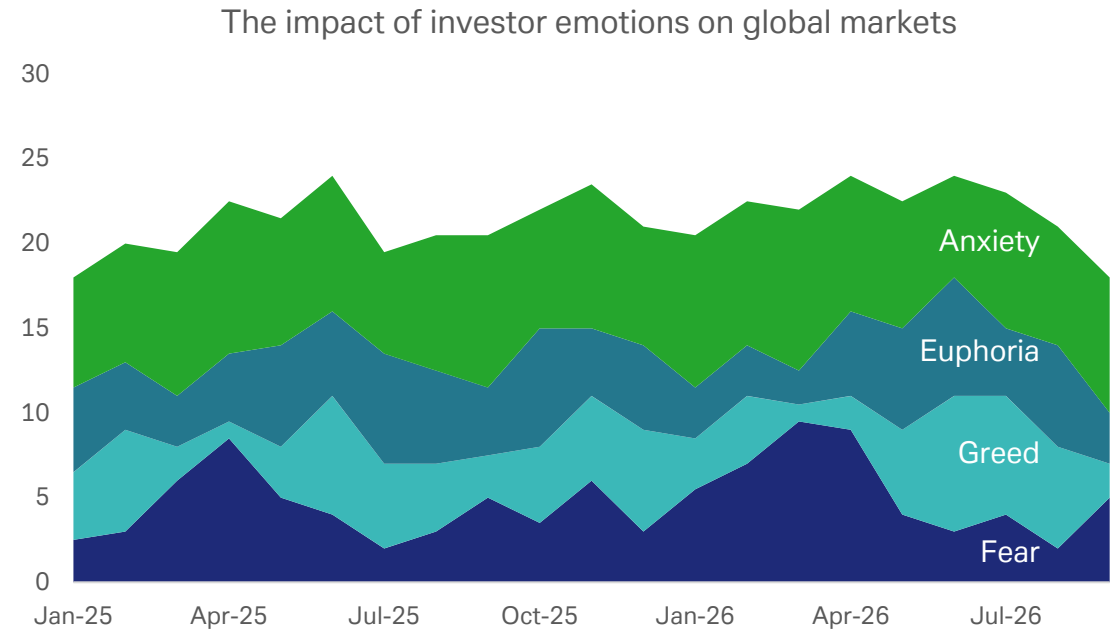
Risk-on periods:
It can be rational to buy into a rising market. Our indicator suggests this is what happened in Q3 as earnings growth continued to be strong.

Fear Indicators:
Sharp, disproportionate reactions to threats, widespread sell-offs on announcements, panic, flight to safety beyond what might be fundamentally warranted.

Emotions cooled as ‘rational’ investing took over

Lower realised volatility in markets in Q3 helped clam emotional market responses. That indicates investors are detaching from the volatile macro and geopolitical backdrop

Emotion dipped markedly in Q3



Source: Factset, Deutsche Bank

Month-by-month

July 2026: A fear-driven sell-off in semiconductor stocks, particularly in Korea, was balanced by optimism from strong earnings reports from major tech firms and a significant drop in oil prices.

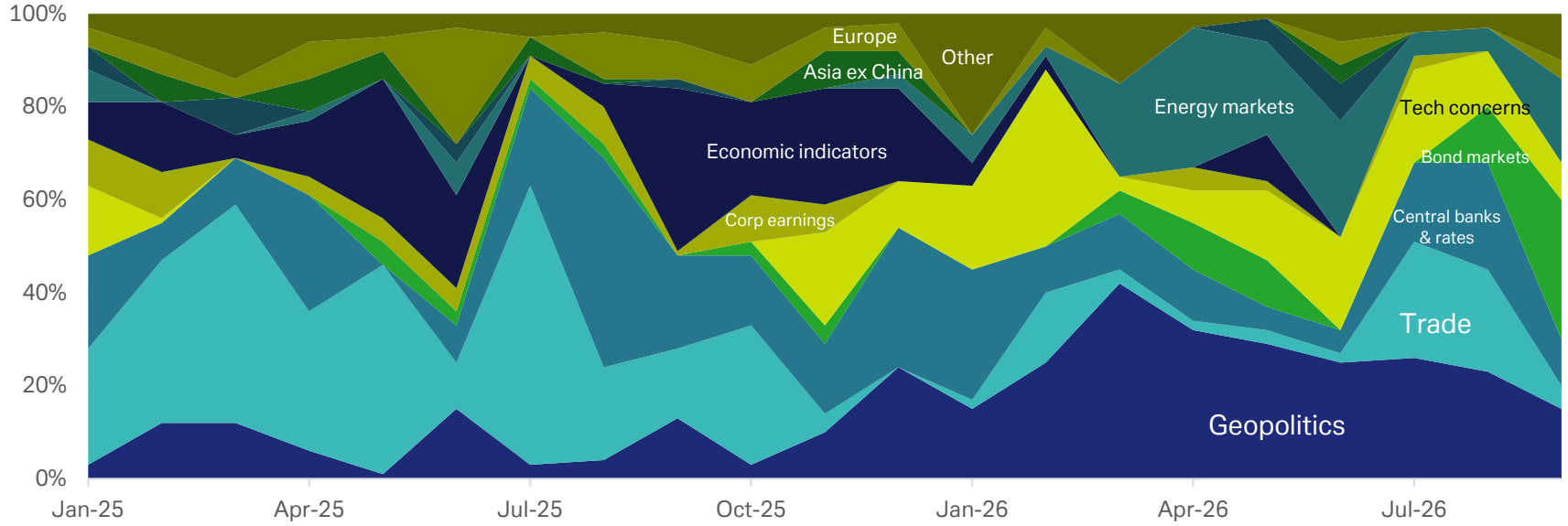
August 2026: An easing in inflation pressures lowered expectations of Fed rate hikes and supported a rally in Treasury markets. Strong corporate earnings season and signs of de-escalation in the Middle East

September 2026: Rising energy prices and renewed uncertainty in the Middle East fuelled a sell-off in global sovereign bonds. Concerns about debt sustainability and hawkish central bank commentary weighed on equity markets despite some encouraging economic signals.

Our AI analysis shows how groups of themes have driven markets over time

Tariffs & 'Liberation day' | Fears of tariff effect on economies | Trade deals propel markets to new highs | Tariff fears subside; focus reverts to 'normal' factors | AI & software worries | Iran war | Tech rebound | Oil jumps + bond sell-off

The themes that impacted markets each month

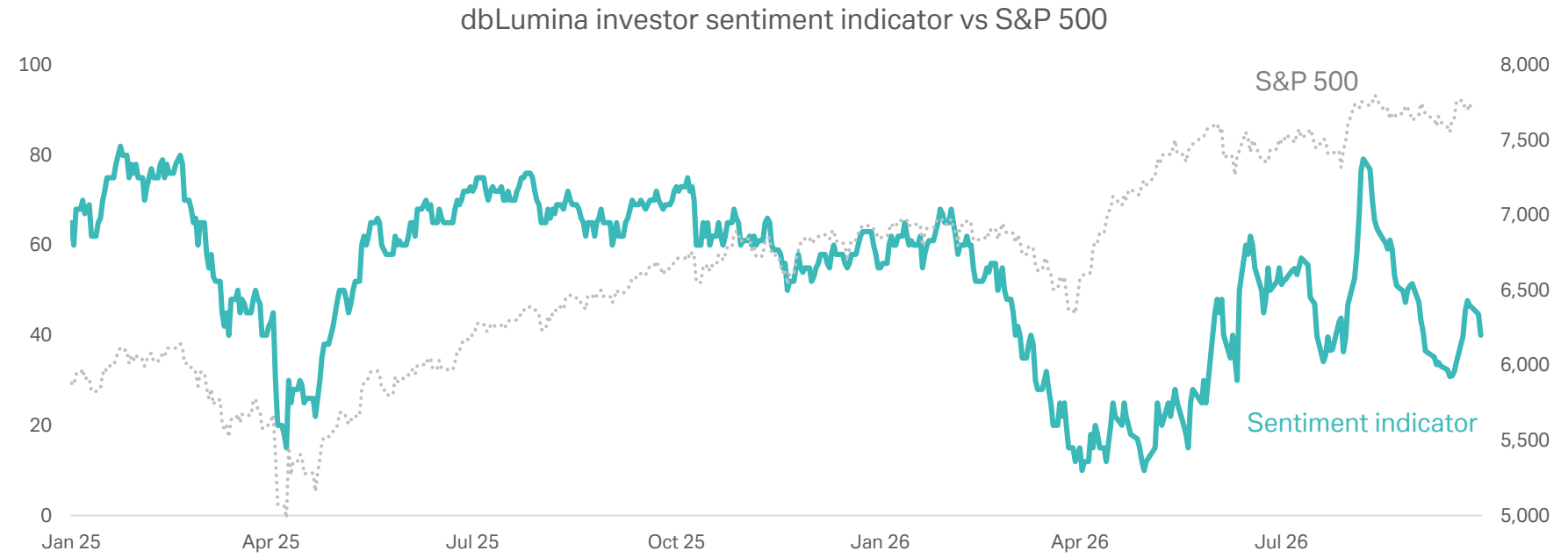


Source: Deutsche Bank

Our gauge of investor sentiment uses AI to interpret macro, economic, and market signals

This is our response to the fact that so many confidence indices today fail to reflect technological and social megatrends that now affect markets. Our sentiment indicator is nowhere near as bad as some others.

The volatile sentiment in Q3 drove some heady jumps and drops in AI-linked equities and, thus, broader markets. A positive sign is that sentiment has dropped below last year's average even though the S&P 500 is broadly steady



Source: Factset, Deutsche Bank

Evolving risks: AI can be more objective than a human. Thus, we ask it to analyse our daily market commentaries and asset prices to determine the top broad-based risks in markets

The top-4 are obvious; but sudden market dislocations happen when non-obvious risks arise. Risks 5-10 are acknowledged but simply not priced in

1 Stagflationary shock from energy prices & geopolitics

2 Sovereign debt crisis & contagion

3 Escalation of the US-Iran conflict

4 AI & technology sector dislocation

5 Central bank policy error

6 US-China geopolitical & trade re-escalation

7 Yield curve inversion & recession signal

8 Fracturing of Western alliances (NATO/G7)

9 Cybersecurity & critical infrastructure attacks

10 US political instability & government shutdown

Emerging risks underestimated by the market – part 1

1 Geopolitical fragmentation of global reserve assets

A reallocation of global sovereign reserve assets is accelerating, spurred by geopolitical uncertainty and the fear of trade and financial sanctions. Notably, central banks are repatriating their physical assets, such as the Dutch and French central banks which have recently moved gold reserves out of New York. Furthermore, a World Gold Council survey reveals that 45% of global central banks plan to increase their gold reserves over the next 12 months.

What the market underestimates: The market views the recent surges in gold prices as temporary, cyclical safe-haven flows. In reality, sovereign institutions are structurally preparing for a more fragmented global trade system where tariff policies and sanctions may make dollar-clearing assets more difficult.

Impact on key factors in our megatrend model:

Interest rates: At a time when the "world is awash in debt", long-term interest rates of some countries may be forced structurally higher to attract alternative private capital.

Inflation: Unexpected dollar depreciation may increase the cost of imported goods

Equity Prices: High-growth US equity valuations, which are highly sensitive to the long-term risk-free rate, may face multiple compression if bond yields are pushed higher.

2 The AI “sameness“ trap and the productivity stagnation

The market currently prices in a material, long-term GDP boost from AI-driven productivity gains. However, some research from the IMF suggests that over-reliance on standardised LLMs can produce a high degree of "sameness" and uniformity in cognitive output. Furthermore, escalating intellectual property and copyright litigation threaten to bifurcate the quality of AI models based strictly on developers' licensing budgets.

What the market underestimates: Some investors assume that the exponential curve of AI hardware capabilities translates directly into exponential economic output. But there is a risk that if AI tools merely standardise corporate processes rather than innovate them, and legal barriers restrict access to premium data, the productivity boom may be less aggressive than expected.

Impact on key factors in our megatrend model:

GDP growth: Structural growth ceiling if AI fails to deliver the multi-percentage point boost to global productivity required to offset aging demographics.

Equity prices: A downward revaluation of the technology and software sector if massive AI infrastructure expenditures fail to generate revenue according to existing expectations.

Emerging risks underestimated by the market – part 2

3 The fragmentation of the EU's political centre

Advanced economies are facing a dual demographic crisis: rapidly aging populations and underfunded pension systems. This is forcing some older people to remain in or re-enter the workforce out of financial necessity rather than choice.

What the market underestimates: Lower unemployment rates are celebrated as a sign of economic strength. However, in structurally ageing societies like Japan and parts of Europe, the growth of the labour force is increasingly driven by part-time, older workers. Furthermore, G20 nations lack adequate, cohesive policy frameworks to address these long-term demographic fiscal drains.

Impact on key factors in our megatrend model:

Inflation: An older workforce working out of necessity has less discretionary spending power, potentially locking economies into persistent, structural demand-side deflationary forces, offset by supply-side wage pressures in highly skilled sectors.

GDP growth: Labour productivity growth that is lower than it might otherwise be as the median age of the workforce climbs, limiting long-term potential GDP.

4 Critical mineral isolationism and 'green' supply chain blockades

The geopolitical battleground is focussed on energy blockades (such as the Strait of Hormuz) but underestimates the weaponisation of critical minerals. China currently holds 'escalation dominance' over rare-earth mineral supply chains. The emerging risk is the proactive restriction of these minerals to Western firms, such as recent bans on supplying rare-earth inputs to some US firms.

What the market underestimates: The market treats rare-earth supply constraints as temporary trade spat negotiation tactics. In reality, as Western nations aggressively push for reshoring and friend-shoring, China is increasingly using its processing monopoly as a long-term strategic blockade to try to slow Western high-tech, defence and the energy transition.

Impact on key factors in our megatrend model:

Inflation: Upwards pressure on inflation as the cost of manufacturing electric vehicles, semiconductors, and advanced electronics surges due to supply blockades and the high capital cost of rebuilding localised processing facilities.

Interest rates: Central banks may be forced to keep policy rates higher to combat the persistent supply-side inflation generated by supply-chain fragmentation.

Emerging themes – part 1

Our dbLumina AI model helps us identify the key emerging themes that have the potential to most impact markets and economies in the short term

1 AI's impact on geopolitics and national security

While the previous focus was on the AI hardware squeeze and commercial competition, the narrative is shifting towards AI's role as a critical component of national security and geopolitical leverage.

- **AI access:** We see this in President Trump's framing of AI development as a geopolitical race ("whoever wins AI wins").
- **AI as a bargaining chip:** Several examples over the last 12 months including the 2025 deal whereby AMD agreed to pay 15% of sales from certain AI chips to the US government in exchange for an export license. Even if these deals change, the trajectory is one where access to AI technology is tied to national economic and security interests.
- **Direct military & security applications:** Several examples including, cyberattacks on critical infrastructure (such as a UK power plant), the use of AI in developing "exceptional" phages to fight E.coli, and the growing field of AI security discussions.

Outlook: Expect more government intervention, the use of AI technology access as a tool in diplomatic negotiations, and a greater focus on the dual-use nature of AI in both commercial and military contexts. This will likely create regulatory and operational uncertainty for tech firms.

2 The fracturing of multilateralism and the rise of "Middle Power Countries"

The trend of reshoring is evolving into a more proactive diplomatic strategy where nations are forming new, often transactional, alliances to navigate a world less dominated by a single superpower.

- **Proactive alliance building**: Canadian Prime Minister Carney has been front and centre making his first international trip to France and the UK in an effort to diversify away from traditional reliance on the US. That follows other key meetings between Japan and India, and more.
- **Bypassing traditional powers:** Greenland signalled its willingness to turn to China for investment if Western countries do not provide support, reinforcing the leverage of some smaller nations.
- **Shifting security arrangements:** The US pushed for a 'reassurance force' in Ukraine, while French President Macron hosted a "coalition of the willing" to discuss Europe's defence without US guarantees. Further examples include the discussion of a Franco-German "nuclear sharing" arrangement.

Outlook: The assumptions of a bipolar world order is giving way to a more complex, multi-nodal system. This will create greater emphasis on regional blocs and issue-specific coalitions. The stability of traditional alliances like NATO and the G7 remain deeply uncertain.

Emerging themes – part 2

These four key themes dbLumina identifies are sometimes ‘known’ by investors, however, they may be underestimated by the market, particularly when there are some potentially series longer-term impacts

3 The return of global bond vigilantes

Previous talk of “sovereign fiscal fragility” has now manifested itself into a severe market reaction. Although fears around government debt and deficits pop up from time to time, the magnitude of September’s bond sell-off has been compounded by oil price concerns

- **Accelerating sovereign sell-off:** Concerns about inflation and debt sustainability are not new but the peaks in bond yields are. Yields on 10-year Treasuries, bunds, and gilts all rose sharply, with bunds hitting their highest level since 2009.
- **Fiscal policy as a direct market catalyst:** The market is increasingly sensitive to fiscal policy announcements. For example, in the UK postponed tax rises, led to a spike in gilt yields. Similarly, France's political turmoil and concerns over its budget deficit caused the Franco-German yield spread to widen significantly.
- **The “awash in debt" narrative:** Treasury Secretary Bessent's comment at the G20 that "The world is awash in debt" signalled that the issue is increasingly top-of-mind for policymakers.

Outlook: Markets will be increasingly quick to punish any perceived fiscal irresponsibility. Expect higher borrowing costs, currency volatility, and increased risk of financial instability, especially for countries with high debt levels.

4 The bifurcation of economies: corporate resilience vs. consumer strain

The strain in the US consumer and labour markets is now diverging sharply from the health of large corporations, reinforcing the K-shaped economy.

- **Strong corporate performance:** Earnings season for Q2 showed remarkably solid global growth, and the resilience of large corporations was a recurring theme that helped support equity markets despite other headwinds.
- **Weak consumer & labour signals:** Economic reports have showed some signs of a weakening labour market, falling consumer confidence, and stress among low-income Americans. The New York Fed's survey also showed households expect a worse financial situation.
- **Policy and market disconnect:** While strong corporate performance depends on a robust economy, the underlying consumer weakness is concerning. Expect the market to continue its twitchy reaction to data, as investors struggle to reconcile these conflicting signals.

Outlook: This theme is potentially the most consequential for the domestic US economy. If corporate profits and equity markets "detach" from the real economy as consumers perceive it, it could lead to significant social and political instability, making economic policy even more difficult to navigate.

The impact of today's themes 1 year ahead:

We asked our AI model, dbLumina, to analyse the impact of the themes for the quarter. The 'positive' and 'negative' indicators are relative to expectations at the beginning of the quarter. For example, a net negative score for the S&P 500 does not mean we expect the index to fall. Rather it means we see the rising probability that returns will be lower than previously expected.

Theme	GDP Growth	S&P 500	Interest Rates[1]	Inflation[1]
1. Heightened geopolitical conflict	Negative	Negative	Higher	Higher
2. Sustained inflationary pressures	Negative	Negative	Strongly higher	Strongly higher
3. Divergence in global monetary policy	Mixed	Volatile	Divergent	Divergent
4. Prominence and risks of AI	Slightly positive	Positive	Neutral to higher	Neutral to higher
5. Shifting global trade & alliances	Neutral to negative	Neutral to negative	Neutral	Neutral
6. Rise of 'Middle Power Countries'	Neutral	Neutral	Neutral	Neutral
7. Political fragmentation in Europe	Slightly negative	Negative	Neutral	Neutral
8. Scrutiny of AI and "Big Tech"	Neutral	Negative	Neutral	Neutral
9. Debt sustainability concerns	Slightly negative	Slightly negative	Slightly higher	Slightly higher
10. Tech and semiconductor volatility	Neutral	Volatile	Neutral	Neutral

	GDP Growth	S&P 500	Interest Rates[2]	Inflation[2]
Sum of positive/negative indicators (positive is +1, negative is -1)	-4	-5	-5	-5

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

Source: Deutsche Bank

The impact of today's themes 3 years ahead:

The medium-term picture is far more optimistic than the short-term picture. The implicit assumption is that the macro and geopolitical volatility that is currently weighing on markets will either resolve itself over the coming 3 years or become normalised enough that the underlying earnings power of the US economy will drive equity markets ahead and, to a lesser extent, economy growth.

Theme	GDP Growth	S&P 500	Interest Rates[1]	Inflation[1]
1. Heightened geopolitical conflict	Slightly negative to neutral	Neutral	Stabilising	Moderating
2. Sustained inflationary pressures	Neutral	Neutral to slightly positive	Structurally higher	Stabilised but higher
3. Divergence in global monetary policy	Converging	Positive	Converging	Converging
4. Prominence and risks of AI	Strongly positive	Strongly positive	Uncertain	Lower
5. Shifting global trade & alliances	Slightly positive	Positive	Neutral	Neutral
6. Rise of 'Middle Power Countries'	Slightly positive	Slightly positive	Neutral	Neutral
7. Political fragmentation in Europe	Uncertain	Uncertain	Neutral	Neutral
8. Scrutiny of AI and "Big Tech"	Neutral	Neutral	Neutral	Neutral
9. Debt sustainability concerns	Strongly negative	Strongly negative	Highly volatile	Highly volatile
10. Tech and semiconductor volatility	Neutral	Less volatile	Neutral	Neutral

	GDP Growth	S&P 500	Interest Rates[2]	Inflation[2]
Sum of positive/negative indicators (positive is +1, negative is -1)	+1	+5	-1	Nil

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

Source: Deutsche Bank

Appendix 1

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Deutsche Bank
Research

The Next 90: Your Geopolitical Risk Radar
Oct–Dec 2026

September 2026

Miha Hribernik | Dr. Helen Belopolsky

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

The Next 90: Your Geopolitical Risk Radar | Oct–Dec 2026

A geopolitical calendar mapping the next 90 days

OCTOBER

3	Latvia	General election
4	Brazil	General election
4	Bosnia and Herzegovina	General election
5-28	Canada	General election in Quebec, municipal elections in British Columbia, Ontario and Manitoba
9-17	Czech Republic	Senate and municipal elections
8-9	EU/China	Trade talks in Beijing; EU deadline for "tangible results"
12-18	IMF/World Bank	Annual meetings of the IMF and WB in Bangkok
15-16	EU	European Council meeting
19	Canada	Alberta independence referendum
20-22	Germany	Berlin Global Dialogue
24	Slovakia	Municipal and regional elections
24	EU/GCC	Second EU-GCC Summit
25	Bulgaria	Presidential election
26-29	China	Fifth Plenum of China's Central Committee
27	Israel	Legislative election
29-30	EU/Canada	EU-Canada Summit
30-31	G20	G20 Foreign Ministers' Meeting in Atlanta

NOVEMBER

3	US	Midterm elections (House, Senate, gubernatorial and state legislatures)
4	South Africa	Municipal elections
4-5	Spain	Ibero-American Summit
7	New Zealand	General election
9	EU	Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings
9-20	COP31	COP31 in Türkiye
10-12	ASEAN	49th ASEAN Summit in Metro Manila
18-19	APEC	APEC Economic Leaders' Meeting in Shenzhen
28	Taiwan	Local elections
28	Palestinian National Authority	Legislative election
30-1 Dec	EU/Argentina	EU-Argentina Investment and Business Forum in Buenos Aires
TBC	US/North Korea	Potential meeting between Donald Trump and Kim Jong-un

DECEMBER

3-4	OSCE	Ministerial Council in Switzerland
4	US	Section 232 tariffs (15% ad valorem) and minimum import prices on polysilicon, derivatives and solar products set to enter into force
5	The Gambia	Presidential election
5-6	Doha Forum	Qatar's flagship dialogue themed "Redefining Global Trust"
10-11	EU	Eurogroup and Economic and Financial Affairs Council (ECOFIN) meetings
13	Haiti	General election
14	EU	Foreign Affairs Council meeting
14-15	G20	G20 Summit in Miami
17-18	EU	European Council Summit
31	OPEC+	Expiry of the current group-wide production framework
TBD	US	US Congress is set to adopt annual-must pass defence legislation
TBD	China	Annual Central Economic Work Conference

Note: The calendar contains selected key events and is not intended to provide an exhaustive list.

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Deutsche Bank
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The Currency Book

September 29, 2026

Introduction

With trillions of dollars, euros and yen changing hands every day, foreign exchange is indisputably the largest and most liquid financial market in the world. Yet despite its size, the currency market is also one of the least understood. Unlike bonds or equities, there is no widely accepted framework for explaining currency moves, which reflect a complex interplay of macroeconomics, capital flows, market structure, investor behaviour and (geo)politics.

Over the past two decades, advances in technology and market infrastructure have transformed how currencies are traded, placing FX at the forefront of broader changes in global finance. Yet because currencies sit at the intersection of national economies, the FX market is also fundamentally connected to international trade and cross-border capital flows. It is therefore increasingly exposed to the forces reshaping the global economic landscape—from geopolitical fragmentation and changing trade relationships to shifts in regulation, payments infrastructure and the architecture of the international monetary system.

This Deutsche Bank Currency Book brings together academic, industry and Deutsche Bank expertise to explain how FX markets function, how exchange rates are determined, how investors seek to generate returns from currencies, and how FX risk can be managed by investors and corporates alike. Our aim throughout the *Book* is to set out the enduring principles driving markets alongside the developments defining FX today. Ultimately, the structure of the FX market is not simply a question of how currencies trade, but of who controls the financial infrastructure through which money moves across borders. The evolution of FX is increasingly intertwined with questions of monetary sovereignty, financial resilience and geopolitical power.

Chapter 1 reviews the structure of the FX market, examining its participants, ecosystem and evolution over time. Chapter 2 discusses exchange rate determination, moving from traditional academic theories to modern macro-financial approaches and practical decision frameworks. Chapter 3 reviews FX as an asset class and the evidence on currency investing, while Chapter 4 considers currency hedging for both investors and corporates. Finally, Chapter 5 discusses the future of FX, including the impact of technology, payment systems and digital assets.

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Important research disclosures located in Appendix 1

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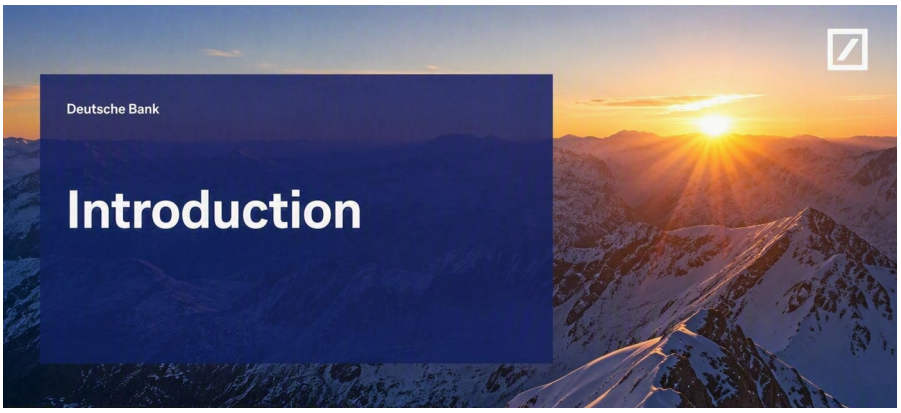
Glossary of terms



Introduction

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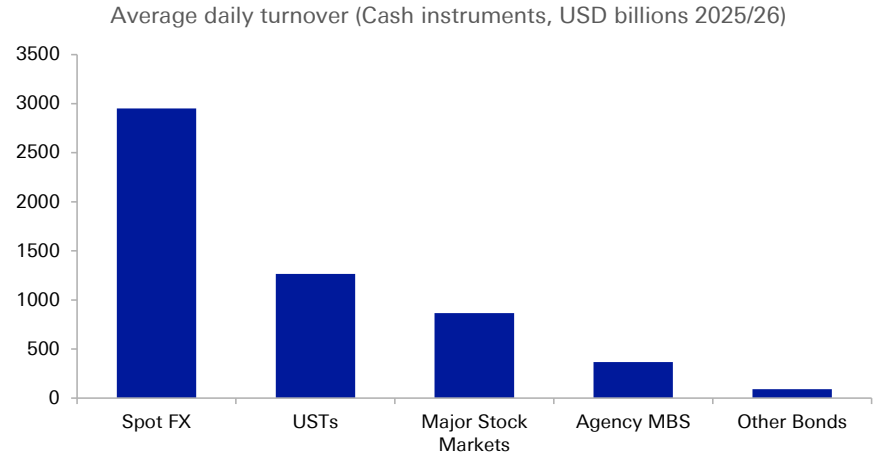
Figure 1: Turnover in spot FX dwarfs other asset classes



Source : Federal Reserve Bank of New York, Bank of England, BIS.
Note: Major Stock Markets covers all US stock Markets, LSE and Nikkei

Introduction

The foreign exchange market is the world's largest and most liquid financial market, but it differs fundamentally from a centralised exchange. Trading takes place across a fragmented, largely over-the-counter network of banks, non-bank liquidity providers, electronic venues and end-users, using instruments ranging from spot and forwards to swaps, futures and options. How do transactions move through this network? Who provides liquidity, who consumes it and what risks arise between execution and settlement? This chapter maps the FX ecosystem from instruments and trading infrastructure to the participants ultimately responsible for currency demand.

1.1 Instruments

The FX market is not one single product. Instead, it is made up of a group of instruments that allow market participants to exchange currencies today, lock in an exchange rate for the future and/or hedge against adverse exchange-rate moves. Below, we outline the main instruments in FX markets and their users.

Spot: This is the simplest and most widely known FX instrument. When market participants refer to "the exchange-rate" or the "spot exchange-rate," they generally mean the price agreed today for exchanging one currency for another on the standard spot settlement date. For most major currency pairs spot settlement is two business days later (T+2).[1] For example, a spot purchase of EUR 1 million against the USD at a EUR/USD exchange-rate of 1.20 means that the buyer receives EUR 1 million in his or her bank account on the settlement date and the seller receives USD 1.2 million from the buyer on the same date.

1 There are exceptions, for example USD/CAD settles T+1.

By convention, the first currency in an exchange-rate quotation is known as the base currency, while the second is the quote currency. The exchange-rate therefore represents the number of units of the quote currency required to purchase one unit of the base currency. In USD/JPY, for example, the USD is the base currency and the exchange-rate indicates the number of Japanese yen required to purchase one US dollar. A USD/JPY exchange-rate of 160 means 1 USD can be exchanged for 160 JPY.

Spot transactions are the most commonly used foreign exchange payments for cross-border payments for goods and services, the purchase of foreign financial assets, as well as portfolio rebalancing and proprietary trading. In 2025, the Bank for International Settlements (BIS) estimates spot turnover at $3trn per day, accounting for 31% of global over-the-counter (OTC) FX turnover.

Forwards and futures: An (outright) forward is similar to a spot transaction, but is instead an agreement to exchange two currencies at a rate (price) agreed today, but with settlement taking place at a date more than T+2 days in the future. They are also traded OTC. Forwards are most commonly used by those who have a known future cash flow and want to remove uncertainty around the future exchange rate. In 2025, the BIS estimates that they account for 19% of global turnover, or $1.8trn per day.

Futures are economically similar to forwards but are standardised contracts traded on exchanges and centrally cleared. This provides transparent pricing and reduces bilateral counterparty risk, but offers less flexibility over contract size, maturity and settlement terms than an OTC forward. Exchange-traded FX futures remain considerably smaller than the OTC FX market.

Non-deliverable forwards (NDFs): An NDF is a cash-settled forward commonly used for currencies where convertibility or offshore deliverability is restricted. Two parties agree today on an exchange rate for a future date, but unlike a standard forward, the underlying currencies are not exchanged. At maturity, the agreed NDF rate is compared with a pre-specified fixing or reference rate. The difference is usually settled in USD. NDFs are commonly used by corporates, banks and investors to hedge or take exposure to currencies subject to capital controls.

Swaps: An FX swap combines two transactions: an exchange of currencies on an initial date and a reverse exchange at a later date, with both exchange rates agreed when the swap is initiated. They are used by institutions that hold cash in one currency (e.g. EUR) but need funding in another (e.g. USD) for investment, hedging or other needs. They are by far the largest FX instrument by turnover, with the BIS estimating turnover of $4trn per day in April 2025, 42% of global FX turnover. A swap is, in effect, a form of collateralised borrowing, with the collateral being cash in a different currency to that of the currency borrowed.

Cross-currency (XCCY) swaps: A cross-currency swap is similar to an FX swap in that principal amounts denominated in two different currencies are typically exchanged at inception and reversed at maturity. The key difference is that a cross-currency swap also involves periodic exchanges of interest payments denominated in the two currencies over the life of the contract. These payments may be fixed or floating. Cross-currency swaps therefore allow market participants to transform both the currency denomination and, where relevant, the interest-rate characteristics of an asset or liability.

For example:

- Suppose a European corporate issues a USD-denominated bond to US-based investors. It receives the bond proceeds in USD, but must periodically pay interest in USD and repay the USD principal when the bond matures. However, because the corporate primarily earns revenue in EUR, it needs to convert these USD liabilities into EUR liabilities to avoid currency and potentially interest-rate risk.
- The corporate removes FX and interest-rate risk by entering into a xccy swap, where, in essence, it converts this USD-liability into a EUR-liability.[2]
- They are commonly used by banks, corporates and investors that need to raise money in one currency but need funds in another. The basis is the extra spread added to one side of the swap to balance unequal demand for funding in the two currencies. See Section 2.11 for more on cross-currency basis and here.

Options: An option gives the holder the right, but not the obligation, to buy or sell one currency against another at a specified exchange rate during a specified period. Options are used to hedge currency risk or speculate on currency movements and are particularly useful when the end-user wants an asymmetric payoff profile. Unlike a forward, an option can limit losses from an adverse currency movement while allowing the holder to retain some or all of the benefit from a favourable movement. In 2025, they represented around 7% of global FX turnover. Options are widely used by corporates and investors for hedging, and by investors and dealers to express directional or volatility views.

The simplest and most widely used instruments are *vanilla options*, which give the holder the right to buy (a call) or sell (a put) a currency at a predetermined strike price. A European-style option can be exercised only at expiry, while an American-style option can be exercised at any point up to expiry. More structured products include barrier options, whose payoff is activated or extinguished if the exchange rate reaches a specified level—for example, knock-in and knock-out options. One-touch and no-touch options pay a fixed amount depending on whether the exchange-rate does, or does not, reach a predetermined level during the life of the option. These instruments are commonly used to express views not just on the direction of a currency pair, but also on the probability, timing and magnitude of exchange-rate moves.

1.2 FX spot trading life cycle and risks

Having discussed the main FX instruments, the next step is to examine how these transactions are executed and settled in practice. The complete trade life cycle, meaning the sequence of events and processes involved in executing a trade, is crucial to understanding how FX transactions work. We therefore describe below the broad steps involved in trading FX, focusing on spot transactions. They are:

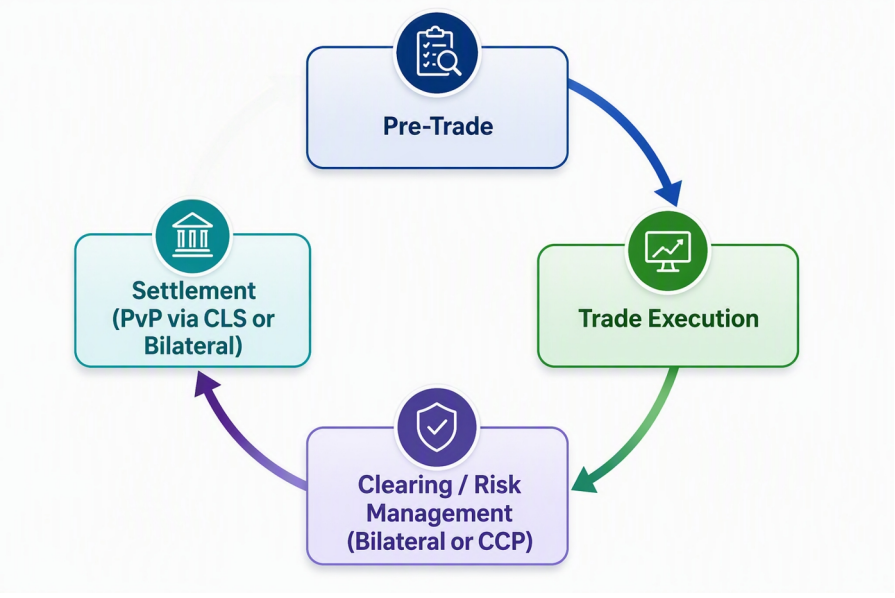
1. **Pre-trade:** This stage encompasses all the necessary preparations before a trade is executed, including client onboarding, platform selection, credit checks, and regulatory compliance.

2 Specifically, it exchanges the USD bond proceeds for EUR at inception, receives USD interest payments (from the swap counterparty to pay U.S. bond investors) and pays EUR interest (to the swap counterparty) over the life of the swap, and receives USD principal (from the swap counterparty to pay U.S. investors) in exchange for EUR principal at maturity. The USD receipts from the swap are used to meet the bond's USD interest and principal payments, effectively converting the USD-denominated bond into a EUR-denominated liability.

2. **Trade execution:** This is the point where the trade is agreed upon and executed at a specific exchange rate, creating a legally binding contract.
3. **Trade clearing:** This stage involves validating trade details and managing counterparty credit risk before settlement, either bilaterally or through a central clearing counterparty (CCP). In centrally cleared FX markets, the CCP becomes the buyer to every seller and the seller to every buyer, reducing direct counterparty exposures between market participants. Specialist FX CCPs, such as LCH's ForexClear, clear both non-deliverable and deliverable FX products. While FX spot transactions can be cleared, they are not subject to mandatory clearing requirements. ForexClear supports 25 non-deliverable forward (NDF) currency pairs, 9 non-deliverable option (NDO) currency pairs, and 8 deliverable (Spot, Forwards and Options) currency pairs, at the time of writing.
4. **Trade settlement:** This final stage represents the completion of the transaction, with the actual exchange of currencies taking place. Settlement for spot trades typically occurs two business days (T+2) after the trade date, although exceptions exist. For example, USD/CAD settles on a T+1 basis.

A Central Clearing Counterparty (CCP), also known as a Central Counterparty, stands between buyers and sellers in financial markets, becoming the buyer to every seller and the seller to every buyer. This mitigates counterparty risk.

Figure 2: FX Spot Trade Life Cycle



Source : Deutsche Bank

Risks surrounding FX spot trading

The over-the-counter (OTC) nature of the FX market, while offering flexibility and adaptability to participant needs, does present certain risks that are often more pronounced compared to exchange-traded markets. Below we discuss some of the key risks in FX.

Credit risk: Credit risk in FX trading, stemming from a counterparty's potential inability to fulfill its obligations, is present throughout the trade lifecycle:

- ***Pre-trade:*** Counterparty creditworthiness is a critical consideration in the

pre-trade stage of FX transactions. A thorough assessment should involve evaluating the potential counterparty's financial strength, credit rating, and reputation. Choosing a creditworthy counterparty significantly reduces the risk of default throughout the trade lifecycle.

- ***Post-trade*** (Clearing and Settlement): While credit risk exists in both the clearing (post-execution/pre-settlement) and settlement stages of a spot FX trade, it primarily materializes during settlement. This stage, involving the actual exchange of currencies, carries the highest risk due to the potential for settlement failure. This risk, often referred to as settlement risk, arises when one party delivers their currency while the counterparty defaults. Cross-border trades are particularly vulnerable due to time zone differences, which can create a window where one party has performed but the other has not. Additionally, mark-to-market risk from FX rate fluctuations between trade and settlement dates can exacerbate settlement risk by increasing potential losses from a default. This is particularly pronounced with forward contracts, where the longer pre-settlement period leads to greater mark-to-market risk and, consequently, higher settlement risk.

Market Risk: This risk in FX trading arises from potential losses due to adverse exchange-rate movements. This is influenced by factors like economic data, geopolitical events, and interest rates. For customers, asymmetric information, where one party has more or better information than another, can lead to inefficient pricing and increased volatility, thereby exacerbating market risk.

- For dealers, a key component of market risk is inventory management risk – the potential for losses from unfavourable price movements in the currencies they hold to facilitate client trading. Effective inventory management, including position limits and hedging, mitigates this risk.

Liquidity Risk: This refers to the overall market's ability to facilitate the buying and selling of a currency pair quickly and efficiently at a desired price. It's influenced by factors such as market depth, trading volume, volatility, and fragmentation.

Operational Risk: This encompasses risks associated with trade processing, settlement, and technology. Errors in trade execution, system failures, and fraud can all lead to significant losses.

Regulatory Risk: This refers to the risk arising from changes in regulations or their interpretation. The evolving regulatory landscape in FX, particularly concerning non-bank dealers and systemic risk, can impact market participants and create uncertainty.

1.3 Market participants

The FX market comprises a variety of participants with different objectives. FX market participants fall into two broad categories: financial and non-financial. Financial participants can act as both liquidity providers and consumers, while non-financial participants act largely as liquidity consumers.

Financial participants: In the FX market, financial participants are primarily banks acting as dealers.

- **Dealers:** These are mainly large commercial banks, investment banks and securities houses that participate in the inter-dealer market for proprietary

trading. [3] and/or for meeting customer demand through market-making.[4]

- **Institutional investors:** These are mutual funds, pension funds, insurance companies and endowment funds, collectively referred to as "real-money investors", which need to buy currencies to satisfy end-user demand for an underlying asset. For instance, a UK pension fund needs to buy USD to purchase US equities. These investors also engage in hedging, proprietary trading and risk management activities.

- **Hedge Funds & Principal Trading Firms (PTFs):** Hedge funds are investment funds that follow a broad range of investment strategies and are generally not subject to the same borrowing and leverage restrictions as traditional investment funds. They are users of FX and derivatives for speculative purposes. PTFs are principal trading firms, such as XTX Markets and Citadel Securities, that hedge or speculate for their own account and also participate in market-making. This category may include high-frequency trading firms that employ high-speed algorithmic trading strategies.

Institutional investors are subject to a variety of (1) regulatory, (2) mandate and (3) process constraints.

***(1) Regulatory constraints** have had an increasingly important influence on cross-border investment since 2008.*

***(2) Mandate constraints** relate to limits placed on investment allocations by a fund's internal rules. These constraints may arise from, among other things, fund design (i.e., fund characteristics, regions, target benchmarks, target returns), founding charters or guidelines (i.e. many endowments have externally imposed constraints on investment allocations and government policy (i.e. Norway's pension fund mandate is determined by the government). These restrictions can materially limit a fund's ability to take currency risk. The simplest mandate distinction is between passive and active funds, with the former following an index and therefore being explicitly required to track its returns.*

***(3) Process constraints** relate to limits placed on the decision-making process. Most investment funds abide by internal risk rules designed to limit exposure to losses during periods of market volatility. These rules can generate value-at-risk (VaR) shocks.*

Non financial institutions: Non-financial participants, such as corporations involved in international trade, retail and the official sector, also primarily consume liquidity.

- **Corporates** are active in currency markets primarily to manage FX exposures [5] arising from international trade, foreign-currency financing, overseas operations and cross-border mergers and acquisitions (M&A).[6]

- **Official sector:** This category includes central banks, sovereign wealth funds, international financial institutions, such as the IMF and BIS, development banks and government agencies. Monetary-policy, financial-stability and mercantilist considerations have all, at various times, motivated central-bank intervention in FX.[7]

- **Other:** This category includes all other market participants, including retail participants. Retail participants engage in FX both to speculate and to meet transactional demand arising from travel, trade or investment.

1.4 Profit versus non-profit seekers

In 2025, daily FX turnover reached $9.5 trillion according to the BIS triennial survey (BIS, 2025) making currency trading the world's largest financial market. Yet, size and liquidity do not necessarily imply efficiency. Unlike most equity or bond transactions, a large share of FX activity serves an underlying transactional purpose. For instance, corporates must convert revenues and meet foreign-currency invoices; institutional investors hedge overseas assets; and dealers intermediate these flows. For these participants, expected FX returns may be secondary, or even irrelevant, to the decision to trade. In this section, we investigate

3 Proprietary trading is a profit-seeking activity that dealers once engaged in more extensively, but which has been severely constrained by regulatory change. Most notably, the Volcker Rule, implemented in 2014-15 under the Dodd-Frank regulations, bans banks from engaging in proprietary trading activities, including many foreign exchange instruments.

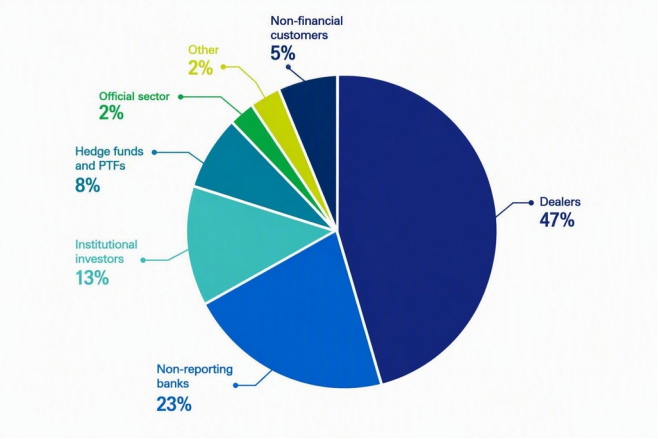
4 Market making: This is an activity that seeks to capture returns by matching buyers and sellers and charging a price for the service through the bid-offer spread. This spread represents the price that liquidity seekers must pay to access the market and is a "frictional" cost of executing transactions. As such, market-making is a liquidity-provision service rather than a source of the 'excess returns' associated with activities such as proprietary trading

5 For more see here

6 Corporates using foreign exchange to hedge future exposures may also seek hedge accounting to

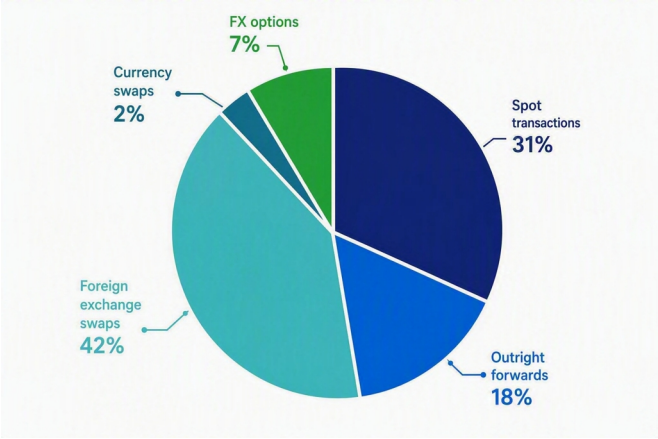
the division of the market between profit-seeking and non-profit-seeking participants. This is the first step in understanding exchange-rate determination.

Figure 3: Daily FX turnover by counterparty



Source : BIS, Deutsche bank

Figure 4: Daily FX turnover by instrument



Source : BIS, Deutsche Bank

Approach 1: The Triennial BIS Survey of FX market volumes

We use the 2025 BIS Triennial Survey to update Deutsche Bank's previous work (Saravelos et al, 2018) and estimate the share of FX turnover initiated with a profit-seeking currency objective. We construct a "realistic scenario", which applies what we deem to be "realistic" assumptions about participants' ability and mandate to take FX risk.[8] We find that profit-seeking activity accounts for approximately 40% of turnover under the realistic scenario.[9]

Financial participants

- **Dealers**: Dealer-to-dealer activity accounts for approximately 47% of total daily FX turnover ($4.44trn) and we classify 10% of dealer flow as profit-seeking ($444bn/day). Given that much of the daily FX turnover reflects market-making, we assume that inter-dealer transactions conducted by voice are not profit-seeking and that only electronic transactions conducted through order-matching systems such as EBS are profit-seeking.

- **Hedge Funds and PTFs**: Hedge Funds and PTFs account for approximately $760bn (8%) of headline daily FX turnover and we classify 90% of this as profit-seeking ($684bn/day). The $684bn/day figure represents the profit-seeking nature of hedge-funds where the remaining 10% reflects the

ensure that the mark-to-market value of their hedge does not introduce unwanted volatility in the income statement. IFRS 9 provided an updated set of standards for hedge accounting, designed to increase flexibility and better link financial reporting to firms' risk management practices. However, hedge accounting still places limitations on the types of instruments and kinds of hedges that can be implemented. For example, to qualify for hedge accounting, corporates must adhere to hedge effectiveness and demonstrate that an economic relationship between the underlying cash flow or asset and the hedge exists, as well as quantitative limits to hedge ratios.

7 Exchange rates are explicitly targeted by some central banks as part of their overall monetary policy framework (for instance the Monetary Authority of Singapore or the Hong Kong Monetary Authority); other central banks have intervened to defend their inflation target or financial stability reasons (most recently, the Swiss National Bank); and others to prevent competitiveness losses in global trade.

expansion of PTFs to act not only as profit-seekers but also as liquidity providers.

Figure 5: Profit-seekers make up 40% of the FX market

Counterparty	Profit-seeking assumptions
Dealers	Mainly market-makers, small profit-seeking **10%**
Hedge Funds and PTFs	All profit-seeking, except PTF liquidity provision **90%**
Non-Reporting Banks	Mainly market-makers, small profit-seeking incentive **10%**
Institutional Investors	Some fixed-income and equity managers take FX risk **30%**
Official Sector	Reserve managers profit seeking **50%**
Corporates	All corporate activity liquidity-seeking **0%**
Other	Retail participation **50%**
Undistributed	Unknown **0%**
Overall	**40%** of daily FX turnover is profit-seeking

Source : BIS, TIC, Federal Reserve Bank of New York, Deutsche Bank

- **Non-Reporting Banks**: Non-reporting banks account for approximately $2.19trn, or 23% of headline daily turnover and we classify 10% of this as profit-seeking ($220bn/day). These largely intermediate for institutional and non-financial customers and apply the same profit-seeking assumptions as we do for dealers.
- **Institutional investors**:Institutional investors account for $2.56tn of daily BIS turnover [10] and we classify 30% of this as profit-seeking. We assume 80% of active fixed-income managers have discretion to take FX risk and 50% of active equity managers have discretion to take FX risk, implying a profit-seeking share of approximately 30%.
- **Official Sector**: The official sector accounts for approximately 1-2% of headline FX turnover and we classify 50% of the turnover as profit-seeking. The motivations within this category vary. Central-bank intervention may pursue monetary-policy, financial-stability or exchange-rate objectives, while reserve managers may also seek to preserve capital and earn returns.

Non-Financial Sector:

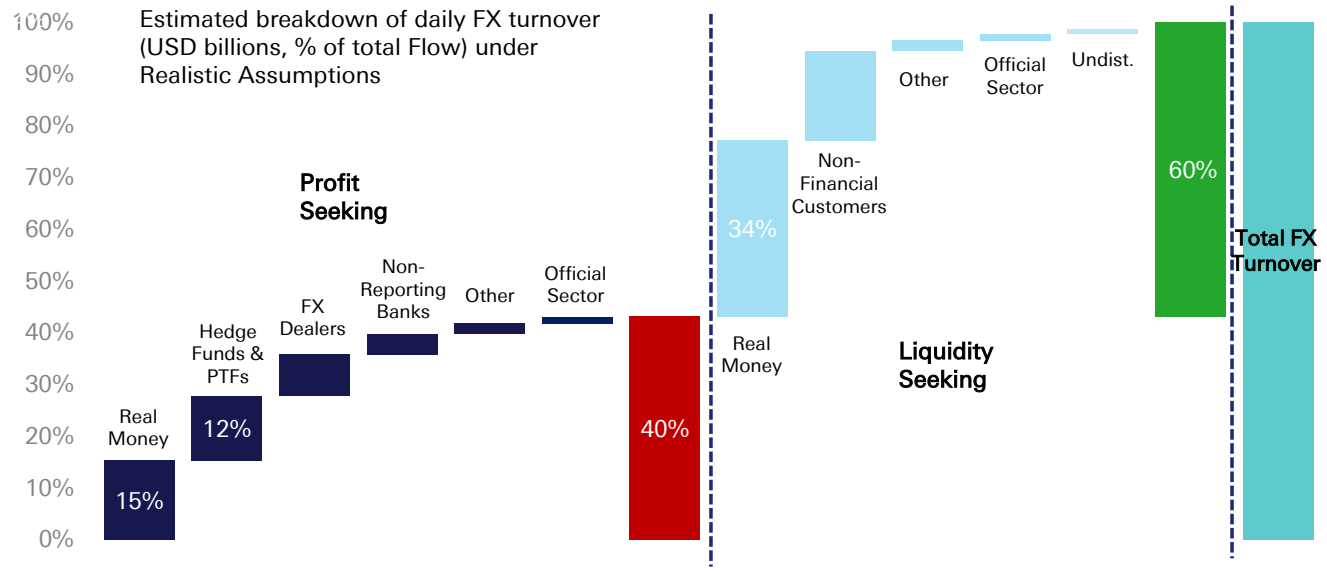
- **Corporates and governments:** Corporates and governments account for 10% of daily FX turnover and we classify 0% of this flow as profit-seeking.[11]

8 An upper-bound scenario, which relaxes those "realistic" assumptions and produces a higher estimate of profit-seeking activity by classifying ambiguous transactions as profit-seeking is outlined in Appendix .
9 See Appendix for full methodology
10 After reallocation flow from intermediated by non-reporting banks, see Appendix for full methodology.

- **Other and Undistributed**: The BIS residual "other" category accounts for approximately 2% of headline turnover. We classify 50% as profit-seeking. [12] The undistributed category accounts for less than 1% of headline daily turnover and we classify this as 0% profit-seeking.

Bringing it all together: Taken together, we estimate that approximately 40% of adjusted FX turnover is profit-seeking, while approximately 60% is liquidity-seeking or transactional.

Figure 6: There are more liquidity seekers than profit seekers in our baseline estimates of FX market size



Source : BIS, TIC, Federal Reserve Bank of New York, Deutsche Bank

Approach 2: Commitment of Traders CFTC data

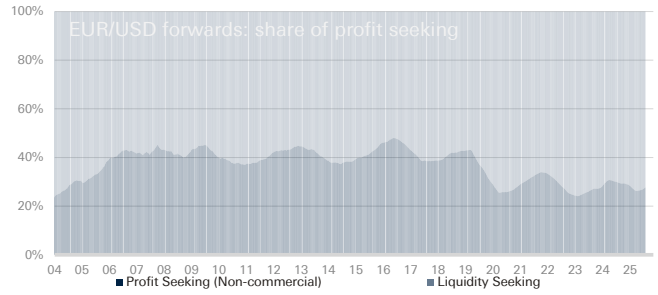
While the BIS data provide the most comprehensive view of the global FX market, the survey is only updated every three years. For a higher-frequency perspective, we therefore complement it with the US Commodity Futures Trading Commission's Commitments of Traders (CoT) report, which is published weekly and provides a breakdown of positions in exchange-traded currency futures and options, principally those traded on the Chicago Mercantile Exchange (CME). The report shows the number of outstanding long and short contracts held by different categories of market participant, based on positions as of each Tuesday. Although the futures market represents only a small subset of total global FX activity—most of which takes place over the counter—the CoT data provide a timely and consistent measure of how speculative and hedging positions are evolving.

11 Corporates use FX to settle foreign-currency revenues and expenses, finance cross-border activity, hedge forecast cash flows and manage earnings volatility. Governments similarly transact to meet an underlying funding or expenditure requirement.

12 Some investors trade currencies to earn returns, while other transactions relate to travel, remittances or purchases of foreign assets

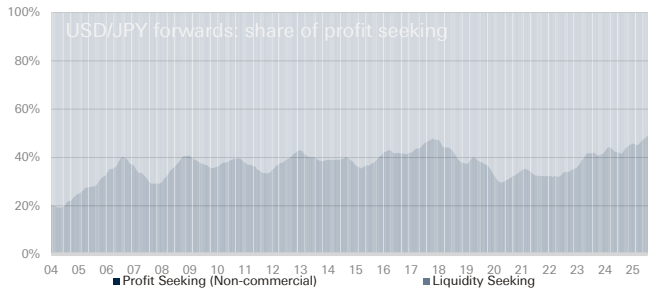
In the traditional or "legacy" CoT report, reportable positions are divided into commercial and non-commercial traders, alongside a residual category of smaller, non-reportable positions. Commercial traders are generally entities that use futures primarily in connection with an underlying business activity and are therefore commonly interpreted as hedgers—for example, corporates or financial institutions managing currency exposures. We classify these positions as liquidity seeking. Non-commercial traders, by contrast, are participants whose futures activity is not classified as commercial hedging and typically includes hedge funds, commodity trading advisers and other leveraged investors. We therefore treat non-commercial positions as profit-seeking. Figure 7 and Figure 8 show how the relative size of the profit-seeking FX market has developed over time. Using one-year rolling averages to reduce the noise of the data, we see that roughly 30-40% of the futures market *(EUR/USD, GBP/USD, USD/JPY)* is profit-seeking.

Figure 7: Profit-seeking activity in EUR futures



Source : CFTC, Bloomberg Finance LP

Figure 8: Profit-seeking activity in JPY futures



Source : CFTC, Bloomberg Finance LP

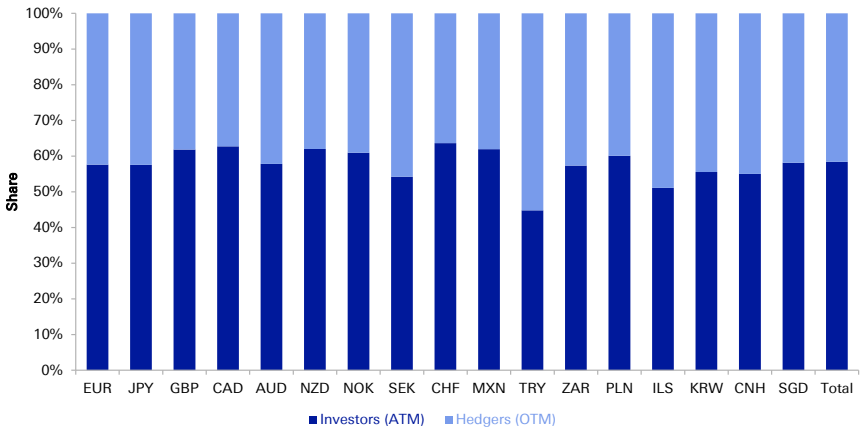
Approach 3: Options market DTCC data

Following the implementation of the Dodd-Frank Act, OTC FX options falling within the US regulatory perimeter are required to be reported to a CFTC-registered swap data repository (SDR). This includes transactions involving US market participants, among others. DTCC operates one of the largest data repositories and therefore captures a substantial share of global institutional FX option activity. We estimate that the DTCC data capture more than 20% of global FX derivative volumes, providing another way of estimating the relative importance of profit- and non-profit-seeking activity in the FX market.

While the data does not provide client granularity, we can back out the likely motivation of the transaction by distinguishing between out-of-the-money and in-the-money option trades. We argue that strikes preferred by the different client groups will likely differ, with hedgers favouring out of the money options (0-25 delta). While speculators focus on directional leverage, the latter often translates to options with strikes nearer to the money (25-75 delta).

Figure 9 shows the split in options volumes for 2025/26 across the two investor categories. We find that option volumes are roughly split equally between investors and hedgers. FX options are instruments that are favoured by hedge funds over corporates and institutional investors, so the results here likely materially overstate the relative importance of profit-maximizing participants in other markets. Again, this shows the FX market is split roughly 50/50 by those who are profit-seeking and those who are not.

Figure 9: Size of profit-seekers in the FX options market



Source : Deutsche Bank , Bloomberg Finance LP, DTCC
**Note: Option volumes captured only for currencies against USD.*

1.5 The FX ecosystem

Following Grover and Saravelos (2025), this section examines the significant changes in FX market structure in recent decades, with a particular focus on the FX spot market, and highlights the key challenges that have emerged as a result of this transformation. To understand how these challenges emerged, it is useful to begin by examining the evolution of the FX spot market.

1.5.1 The Evolution of FX Trading: From Voice to Electronic

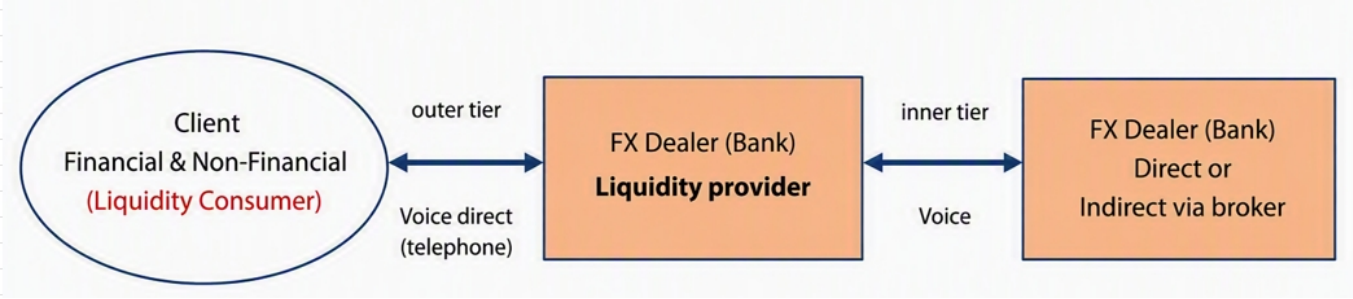
We analyse the evolution of the FX spot market, focusing on the shift from a pre-electronification era of voice-based trading and bank-centric risk management to the phased electronification of interdealer and dealer-client segments. This transformation has led to the emergence of diverse execution styles and trading venues, which will be explored to provide a comprehensive overview of the changing FX landscape and its implications for risk mitigation. The discussion begins with an examination of the pre-electronification era.

1. Voice trading: The age of the telephone

Early foreign exchange (FX) trading was a two-tiered system conducted over the phone (Figure 10). Clients contacted dealers (outer tier) directly to inquire about bid and ask prices. Dealers profited from the difference between these prices (the bid-ask spread). Dealers further traded amongst themselves (the "inter-dealer market" - inner tier) to manage inventory imbalances arising from trading with clients. They did this either directly or anonymously through voice brokers.

In this two-tier system, *banks* acted as the sole liquidity providers and risk bearers in the FX market. Inter-dealer trading, used for hedging and inventory management, formed a significant portion of overall trading volume.

Figure 10: The age of the telephone



Source : Deutsche Bank

2. Rise of the machines

The first wave of electronic trading in FX markets primarily affected the inter-dealer market, replacing phone-based trading with computer systems while leaving the dealer-client relationship unchanged. Here, we first discuss the main innovations in the inter-dealer market and then move on to the dealer-client segment.

(a) Electronification of inter-dealer market

Two major innovations in the trading process facilitated by technological advances led to major transformations in the inter-dealer market. They are:

- The **introduction of electronic brokers**, Reuters/Refinitiv (now LSEG) and EBS, in the early 1990s brought a centralized, anonymous limit order book to FX trading, similar to equity market exchanges. This increased the efficiency of price discovery and inventory risk management. These two platforms are still considered primary trading venues.
- Large banks increasingly **internalized trades** (settling within their own books) in the early to mid-2000s. This reduced their need to manage inventory risk through inter-dealer platforms. Figure 11 demonstrates this shift in FX market dynamics, showing a decline in dealer-reported spot trading from roughly 70% in the 1990s to 40% recently. For the largest dealers, internalization now accounts for an estimated 80% or more of major currency pair trades (Moore et al., 2016; Schrimpf and Sushko, 2019).

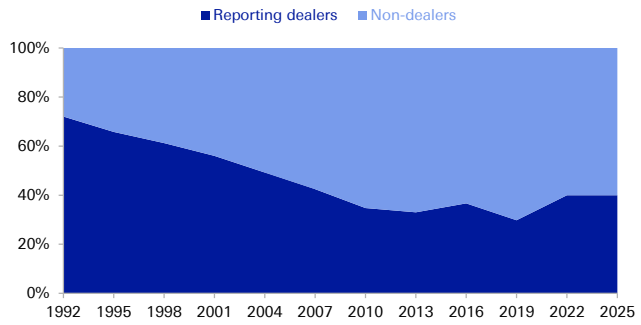
The two innovations (electronic brokers and trade internalisation) primarily impacted the inter-dealer market, improving efficiency and narrowing bid-ask spreads (King et. al, 2012). Thus, resulting in a more efficient inventory risk management (Figure 12).

The benefits of electronification were not limited to the trading process. It also helped FX markets address **settlement risk** in the late 1990s. **Settlement risk (a form of credit risk)**, also known as "Herstatt risk," arose due to cross-border currency transfers being limited by central bank operating hours. This created a window where one party to a transaction could default after the other had already settled.

Settlement risk posed a significant challenge. The potential for a default after one leg of an FX transaction had settled created a substantial credit exposure for banks. This exposure could be particularly large for major dealers, where transactions with

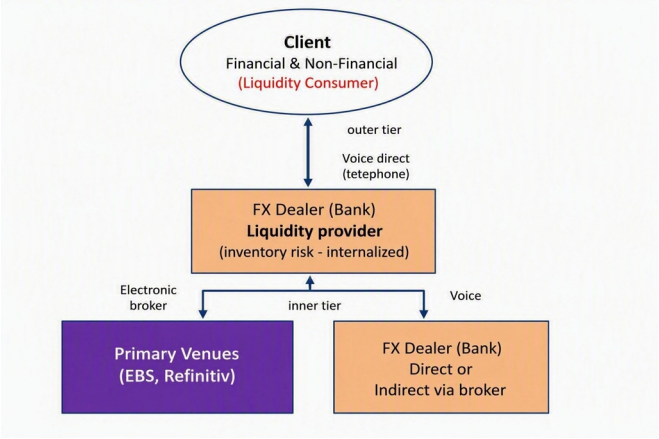
even a single counterparty could exceed the bank's capital.

Figure 11: Drop in dealer share of spot trading as a result of internalization



Source : Deutsche Bank, BIS Triennial Survey

Figure 12: Rise of the machines Part I – electronification of the inter-dealer



Source : Deutsche Bank

To address this, a group of banks developed a new electronic settlement system – the **Continuous Linked Settlement (CLS) Bank**, which began operating in 2002. CLS mitigates settlement risk by simultaneously settling both sides of a trade and by netting payments between member banks.

This netting process drastically reduces the actual funds transferred for settlement to just 4% of the total traded value. *More than $14 trillion in gross financial obligations were settled on an average day in April 2025, with just over one-third (36%) settled through PvP systems, predominantly CLS, eliminating FX settlement risk for those transactions.*

(b) Electronification of the dealer-client segment

Electronic trading in the inter-dealer market increased efficiency and narrowed bid-ask spreads within that market. However, the dealer-client segment remained largely unchanged, and did not experience the same efficiency gains and reduced bid-ask spreads as the inter-dealer market. This resulted in a surge of new electronic trading platforms specifically designed for clients.

With the electronification of the FX market, new liquidity providers emerged, aiming to reduce market-related liquidity risk and reshape the trading landscape. These new players, enabled by technological advancements, include:

1. Multi-dealer platforms (MDP), emerging in the early 2000s, allowed clients to request for quotes (RFQs) from multiple dealers at once. This streamlined the price discovery process, enabling clients to compare pricing from various dealers through a single platform rather than individual phone calls. The increased competition and transparency benefited clients. Example of MDPs include but are

not limited to FXconnect, FXall, 360T.

2. Single dealer platforms (SDP): Many banks developed their own single-dealer platforms (SDPs) to facilitate direct electronic trading with clients. These SDPs largely replaced older communication methods such as telephone and telex for bilateral trading. Examples include J.P. Morgan eXecute, UBS Neo, and Citi Velocity.

Financial and non-financial consumers of FX (liquidity consumers) can access both multi-dealer platforms (MDPs) and single-dealer platforms (SDPs) for liquidity. Both platform types operate using a request for quote (RFQ) execution protocol and full pre-trade identity disclosure.

A key distinction, however, is that MDPs primarily facilitate connections between liquidity consumers (e.g., asset managers, hedge funds, corporations) and liquidity providers (mainly banks). They do not bear the risk of inventory imbalances stemming from client trades. Conversely, SDPs are integrated within individual bank infrastructures and, consequently, are part of the bank's broader risk management function.

3. Prime-brokerage: The emergence of prime brokerage (PB) arrangements in the late 1990s marked a significant shift in FX market structure, allowing clients such as hedge funds to access the inter-dealer market directly or through electronic brokers.Under a prime-brokerage arrangement, a large bank effectively lends its credit standing and trading relationships to a client—typically a hedge fund, asset manager or other active non-bank institution. Rather than having to establish separate bilateral credit lines and trading documentation with every dealer it wishes to face, the client establishes a primary credit relationship with a single prime broker. The prime broker sets trading limits, monitors the client's exposures, requires collateral or margin where appropriate and charges a fee, typically linked to trading volume and the amount of credit consumed.

Request for Quote (RFQ): It's a request from a liquidity consumer to a liquidity provider (like a bank) asking for the price at which they are willing to buy or sell a specific currency pair. The RFQ will typically specify the currency pair, the amount to be exchanged, and potentially the desired settlement date. The FX provider then responds with a quote, which includes the exchange rate they are offering.

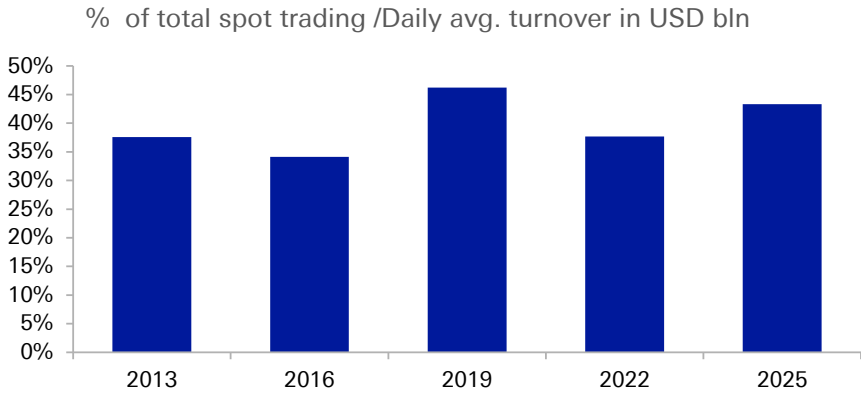
This structure allows hedge funds to access liquidity that was historically available mainly to banks in the interdealer market. A hedge fund can execute an FX trade with one of many approved dealing banks, or through an electronic venue linking multiple dealers, using the credit of its prime broker. The executing dealer therefore does not need to extend a separate bilateral credit line to the hedge fund itself: subject to pre-agreed limits, it accepts the prime broker as the ultimate credit counterparty. Once the client executes the trade, it is "given up" to the prime broker. In legal and settlement terms, the transaction is generally transformed into two matching trades: one between the executing dealer and the prime broker, and another between the prime broker and the hedge fund client. The prime broker consequently stands between the client and the wider dealer network from a credit perspective, even though it need not be involved in choosing the price or executing the original trade.

Prime brokerage therefore separates execution from credit provision. A hedge fund can seek the best price across many banks and electronic liquidity pools while concentrating its credit, collateral and settlement relationship with one or a small number of prime brokers. Before PB became widespread, a hedge fund's ability to trade with a dealer depended much more directly on whether that dealer was willing to establish a bilateral credit relationship with it, limiting the number of counterparties it could access and often preventing it from trading on the same electronic venues or at the same prices as large banks. PB substantially lowered this

barrier and helped non-bank institutions become much more active participants in what had previously been a predominantly bank-to-bank market.

Prime brokerage also generates important operational and balance-sheet efficiencies. Because trades with multiple executing dealers are ultimately consolidated at the prime broker, clients can net exposures across counterparties, reducing the amount of collateral and settlement infrastructure required. PB arrangements also facilitate access to a wider range of liquidity providers—including smaller and specialist banks—and support automated trade matching, confirmation and straight-through processing. At the same time, the prime broker assumes significant counterparty and operational risk: it is responsible to executing dealers for trades conducted under its name and must therefore impose credit limits and monitor the client's positions, leverage and trading activity in real time. Prime brokerage thus does not eliminate credit risk; rather, it centralises it at the prime broker while allowing execution to be dispersed across the wider market. Today, a substantial share of spot FX trading is conducted through prime brokerage arrangements (Figure 13).

Figure 13: Around 40% of total spot trading is done via prime brokerage

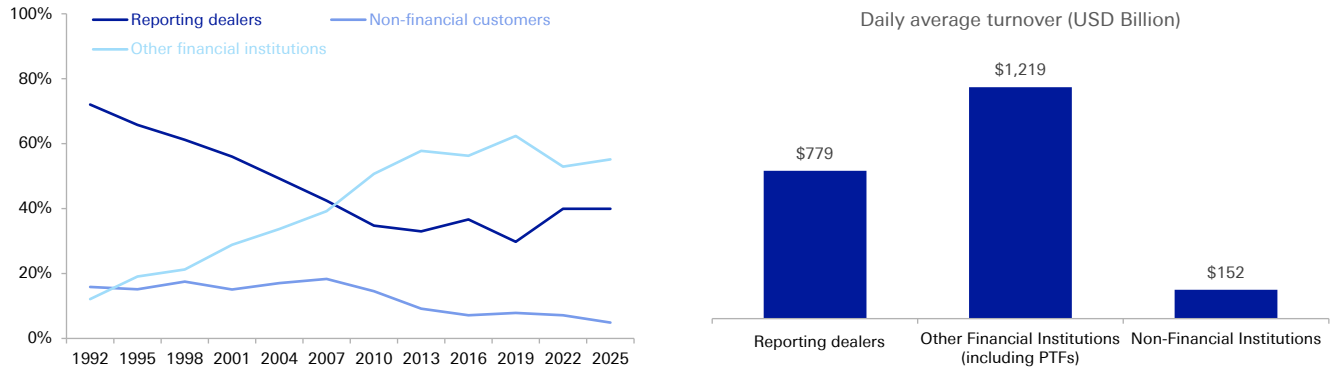


Source : Deutsche Bank, BIS Triennial Survey

4. Principal trading firms (non-bank dealers): Traditionally, banks acted as the primary dealers, or liquidity providers, in the decentralised OTC FX market. However, prime brokerage facilitated client access to primary FX trading venues (EBS/Reuters). This allowed principal trading firms (PTFs), such as XTX Markets and Citadel Securities, to leverage their technological capabilities for electronic market-making, becoming key liquidity providers and serving a wide range of clients, from regional banks to institutional investors.

While Principal Trading Firms (PTFs) are key FX market liquidity providers, they also engage in proprietary trading (investing, hedging, speculating) including high-frequency trading (HFT). HFT firms utilise sophisticated algorithms to execute a large number of rapid trades, often holding positions for extremely short durations. In this capacity, some PTF firms also act as liquidity consumers. This dual role has contributed to the growth of algorithmic traders in FX, impacting inter-dealer trading and the rise of other financial institutions' trading, as noted by BIS Triennial Surveys (Figure 14).

Figure 14: Increase in financial institution spot trading due to the emergence of principal trading firms (algorithmic firms)



Source : Deutsche Bank, BIS Triennial Survey

5. Electronic communication networks (ECNs): While prime brokerage has facilitated client access to primary FX trading venues (EBS/Reuters), numerous secondary venues, such as Currenex and LMAX, have also emerged as a result of competition driven by technological advances. This has enabled clients (non-banks) to both provide and consume liquidity through limit orders or RFS/RFQ model.

Beyond these venues, futures exchanges and dark pools are other alternative liquidity sources. Exchanges utilise centralized limit order books with standardised products, while dark pools operate with hidden order books, matching buy and sell interest at a venue-calculated midpoint. BGC MidFX and FXSpotStream(FSS) MidMatch are prominent examples of FX dark pools.

The rise of secondary and alternative trading venues has also changed how banks manage inventory risk. Banks also utilise multiple venues for hedging client trades. Crucially, the inter-dealer market is no longer exclusive to banks; some clients now also provide liquidity through primary, secondary, and alternative venues.

A Request for Stream (RFS) allows a liquidity consumer to receive a continuous flow of quotes for a specified size and duration, such as a trading day. This functions as a continuous Request for Quote (RFQ), providing real-time price updates for the desired transaction.

1.5.2 Execution styles: voices versus electronic trading

This section explores the various FX execution styles, adhering to the Bank for International Settlements (BIS) definitions for the dealer-client segment and the inter-dealer segment.

Outer tier: Dealer-client segment

The two main styles of execution in the dealer-client (outer tier) segment are: Voice and Electronic (BIS Triennial Survey). Each of these can further be classified as direct or indirect. As per BIS, they are defined as follows:

1. Voice-direct (telephone): Trades originated in person, by phone, by telefax or through general messaging systems (eg Outlook, Gmail or Yahoo mail) regardless of how trades are subsequently matched, and not intermediated by a third party.

2. Voice-indirect (prime-broker): Trade agreed by a voice method and intermediated by a third party (eg a voice broker).
3. Electronic-direct (SDP): Trades executed via an electronic trading system, not intermediated by a third party. These include transactions originated through specific messaging systems that are part of trading platforms. This covers single dealer platforms. It also includes direct API streams: Bloomberg Chat.
4. Electronic-indirect (MDP/ECN): Trades executed via an electronic medium, a third party electronic platform (eg via a matching system). Here, we have multiple platforms that stream prices (RFS/RFQ), run a central limit order book, or matching buy and sell interest at a venue-calculated midpoint. Figure 15 highlights the different platforms available where multi dealer platforms are primary RFS/RFQ focussed while ECNs do both live streaming and matching.

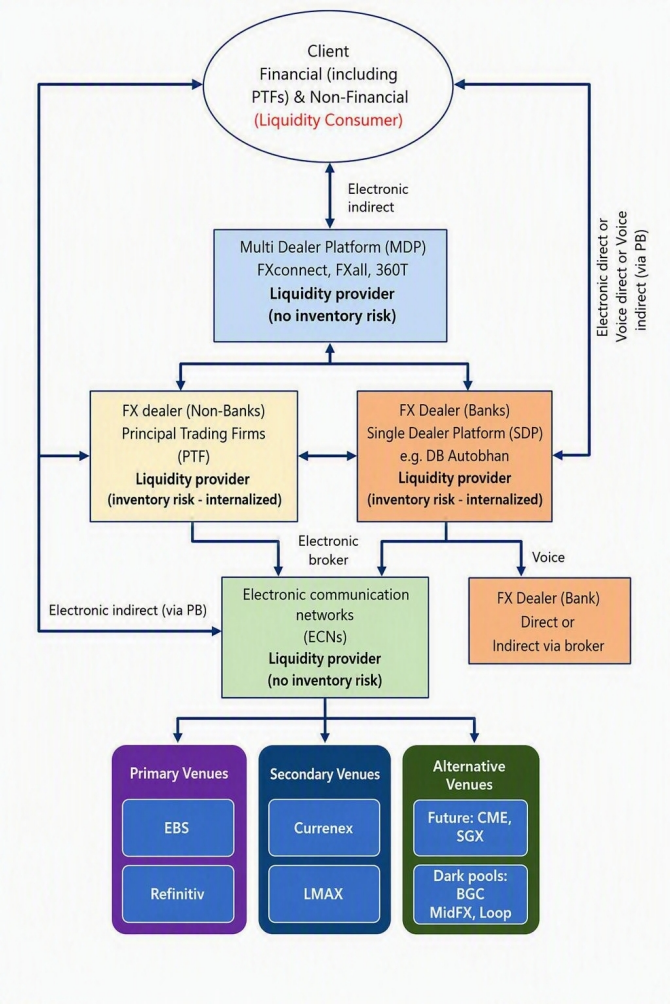
A Request for Stream (RFS) allows a liquidity consumer to receive a continuous flow of quotes for a specified size and duration, such as a trading day. This functions as a continuous Request for Quote (RFQ), providing real-time price updates for the desired transaction.

Single dealer platform (SDP): Bank developed single-dealer platforms (SDPs) to facilitate direct electronic trading with clients.

Multi-dealer platform (MDP): Allows clients to request quotes (RFQs) from multiple dealers at once. Example of MDPs include but not limited to FXconnect, FXall, 360T.

Electronic Communication Network (ECN): Primary, secondary and alternative trading venues are collectively referred to as ECN.

Figure 15: Rise of the machines Part II – electronification of dealer-client segment – current market structure

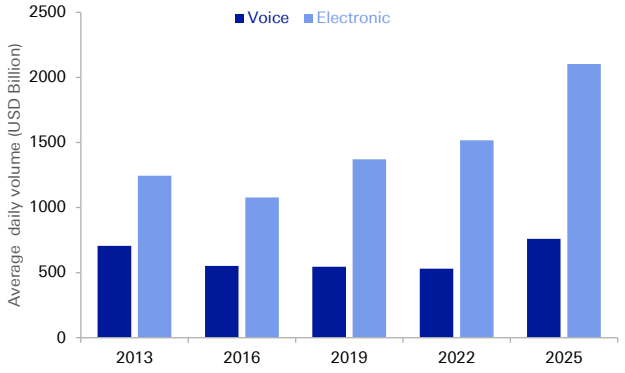


Source : Deutsche Bank

Figure 16 highlights the impact of electronification on FX markets, revealing a

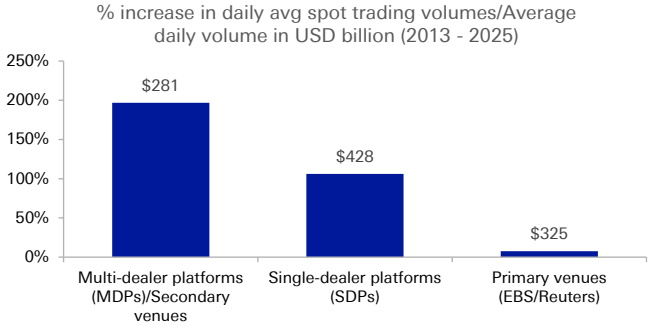
substantial increase in electronic trading in recent years. Notably, the growth of third-party venues has significantly outpaced that of single-bank dealer platforms (Figure 17). These third-party venues provide price streaming aggregation technology and typically charge a pre-trade brokerage fee to liquidity providers.

Figure 16: Share of FX spot trading using electronic methods has been on the rise



Source :Deutsche Bank, BIS Triennial Survey

Figure 17: More than 100% growth in e-spot trading via third party intermediaries (MDPs/Secondary Venues)



Source :Deutsche Bank, BIS Triennial Survey

Inner tier: Inter-dealer segment

For trades that are not internalized (settled across own books), dealers can choose between voice or electronic trading to manage remaining inventory. While traditionally relying on primary venues like EBS/Reuters, they now also have the option of hedging through secondary or alternative venues. This expanded choice, along with increased internalisation, contributes to the observed decline in inter-dealer platform activity (Figure 17).

1.5.3 Trading Venues: Anonymity and Liquidity

Different trading venues can be distinguished based on their offerings in terms of two key characteristics: pre-trade anonymity and liquidity firmness (Chaboud et al., 2023). Pre-trade anonymity refers to the degree to which counterparties remain unknown to each other before a trade is executed. Liquidity firmness, on the other hand, describes the certainty of execution at a quoted price.

Pre-trade anonymity: The level of anonymity can range from fully disclosed to fully anonymous. Multi-dealer platforms (MDPs), acting as aggregators of dealer liquidity, typically operate on a fully disclosed basis. Clients, both financial institutions and corporations, submit requests for quote (RFQs) via the platform to multiple dealers, revealing their identity before trade execution. Similarly, bank-owned single-dealer platforms also function with full disclosure, as the bank dealer knows the identity of the client submitting the RFQ.

At the other end, primary venues running a centralized limit order book (such as EBS/Reuters) generally operate on a fully anonymous basis. The displayed orders do not show a source and all participants can be matched based on the typical price and time priority rules of a CLOB. However, it is important to note that unlike exchanges, this still preserves bilateral relationship of an OTC market structure

since only those that have established bilateral credit relationships, either directly or via their PBs can participate on such venues. Exchange venues, on the other hand, provide the highest possible level of pre-trade (and post-trade) anonymity, basically by definition, as the exchange is the counterparty to each trade.

This anonymity stems from the exchange acting as the central counterparty to every trade, effectively shielding the identities of the ultimate buyers and sellers. This characteristic distinguishes exchange-based trading from other FX venues and contributes to a distinct trading environment.

Secondary FX trading venues occupy a middle ground between the full disclosure of MDPs/SDPs and the complete anonymity of primary venues/exchanges. They offer partial pre-trade disclosure, often employing "tags" to identify counterparties based on specific characteristics or trading styles. This allows participants to select counterparties based on factors like execution quality, creating a tiered system of interaction.

Liquidity firmness: The firmness of liquidity in FX trading, meaning the certainty of execution at a quoted price, is significantly influenced by the practice of "last look." This practice allows liquidity providers (LPs) a brief window, typically milliseconds, to review a trade request before accepting or rejecting it (Oomen, 2016; Cartea, Jaimungal, and Walton, 2019).

Last look was initially developed to address challenges posed by network latencies, platform multiplicity, and varying technological sophistication among market participants. Its acceptable use includes near-instantaneous checks for credit availability and price validity. However, more controversial applications involve imposing additional hold times before trade confirmation.

Liquidity on primary venues is generally considered firm, as last look is prohibited. Conversely, last look is widely employed on secondary venues and universally available on MDPs and SDPs, where LPs retain the option to use it. Consequently, liquidity on these platforms is viewed as less firm, introducing uncertainty regarding trade execution upon hitting a quoted price.

A Request for Stream (RFS) allows a liquidity consumer to receive a continuous flow of quotes for a specified size and duration, such as a trading day. This functions as a continuous Request for Quote (RFQ), providing real-time price updates for the desired transaction.

Single dealer platform (SDP): Bank developed single-dealer platforms (SDPs) to facilitate direct electronic trading with clients.

Multi-dealer platform (MDP): Allows clients to request quotes (RFQs) from multiple dealers at once. Example of MDPs include but not limited to FXconnect, FXall, 360T.

Centralised limit order book (CLOB): It is a centralised electronic trading platform used to match buy and sell orders.

1.5.4 Current market structure challenges

The adoption of electronic trading has generally enhanced FX market quality. Benefits include increased pre-trade transparency, improved operational efficiency through electronic clearing and settlement, and tighter bid-ask spreads and lower intraday volatility in the spot market.

Figure 18 and Figure 19, which track G10 FX high-frequency realized volatility and top-of-the-book spreads respectively, both calculated using one-minute mid-price changes, indicate a downward trend in these metrics. This suggests an improvement in FX market quality over time. The sustained decrease in spreads over time reflects intensified competition for client business in the FX market. The proliferation of trading venues has fostered a competitive environment among liquidity providers, driving them towards consistently narrower, approaching zero, spreads.

Single dealer platform (SDP): Bank developed single-dealer platforms (SDPs) to facilitate direct electronic trading with clients.

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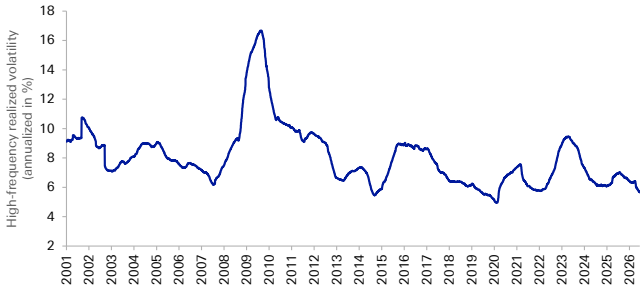
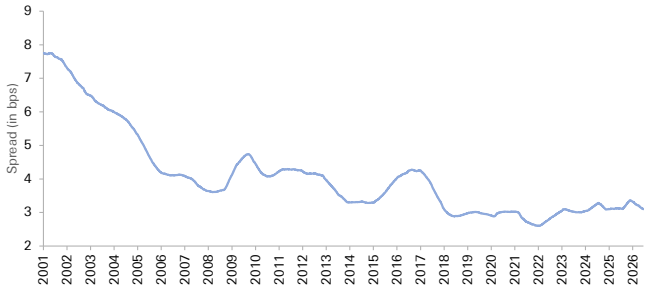
Last look: This practice allows liquidity providers (LPs) a brief window, typically milliseconds, to review a trade request before accepting or rejecting it.

Figure 18: High-frequency G10 FX vol has fallen over time



*Source : Deutsche Bank, Reuters. Daily HF realised vol= square root of sum of squared intraday price changes. Annual vol= Daily vol * sqrt(252)*

Figure 19: Top of the book spread has also fallen over time



Source : Deutsche Bank, Reuters
Spread=(Ask-Bid)/(Ask+Mid)/2

However, the current structure is not without its flaws. We highlight a few key areas of research for our future publications.

The FX Liquidity Mirage: Myth or Reality?

The fragmentation of the FX market, driven by the decline of inter-dealer electronic brokerage and the rise of diverse trading venues, particularly in the dealer-to-customer segment, presents significant challenges for liquidity assessment. The proliferation of platforms can create a deceptive appearance of increased liquidity, a "liquidity mirage," as the same liquidity might be reflected across multiple interconnected venues. Executing a trade on one platform can trigger a simultaneous reduction in liquidity on others, leading to an overestimation of available liquidity when simply aggregating displayed market depth.

Further, accessing liquidity across this fragmented landscape can be technologically challenging and costly due to latency issues and access fees. While aggregated depth might suggest the feasibility of executing a large order, attempting to do so could significantly impact prices as liquidity is tapped across various platforms.

Accurately measuring FX liquidity requires a more sophisticated approach than simply aggregating depth. Metrics like effective spreads (inclusive of slippage and fees) and price impact offer a more realistic representation of transaction costs and market depth. Analysing volume distribution across venues can help identify concentrations of genuine liquidity, but practical constraints such as technological limitations and access fees must also be factored into the assessment.

Electronification of markets and flash crashes

The increasing attention given to flash crashes in recent years has coincided with the electronification of financial markets. Computer algorithms are now used to automatically make trading decisions, submit orders, and manage those orders after submission. Algorithmic trading (AT) now contributes to more than 70% of trading volumes in equity and FX markets (Hendershott, 2012 and BIS, 2018). Recent flash crashes have triggered an important policy debate over the merits of algorithmic trading in financial markets. As argued in Deutsche Bank research (Grover et al., 2019), establishing a direct link between flash crashes and electronification is far from straightforward. First, there is no universal agreement

Algorithmic trading: Automated transactions where a computer algorithm decides the order-submission strategy.

on what constitutes a flash crash event, though recently Christensen et al. (2017) have made some important contributions on this topic. Second, limited high-frequency data makes it difficult to identify flash crash events going back in time, in turn making it difficult to compare the frequency of such events with changes in market structure. Third, there is a risk of confounding the presence of flash crashes and market electronification. Market automation allows trading to take place during highly illiquid hours with limited human presence, raising the question of whether flash crashes as a phenomenon are in any case only possible due to the electronified nature of the market - the alternative in previous eras being a complete market breakdown where no trades took place.

New intermediaries and Bank dealers: Disruption or Evolution?

Changes in the composition of market intermediaries in the FX spot market also raise interesting research questions.

1. Third party multi-dealer platforms: Multi-dealer platforms (MDPs) offer price aggregation technology and charge liquidity providers a pre-trade brokerage fee, enabling clients to access liquidity from multiple dealers. However, MDPs do not assume inventory risk. Increased competition among dealers, driven in part by MDPs, has compressed spreads, reducing bank dealer profit margins. This presents a challenge for banks, who face declining profits while still incurring MDP brokerage fees and bearing inventory risk from client trades executed through these platforms. Thus, potentially reducing the bank's risk-taking capacity.

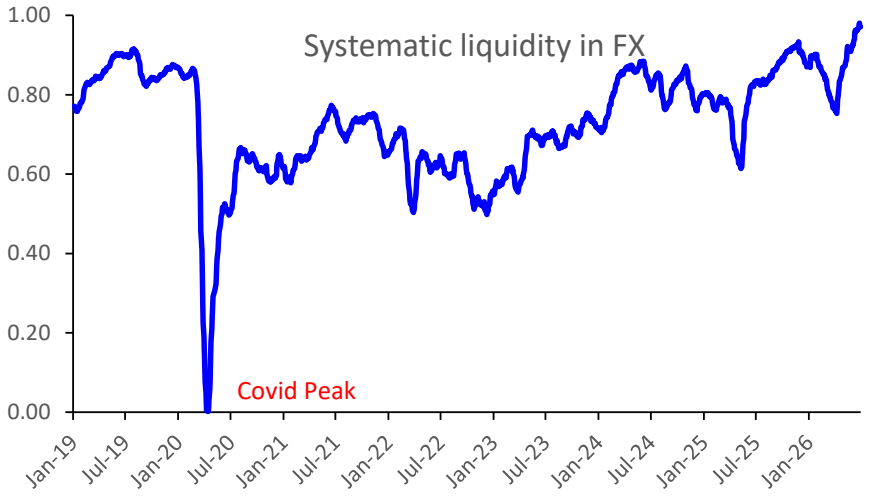
2. Non-bank firms: Bank dealers now face competition from principal trading firms (PTFs) in providing FX liquidity. These lightly capitalised, non-bank PTFs leverage speed and cross-asset correlations to manage their risk. The rise of multiple trading venues, in contrast to the pre-electronification era where banks primarily relied on platforms like EBS/Reuters for inventory management, adds another layer of complexity. This fragmentation makes it challenging for banks to discern true liquidity from the "liquidity mirage" of aggregated, but potentially inaccessible, depth across various platforms.

Has the risk-bearing capacity of FX market intermediaries, in total, changed as a result of (1) and (2)? Has this affected market volatility or the frequency of flash events?

Multi-dealer platform (MDP): Allows clients to request quotes (RFQs) from multiple dealers at once. Example of MDPs include but not limited to FXconnect, FXall, 360T.

For example, during the initial stages of the Covid crisis in March 2020 (Figure 20), top of the book bid-ask spreads in the FX market widened significantly. For instance, the spread for USD/JPY doubled from 1 basis point (bp) to 2 bps, while the spread for GBP/USD increased from 1.5 bps to 3 bps over a 10-day period see Grover, (2020).

Figure 20: FX liquidity dropped significantly during Covid Peak (March 2020)



Source : Deutsche Bank
Systematic FX Liquidity is computed as scaled first principal component of Z-scores across measures and currencies. Here, liquidity measures include but not limited to top of book spreads, order book depth, price impact etc. A higher score implies higher liquidity. For more details, please see FX Liquidity Snapshot: Introducing the DB FX liquidity monitor

New intermediaries: Is FX Price Discovery Broken?

FX market fragmentation raises key questions about price discovery. Have primary CLOBs, despite lower trading volumes, retained their central role? Or has it shifted, potentially to futures markets, given the smaller CLOB size and the increased influence of PTFs using futures-based spot pricing? Analysing latency, order flow, market impact, and correlations across venues is crucial to understanding price discovery in this evolving landscape.

Clearing in FX: Balancing Benefits and Risks in the Modern Market

Regulations like the Standardised Approach for Counterparty Credit Risk (SA-CCR) and the Uncleared Margin Rules (UMR) are increasing capital and margin requirements for non-cleared FX derivatives, making bilateral trading more costly and complex. This is driving some market participants to consider clearing a portion of their FX exposure through CCPs.

Besides exchange-based clearing for standardised FX products using CLOBs, initiatives are emerging to facilitate clearing for bilateral trades like block trades and Exchange For Physical (EFPs). Block trades allow private negotiation of FX futures/options with later clearing house reporting, subject to size/time thresholds. EFPs enable migration of OTC FX exposures to cleared positions, enhancing risk management, operational efficiency, and access to diverse liquidity pools. These initiatives broaden the scope of FX clearing. For more details, please see Grover (2025).

The potential advantages of increased FX clearing include enhanced operational efficiency, potential margin benefits compared to some bilateral arrangements under SA-CCR, expanded counterparty access, and regulatory support. However, drawbacks remain, including increased clearing fees, reduced flexibility in trade

Multi-dealer platform (MDP): Allows clients to request quotes (RFQs) from multiple dealers at once. Example of MDPs include, but not limited to FXconnect, FXall, 360T.

Centralized limit order book (CLOB): It is a centralised electronic trading platform used to match buy and sell orders.

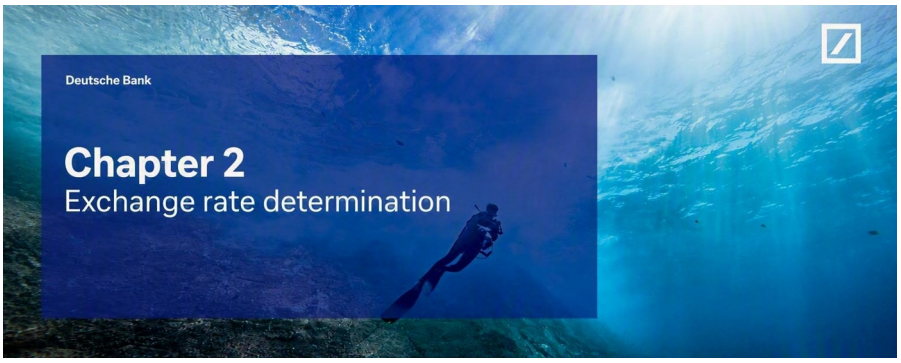
Central Counterparty (CCP): It is a financial market infrastructure that reduces bilateral counterparty risk by acting as the intermediary to both sides of a trade and managing the resulting credit exposure through collateral and default-management mechanisms.

CLS is an independent multi-currency settlement system for FX transactions.

execution and settlement, potential market fragmentation from diverse clearing venues, and the operational challenges of adapting to new technologies and processes. The future of FX clearing depends on the evolving balance between these advantages and disadvantages.

Is FX Settlement Still a Concern?

Settlement risk (Herstatt risk) is the risk of paying out a currency in an FX transaction without receiving the purchased currency. This arises from separate payments in different currencies, often settling across time zones. PvP (Payment versus Payment) mitigates this risk by ensuring simultaneous currency exchange, eliminating the possibility of one party paying while the other defaults. PvP is crucial for FX transactions, especially those settling across time zones, where settlement failures are more likely. CLS operates a PvP settlement service for eligible FX transactions. However, not all FX trades settle through CLS. CLS settlement is not universal due to currency eligibility (not all currencies are included), cost (membership and transaction fees), operational requirements, the availability of alternative methods (like bilateral netting), and internal settlement of intra-group transactions.



Introduction

Exchange rates are famously difficult to forecast. Why can currencies move sharply when macroeconomic fundamentals appear unchanged? Why do apparently compelling exchange-rate models so often fail out of sample? This chapter provides an overview of FX drivers over the short, medium, and long-run horizon. We rely on academic evidence, Deutsche Bank research and practitioner experience to show that while there is no single, universally reliable model of exchange-rate determination, there is a compelling set of frameworks and drivers that can help explain and forecast exchange rate movements over time.

2.1 Why the efficient market hypothesis breaks down in FX

As the world's largest and most liquid financial market, foreign exchange is often regarded as a particularly strong candidate for satisfying the efficient market hypothesis. This states that the current exchange rate fully reflects all publicly available information and it should not be possible to use that information systematically to earn abnormal risk-adjusted profits. In his seminal paper Fama (1970) identified three conditions that would be sufficient to ensure market efficiency:

> *(a) there are no transaction costs;*
>
> *(b) all available information is freely available to all market participants; and*
>
> *(c) all participants agree on the implications of that information for current prices and the underlying model of the economy.*

Omitted from these conditions is, however, a fourth and critically important one that is only left implicit in Fama's paper:

(d) a sufficient number of market participants are profit-seeking and are able to move the market rapidly to equilibrium valuations.

FX market structure itself provides perhaps the most compelling reason why the mechanism supporting market efficiency in FX may be significantly weaker than is commonly assumed. Earlier in this book we presented evidence from several sources suggesting that FX turnover is divided roughly evenly between transactions motivated by profit and transactions undertaken for commercial, hedging, institutional or policy purposes. Moreover, with the expansion of passive investment and the increasing importance of regulatory, mandate and balance sheet constraints, the relative importance of unconstrained, return-seeking risk capital may be declining.

A high share of non-profit-motivated or price-insensitive demand has important implications. If transactional flows are large relative to the capital available to arbitrageurs, prices may then reflect not only expectations of future fundamentals, but also hedging demand, corporate cash flows, reserve-management decisions, benchmark rebalancing, regulation and the balance sheet capacity of intermediaries. These forces will generate persistent deviations from conventional measures of fundamental value and predictable returns as compensation for providing liquidity and bearing inventory or balance sheet risk.

Note that the market inefficiency argument does not require non-speculative participants to be irrational or poorly informed. It rests on the simpler observation that, unlike equities and bonds, currencies are required to complete a broad range of underlying economic transactions unrelated to FX profit-seeking. Other participants may seek to maximise returns but be constrained by leverage limits, risk budgets, benchmarks, regulation or investment mandates. The presence of such participants weakens the practical arbitrage mechanism through which information is presumed to be incorporated into prices. Where non-profit-motivated flows are large and the capital of informed speculators is limited, exchange-rate determination is likely to depend on a broader and more distinctive set of forces than in traditional asset markets.

2.2 Underlying principles of exchange rate determination

First, like all asset prices, the exchange rate is forward-looking: today's level embeds expectations about the entire future path of fundamentals and of risk premia. Engel and West (2005) formalised this, showing that in a rational-expectations present-value model when fundamentals are shifting over time the exchange rate will behave almost exactly like a random walk under certain conditions. This is not because fundamentals are irrelevant, but because the exchange rate optimally aggregates expected future fundamentals. This reconciles the empirical failure of fundamentals to *forecast* currencies with the belief that currencies are ultimately driven by fundamentals.

Second, exchange rates are not purely driven by macroeconomic but also financial drivers. In the traditional academic paradigm, the exchange rate equilibrated goods and money markets. In the modern paradigm, it is set at the margin by financial intermediaries and investors operating in imperfect, segmented capital markets subject to balance sheet constraints and time-varying risk-bearing capacity. This shift, associated with Gabaix and Maggiori (2015), Itskhoki and Mukhin (2021), Jiang, Krishnamurthy and Lustig (2021) and the broader intermediary-asset-pricing literature explains not only long-standing "puzzles" of exchange-rate determination but also begins to explain phenomena like the cross-currency basis.

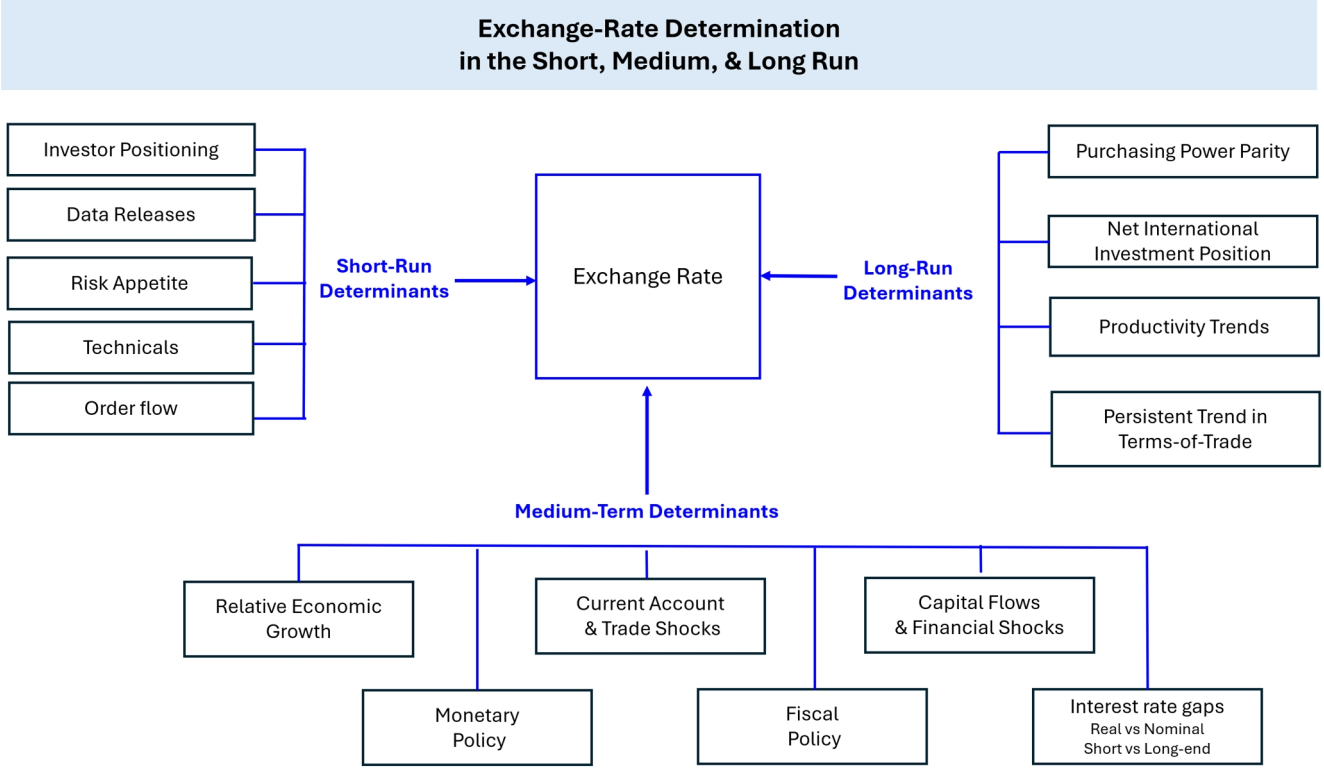
2.3 Time horizons of exchange rate determination

Predicting exchange rates is famously difficult. A helpful organising principle, and the one that structures the remainder of this chapter, is to think about exchange rate determination across three horizons:

- In the **short run** (intraday to a few weeks), exchange rates are dominated by order flow, positioning, sentiment, risk appetite, data releases and technical/trend-following behaviour.
- In the **medium run** (months to a couple of years), cyclical macro forces — interest rate differentials, the monetary policy cycle, relative growth, the current account and capital flows pull currencies around their long-run anchors.
- In the **long run** (several years and beyond), fundamentals reassert themselves: relative prices (PPP), productivity (Balassa-Samuelson), terms of trade, and net international investment positions shape the equilibrium real exchange rate.

In the next few sections we will proceed by horizon and by mechanism. We begin with the long-run anchors (PPP, productivity, terms of trade, external balances and equilibrium-exchange-rate concepts), then turn to the medium-run parity conditions and macro models, then to the short-run microstructure of the market (order flow, positioning, options and technical analysis). We then review forecasting performance tests, before moving on to modern financial theories of the exchange rate, central bank intervention, currency regimes and crises, and finally a set of frontier topics — machine learning and digital currencies.

Figure 21: Overview of Exchange Rate Determination



Source : Deutsche Bank

2.4 Purchasing Power Parity and the long-run real exchange rate

The concept

Purchasing power parity is the oldest and most intuitive theory of the exchange rate.

- *Absolute* PPP says that once converted into a common currency, a representative basket of goods should cost the same everywhere; the nominal exchange rate should equal the ratio of national price levels. This absolute PPP form has little empirical support for reasons discussed later.
- *Relative* PPP says that the change in the exchange rate should offset the difference in inflation rates, so that the real exchange rate (the nominal rate adjusted for relative prices) is stationary and mean-reverting around a constant.

The real exchange rate can be written simply as:

$$q = s + p^* - p$$

Where s is the nominal exchange rate (domestic price of foreign currency, in logs), p is the domestic price level and p* the foreign price level. PPP holds in the long run if q reverts to a constant mean.

The PPP puzzle and the half-life debate

The central empirical problem, crystallised by Rogoff (1996) as the "PPP puzzle", is FX deviations from PPP are highly persistent and volatile. In Rogoff's (1996, p. 648) own words, "consensus estimates for the rate at which PPP deviations damp... suggest a half-life of three to five years, seemingly far too long to be explained by nominal rigidities." Such slow reversion is very hard to reconcile with nominal price stickiness that lasts one to two years; something other than sticky goods prices must be driving the persistence.

The survey by Taylor and Taylor (2004, Journal of Economic Perspectives) documents how the profession's view on PPP has evolved, concluding that by the early 2000s "the idea of long-run PPP now enjoys perhaps its strongest support in more than thirty years." Two methodological advances drove this:

1. Panel and long-horizon data greatly increased the power to reject the random-walk hypothesis and detect mean reversion.
2. More importantly, nonlinear mean reversion modelling is used: because arbitrage in goods markets is subject to transaction costs, trade barriers and large non-traded sectors, the real exchange rate reverts faster the further it is from equilibrium, and can wander almost freely within a "band of inaction". Once nonlinearity is modelled, estimated half-lives fall and PPP reversion looks far more consistent with economic theory. The more recent survey by Vo and Vo (2023, Journal of Economic Surveys) confirms the long-run validity of PPP and a 3-5 year half-life.

Practical implications and DB house research

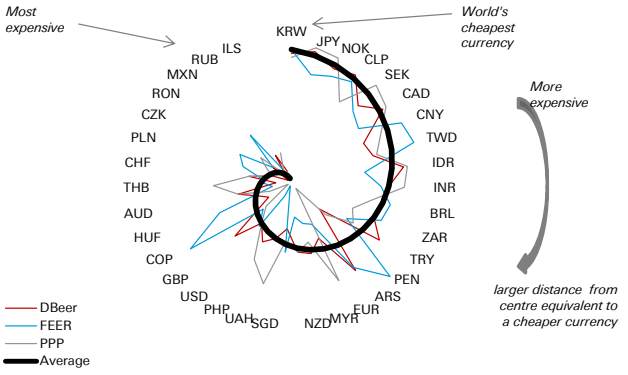
For investors, the operational lesson is that PPP is a poor timing tool but a powerful valuation anchor. Deviations can persist for years, an overvalued currency can become more overvalued, but they do eventually correct, and the correction is exploitable if the holding period is long enough. This is precisely what Deutsche

Bank's own back-testing confirms.

Grover (2025), in a study of long-term FX value investing, finds that PPP-based valuation signals generate economically significant mean-reversion signals whose performance improves as the holding period lengthens, peaking at around 24 months - consistent with a PPP misalignment half-life in excess of two years, echoing Lothian and Taylor (2000), Ricci et al. (2013) and Winkler (2017).

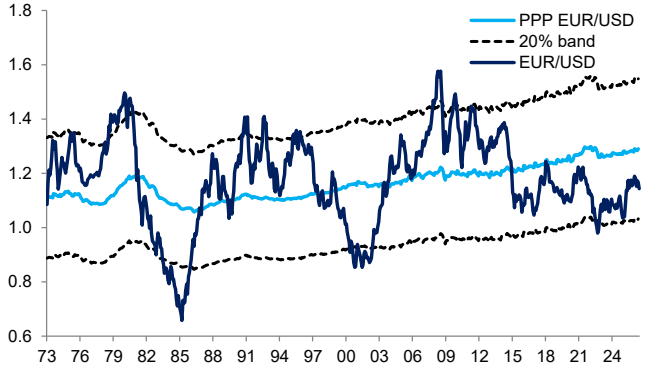
A cross-sectional portfolio that goes long the most undervalued and short the most overvalued currencies on a 12-month hold delivers a Sharpe ratio of 0.60 on the PPP signal after costs, with PPP working best in G10. Volatility-adjusted valuation signals outperform simple sign-based signals, particularly in G10. Notably, Grover (2025) also finds that valuation models underperformed over the two years to early 2025, especially in G10, plausibly because cyclical drivers and extreme macro divergence (aggressive and desynchronised hiking cycles) dominated valuation over that window, a reminder that valuation is a long-horizon strategy that can suffer extended drawdowns.

Figure 22: Valuations ranked



Source : Haver Analytics, Deutsche Bank

Figure 23: EURUSD versus PPP



Source : Refinitiv Workspace Datastream, Deutsche Bank Research

Relative PPP does not imply that real exchange rates must remain constant over all horizons. Even when nominal exchange rates broadly adjust to differences in national inflation rates, equilibrium real exchange rates can drift as the relative price of domestic and foreign consumption baskets changes. Such movements may reflect persistent differences in productivity growth between tradable and non-tradable sectors, shifts in the terms of trade, changes in trade barriers and consumption preferences, pricing-to-market, or movements in international capital flows and risk premia.

Real exchange rate appreciation or depreciation should therefore not automatically be interpreted as a deviation from equilibrium or evidence against PPP; it may instead represent an adjustment in the equilibrium real exchange rate itself. The best-known explanation for such structural movements is the Balassa–Samuelson effect, to which we turn in the following section, before considering other forces that can generate persistent cross-country differences in prices and real exchange rates.

2.5 Productivity, the Balassa-Samuelson effect and terms of trade

Balassa-Samuelson

Absolute PPP based on broad consumer-price baskets fails partly because those baskets include large non-traded components (services, housing). These components cannot be arbitraged over borders. The Balassa-Samuelson (BS) hypothesis provides the classic explanation for systematic trend deviations from absolute PPP.

The logic is that countries that experience faster productivity growth in their traded-goods sector can pay higher wages there; wage equalisation across sectors pushes up wages, and hence prices, in the non-traded sector; this raises the country's overall price level and appreciates its real exchange rate. Rich, high productivity countries therefore have higher price levels and currencies that look "overvalued" on a naive PPP metric but are in fact in equilibrium. Productivity convergence should thus produce trend real appreciation in catching-up economies, or even in developed economies with different productivity rates.

The empirical record on Balassa-Samuelson (BS) is mixed. Egert, Halpern and MacDonald (2006) and the survey by Tica and Druzic (2006) document that the hypothesis has been tested dozens of times across scores of countries. They find that the evidence is strongest for the intermediate links in the mechanism: richer or more productive economies have higher relative prices, and faster tradable-sector productivity growth is often associated with rising non-tradable prices. Cross-sectional tests are overwhelmingly supportive, while panel and time-series results are more heterogeneous and sensitive to the countries, periods and econometric specifications used.

Evidence for the full proposition, that productivity differentials reliably explain aggregate real exchange-rate appreciation, is considerably weaker, in part because purchasing-power parity often fails even within the tradable sector. Productivity catch-up tends to raise relative service prices and can generate real appreciation, but its magnitude is highly uncertain and it is rarely sufficient on its own to explain observed exchange-rate movements.

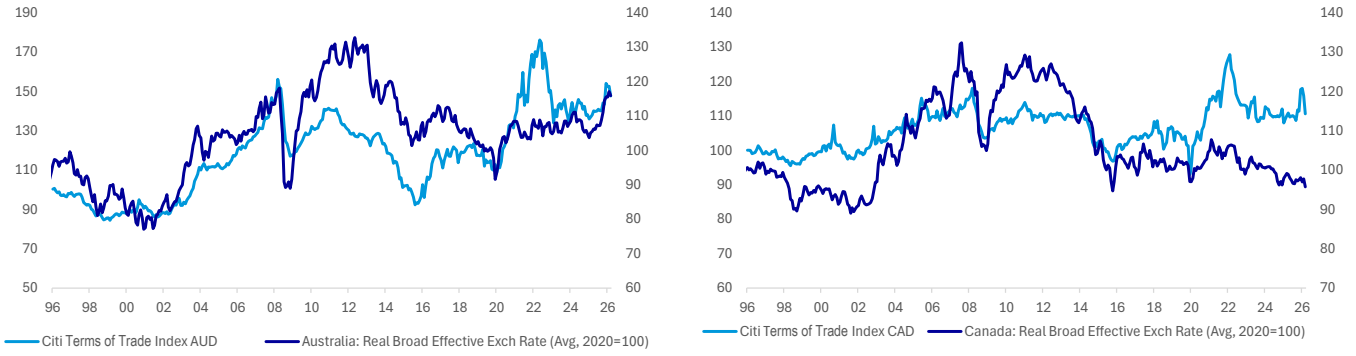
Terms of trade and commodity currencies

For commodity-exporting economies, the terms of trade, the price of exports relative to imports, can be a more important driver of the real exchange rate than productivity. A rise in the world price of a country's main export commodity increases the purchasing power of its exports and raises national income and wealth. Higher income increases domestic demand, including for non-tradable goods and services whose prices are determined locally. Because the prices of internationally traded goods are more tightly anchored by world markets, the resulting rise in the relative price of non-tradables produces a real exchange-rate appreciation.

This mechanism is sometimes associated with what is known as "Dutch disease". In the classic framework, a commodity boom generates a spending effect, as higher national income raises demand and prices in the non-traded sector, and potentially a resource-movement effect, as labour and capital shift toward the booming commodity sector and away from other tradable industries. Both

channels tend to raise the relative price of non-tradables and appreciate the real exchange rate. The resulting currency appreciation can in turn reduce the competitiveness of other non-commodity exporters in the country.

Figure 24: Commodity terms of trade vs. commodity-currency real exchange rates for AUD and CAD



Source : Deutsche Bank

Terms of trade commodity shocks should be distinguished from the Balassa–Samuelson effect. Balassa–Samuelson attributes real appreciation to relatively rapid productivity growth in the tradable sector: higher tradable-sector productivity raises wages, which spill over into the non-tradable sector, increasing non-tradable prices and appreciating the real exchange-rate. A commodity terms-of-trade shock can produce a similar observable outcome but originates from an improvement in export prices and national income rather than faster productivity.

Chen and Rogoff (2003) formalised the empirical importance of this mechanism for "commodity currencies," showing that the real exchange-rates of major commodity exporters such as Australia, New Zealand and Canada have strong long-run relationships with world commodity prices.

A striking and much-cited result is the forecasting reversal in Chen, Rogoff and Rossi (2010, Quarterly Journal of Economics): because exchange rates are forward-looking asset prices that embed expectations about future fundamentals, commodity-exporter exchange rates actually help predict world commodity prices, more reliably than the reverse. The follow-up by Ferraro, Rogoff and Rossi (2015, Journal of International Money and Finance) sharpens the caveat for practitioners: the relationship between oil prices and the Canadian dollar is robust and strong at the daily (contemporaneous) frequency but is fragile and largely disappears at the monthly and quarterly horizons and out of sample. Commodity terms of trade are a real-time co-movement story more than a forecasting tool. Terms of trade and the BS effect are incorporated into our DBeer model, below.

2.6 External balances, net foreign assets and equilibrium exchange rate concepts

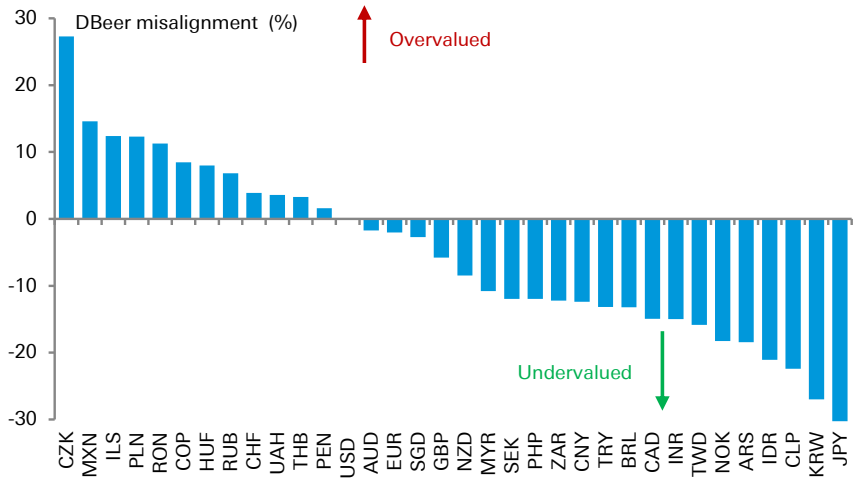
The flow and stock approaches

Two related traditions link the exchange rate to a country's external position. The flow (current account) approach holds that a currency must adjust to bring the current account toward a sustainable level: persistent deficits require a weaker real exchange rate to restore competitiveness. The stock (portfolio/net-foreign-asset) approach emphasies that a country running deficits accumulates external liabilities, and the real exchange rate must eventually be weak enough to generate the trade surpluses that service them or strong enough, for creditor nations, to permit the running down of surpluses. The policy and sell-side community operationalises these ideas through equilibrium exchange-rate models, of which there are three main families:

- The **FEER** (**Fundamental Equilibrium Exchange Rate),** (Williamson 1994) and the closely related DEER (Desired EER) are normative, macroeconomic-balance concepts: they identify the real effective exchange rate consistent with the economy achieving internal balance (full employment, target inflation) and external balance (a "sustainable" current account) over the medium term. FEER requires an assumption about a current account equilibrium and trade elasticities but is largely model-free about the adjustment path. Such an approach is used as part of our FX Valuation Snapshot.
- The **BEER** (**Behavioural Equilibrium Exchange Rate)**, Clark and MacDonald 1998) is essentially an econometric model. It regresses the observed real exchange rate on a set of fundamentals, net foreign assets, productivity/relative prices (Balassa-Samuelson), the terms of trade, and real interest-rate differentials, and defines the equilibrium as the fitted value, with the residual measuring misalignment. One of our proprietary valuation frameworks use a BEER model, called DBeer. It uses terms of trade, a productivity ratio and trade openness. Grover (2025) shows it produces mean-reversion signals and that, in contrast to PPP, DBeer performs better in emerging markets, where idiosyncratic and terms-of-trade dynamics dominate, while PPP dominates in G10.

- The **IMF's External Balance Assessment (EBA)**, introduced in 2012, is not a separate exchange rate theory so much as a framework that combines three approaches. First, its principal current account model estimates an underlying current account balance and compares it with a "norm" consistent with economic fundamentals and policies set at their desired levels. The resulting current account gap is then converted, using estimated trade elasticities, into the real-exchange-rate adjustment that would be required to close it. In this respect, the operational procedure is similar to a FEER calculation, although the current-account norm is itself estimated econometrically rather than imposed entirely outside the model. Second, EBA also contains direct REER panel regressions that are closer in spirit to the BEER approach, estimating the exchange rate implied by fundamentals and desired policy settings. A third, external-sustainability approach calculates the current-account balance needed to stabilise or move the net foreign asset position towards an appropriate benchmark. IMF staff use the three approaches as cross-checks and combine them with country-specific evidence and judgement to arrive at an overall assessment of the current-account and REER gaps.

Figure 25: Dbeer estimated over/undervaluation (%) versus USD across G10 and major EM currencies



Source : Haver Analytics, Deutsche Bank

2.7 Medium-run macro models

The monetary model of exchange rates

The monetary approach treats the exchange rate as the relative price of two monies, determined by money supplies, output (money demand) and interest rates. In Dornbusch's (1976) sticky-price version this is how exchange rate overshooting is generated. Because goods prices are sticky, a monetary expansion causes the exchange rate to depreciate more than proportionately on impact before reverting, explaining why exchange rates are more volatile than the underlying fundamentals.

The original monetary model dominated academic thinking in the 1970s but was challenged by the forecasting failure documented by Meese and Rogoff (1983). It nonetheless remains a useful framework for thinking about the exchange rate consequences of relative monetary policy and the interest rate differential is a key input used by FX market practitioners to this day.

Covered and uncovered interest parity, carry trades in FX

The relationship between interest rate differentials and exchange rates is one of the most widely studied in academic research and has given birth to what are known as the two "parity conditions".

- Covered interest parity (CIP) is an arbitrage relationship linking spot and forward exchange rates to interest rates across currencies. It states that an investor should earn the same return from investing in domestic cash as from converting funds into a foreign currency, investing at the foreign interest rate, and simultaneously using a forward contract to lock in the exchange rate at which the proceeds will be converted back. If the two hedged returns differed, investors could borrow in the lower-returning market and invest in the higher-returning one, using the forward market to eliminate exchange-rate risk, until the arbitrage opportunity disappeared.

CIP held almost exactly in the data until 2008 (see later section).

- Uncovered interest parity (UIP) is a looser condition: it says the expected change in the exchange rate should equal the interest rate differential, so that a high-interest-rate currency should be expected to depreciate by exactly enough to offset its yield advantage. UIP is tested through the Fama (1984) regression of realised exchange-rate changes on the interest differential (or forward premium):

$$\Delta s = \alpha + \beta(i - i^*) + \varepsilon$$

Where Δs is the change in the base currency and i is the base-currency interest rate.

UIP predicts β = 1. The overwhelming empirical finding for decades was that the estimated β was not only below one but frequently negative. This is what is known as the forward premium puzzle in academia or as the "carry trade" in daily market practice: high-yield currencies tend, on average, to appreciate rather than depreciate, which is exactly what makes the carry trade profitable. This is one of the most robust anomalies in international finance.

Figure 26: Illustration of FX carry trade

Source :Deutsche Bank

The literature offers several competing explanations for the existence of carry trades. One interpretation is that carry returns represent compensation for bearing state-dependence or crash risk rather than a free arbitrage. Brunnermeier, Nagel and Pedersen (2009) show that carry returns are negatively skewed: high-yielding currencies can fall sharply and funding currencies appreciate when risk appetite and market liquidity deteriorate. Such episodes can trigger simultaneous unwinding of leveraged positions, amplifying exchange-rate movements. Menkhoff, Sarno, Schmeling and Schrimpf (2012) similarly find that high interest rate currencies tend to perform poorly following unexpected increases in global FX volatility, while low interest rate funding currencies provide insurance in precisely those adverse states.

The broad conclusion is therefore more nuanced than a simple rejection of UIP. Carry trades have historically earned positive average returns, but those returns are negatively skewed, perform poorly during volatility and liquidity shocks, and appear partly related to global risk exposure. At the same time, conventional models have struggled to explain fully the magnitude and time variation of the returns.

Taylor-rule fundamentals and interest rate differentials

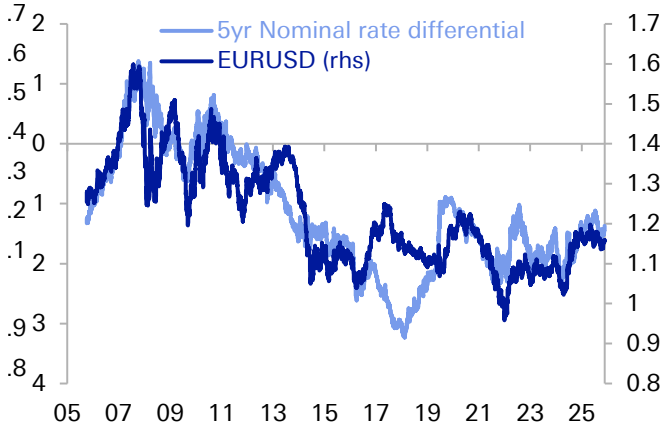
In its most recent form, monetary modelling of exchange rates replaces money supplies with the monetary policy reaction function and interest rates. If central banks set interest rates according to a Taylor rule (responding to inflation and the output gap), then the interest differential and hence, via UIP, the expected exchange rate change can be expressed in terms of relative inflation and output gaps. Engel and West (2005), Molodtsova and Papell (2009) and Rossi (2013) showed that Taylor-rule fundamentals can help model exchange rates. With Rossi finding that Taylor-rule and net-foreign-asset fundamentals as the most promising predictors. The strong academic support of monetary exchange-rate modelling is reflected in the strong emphasis on interest rate differentials in market practitioner analysis.

Relative interest rates remain among the most closely watched drivers of exchange rates among market participants, but the distinction between nominal and real differentials is critical:

- Nominal interest-rate differentials determine the carry earned on an unhedged currency position and therefore matter directly for short-run portfolio allocation; in practice, currencies often respond most strongly to revisions in the expected path of policy rates rather than to the current policy rate itself.
- For medium-run valuation, however, the more relevant concept is the expected ex ante real interest-rate differential**,** the nominal rate adjusted for expected inflation, because it measures the relative return on purchasing power. A high nominal yield generated by tighter monetary policy and a higher real rate should, other things equal, support a currency, whereas the same nominal yield caused by high expected inflation may provide little support and can instead be associated with depreciation.

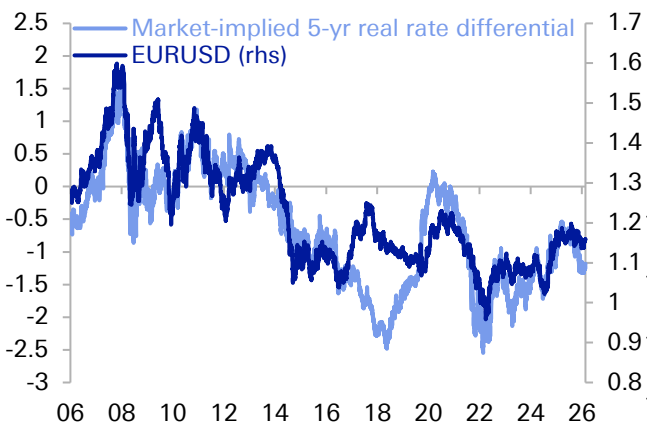
This distinction is central to Frankel's (1979) real-interest-differential model, although the empirical relationship is not stable. Meese and Rogoff (1988) find little robust connection between real exchange rates and real-rate differentials, while Baxter (1994) found a stronger relationship at business-cycle and lower frequencies. The practical conclusion is that nominal differentials matter most for carry and near-term policy pricing, while expected real rate differentials are the more meaningful, though still imperfect, guide to medium-run exchange-rate valuation.

Figure 27: EUR/USD and the nominal interest rate differential



Source : Deutsche Bank, Bloomberg Finance LP

Figure 28: EUR/USD and the real interest rate differential



Source : Deutsche Bank, Bloomberg Finance LP

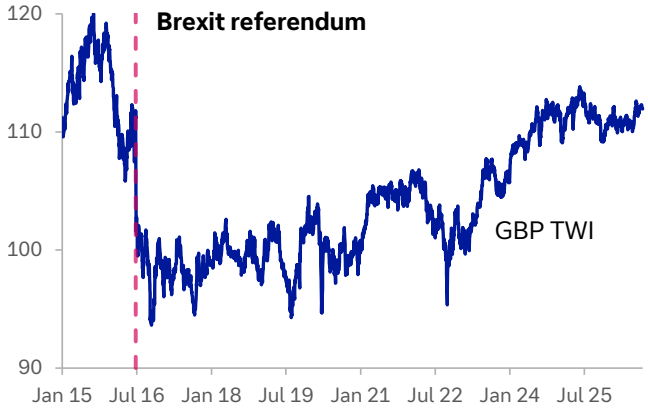
Current accounts, capital flows and the basic balance

Current accounts provide one of the clearest medium-run anchors for exchange rates because they summarise the gap between national saving and domestic investment. Persistent current-account deficits imply that domestic consumption exceeds output and must be financed by net capital inflows, while persistent surpluses imply the opposite. In the medium run, those external positions cannot drift without consequence: a weaker net foreign asset position must eventually be stabilised through stronger future net exports, favourable valuation effects on the external portfolio, or both, and the exchange rate is one of the key relative prices through which that adjustment is delivered (Chinn and Prasad, 2003; Gourinchas and Rey, 2007).

The composition of capital flows matters as much as the headline balance. Portfolio equity and bond flows are typically more sensitive to relative returns, hedging demand and global risk appetite, so marginal reallocations in these categories can produce sizeable currency moves. FDI is usually the stickier component of external financing, reflecting the partial illiquidity of direct ownership. For FX, a deficit financed by long-horizon FDI is therefore very different from one financed by return-seeking portfolio debt or equity inflows that can reverse quickly when relative yields or risk sentiment change (Hau and Rey, 2006; Albuquerque, 2003; Gabaix and Maggiori, 2015; Greenwood et al., 2023).

Large changes in the trading regime can also shift the expected path of the external balance abruptly. Events such as Brexit, the imposition of tariffs or a major trade agreement can alter expected export demand, import costs, investment and the location of production, with potentially significant consequences for the equilibrium exchange rate. Importantly, currencies are forward-looking asset prices: they need not wait for a trade shock to appear in realised current account data. Once markets assign a sufficiently high probability to a future change in trade arrangements, the exchange rate can adjust immediately in anticipation of its expected effects on future trade flows, investment and national income. The sizeable adjustment in sterling around the 2016 Brexit referendum is a prominent example of the exchange rate moving well before the full effects of the new trading relationship became visible in the balance of payments data.

Figure 29: GBP moved well before the full effects of Brexit took place

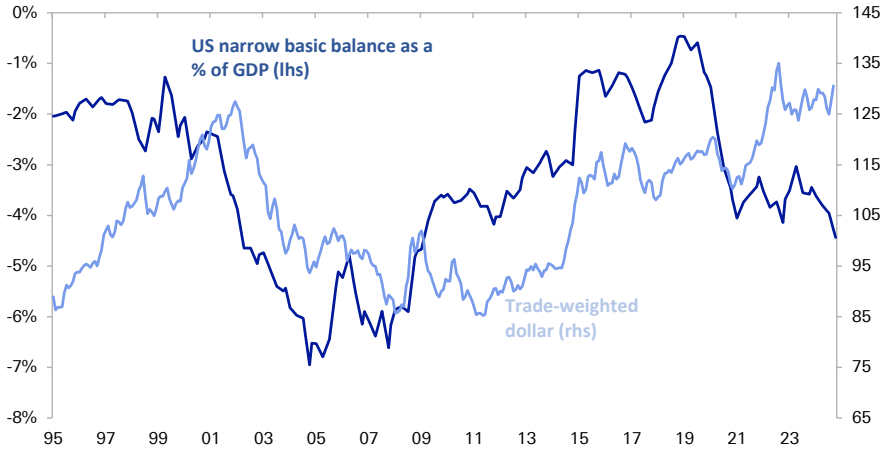


Source : BIS, Bloomberg Finance LP, Deutsche Bank

For medium-run FX analysis, the key issue is whether saving-investment trends and the composition of external financing are moving in a currency supportive or negative direction. Economies with chronically low saving relative to investment are more likely, over time, to require a weaker real exchange rate to generate the external adjustment needed to service a larger stock of foreign liabilities, whereas economies with excess saving and stronger external balance sheets can sustain firmer currencies for longer (Chinn and Prasad, 2003; Gourinchas and Rey, 2007; Lane and Milesi-Ferretti, 2004).

Practitioners often construct measures of the basic balance to assess shifts in external financing and the underlying demand for a currency. A narrow measure typically combines the current account balance with net foreign direct investment, on the grounds that these are among the slowest-moving components of the balance of payments. Broader measures may also include longer-term portfolio flows, particularly bond and equity investment. These indicators are useful but incomplete, because other financial flows can at times dominate exchange rate dynamics. This is especially true for currencies used extensively as funding currencies: during the 2000s, for example, large-scale borrowing in the low-yielding yen generated substantial capital outflows, while the unwinding of those positions periodically produced sharp yen appreciation.

Figure 30: Basic balance metrics are loosely correlated to currencies



Source : Deutsche Bank, Bloomberg Finance LP

The importance of hedge ratios as a medium-run FX driver

Alongside the current account and portfolio flow channels discussed above, the hedging decision on existing foreign asset stocks may become a first-order driver of exchange rates**.** The mechanism is a stock adjustment one: when a non-US pension fund or insurer holds unhedged US securities, a decision to raise its hedge ratio does not require selling the underlying bond or equity, but it does generate a flow of dollar sales to enact the forward hedge that is then rolled in perpetuity. Because the pool of cross-border foreign ownership of assets is very large, even modest shifts in the desired aggregate hedge ratios of foreign investors can dominate other flows in the balance of payments. The medium-run character of the channel comes from its persistence: hedge ratios are sticky, adjusted slowly and in cycles rather than day-to-day.

The incentive to hedge is governed largely by the cost of hedging, which for a rolling short-dated hedge would tend to be in the 3m to 1-year tenor and approximated by the front-end interest rate differential plus the cross-currency basis. A wide positive rate differential between the US and the rest of the world makes selling the dollar forward expensive, it subtracts directly from the hedged return and, when the hedged yield on a US bond turns negative, removes the entire rationale for holding the asset in hedged form. This is precisely why hedge ratios fell across the developed world as US front-end rates rose after 2022, leaving investors structurally under-hedged and dollar exposure near historical highs (Gopal, Baker and Dynan, 2025).

The relationship is not mechanical, however and after the US administration's trade actions in 2025 investors raised dollar hedges sharply in response to surprise shifts in US economic and foreign policy (Shin, Wooldridge and Xia, 2025 and Saravelos, 2025). The practical conclusion is that hedge ratio dynamics are best read as a joint function of the short-end rate differential (the cost) and broader shifts in underlying currency views (Grover and Harvey, 2024).

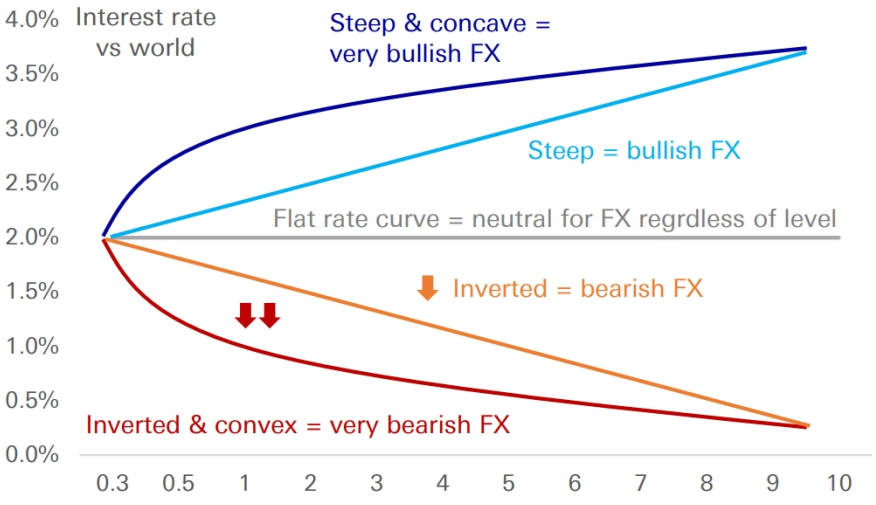
The yield curve and FX

The shape of the yield curve carries independent information for the medium-run exchange rate, and does so through two complementary channels. The first is

predictive. Because the term structure embeds the market's expectations of future short rates, growth, inflation and policy risk (term premium), cross-country differences in the slope of the curve can help forecast currency returns at longer-horizons. Ang and Chen (2010) show that term spreads and changes in interest rates significantly predict excess FX returns for up to twelve months, and that portfolios built on these signals carry low skewness and little correlation with the carry trade.

Winkler (2022) breaks down term structures into three curve factors: level, slope and curvature. He shows that the dollar depends more on the slope and curvature of the US rate curve than its level. A steep and hump-shaped rate curve relative to the rest of the world, implying a young business cycle and rising real rates, predicts a stronger dollar, and vice versa. The practical reading is that a relatively steep domestic curve is a signal of expected future easing or of a term premium demanded for macro risk, both of which are currency-negative over the medium run.

Figure 31: FX implications of the shape of the rate curve relative to the world



Source : Deutsche Bank

The second channel via which the yield curve affects FX is a flow mechanism that connects directly to the hedging discussion above. Nenova, Schrimpf and Shin (2025) show that the relative slope of yield curves governs where hedged bond capital is allocated: a steeper curve draws foreign bond inflows and, with them, a rise in FX swap hedging volumes because hedging is cheap. Flatter curves reduce the attractiveness of FX hedging encouraging more un-hedged inflow.

2.8 Short-run determination I: Order flow and market microstructure

Order flow and market microstructure

Because macro fundamentals are near-useless at high frequency, the short-run behaviour of exchange rates is the domain of market microstructure, the study of how the trading process itself conveys information and moves prices. The initial insight, due to Evans and Lyons (2002), is that order flow (signed transaction volume — the net of buyer-initiated and seller-initiated trades) is the proximate driver of exchange rates and the channel through which dispersed information about fundamentals, risk appetite and hedging demand is aggregated into price. Order flow explains a large share of daily and intraday exchange-rate variation, more than macro news alone.

The most important modern contribution is Menkhoff, Sarno, Schmeling and Schrimpf (2016), where using a large dealer's disaggregated customer order flow, they show that order flow is highly informative about future exchange rates. Crucially, different customer segments (asset managers, hedge funds, corporates, private clients) differ markedly in their trading styles, information content and risk exposures. Financial and non-financial customers tend to trade in opposite directions, so the dealers intermediate risk sharing between them, and the information in financial-customer flow has a persistent, longer-term impact on returns consistent with it carrying fundamental information.

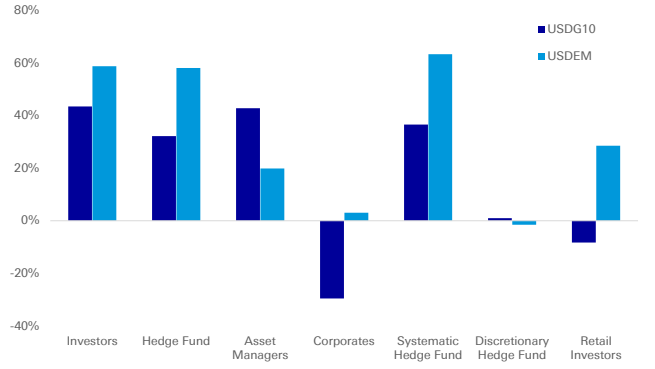
DB house research: CORAX and the anatomy of FX flows

Deutsche Bank's CORAX indicators (Categorised ORderflow from Autobahn FX), derived from the bank's globally aggregated electronic spot flow, allow this literature to be applied directly. Grover (2021), building explicitly on Menkhoff et al. (2016), studies discretionary versus systematic hedge-fund flows across G10 and liquid EM from January 2016. The findings largely mirror the results of the academic literature:

- Both discretionary and systematic hedge-fund flows are positively correlated with contemporaneous spot, the relationship being stronger for systematic funds.
- Looking forward, discretionary investor flows in G10 positively predict spot for up to about ten days, turning negative at longer horizons — while systematic flows provide more of a mean reversion signal. In EM, discretionary flows show strong mean reversion.
- Across the broader client taxonomy, real-money flows are positively correlated with contemporaneous and one-day-ahead spot; corporate flows are negatively correlated contemporaneously but positively related to future spot in the near-to-medium term in G10; and retail flows are negatively correlated (contrarian in the near term, becoming trend-following at the medium term).

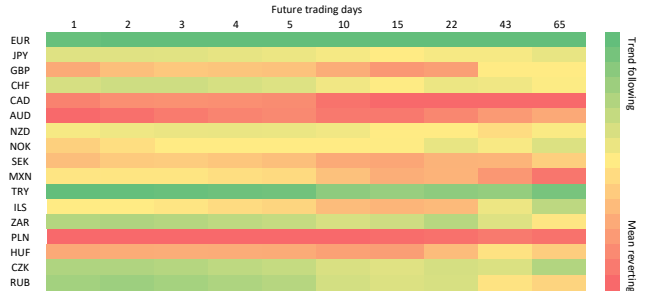
Flows are powerful contemporaneous and short-horizon indicators, and the sign of their predictive content depends on which client segment is trading and over what horizon.

Figure 32: Systematic HF flows show a higher degree of contemporaneous correlation than other client categories



Source : Deutsche Bank

Figure 33: Correlation between CORAX systematic HF flows and future spot move



Source : Deutsche Bank

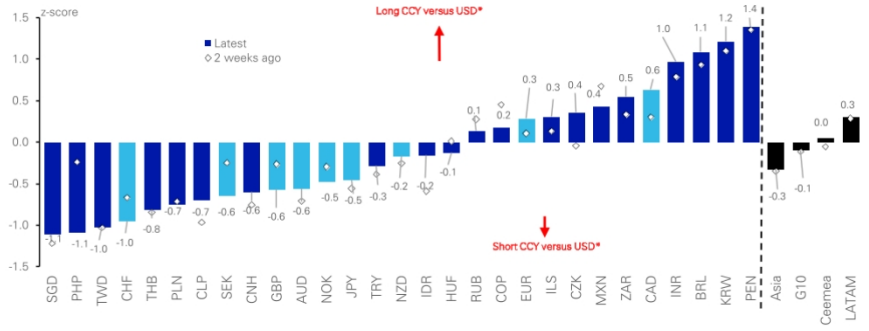
2.9 Short-run determination II: Positioning, options flow, sentiment, data releases and technical analysis

Positioning

Because currencies can overshoot and trend, measures of market positioning are valuable, particularly as contrarian signals at extremes. Positioning refers to the stock of existing long and short currency exposures held by investors: a market is described as "long" a currency when investors have, in aggregate, built positions that benefit from its appreciation, and "short" when they are positioned for depreciation. This differs from flows, which measure new buying and selling over a period; positioning instead captures the cumulative result of past flows and therefore indicates how much directional risk is already embedded in investor portfolios.

Positioning matters because the impact of new information depends partly on how investors are already positioned. When a currency has been rising for some time, investors may progressively add to long positions, reinforcing the trend. But once positioning becomes crowded, fewer investors remain to initiate additional longs, while a negative surprise can force existing holders to unwind positions simultaneously. Stop-losses, margin constraints and systematic trading strategies can amplify this process, turning an initially modest reversal into a sharp move. Conversely, an extremely short currency can rally strongly even on modestly positive news as investors rush to cover their positions. Positioning is therefore most useful not as a mechanical timing indicator, but as a measure of market vulnerability and asymmetry: stretched positioning raises the probability that an adverse surprise generates a disproportionately large price response.

Figure 34: DB EPIC+ positioning (as of 01-Sep-2026)



*Source : Deutsche Bank, Bloomberg Finance LP, DTCC, EBS, Reuters, IMM, CORAX Note: 1) * Long (or short) versus EUR for CZK, HUF, PLN, CHF, NOK and SEK. 2) The black bars on the right-hand side are average positioning levels for G10, Asia, CEEMEA and LatAm. 3) The y-axis shows readings of positioning index in a z-score format, which is an average of rolling one year z-scores for individual sub-indicators. Thus, the long or short of each currency is relative to its one year's average instead of absolute level.*

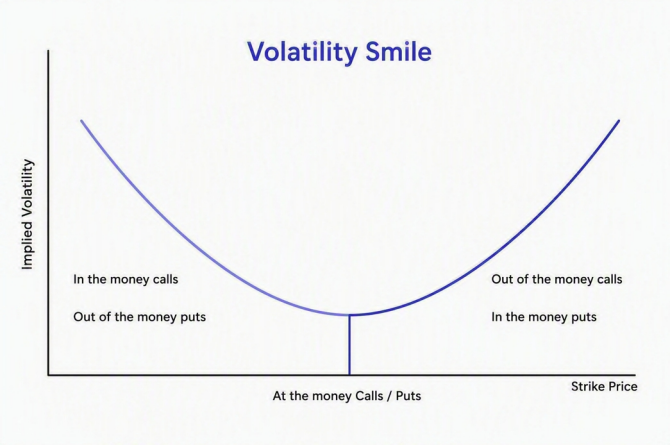
The classic public gauges are the CFTC/IMM speculative futures positions, which provide weekly data on long and short positions held by different investor categories, and dealer or investor sentiment surveys. These measures capture only parts of the global FX market, however, and practitioners therefore supplement them with other metrics.

Deutsche Bank's EPIC+ framework aggregates positioning across both direct measures (CFTC/IMM) and indirect ones (CORAX & spot flows, inter-dealer order-book data, risk reversals, and the DTCC-derived Skew Volume/COFFEE options indicators). Grover, Nandy, Charles and Xu (2024) evaluate the live, out-of-sample performance of position-fading (contrarian) strategies since the EPIC+ launch in January 2021 and find them still profitable for most pairs over the medium term: an equal-weighted position-fading portfolio delivered a Sharpe ratio of 0.54 after costs (0.75 excluding EUR, CNH and THB, which are better traded as position-following), while a cross-sectional long-short portfolio that fades excessive positioning delivered a Sharpe ratio of 1.2 after costs (against 1.7 in backtest). Importantly, performance dips when FX volatility is high and the dollar is trending strongly — contrarian signals work best in less-trending markets. Risk reversals, CORAX and IMM are the most effective inputs for spotting fading opportunities in DM; the COFFEE options-skew indicator and spot imbalance work better in EM.

Options market information

Given the size and depth of the currency option market, option prices can contain information about market expectations and risk sentiment that is not fully visible in spot or forward rates. Implied volatility measures the market price of protection against exchange rate movements, while the volatility smile provides information about the perceived balance of tail risks.

Figure 35: Simplistic depiction of a volatility smile



Source : Investopedia, Deutsche Bank

The most widely followed measure of this asymmetry is the risk reversal, usually calculated as the implied volatility of an out-of-the-money currency call minus that of an equivalent-delta currency put. A positive risk reversal means that calls on the base currency are more expensive than comparable puts, indicating greater demand for protection against, or exposure to, base-currency appreciation. Risk reversals should not, however, be interpreted as pure forecasts: they reflect the market's risk-neutral distribution and therefore combine expected outcomes with risk premia, hedging demand and position imbalances. More generally, option prices can be used to recover information about the market-implied distribution of future exchange rates, including its volatility, skewness and tail probabilities (Campa, Chang and Reider, 1998).

Options can also affect the exchange rate mechanically through the hedging activity they generate. Dealers typically manage the delta of their option books (the change to the value of their positions driven by spot exchange-rate movements) by buying or selling the underlying currency as spot moves. The direction of these flows depends importantly on the market's aggregate gamma, which measures how rapidly an option portfolio's delta changes with the exchange rate. Dealers who are long gamma generally sell the currency after it rises and buy it after it falls, creating countercyclical flows that can dampen volatility and sometimes pull spot towards large strikes around expiry. Dealers who are short gamma must instead buy as the currency rises and sell as it falls, producing procyclical flows that can amplify an existing move. Empirical evidence using trade-repository data confirms that negative market-maker gamma can increase spot-market volatility, while positive gamma has the opposite effect (Anderegg, Ulmann and Sornette, 2022).

These effects can become especially pronounced around barrier options, which are activated or extinguished when the exchange rate reaches a predetermined level. As spot approaches a barrier, the option's delta and gamma may change sharply, requiring increasingly large hedge adjustments. Investors with an interest in preventing the barrier from being triggered may also trade defensively around the level. If the barrier is ultimately breached, the associated option exposure and hedges can disappear or change abruptly, generating a sudden reversal of previous hedging flows or accelerating the move through the level. The precise direction depends on whether the option is a knock-in or knock-out, whether it is a call or put,

and which side of the transaction dealers hold. Consequently, large strikes, expiries and barriers do not provide deterministic support or resistance, but they can materially alter the short-run elasticity of market flows and help explain why exchange-rates occasionally become unusually stable near particular levels.

Figure 36: Simplistic depiction of a knock-out call

Source : Financialtrading.com Deutsche Bank

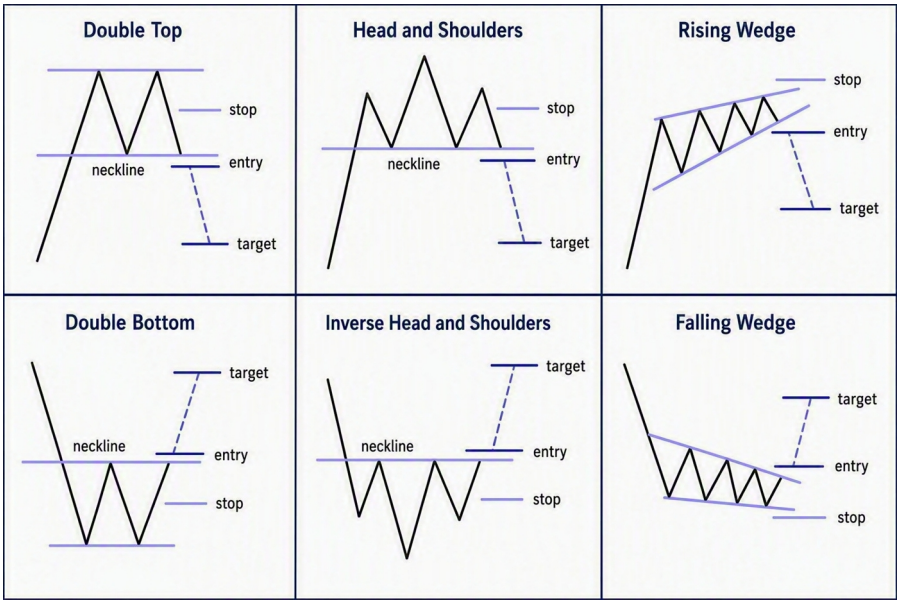
Weng and Grover (2017) introduce Deutsche Bank's Skew Volume Indicators, built from DTCC data (the DTCC registers almost all US currency-option trades, over a fifth of global derivative volumes). The key innovation is separating option transactions into investors (at-the-money options, 25-75 delta, the SVA indicator), versus hedgers (out-of-the-money options, 0-25 delta. the SVO indicator). The findings are tradeable: investor flows positively lead currency moves (it pays to follow investors), whereas hedging flows negatively lead returns (it pays to fade hedgers). A momentum strategy on SVA earned a Sharpe ratio of 0.83, a contrarian strategy on SVO a Sharpe of 0.67, and a blended strategy 0.96 in their sample.

Technical analysis and trend-following

Technical analysis refers to forecasting future price movements from patterns in past market data rather than from an explicit model of economic fundamentals**.** In FX, this typically means analysing the history of exchange rates and sometimes volume, volatility or positioning to identify trends, momentum, turning points and important price levels.

Common techniques include moving averages, which generate signals when a short-term price trend crosses a longer-term one; momentum and oscillator indicators, which attempt to identify overbought or oversold conditions; and support and resistance levels, where previous highs, lows or other widely watched prices are expected to influence future trading behaviour. More sophisticated systematic versions of technical analysis translate these ideas into mechanical trend-following or breakout rules.

Figure 37: Examples of technical analysis



Source : Aimarrow.com, Deutsche Bank

The underlying premise is not necessarily that past prices contain fundamental information, but that investor behaviour, herding, stop-loss activity and institutional trading practices can generate persistence or recurring patterns in prices. Technical analysis is pervasive in FX markets, with surveys finding widespread use among FX dealers and professional fund managers, associated with the belief that psychological and non-fundamental factors can influence prices (Gehrig and Menkhoff, 2006; Menkhoff, 2010).

There is also a substantial literature showing that simple technical trading rules have historically generated statistically and economically significant returns in FX. Levich and Thomas (1993), using bootstrap tests on currency futures, find that moving-average and filter rules produced profits that were difficult to reconcile with simple random exchange-rate processes. Neely, Weller and Dittmar (1997) similarly find significant out-of-sample excess returns across several major exchange rates using rules selected by genetic-programming techniques. Menkhoff and Taylor (2007) survey this literature and evaluate four broad explanations for the use of technical analysis: irrational behaviour, exploitation of central bank intervention, efficient processing of dispersed information, and the ability of technical signals to capture non-fundamental influences on exchange rates. They regard the final explanation as the most persuasive. In this interpretation, technical analysis is useful because FX prices can be driven for extended periods by order flow, positioning, herding and other market dynamics that are not immediately captured by standard macroeconomic models.

News, Announcements and High-Frequency Dynamics

A distinct microstructure literature studies how exchange rates react to scheduled macroeconomic news at high frequency. The work of Andersen, Bollerslev, Diebold and Vega (2003) showed, using real-time data, that exchange rates respond within seconds to the surprise component of macro announcements (relative to survey

expectations) and that announcement surprises explain a meaningful share of high-frequency volatility. This body of work reconciles the "fundamentals don't matter" result with intuition: fundamentals do move exchange rates, but they do so in sharp, news-driven jumps around data releases and central-bank communications, which are hard to see in monthly data. It also underpins the modern reality that the largest intraday FX moves cluster around US payrolls, CPI and FOMC decisions.

Risk sentiment and impact on FX

Risk sentiment is an important cross-cutting driver of exchange rates because it changes investors' willingness and capacity to hold leveraged or risky currency positions. During periods of strong risk appetite and low volatility, investors tend to borrow in low-yielding funding currencies and invest in higher-yielding currencies, supporting carry-trade returns. When volatility rises or market liquidity deteriorates, however, losses, tighter funding constraints and margin calls can trigger a simultaneous unwinding of these positions: high-yielding and relatively illiquid currencies are sold, while funding currencies are repurchased, amplifying the initial exchange-rate move.

Brunnermeier, Nagel and Pedersen (2009) document the resulting negative skewness and crash risk of carry trades, while Menkhoff, Sarno, Schmeling and Schrimpf (2012) show that high-yield currencies perform poorly following increases in global FX volatility. Risk appetite is generally perceived to be negatively correlated with the broad US dollar: the dollar often strengthens when global risk sentiment deteriorates, reflecting its central role in international funding, liquidity and safe-asset demand. The relationship between the dollar and risk appetite is however not stable over time, with shocks that disproportionately impact the US such as the bursting of the 2000 dot-com bubble often associated with a weaker dollar.

2.10 Forecasting performance and the Meese-Rogoff Puzzle

Meese and Rogoff (1983), showed that the structural exchange-rate models of the day, monetary, portfolio-balance, could not out-forecast a naive random walk out of sample, even when fed the realised (future) values of the fundamentals. Four decades on, the Meese-Rogoff puzzle remains the benchmark against which every forecasting claim is measured, and the random walk without drift has proven extraordinarily hard to beat at horizons under a year.

One of the most useful modern surveys on exchange-rate forecasting is Rossi (2013, Journal of Economic Literature). Her review of the post-1983 literature reaches a mixed verdict: predictability does exist, but it is fragile and conditional. Parameter instability is one of the biggest challenges to systematic exchange-rate modelling: it is not only that fundamentals change over time, but the relationship between these and exchange rates changes too.

2.11 Modern theories of the exchange rate

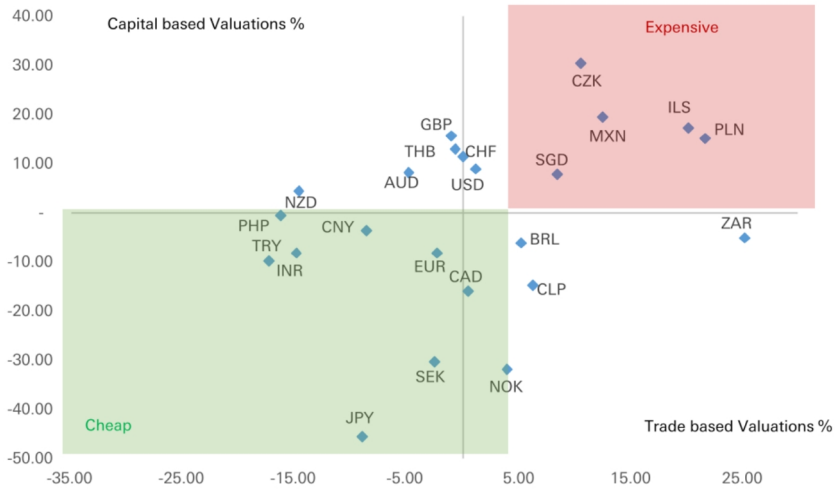
Exchange rates dominated by financial markets

With financial markets outpacing the real economy in size and growth, Gabaix and Maggiori (2015) provide a theory in which exchange rates are determined by capital flows intermediated through imperfect financial markets. Gabaix and Maggiori introduce financial intermediaries (primarily banks) with limited risk-bearing capacity into their model. When, for example, investors in one country increase their demand for foreign assets, intermediaries must take the opposite currency exposure to clear the market. As their balance sheets become more exposed, they require greater compensation for bearing that risk, and the exchange rate moves sufficiently to induce them to absorb the imbalance. The larger the required flow relative to intermediaries' risk-bearing capacity, the larger the resulting exchange-rate adjustment. This provides an important explanation for why exchange rates can move substantially even when trade flows or conventional macroeconomic fundamentals change little. Shifts in international portfolio preferences, financial regulation, intermediary balance sheet capacity or global risk appetite can all generate exchange-rate movements by changing the willingness of financial institutions to intermediate the resulting flows.

The framework is therefore particularly useful for understanding episodes such as the 2008 global financial crisis, when large exchange-rate movements were associated with deleveraging, balance-sheet constraints cross-border portfolio adjustment rather than simply changes in goods trade. A closely related perspective is provided by Rey (2015), who argues that the US dollar sits at the core of the global financial cycle. The large pool of dollars sitting outside the US banking system means easier US financial conditions are associated with expanding cross-border leverage and capital flows and, typically, a weaker dollar. Conversely, tighter financial conditions and rising global risk aversion trigger deleveraging, repatriation and demand for dollar liquidity, tending to strengthen the dollar. The dollar therefore acts not simply as the relative price of US and foreign goods, but as a key price through which changes in global financial conditions and risk-bearing capacity are transmitted internationally.

Conventional PPP and real effective exchange rate measures weight currencies according to bilateral trade, implicitly reflecting the importance of different countries in goods and services markets. This financial-market perspective is the motivation behind the Deutsche Bank's Capital-weighted PPP model. A capital-weighted approach instead gives greater weight to currencies against which a country has large cross-border financial exposures, recognising that the relevant counterparties for exchange rate determination may increasingly be determined by the geography of portfolios rather than trade. Our capital-based valuation measures therefore construct a real effective financial exchange rate using capital weights and compare its level with a historical equilibrium in much the same way that conventional PPP compares a trade-weighted real exchange rate with its long-run norm. For countries with very large external asset and liability positions, this can yield materially different measures of currency valuation from traditional trade-weighted PPP and provides a useful complement in a world where gross capital flows substantially exceed the flows associated with international trade.

Figure 38: Scatter plot of current capital and trade-based valuations



Source : Haver Analytics, Deutsche Bank

Convenience yields and safe-asset demand for dollars

A complementary financial mechanism, specific to the dollar, is the convenience yield on safe dollar assets. Jiang, Krishnamurthy and Lustig (2021) develop a theory in which the dollar's value reflects the convenience (safety and liquidity) services that global investors derive from holding US Treasuries. This convenience yield is measurable via the Treasury basis (the yield gap between US government bonds and currency-hedged foreign government bonds); when it widens, the dollar appreciates immediately and then depreciates subsequently. In their words, "the variation in the Treasury basis accounts for 25 percent of the quarterly variation in the dollar between 1988 and 2017," and "on average, foreign investors earn an extra convenience yield of 2% on Treasury holdings," 90% of which is directly attributable to the dollar denomination.

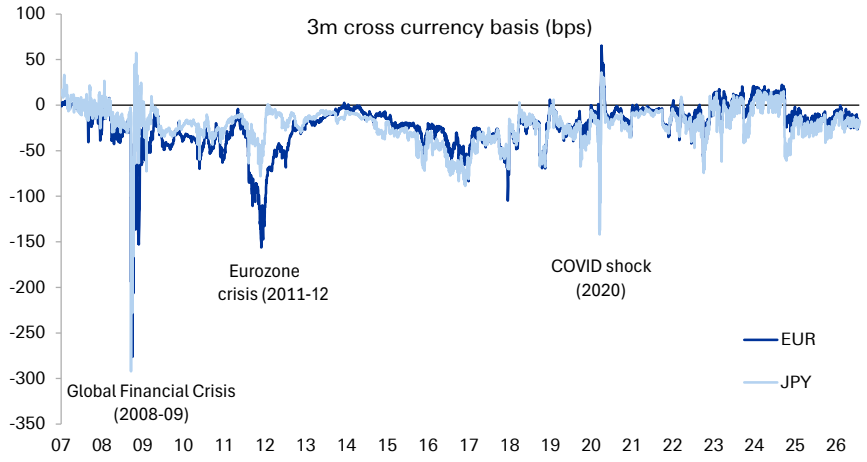
Covered Interest Parity Deviations and the Cross-Currency Basis

Among the most consequential post-crisis developments is the breakdown of covered interest parity. CIP is a pure no-arbitrage identity and held almost perfectly before 2008. Since the crisis, persistent, systematic deviations have opened up, measured by the cross-currency basis: the extra cost, over and above the interest differential, of borrowing dollars synthetically through the FX swap market. Du, Tepper and Verdelhan (2018) document that these deviations are large, persistent and systematic, and are not explained by credit risk or transaction costs alone. They are strongest for contracts that sit on bank balance sheets at quarter-ends, pointing to post-crisis banking regulation (leverage ratios, risk-weighted capital) as the cause: the balance-sheet cost of arbitrage means dealers no longer close the gap. The basis is typically negative for currencies whose investors demand dollar hedges (JPY, EUR), reflecting a structural shortage of dollar funding. See more here.

The macro-financial significance was crystallised by Avdjiev, Du, Koch and Shin (2019), who document a "triangular" relationship: a stronger broad dollar goes hand-in-hand with a higher cost of dollar funding and a contraction in cross-border dollar bank lending. This links the cross-currency basis (a microstructure/funding

variable) directly to the broad-dollar cycle and global credit conditions.

Figure 39: Cross currency basis for EUR and JPY



Source : Deutsche Bank calculations, Bloomberg Finance LP

The Dominant Currency Paradigm and Invoicing

The Dominant Currency Paradigm (DCP) of Gopinath, Boz, Casas, Díez, Gourinchas and Plagborg-Møller (2020) and alters the textbook framework in which prices are set either in the producer's currency or the destination's. Using a dataset of more than 2,500 country pairs covering 91% of world trade, they show that the vast majority of world trade is invoiced in a small number of dominant currencies, overwhelmingly the US dollar, and that those invoice prices are sticky in dollars (Sachdeva, 2026).

This has important implications: exchange rate movements against the dollar, rather than bilateral rates, drive trade prices and quantities and expenditure switching operates weakly and asymmetrically. The dollar therefore exerts a systemic influence on global trade and inflation that no other currency matches, reinforcing the demand for dollar invoicing, dollar funding and dollar reserves in a self-sustaining equilibrium. Combined with the safe-asset/convenience-yield channel and the funding/basis channel, this literature completes the picture of dollar dominance as a structural feature of the international monetary system.

2.12 Central bank intervention

Official intervention in the FX market, the purchase or sale of foreign currency by the central bank, is a long-standing but contested policy tool. The theoretical channels are the portfolio-balance channel (intervention changes the relative supply of assets), the signalling channel (intervention reveals information about future policy) and, more recently, an order-flow/microstructure channel (intervention

affects price like any large order). Sterilised intervention[14] by advanced economies was long viewed sceptically. The best modern evidence is more encouraging. Fratzscher, Gloede, Menkhoff, Sarno and Stöhr (2019) using a novel daily dataset covering 33 countries from 1995–2011, find that "intervention is widely used and an effective policy tool, with a success rate in excess of 80 percent under some criteria." It works best at smoothing the exchange-rate path and defending narrow bands; moving the level of a floating currency requires that interventions be large, public and supported by communication.

Menkhoff (2013) surveys the specific and generally more favourable evidence for emerging markets, where intervention and reserve management are pervasive. Recent episodes illustrate the spectrum: the Swiss National Bank's EUR/CHF floor (2011–15) and its abrupt abandonment in January 2015; large-scale Japanese intervention to slow yen depreciation in 2022 and 2024; and routine EM intervention. Intervention is most effective when it works with fundamentals and market positioning rather than against them.

2.13 Currency regimes and emerging market crises

The distinction between fixed and floating exchange rates is in practice a continuum. The IMF's de facto classification distinguishes ten regimes according to the strength of the authorities' commitment to an exchange-rate level or path, the observed behaviour of the currency and the extent of official intervention. At one extreme, dollarisation and currency boards largely eliminate monetary autonomy; at the other, freely floating currencies are almost entirely market determined. Between them sit a range of pegs, bands, crawling regimes and managed arrangements that trade greater exchange-rate stability for progressively less monetary-policy independence.

14 Intervention by a central bank to buy or sell a currency changes the currency's money supply. A sterilization involves an offsetting transaction by the central bank in the domestic money market to remove or add back the liquidity.

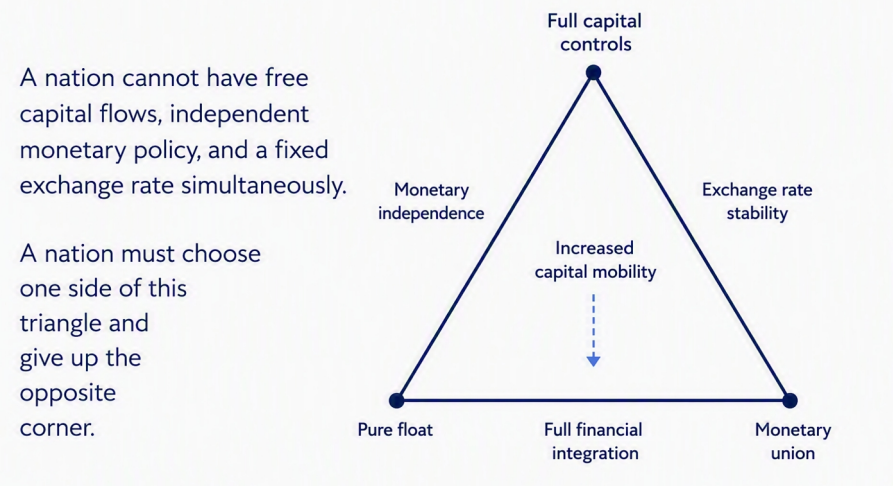
Figure 40: Exchange rate regimes according to the IMF

Broad FX regime	IMF category	What it means	Main IMF classification criteria
Hard peg	**No separate legal tender**	The country uses another country's currency as its legal tender—for example, through formal dollarisation—so there is no independent domestic exchange rate.	Authorities' de jure arrangement is confirmed; another country's currency is the sole legal tender. This effectively eliminates independent national monetary policy.
Hard peg	**Currency board**	Domestic currency exists, but its value is rigidly fixed to an anchor currency and its issuance is constrained by the availability of foreign reserves.	There is an explicit legal commitment to exchange domestic currency for a specified foreign currency at a fixed rate. Domestic currency is normally backed by foreign assets and the issuing authority faces tight restrictions on discretionary monetary policy.
Soft peg	**Conventional pegged arrangement**	The currency is formally pegged to another currency or a basket, but unlike a currency board the peg can ultimately be changed and the central bank retains more policy flexibility.	Authorities formally announce or notify the IMF of the peg and stand ready to defend it through direct or indirect intervention. The rate must generally stay within ±1% of a central parity, or within a total 2% range, for at least six months.
Soft peg	**Stabilized arrangement**	A de facto peg: the authorities may not formally announce a peg, but the exchange rate behaves as though one exists.	The spot rate remains within a 2% margin for six months or more relative to an identifiable currency or basket, and the stability is judged to result from official action or structural market rigidities. Unlike a conventional peg, there need not be an announced commitment.
Soft / intermediate	**Crawling peg**	Instead of fixing the currency permanently at one level, the authorities periodically adjust the central rate in small steps.	A formally announced rule determines the crawl—for example a fixed depreciation rate or adjustments based on inflation differentials or other indicators. The rule and its parameters are public or notified to the IMF.
Soft / intermediate	**Crawl-like arrangement**	A de facto crawling peg: the currency follows a relatively smooth trend even though the authorities have not formally committed to a crawling rule.	The rate remains within a 2% margin around a statistically identified trend for at least six months and cannot qualify as floating. Generally the trend must be sufficiently persistent; an annualised movement of at least 1% can qualify if appreciation or depreciation is sufficiently monotonic and continuous.
Soft / intermediate	**Pegged within horizontal bands**	The authorities commit to keeping the exchange rate inside a band around a central parity rather than at one precise level. The band gives monetary policy somewhat more room than a conventional peg.	The central rate and band are formally announced or notified to the IMF. The permitted fluctuation is at least ±1% around the central rate, or the distance between the maximum and minimum permitted exchange rates exceeds 2%.
Floating	**Floating**	The exchange rate is primarily determined by market supply and demand. The central bank can still intervene to dampen volatility, but does not defend a particular exchange-rate level.	The rate is largely market determined and has no ascertainable or predictable path. Intervention may moderate excessive movements, but targeting a particular exchange-rate level is inconsistent with the classification.
Floating	**Free floating**	The strongest form of floating: the market determines the exchange rate and official intervention is very rare.	Must first meet the criteria for floating. Intervention should occur only exceptionally to address disorderly markets; under the IMF methodology it is limited to no more than three episodes in the previous six months, each lasting no more than three business days.
Residual / managed	**Other managed arrangement**	The authorities materially influence the exchange rate, but its behaviour does not fit any of the IMF's peg, crawl or floating definitions. This can include regimes where the intervention strategy or exchange-rate objective changes over time.	Residual category used when none of the other definitions is satisfied; regimes characterised by frequent changes in exchange-rate policy can fall here.

Source : Deutsche Bank, IMF

The exchange-rate regime shapes how shocks are absorbed and how vulnerable a currency is to crisis. The "Impossible Trinity" (fixed rate, free capital mobility, monetary autonomy: choose two) remains the organising principle.

Figure 41: The Impossible Trinity



Source : Feridun, M. (2014), Deutsche Bank

The literature on early warning systems for emerging-market currency crises identifies recurrent symptoms: real overvaluation, weak growth, adverse terms-of-trade shocks, deteriorating current accounts, excessive domestic credit expansion, banking-system fragility, large fiscal deficits, high ratios of short-term external liabilities (and of M2) to reserves, and reserve losses (Frankel and Saravelos, 2012). Successive generations of crisis models – first-generation (fiscal/monetary fundamentals), second-generation (self-fulfilling expectations and multiple equilibria), and third-generation (balance-sheet and banking/currency-mismatch channels, motivated by the 1997–98 Asian crisis) — map onto these symptoms.

2.14 Frontier topics: machine learning and digital currencies

The application of machine learning (ML), random forests, gradient boosting, neural networks, and now large language models for news processing, to exchange-rate forecasting has grown significantly in recent years. Several studies report ML models beating the random walk in directional (sign) forecasting and when many predictors and time effects are combined, but the improvements are typically small and concentrated in specific regimes.

Rigorous recent work continues to find that ML models frequently fail to beat a random walk on error-magnitude (RMSE) criteria under strict out-of-sample, expanding-window evaluation — the Meese-Rogoff result proves robust even to modern methods (Pfahler, 2022 and Jackson & Magkonis (2024). ML is most promising for (i) directional and volatility forecasting, (ii) exploiting high-frequency and alternative data (order flow, news, positioning), and (iii) nonlinear combination of the fundamentals and factors described in this section rather than as a magic

solution to level forecasting.

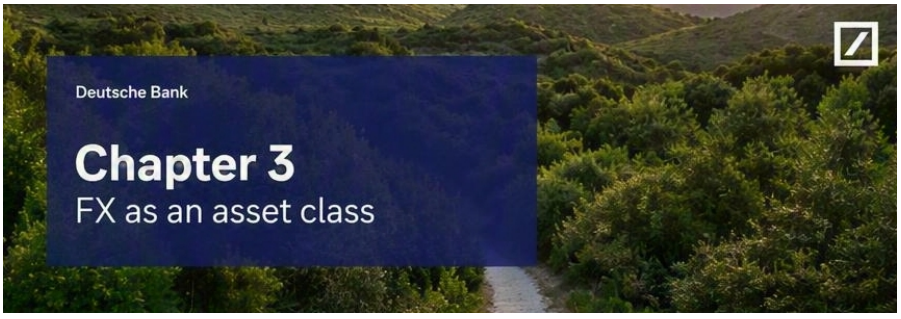
2.15 Cryptocurrencies, stablecoins and CBDCs

Digital assets are increasingly relevant to FX. Dollar-pegged stablecoins are emerging as a potentially meaningful conduit for dollar demand, particularly in emerging markets with weak currencies or capital controls, effectively extending dollar dominance. A recent Deutsche Bank study has expanded into this topic in more detail (Sachdeva and Saravelos, 2025). Central-bank digital currencies (CBDCs) and cross-border CBDC projects could, over time, alter FX-market plumbing, settlement and the mechanics of intervention, though their macro impact on exchange-rate determination remains speculative as of 2026. Cryptocurrencies themselves remain too volatile and too small to function as genuine exchange rate anchors, but the infrastructure they have spawned — 24/7 settlement, tokenised deposits — is beginning to matter for market microstructure.

2.16 Bringing it all together

No single model of exchange rate determination is right all the time; the challenge lies in matching the tool to the horizon and the regime. The evidence assembled here supports a staged approach to exchange rate determination.

- First, *anchor exchange rates with valuation (strategic, 3 years+)*. Use PPP, BEER and FEER to establish a fair-value range and the direction of long-run misalignment. Prefer PPP for G10 and DBeer for EM.
- Second, *align FX forecasting with the economic cycle (cyclical, 1–2 years).* Use interest rate differentials, both level (carry) and change, the monetary policy cycle and Taylor rule fundamentals for medium-run direction. Take account of shifts in the balance of payments, both on the current account and the capital flow side of the ledger.
- Third, *flow and positioning matter (tactical overlay).* Use order flow, options flow and positioning as high-frequency short-horizon mean-reversion signals.
- Finally, be aware of regime changes**.** This can take the form of a change in exchange rate regime from a peg to a float, an underlying abrupt shift in macroeconomic fundamentals (Brexit) or a large financial shock that shifts the market to a new equilibrium (2008 Financial Crisis or Eurozone crisis).



Introduction

Currency investing starts from a different premise to equities or bonds: a currency has no stand-alone expected return and every exposure is inherently relative, involving a long position in one currency and a short position in another, typically implemented through spot, forwards or options. Do investors make money from trading FX? Can FX be treated as an investable asset class? Our first chapter demonstrated the importance of non-profit-seeking participants in the FX market. This weakens the application of the efficient-market hypothesis and creates scope for excess returns. Our second chapter introduced the exchange-rate frameworks that can inform an investment approach to currencies. This chapter turns from theory to real-world evidence. We examine whether systematic currency factors such as carry, momentum and value have generated persistent returns, whether quantitative investors can capture them after costs and whether discretionary currency managers generate alpha beyond these systematic exposures.

3.1 FX "beta" and factor investing

The poor out-of-sample performance of models designed to forecast exchange rate levels, together with the persistent empirical rejection of uncovered interest parity (UIP) has generated a large investor ecosystem focused on harvesting currency risk premia via systematic strategies and indices across baskets of currencies rather than individual exchange rates. Three investment-style factors have become canonical:

- *Carry* involves buying high-interest-rate currencies and funding the position by selling low-interest-rate currencies. Its profitability is the investment counterpart of the forward-premium puzzle: high-yielding currencies have not, on average, depreciated sufficiently to eliminate their interest-rate advantage. Carry returns are often interpreted as compensation for exposure to global risk, since high-yielding currencies

tend to underperform during episodes of rising volatility, deteriorating liquidity and widespread deleveraging (Lustig, Roussanov and Verdelhan, 2011; Menkhoff et al., 2012a).

- *Momentum* involves buying currencies with strong recent excess returns—the "winners"—and selling recent "losers." Menkhoff, Sarno, Schmeling and Schrimpf (2012b) document substantial cross-sectional momentum in currencies. Proposed explanations include the gradual incorporation of information into exchange rates, investor underreaction and overreaction, and limits to arbitrage, although transaction costs reduce the returns available to implementable strategies.
- *Value*, sometimes described as a valuation or mean-reversion strategy, involves buying currencies judged to be undervalued and selling those judged to be overvalued relative to an equilibrium benchmark such as purchasing-power parity or a behavioural equilibrium exchange rate model. Value signals generally operate over longer horizons than carry or momentum, reflecting the slow tendency of real exchange rates to revert towards equilibrium. Evidence for currency value strategies forms part of the broader finding that value and momentum premia appear across several asset classes (Asness, Moskowitz and Pedersen, 2013). Practitioner implementations include Deutsche Bank's PPP and DBeer frameworks (Grover, 2025).

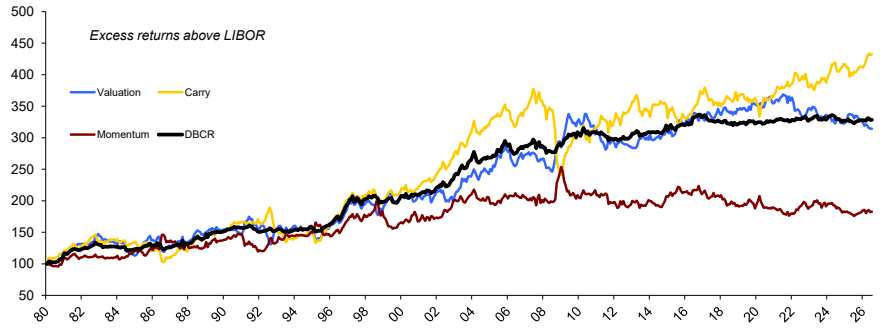
One of the earliest practical attempts to package these systematic returns into a currency "beta" equivalent to the S&P 500 for equities was the Deutsche Bank Currency Returns (DBCR) Index, introduced in 2007. The index equally weights transparent, rules-based carry, momentum and valuation strategies across liquid developed-market currencies and was designed to be investable after transaction costs. In its original backtest over 1980–2006, DBCR generated an annualised excess return of approximately 4 per cent, volatility of 5.2 per cent and a Sharpe ratio of 0.77, with a maximum drawdown of 11 per cent. Its relatively low correlations with global bonds and equities illustrated the potential diversification benefits of combining systematic currency exposure with traditional asset classes (Hafeez, 2007).

A subsequent reassessment by Saravelos et al. (2018) found that the risk-adjusted returns of the combined carry, momentum and value approach had remained resilient after the global financial crisis. Over the longer sample beginning in the 1980s, systematic currency returns were found to be comparable on a risk-adjusted basis with those of global bond and equity portfolios, while adding FX exposure to a conventional portfolio reduced historical volatility and drawdowns.

These benefits arise partly because the component factors perform well in different environments: momentum can offset some of the losses suffered by carry and value during sharp regime changes, while valuation may benefit from the eventual reversal of prolonged trends.

Factor returns nevertheless remain cyclical and can be exposed to crash risk, crowding, changing monetary regimes and implementation costs; they should therefore be understood as compensation for bearing a collection of risks rather than as mechanical arbitrage profits.

Figure 42: DBCR returns vs risk-free-rate



Source : Deutsche Bank

3.2 Systematic and quant-based investors

Systematic currency and macro managers construct portfolios using predefined, rules-based models that are more advanced than the drivers highlighted above. These may combine price-based signals, such as trend, momentum and volatility, with macroeconomic and valuation variables, including interest-rate differentials, purchasing power parity, external balances and policy expectations. Traditional factor strategies, econometric models and more recent machine-learning approaches all fall within this category. Their defining feature is the consistent translation of observable data into positions through a repeatable portfolio-construction and risk-management process. Commodity Trading Advisors, or CTAs, are an important subset of systematic macro managers. Despite their name, CTAs are not confined to commodities: they typically trade diversified portfolios of equity-index, government-bond, interest-rate, commodity and currency markets. Their currency positions are therefore usually part of a broader cross-asset portfolio rather than a dedicated FX mandate.

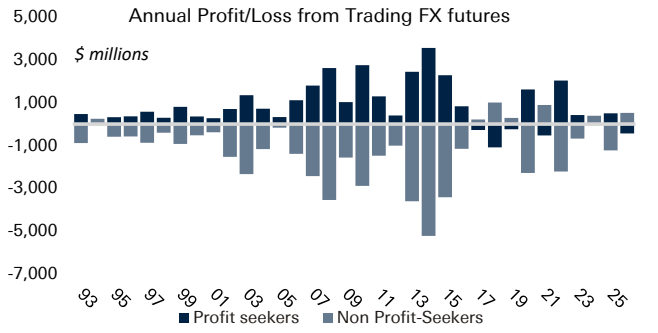
CTAs commonly implement currency exposure through exchange-traded futures. Futures are standardised, centrally cleared and highly liquid in the major currency contracts, while allowing investors to take long or short positions with relatively little initial capital through margining.

We thus return to the CFTC weekly commitment of traders report to produce an estimate of returns from profit-seeking activity from quant-based models using the methodology of Kearns and Manners (2004). As noted above, the IMM report gives the FX futures position of commercial and non-commercial every Tuesday. Given this information, it is a relatively straightforward exercise to calculate the profit or loss made every week by the respective parties.

We find that profit seekers make billions of dollars trading FX futures at the expense of hedgers almost every year since 1993. [16]. The breakdown for the individual currency pairs is also as expected, with annualised losses for non-profit seekers highest in EUR and JPY.

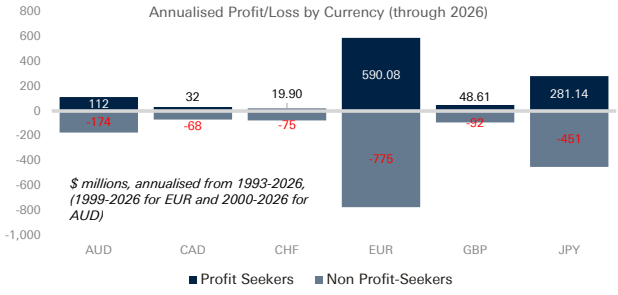
16 (We use dollar rates for CAD, CHF, EUR (since 1999), GBP, AUD (since 2000) and JPY)

Figure 43: CFTC data shows that speculators consistently make money



Source : CFTC, Bloomberg Finance LP, Deutsche Bank

Figure 44: Profit-seekers consistently make money across all currencies



Source : CFTC, Bloomberg Finance LP, Deutsche Bank

3.3 Discretionary currency manager "alpha"

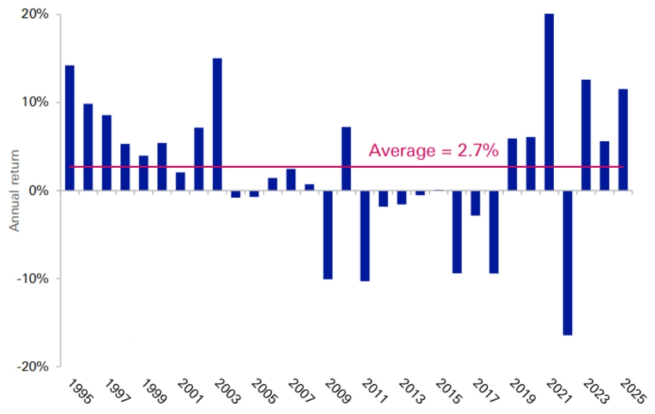
FX has been one of the asset classes of choice for macro investors since the 1980s so in contrast to other asset classes, we have a long history of "active" manager returns too. Discretionary macro investors take currency positions as part of a broader assessment of the global economic and policy environment. Their decisions may reflect views on monetary-policy divergence, fiscal policy, political risk, capital flows, commodity prices or shifts in the international monetary system.

Unlike purely systematic strategies, discretionary investors can place greater weight on events that are difficult to incorporate into historical models, including elections, policy regime changes and geopolitical shocks. This flexibility can be valuable around turning points, but performance depends heavily on the quality of the investor's judgement, timing and risk management.

How have these returns behaved? We take the average of the Barclays, Parker, Hedge Fund Research and Bloomberg macro indices - where returns are available - to come up with an estimate of active manager returns. We show that since 1995, the annual return has been 2.7% on average.

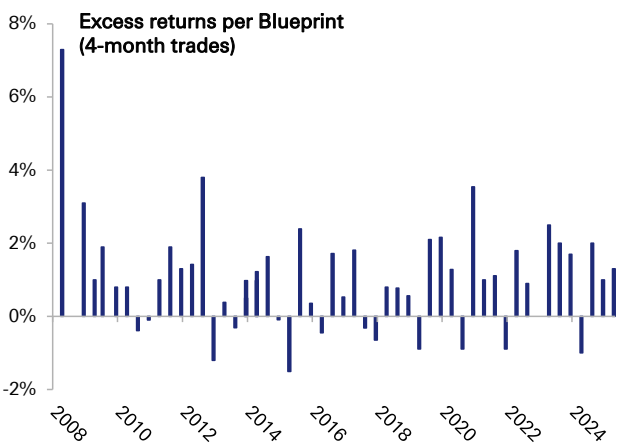
Finally, the Deutsche Bank research team has been producing a tri-annual publication of top trade "FX Blueprint" recommendations since 2008, intended to replicate the trade recommendations that would appeal to a "traditional" macro hedge fund. The returns of this discretionary macro portfolio are shown below and have also been positive over time.

Figure 45: Active FX manager currency returns



Source : Deutsche Bank, Bloomberg Finance LP [Average returns of Barclays, Parker, HFR and Bloomberg Finance LP FX macro indices]

Figure 46: DBFX Research trade recommendations



Source : Deutsche Bank, Bloomberg Finance LP

Introduction

Currency exposure is unavoidable whenever assets, liabilities or cash flows are denominated in a foreign currency. Yet the decision to hedge is not as simple as choosing between being hedged and unhedged. What is the objective of the hedge? How much risk should be removed, over what horizon, with what instrument and at what cost? Every hedge involves a trade-off. It can reduce uncertainty around portfolio returns, revenues or costs, but it also incurs direct costs through forward points, option premia and transaction expenses, as well as an opportunity cost if the currency subsequently moves in the hedger's favour. This chapter develops a framework for making those decisions. Rather than treating FX hedging as a standalone transaction, it considers currency risk within the investor's total portfolio or the corporate's full set of exposures. It examines passive and dynamic hedge ratios, currency overlay programmes, cash-flow and balance-sheet hedging, and the additional complications created by restricted emerging market currencies. The central argument is that there is no universally optimal hedge ratio: the appropriate strategy depends on the hedger's objective, constraints, underlying exposures and tolerance for cost and risk.

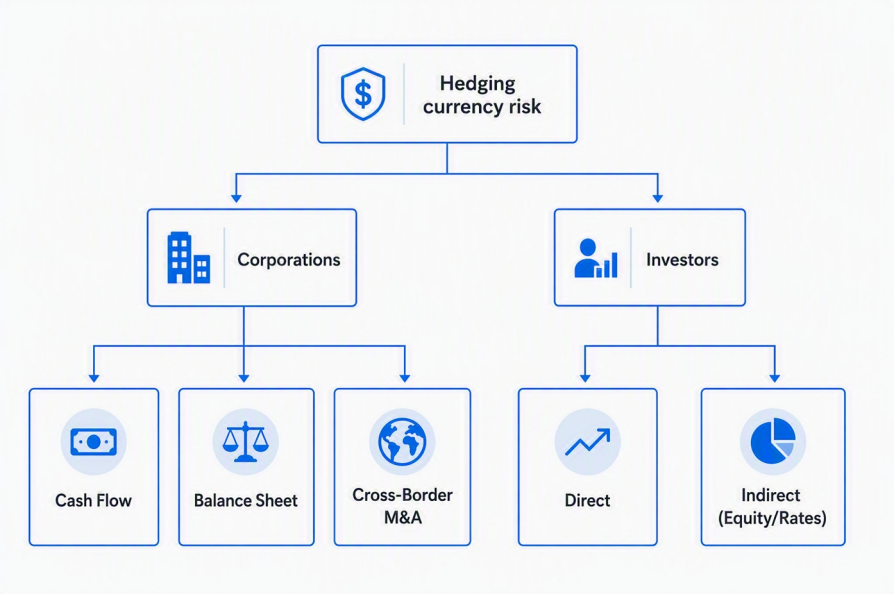
4.1 The problem of defining the hedging "objective"

So far we have looked at currency returns from the perspective of an active investor. The objective here is clear: maximise returns. But what about market participants where FX risk is an unwanted by-product of underlying economic exposure. While in theory simple to understand, a hedging objective is far more complicated to implement in practice. The first challenge is that a hedging objective is not well-defined. An investor aims to maximise returns; a "hedger" aims to offset underlying

currency exposure, but to what end? The most frequent objective of hedging is to minimise earnings volatility for a corporate or returns volatility for a passive investor versus a benchmark. In practice, this objective faces multiple difficulties.

First, particularly for corporates, there is no widely accepted and well-defined "benchmark" against which to measure the success or cost of a hedging decision. Second, the objective of earnings volatility minimisation may not be achievable because of imperfect foresight: cashflows and balance sheet exposure are not infinitely predictable and hedgeable into the future. Third, hedging involves a cost. There is the direct cost that needs to be paid to purchase the underlying hedging instrument, usually in the form of forward points. There is also the indirect cost of the "counter-factual" exchange-rate that would have prevailed had the underlying exposure not been hedged. The simple point is that there is an opportunity cost to hedging; and that the volatility minimisation or other objective needs to be set in the context of this cost.

Figure 47: The broad landscape of hedging currency risk



Source : Deutsche Bank

4.2 Currency hedging for investors

Institutions with global investments face unavoidable currency exposure and therefore currency risk. A EUR-based investor holding US equities, for example, is exposed not only to US equity market risk but also to movements in EUR/USD. Because exchange rates can be volatile, economically meaningful and correlated with the underlying asset exposure, FX can materially affect overall portfolio risk and returns.

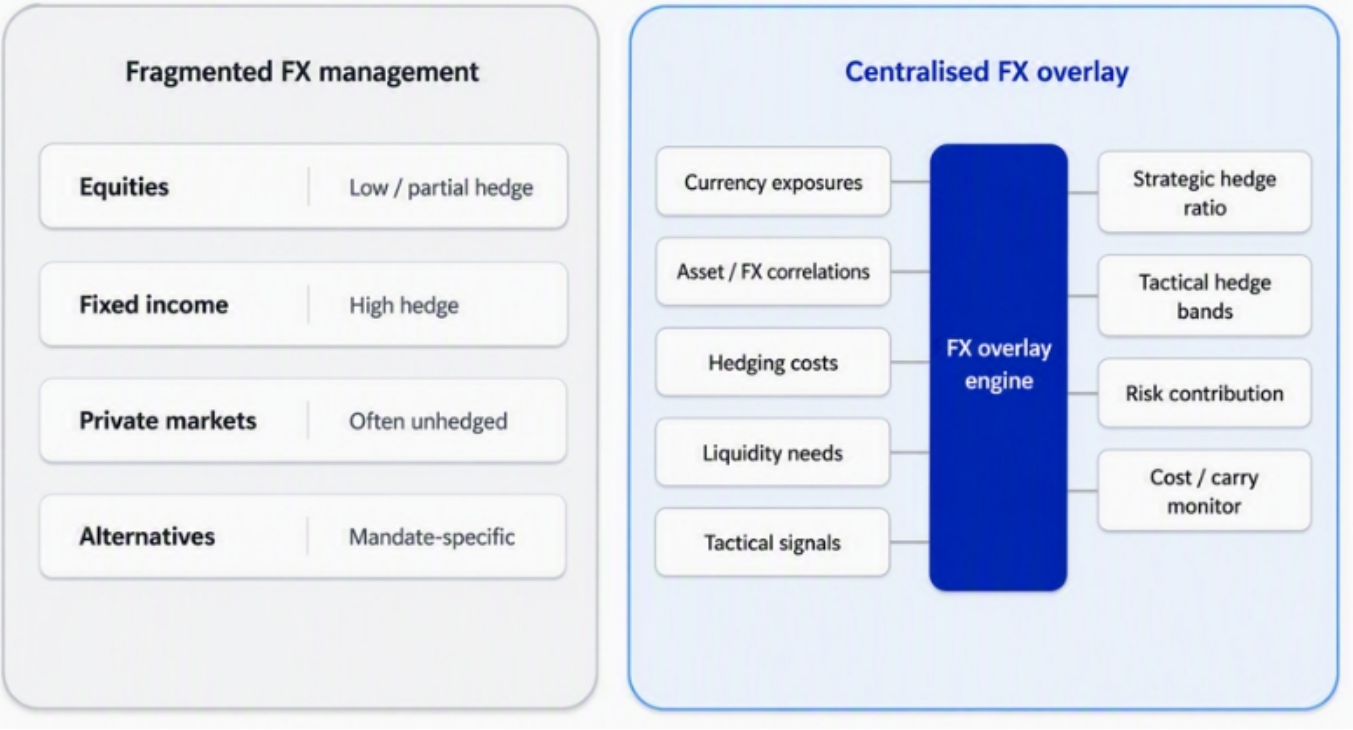
From static hedging to portfolio-aware hedging

Traditional FX hedging has often been managed in a siloed way: investors may leave exposures unhedged, fully hedge them, or apply a fixed hedge ratio such as 50%

across the portfolio. While simple to implement, this approach can overlook important interactions between currencies, asset classes and portfolio objectives.

A more integrated approach treats currency exposure as part of the total portfolio rather than as a standalone transaction cost. This allows investors to assess whether FX should be hedged to reduce risk, retained as a potential source of return, or managed dynamically through an overlay strategy.

Figure 48: Currency risk – the underrated opportunity under the total portfolio approach



Source : Deutsche Bank, Grover and Anand (2026)

Dynamic FX hedging and total-portfolio risk

Currency risk management is increasingly being viewed through a holistic portfolio lens. Grover and Anand (2026) argue that currency exposure should be assessed within the full portfolio rather than treated as a standalone overlay.

The hedging decision can be viewed through three key considerations:

- How FX contributes to total portfolio risk.
- The cost or benefit of hedging, particularly through forward carry.
- The expected direction of spot, informed by tactical signals such as carry, momentum and valuation.

Cost of carry is only one part of the decision. The realised cost or benefit of hedging ultimately depends on how spot evolves over the hedging period. Grover and Anand

(2026) find that tactical hedging can help reduce the opportunity cost of hedging by identifying periods when currency returns are more likely to support portfolio performance. The main conclusions of their paper include:

- Dynamic hedging strategies can deliver stronger risk-adjusted outcomes than static hedge ratios.
- Investors generally benefit more from prioritising portfolio volatility reduction than maximising FX carry.
- Incorporating a view on future spot movements through currency signals can improve risk-adjusted returns.
- Tactical signals can help reduce the opportunity cost of hedging by selectively participating in favourable currency moves.

Why currency risk is moving up the agenda

Currency risk has often been overlooked or managed in silos, but investor interest in more active FX management is increasing as institutions adopt a Total Portfolio Approach (TPA) to investing. Choosing not to hedge is itself a risk decision and, if implemented passively, can materially influence portfolio outcomes. While bond portfolios have traditionally been hedged more actively than equity portfolios, many institutional investors are increasingly exploring centralised management of FX exposures across the full portfolio.

Grover and Anand (2026) identify key themes in institutional FX risk management:

1. **Defining the objective**: A fundamental question is whether currency risk should be managed primarily to enhance returns or to improve portfolio risk management. While FX can generate returns in certain market environments, a hedging approach should aim to design programmes that improve portfolio stability while preserving returns where possible.
2. **Benchmark-aware hedging:** An important theme in currency risk management is how to manage currency risk without straying too far from benchmark performance. Many funds are measured against market indices that are unhedged—for example, equity portfolios often track the MSCI World. Another common reference point is peer performance. These practical constraints can be incorporated into a total portfolio approach framework and may still allow for efficiency gains over standard approaches.
3. **Hedging instruments:** While forwards remain the main tool for managing currency risk, options are becoming more widely used. Their embedded cost efficiencies have made them increasingly attractive as a secondary instrument. Funds that must maintain a fully hedged currency exposure are also using options to introduce more flexibility and improve efficiency.
4. **Looking beyond direct currency holdings:** A holistic view of currency risk extends beyond the currencies directly held within a portfolio. For example, if a fund owns a US stock whose business is materially exposed to the Japanese yen, the fund may also have indirect yen exposure. While there is no simple way to measure this, one possible approach is to review company annual and quarterly reports to estimate those exposures.
5. **Private markets and uncertain cash flows:** Currency management is more difficult in private markets because the timing and size of cash flows are uncertain. One emerging approach is to model expected private-asset

distributions under different scenarios, using long-term currency volatilities and correlations, and then design hedging strategies that balance risk reduction against the carry cost of forwards and options.

6. **Cash-flow and liquidity management:** A key practical issue in hedging is managing margin calls, especially when the hedge moves against the hedger. For example, if a currency someone is hedging against starts to strengthen, the hedged position can experience large swings. Solutions are emerging to help clients manage the resulting liquidity needs.

Currency overlay management

Many of these hedging approaches are implemented through currency overlay programmes managed by specialist managers. These firms specialise in managing the foreign exchange exposures that arise from international investment portfolios. Their traditional role has been to implement hedging programmes for institutional investors such as pension funds, insurance companies and asset managers, helping to reduce the impact of currency fluctuations on portfolio returns. These programmes are typically managed separately from the underlying investment portfolio. Systematic currency overlay programmes can range from simple passive hedging to more active approaches. Passive programmes maintain a predefined hedge ratio across foreign currency exposures, while dynamic programmes adjust hedge ratios according to predefined signals such as valuation, momentum or carry. Some overlay managers also run currency alpha mandates, taking active currency positions with the objective of generating returns.

4.3 Currency hedging for corporates

While many of the principles of dynamic FX hedging also apply to corporates, the underlying hedging problem is different. Institutional investors typically hedge currency exposure embedded within assets whose value changes over time. Because the asset value itself fluctuates, a 100% hedge does not necessarily eliminate risk and may even increase volatility if currency and asset returns naturally offset one another. The objective is therefore to manage FX risk as part of the total portfolio.

Corporates, by contrast, are usually hedging known or forecast foreign-currency cash flows, revenues, costs, assets or liabilities. Because the size of the underlying exposure is often known in advance, a corporate can in principle hedge a much larger proportion of its FX risk. The challenge is therefore less about portfolio optimisation and more about balancing risk reduction against hedging costs and opportunity costs. While investors ask how much currency risk should be retained within a portfolio, corporates are typically focused on how much uncertainty around future cash flows should be eliminated and at what cost.

For corporates, the most common source of FX risk arises when purchases, sales, assets or liabilities are denominated in a currency other than the entity's functional currency. The main corporate hedging strategies are cash flow hedging, balance sheet hedging and, less frequently, deal-contingent hedging linked to transactions such as cross-border M&A.

- **Cash flow hedging**: This is one of the most common hedging strategies that require hedging currency risk when the underlying item is not in the entity's functional currency. For example, a sale that has been forecast but has not yet happened does not get recorded on the income statement of the company. These types of transactions that are forecast but not recorded

require a cash flow hedge. In this case, hedge accounting is important as the hedge's mark to market does not affect net income until the underlying transaction is recorded in earnings. It does not create profit/loss volatility during the hedging period. Hedging strategies that are commonly used to hedge future cash flow are the purchase or sale of FX forwards or net purchased options. Let's say a US-based company is contracted to manufacture tyres for a German automaker. This results in a predictable stream of euro cashflows. However, given the company's cost base is in US dollars, it may not be able to protect its revenue margin if euro depreciates considerably against the dollar. The company here then opts for a cash flow hedge to protect its margins.

- **Balance sheet hedging**: In the example above, the company is hedging a transaction which has not yet been recognised as a sale or expense on the income statement. Now say the manufacturer decides to shift its cost base to Europe by building a factory in Europe. Now, the parent company has a EUR asset on its USD balance sheet. Once they start billing out of their EUR subsidiary, for accounting purposes a regular balance sheet hedge may be required as the parent company's functional currency is still USD. In summary, this type of hedging hedges assets and liabilities on corporate balance sheets that are not in the entity's functional currency. A typical balance sheet hedging strategy may be rolling short dated forwards.
- **Deal-contingent and M&A hedging**: Continuing with the example, say the US company instead chooses to merge or acquire a European company. Thus, the cross-border merger and acquisition (M&A) now creates a need for hedging the foreign currency. In certain respects, this type of hedge is conceptually similar to cash flow hedges, as it is based on a potentially unpredictable transaction, but with additional complexities, including whether the underlying transaction is in stock or cash.

The corporate hedging decision

The "bottom-up" hedging decision involves a large number of parameters the optimal mix of which is unique for each firm. While it is true that many firms share the same basic exposure profile, each firm's strategy can vary dramatically depending on its objectives and constraints. Empirical investigations into the hedging practices of multinational firms show that hedging decisions are influenced by a large number of case-specific factors. These include accounting treatment, competitive price concerns and recent hedging outcomes. Overall empirical studies of corporate risk management suggest that a single, clear and value-maximising policy for risk management is not obvious (Brown, GW 2001). The micro hedging decision will also vary depending on market conditions. In all, the evidence cited above argues against recommending a universally applicable hedging programme for all firms.

The "top-down" elements of the hedging decision relate to the market drivers of FX that we identified in the earlier sections of the report. Unlike the multiple parameters involved in the micro hedging decision, the macro decision boils down to market variables: where does the Treasurer or CFO expect the exchange rate(s) to be over the relevant hedging period and how does this relate to the underlying exposure of the firm? One approach is for the hedger to make a discretionary decision based on his or her underlying assessment of market drivers. Another is to adopt a rules-based approach based on the systematic excess returns we have identified in the FX market. Such an approach would be no different to the dynamic hedging strategy employed by a passive "profit-constrained" investor discussed in the section above.

Figure 49: The Hedging Decision

Source : Deutsche Bank

Dynamic FX hedging for corporates

Dynamic FX hedging differs for corporates and institutional investors. Corporates typically hedge known exposures tied to contracted or forecast cash flows, rather than currency risk embedded in assets whose value can grow over time. In principle, they can remove much of this risk by hedging 100% of the identified exposure. However, a full hedge can carry opportunity costs. For multi-currency exposures, it may overlook diversification benefits when currency moves partly offset one another. It can also lock in hedging costs before the full-period outcome is known, potentially causing corporates, like institutional investors, to forgo gains from favourable currency moves. For corporates, the hedging framework should aim to reduce these opportunity costs without materially increasing risk.

For companies with a single-currency exposure, signal-based hedging frameworks can be useful. Rather than relying solely on forward points, hedge decisions can incorporate carry, momentum and valuation signals to assess whether hedging is likely to be more or less beneficial. Grover (2024) suggests this can reduce the opportunity cost of fully hedging, or of hedging based only on current forward points.

For companies with multiple currency exposures, hedging becomes more portfolio-like. Managing the joint risk of the currency basket through Value-at-Risk or cross-currency correlation analysis can help capture diversification benefits that a full-hedge approach may miss. Grover and Nandy (2025) show that, over recent decades, a holistic FX framework has delivered lower-cost and higher-return hedging outcomes than fully hedging each exposure. This result is particularly clear for euro-based exporters, where carry costs have been higher and correlations between euro crosses have been lower, increasing the value of a portfolio-based approach. Overall, corporate hedging is best approached as a two-step process:

- Manage risk across the full currency exposure, particularly where diversification between currencies exists.
- Use currency signals such as carry, momentum and valuation to reduce the opportunity cost of hedging and retain some participation in favourable currency moves.

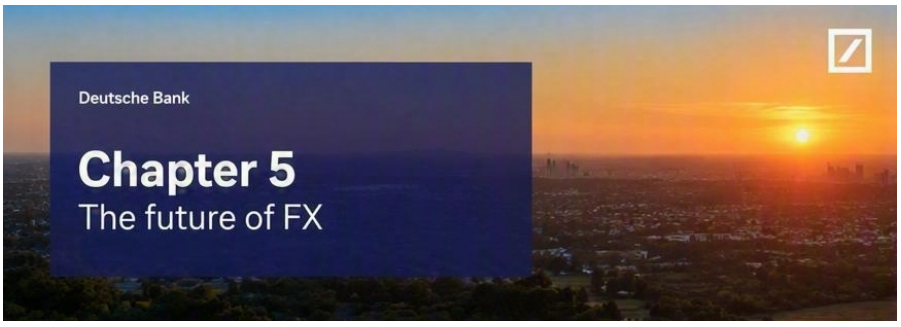
This approach recognises that, while corporates and investors start from different objectives, both can benefit from combining strategic risk management with dynamic currency signals. The difference is that institutional investors seek to optimise FX exposure within a broader investment portfolio, whereas corporates seek to minimise uncertainty around future cash flows while controlling the cost of doing so.

4.4 Hedging emerging market currencies

While the framework above is applicable to both G10 and EM currencies, some EM currencies require additional consideration when hedging. In particular, currencies that are subject to current account restrictions, capital controls, or both operate under different market constraints. We refer to these as restricted markets. Many of these currencies trade offshore through non-deliverable forwards (NDFs), and foreign institutional investors and corporations have historically relied on NDFs to hedge their currency exposures.

However, offshore markets are not always the most cost-effective venue for hedging. On average, onshore forward curves have traded around 40 basis points below offshore curves over the past decade. That creates a potential opportunity for investors. By accessing onshore markets, they may be able to achieve lower hedging costs. For currencies, where operational challenges exist, tactical hedging via cheaper onshore forward markets can further improve hedging efficiency (Grover, 2026 and Harvey et al., 2025).

Another important consideration when hedging EM currencies is the cross-currency basis, which represents a hidden component of hedging costs for corporate and institutional investors (Harvey and Friedemann, 2023). Although it has attracted significant attention since the 2008-09 financial crisis, its role in emerging markets is often overlooked despite being larger and more volatile than in many developed markets. The cross-currency basis can create either costs or opportunities for hedgers and issuers. While instruments such as cross-currency basis swaps can be used to manage these exposures, liquidity and market access vary considerably across EM currencies, and usage is typically concentrated at longer tenors. As most shorter-dated EM hedging remains centred on FX forwards and swaps, investors should remain aware of the cross-currency basis when assessing the true cost of hedging.



Introduction

The foreign exchange market has always evolved alongside the technologies, institutions and regulations through which money moves across borders. Payment systems determine which currencies are easiest to use, where liquidity accumulates and who controls the infrastructure supporting cross-border transactions. This final chapter considers how technology, regulation and geopolitics may reshape FX market structure, payments and currency demand.

5.1 Technology and market structure

While electronification aimed to improve the FX market by reducing risk and increasing efficiency, it has inadvertently introduced greater complexity and fragmentation. This has resulted in a more decentralised and dispersed market structure, increasing operational risks such as vulnerability to cyberattacks, even in a traditionally liquid market like FX. Despite these challenges, technological innovation continues to drive change. Distributed ledger technology (DLT), with its decentralised and shared database, offers potential benefits like improved efficiency, transparency, and reduced settlement risk. Furthermore, automation promises cost reductions and efficiency gains in trading processes, while AI and machine learning are being increasingly applied to price prediction, risk management, and fraud detection. The interplay between these technological advancements and the evolving market structure will continue to shape the future of the FX market.

5.2 Stablecoins and payment rails as the new geopolitical battle lines

The dollar is hugely dominant in FX markets today. According to BIS, 88% of FX transactions have a USD leg. The USD acts as a global vehicle currency. It offers the deepest pools of liquidity which makes it cheaper to trade third currencies via their USD crosses. The USD accounts for 50% of messages exchanged on SWIFT for cross-border payment, and its share in trade invoicing far exceeds the US' economic share of global trade. Cross-border USD payments today are mostly

intermediated through the USD correspondent banking system. Global banks are able to transact in USD with each other because they hold accounts with large correspondent banks that settle with each other via the Fed balance sheet in wholesale central bank USD.

But Sachdeva and Saravelos (2025) show that the dollar correspondent banking has been at growing risk of disruption from faster distributed ledger technology, geopolitical competition, alternative payment systems, and underserved corridors in the Global South. Correspondent banking still features slow settlement times, high fees, and fractured access. Many parts of the Global South have declining access to correspondent bank relationships due to growing AML/CFT. Combined with the use of sanctions that cut certain countries from SWIFT, there has been interest in developing alternative payment rails, particularly by BRICS. China has developed CIPS as an equivalent RMB correspondent banking system. China has also been a core member of Project mBridge that explores cross-border CBDC payments on a distributed ledger.

Keeping the dollar dominant in payments and transactions will be of significant importance to the US in a world of heightened geopolitical competition. Remaining the preferred currency of payment is closely linked to being the currency in which external surpluses are built. The world's willingness to save in USD in turn lowers the cost of USD financing which is a big advantage for the US government and corporates. To support the dollar's role in payments in the future, the US appears to be turning to the power of distributed ledger technology and a private form of money: stablecoins.

Stablecoins are new forms of private digital money tokens hosted on blockchains that maintain a peg to a fiat currency. Under GENIUS Act regulation, USD stablecoins will be fully backed by liquid, short-term USD reserves like T-bills to help maintain their 1:1 peg with the USD. 99% of stablecoins outstanding today are linked to the USD. They are issued and traded on distributed ledgers, making them accessible anywhere through the internet. Their on-chain attributes mean stablecoins offer the possibility of near-instant settlement - where all aspects of monetary value transfer from messaging to reconciliation happen together in the form of a ledger update. Stablecoins do however disintermediate the Fed from the settlement process for international payments in USD. Europe and China have taken a very different approach thus far to digital money, concentrating more efforts on central bank digital currencies as their response to the future of cross-border payments.

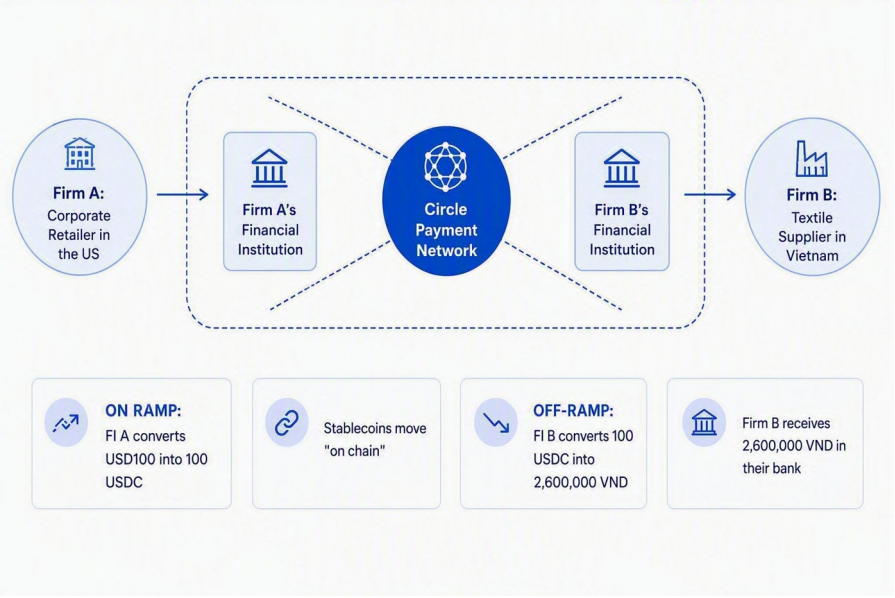
5.3 Tokenization and programmability

Stablecoins are arguably just a subset of a broader trend towards tokenisation that could open many new possibilities for how FX markets operate in the future. Tokenisation is the process of creating claims on underlying money or assets, that can then be represented as digital tokens and made transferrable on a shared infrastructure. Stablecoins are a new type of tokenized private money, backed by cash, bank deposits or bills. But tokenisation also extends into existing commercial bank deposits, central bank money, and traditional assets like equities, money markets and bonds. In the earlier section we covered how stablecoins might help with cross-border payments, by offering 24/7 infrastructure to move USD that does not rely on traditional banking relationships and hours. But tokenized money can also help with other aspects of the FX settlement process - the exchange of currencies for each other, or for assets. Because FX trading takes place primarily in the over-the-counter market, once a trade is executed and enters the post-trade

process there is settlement risk that is created: payment-versus-payment mechanisms such as CLS or pre-settlement netting reduce the risk to some extent, but tokenized money could offer another feature that addresses it more directly: programmability.

Programmability allows certain automated rules and conditions to be embedded into money processes. This is often implemented through smart contracts that execute transactions when predefined conditions are met, helping improve timing frictions, settlement risk, or the need to pre fund and trap liquidity to meet payment obligations. In a programmable world, a payment can be released only if a corresponding asset is transferred, or an FX transaction can settle only if both currency legs are ready for delivery. This technology can thus support more synchronised settlement processes, linking cash, securities and collateral movements through predefined rules and automated execution. While tokenised money can improve payments and settlement on a standalone basis, its full potential may emerge when combined with increasingly tokenised securities.

Figure 50: How stablecoins might change a cross-border payment



Source : Deutsche Bank, Circle Payment Network White Paper

5.4 Concluding thoughts

The foreign exchange market is one of the most innovative and rapidly evolving financial markets in the world. Over the past two decades, advances in technology, electronic trading, data and market infrastructure have transformed how currencies are traded, priced and managed, placing FX at the forefront of broader changes in global finance. Yet because currencies sit at the intersection of national economies, the FX market remains fundamentally connected to international trade and cross-border investment. It is therefore increasingly exposed to the forces reshaping the global economic landscape—from geopolitical fragmentation and changing trade relationships to shifts in regulation, payments infrastructure and the architecture of the international monetary system. Ultimately, the structure of the FX market is not simply a question of how currencies trade, but of who provides liquidity, intermediates capital and controls the financial infrastructure through

which money moves across borders. In that sense, the evolution of FX is increasingly intertwined with questions of monetary sovereignty, financial resilience and geopolitical power. One thing is therefore clear: understanding what moves and shapes exchange rates, and the market through which they are traded, is more important than ever.

Appendix

Profit vs non-profit seekers in the FX market

We use the 2025 BIS Triennial Survey to update previous work and estimate the share of FX turnover initiated with a profit-seeking currency objective. We construct two estimates:

- A realistic scenario, applying what we deem realistic assumptions about participants' ability and mandate to take FX risk;
- And an upper-bound scenario, which relaxes those assumptions by classifying ambiguous transactions as profit-seeking.

We find profit-seeking activity accounts for approximately 40% of turnover under the realistic scenario and 60% under the upper-bound scenario.

Financial participants

Dealers:

Dealer-to-dealer activity accounts for approximately 47% of total daily FX turnover (~$4.44tn). However, treating all inter-dealer activity as speculative would materially overstate profit-seeking demand. Much of this turnover reflects market-making: dealers absorb client flow, manage the resulting inventory and redistribute risk across the market. In other words, market-making activity provides liquidity to the FX market and should not be interpreted as reflecting speculative objectives, especially given regulatory change since 2008.[17]

In order to find the percentage of dealers who engage in profit-seeking behaviour, we assume that inter-dealer transactions conducted by voice are not profit-seeking and that only electronic transactions conducted through order-matching systems such as EBS are profit-seeking. [18]

By doing this we find that:

- 20% of dealer flow, approximately $889bn per day, goes through electronic broking systems, as such, we class all of this as profit-seeking under our "upper-bound" scenario. Given the impact of Dodd-Frank and the Volcker rule on banks' proprietary activities we believe this to be a relaxed assumption.
- Therefore, under a more realistic set of assumptions we classify half of this

17 At the same time, post-global-financial-crisis regulation has increased the balance-sheet cost of some forms of dealer intermediation and arbitrage. Basel III introduced tighter risk-based capital requirements alongside new liquidity and leverage constraints, including a leverage ratio that captures both on- and off-balance-sheet exposures and derivative positions. These constraints can affect banks' willingness and capacity to deploy balance sheets to market-making and arbitrage activities. One important manifestation has been the persistent, although time-varying, deviations from covered interest parity (CIP) observed since the global financial crisis, reflected in the cross-currency basis. See Borio et al., "Covered Interest Parity Lost: Understanding the Cross-Currency Basis," BIS Quarterly Review, September 2016

18 More specifically, we use data from the April 2017 Federal Reserve Bank of New York FX Survey and data from the 2025 FXJSC FX turnover survey. In the FRBNY paper, trade execution method is split

flow, 10% of dealer flow as profit-seeking ($444bn/day). To account for the decline in bank proprietary trading and the importance of inventory management within inter-dealer activity

Hedge Funds and PTFs

Hedge Funds and PTFs account for approximately $760bn (8%) of headline daily FX turnover and are reported together in the BIS data. Traditionally, banks acted as the primary dealers, or liquidity providers, in the decentralised OTC FX market. However, the expansion of prime brokerage, which has facilitated client access to primary FX trading venues such as EBS and Reuters, has allowed principal trading firms to act not only as profit-seekers but also as liquidity providers.

We assume that:

- All hedge-fund activity is profit-seeking. Hedge funds generally have a mandate to take FX risk, meaning that even transactions associated with purchases and sales of foreign assets may incorporate an active currency view.
- Because the BIS data does not disaggregate the dual role of PTFs (liquidity providing and profit-seeking) we assume liquidity provision is 10% of the BIS' headline "hedge fund & PTFs" $760bn figure. PTFs, we assume that 90% of headline activity is profit-seeking and 10% reflects liquidity provision. [19]

Therefore, under both the realistic and upper-bound scenario we assume that $684bn (90% of their daily turnover) of hedge fund and PTF activity is profit-seeking.

Non-Reporting Banks

Non-reporting banks account for approximately $2.19tn, or 23% of headline daily turnover. However, these institutions frequently intermediate transactions for institutional investors and non-financial customers rather than representing a separate source of final currency demand.

We assume that:

- Under our realistic (upper-bound) scenario, we assume that 90% (80%) of this daily flow represents the fact that non-reporting banks act as an intermediary for institutional and other non-financial customers. Therefore, we distribute the 90%, $1.97tn of daily turnover proportionately, across institutional investors and other non-financial customers. [20]
- We therefore assume that under the realistic (upper-bound) scenario 10% (20%) of non-reporting bank flow is profit-seeking, in line with our dealer

across two dimensions - voice/electronic and direct/indirect. We assume all voice transactions are liquidity seeking, as well as direct electronic transactions that are conducted through single-bank electronic trading systems. Transactions executed via voice are highly unlikely to be speculative in nature since they typically involve large-ticket customer flow. Foreign exchange desks at banks would be unlikely to be executing large proprietary transactions of such size which would typically take place on electronic matching systems such as EBS. More recent data from the Bank of England finds that 20% of reporting dealer flow goes through these electronic broking systems. We apply the final 20% figure to the worldwide BIS survey from 2025 (10% under our more realistic set of assumptions) to come to an estimate.

19 Our first assumption is that liquidity provision by PTFs is largely dominated by the spot market. We therefore focus on Hedge fund and PTF volumes from spot trading. According to BIS 45% of Hedge

assumptions.[21]

Institutional investors:

Institutional investors account for $1.26tn of daily BIS turnover and present the most difficult classification challenge. After reallocating the relevant flow intermediated by non-reporting banks, their estimated turnover increases to $2.56tn under the realistic scenario.

We assume that:

- First, given that BIS data does not divide institutional-investor turnover according to the underlying asset class we instead use US Treasury International Capital (TIC) data on gross cross-border purchases and sales of securities to estimate the relative contribution of fixed-income and equity portfolios. [22] This adjustment produces an estimated asset-allocation split of 63% equities and 37% fixed income.
- Second, considering this asset allocation split, we use Morning Star and ICI data on the share of passive versus active investment funds. We estimate that approximately 45% of equity assets and 60% of fixed-income assets are actively managed. Passive investors consist of index or benchmark-tracking funds and we assume they are active in FX on a transactional non-profit-seeking basis. Active investors aim to outperform a stated benchmark or offer unconstrained positive returns. In practice, this type of investor would still face limits on the amount of exchange-rate risk that can be taken but for the purposes of this exercise we assume all "active" investor flow is profit-seeking in FX.

Putting this together we find that:

Using the US TIC data, it implies an asset allocation split of 66% equity and 34% fixed income split. As per data from Morningstar and ICI, 45% of those equities will be actively managed, compared with 60% of those fixed income holdings.

- **Under our upper-bound scenario**, we assume that all of these actively managed holdings are with firms that are mandated to take FX risk. This implies that approximately 50% of institutional-investor turnover, or around $1.28tn per day, is profit-seeking.
- Under the **realistic assumption**, active management of the underlying portfolio does not necessarily imply an active currency mandate. An investor may actively select foreign bonds or equities while remaining required to hedge the associated FX exposure mechanically. So, instead of assuming all actively managed equities and fixed income assets are with funds that are mandated to take FX risk, we assume 80% of active fixed-income managers have discretion to take FX risk and 50% of active equity

Fund and PTF volume comes from spot trading. We assume that this is split 50/50 between Hedge Funds, an upper-bound estimate. Further, we also find that 30% of PTF trading can be attributed to market making. Together, these assumptions leave us with an estimate of approximately 10% liquidity provision.

20 We do not allocate a share to hedge funds or the official sector, which are more likely to transact directly with large reporting dealers.

21 Recent evidence shows that a proportion of non-reporting bank activity is growing https://www.greenwich.com/market-structure-technology/behind-market-structure-future-nblp-landscape, as is their share of profit-seeking activity. As a result of their growing importance, we apply the same split as we do for dealers.

22 TIC data reports (1) monthly purchases and sales of domestic and foreign securities by security type

managers have discretion to take FX risk. This implies a profit-seeking share of approximately 30% under the realistic scenario. Applied to $2.56tn of institutional-investor turnover, this would equal approximately $800bn per day.

Official Sector

The official sector includes central banks, sovereign wealth funds and international financial institutions. Its activity accounts for approximately 1%–2% of headline FX turnover. The motivations within this category vary. Central-bank intervention may pursue monetary-policy, financial-stability or exchange-rate objectives, while reserve managers may also seek to preserve capital and earn returns.

We therefore divide official-sector activity evenly between the two categories under both realistic and upper-bound scenarios.

- We classify 50% of the turnover as profit-seeking. Because official-sector activity represents a small share of aggregate turnover, the final result is relatively insensitive to this assumption.

Non-Financial Sector:

Corporates and governments

After allocating non-reporting-bank turnover, estimated non-financial-customer activity is equivalent to around $952 bn per day in the realistic scenario and $896 bn in the upper-bound scenario.

We assume that:

- Corporates use FX to settle foreign-currency revenues and expenses, finance cross-border activity, hedge forecast cash flows and manage earnings volatility. Governments similarly transact to meet an underlying funding or expenditure requirement. We classify all non-financial-customer turnover as liquidity-seeking under both scenarios.

Other

The BIS residual "other" category accounts for approximately 2% of headline turnover. It includes activity for which the underlying participant or motivation cannot be identified precisely, including some retail-driven flow. Retail and other residual transactions can serve both speculative and transactional purposes.

Some investors trade currencies to earn returns, while other transactions relate to travel, remittances or purchases of foreign assets. We assume that:

- We classify 100% as profit-seeking under the upper-bound scenario and 50% as profit-seeking under the realistic scenario.

Undistributed

(Table 4). and; (2) foreign holdings of US securities, split between official and private investors and by asset class (Table B). Because we assume official-sector investors have different profit-seeking incentives from private institutional investors, we exclude their activity from the asset-class split. When doing so we find the proportion of institutional investors transacting in fixed income vs equity markets.

Final residual category accounts for less than 1% of headline turnover. Because there is insufficient information to identify its underlying motivation, we classify this turnover as

- 100% liquidity-seeking rather than assume that it represents directional risk-taking. The treatment has little effect on the aggregate result because the category is very small.

Bringing it all together:

Taken together, **under our realistic assumptions** the current model estimates that approximately 40% of adjusted FX turnover is profit-seeking, while approximately 60% is liquidity-seeking or transactional.

Under the upper-bound assumptions, the split is approximately 60% profit-seeking and 40% liquidity-seeking.

Glossary of terms

A

- Algorithmic trading (AT) — Automated trading in which a computer algorithm decides the order-submission strategy.
- Alpha — Return generated by skill or active judgement in excess of a benchmark or of systematic factor exposure, as pursued by discretionary currency managers.
- American-style option — An option that can be exercised at any point up to expiry.
- At-the-money (ATM) — An option whose strike is at or near the prevailing spot/forward rate.

B

- Balance sheet hedge — A corporate hedge of assets or liabilities on the balance sheet that are denominated in a currency other than the entity's functional currency.
- Balassa–Samuelson effect — Theory that faster productivity growth in the tradable sector raises wages and non-tradable prices, generating real exchange-rate appreciation.
- Barrier option — An option whose payoff is activated (knock-in) or extinguished (knock-out) when spot reaches a specified level.
- Basel III — Post-crisis bank regulation introducing tighter capital, liquidity and leverage requirements that raise the balance-sheet cost of dealer intermediation and arbitrage.
- Basic balance — A measure combining the current account with the slowest-moving financing flows (net FDI, and sometimes long-term portfolio flows) to gauge underlying currency demand.
- Basis point (bp) — One hundredth of a percentage point (0.01%).
- Behavioural Equilibrium exchange-rate (BEER) — An equilibrium model estimating the exchange-rate implied by economic fundamentals through panel regression.
- Beta (FX beta) — The systematic, rules-based return available from generic currency factor exposures such as carry, momentum and value.
- Bid-ask (bid-offer) spread — The difference between the price at which a dealer buys (bid) and sells (offer); a frictional cost of trading and the market maker's compensation.

C

- Call option — The right, but not the obligation, to buy one currency against another at a predetermined strike price.
- Capital controls — Official restrictions on cross-border capital movements, common in restricted and emerging-market currencies.
- Capital-weighted PPP — A valuation approach that weights currencies by cross-border financial exposures rather than by trade, reflecting the

geography of portfolios.

- Carry / carry trade — Borrowing a low-yielding currency to invest in a higher-yielding one to earn the interest differential; historically profitable but with negative skew and crash risk.
- Central Bank Digital Currency (CBDC) — A digital form of central-bank money.
- Central Counterparty / Central Clearing Counterparty (CCP) — Financial market infrastructure that interposes itself between buyer and seller, becoming the buyer to every seller and the seller to every buyer, reducing counterparty risk.
- Commodity currencies — Currencies of major commodity exporters (e.g. AUD, NZD, CAD) whose real exchange-rates have strong long-run relationships with world commodity prices.
- Commodity Futures Trading Commission (CFTC) — The US derivatives regulator; source of the IMM speculative-positioning data used to gauge FX positioning.
- Commodity Trading Advisor (CTA) — A systematic macro manager trading diversified futures across asset classes, including currencies, typically via exchange-traded futures.
- Continuous Linked Settlement (CLS) — An independent multi-currency payment-versus-payment settlement system that mitigates FX settlement risk by settling both legs simultaneously and netting payments.
- Convenience yield — The safety and liquidity premium global investors derive from holding US Treasuries, which supports the dollar; measurable via the Treasury basis.
- CORAX — Categorised ORderflow from Autobahn FX — Deutsche Bank's categorised indicators built from globally aggregated electronic spot order flow.
- Covered Interest Parity (CIP) — The no-arbitrage condition that the forward premium/discount equals the interest-rate differential; held almost exactly until 2008.
- Cross-currency basis — The extra spread, over and above the interest differential, of borrowing dollars synthetically through the FX swap market; reflects structural dollar-funding demand.
- Cross-currency swap (XCCY) — A swap that exchanges principal in two currencies at inception and maturity and also involves periodic interest payments in each currency over the life of the contract.
- Cryptocurrency — A digital asset secured through cryptographic and distributed-ledger technologies, such as Bitcoin.
- Currency board — A hard-peg regime in which domestic currency issuance is rigidly backed by foreign reserves at a fixed rate, tightly constraining monetary policy.
- Currency overlay — A programme, often run by specialist managers, to manage the FX exposures arising from an international investment portfolio, separately from the underlying assets.
- Current account — The balance of a country's trade and income flows; a key medium-run anchor summarising the gap between national saving and investment.

D

- Dark pool — A venue with hidden order books that matches buy and sell interest at a venue-calculated midpoint (e.g. BGC MidFX, FXSpotStream MidMatch).
- Deal-contingent / M&A hedge — A hedge linked to a potentially unpredictable transaction such as a cross-border merger or acquisition.
- Dealer — A large commercial or investment bank (or securities house) that makes markets and trades in the inter-dealer market for proprietary purposes and to meet customer demand.
- Delta — The change in an option's value for a change in spot; also used to describe moneyness (e.g. 0–25 delta is out-of-the-money).
- Depository Trust & Clearing Corporation (DTCC) — Operator of a major swap data repository that captures a substantial share of global institutional FX option activity.
- Deutsche Bank Currency Returns Index (DBCR) — A 2007 index packaging transparent, rules-based carry, momentum and valuation strategies into an investable currency 'beta'.
- Deutsche Bank Equilibrium exchange-rate (DBeer) — DB's proprietary behavioural-equilibrium model using the terms of trade, a productivity ratio and trade openness; performs better in emerging markets.
- Discretionary manager — An investor who takes active currency positions based on judgement rather than a purely systematic, rules-based process.
- Dodd-Frank — The US post-crisis regulatory framework; includes the Volcker Rule and OTC-derivative reporting requirements.
- Dollarisation — Use of another country's currency as legal tender, which largely eliminates independent national monetary policy.
- Dornbusch overshooting — Model in which sticky goods prices cause the exchange-rate to overshoot its long-run level on impact before reverting, explaining excess FX volatility.
- Drawdown — The peak-to-trough decline in the value of a strategy or portfolio.
- Dutch disease — A situation in which a commodity boom raises the real exchange-rate and erodes the competitiveness of a country's other tradable sectors.

E

- EBS — A primary inter-dealer electronic broking venue, historically central to banks' inventory management.
- Effective spread — A transaction-cost measure inclusive of slippage and fees, giving a more realistic picture of liquidity than quoted depth alone.
- Efficient Market Hypothesis (EMH) — The theory that asset prices fully reflect available information.
- Electronic Communication Network (ECN) — A third-party electronic venue that both streams prices and matches buy and sell orders.
- Electronification — The shift of FX trading from voice to electronic execution across both the inter-dealer and dealer-client segments.
- Emerging markets (EM) — Developing-economy currencies, often subject

to capital controls, higher basis volatility and offshore NDF trading.

- EPIC+ — Deutsche Bank's positioning framework aggregating direct measures (CFTC/IMM) and indirect measures (CORAX, inter-dealer orderbook, risk reversals, DTCC options).
- Equilibrium exchange-rate — The exchange-rate consistent with economic fundamentals and internal/external balance; estimated via frameworks such as FEER, BEER and DBeer.
- European-style option — An option that can be exercised only at expiry.
- External Balance Assessment (EBA) — The IMF methodology that estimates current-account and REER 'gaps' using several complementary approaches.

F

- Fama regression — A test of UIP that regresses realised exchange-rate changes on the interest differential (or forward premium); typically yields a slope below one — the forward-premium puzzle.
- Flash crash — A sudden, severe and short-lived price dislocation, increasingly studied alongside the electronification of markets.
- Foreign Direct Investment (FDI) — The stickier, long-horizon component of external financing, reflecting the partial illiquidity of direct ownership.
- Forward (FX forward) — A contract to exchange currencies at a future date at an exchange-rate agreed today.
- Forward points — The forward premium or discount added to spot; the direct carry cost or benefit of hedging with a forward.
- Forward premium puzzle — The robust empirical finding that high-yield currencies tend on average to appreciate rather than depreciate, making the carry trade profitable and contradicting UIP.
- Functional currency — The primary currency in which an entity operates and reports; FX risk arises from items denominated in other currencies.
- Fundamental Equilibrium exchange-rate (FEER) — The real exchange-rate consistent with the economy operating at internal and external balance.
- FX swap — A contract combining a spot exchange of currencies with a reverse exchange at a later date; the largest FX instrument by turnover and effectively collateralised borrowing.

G

- G10 — The group of major, most liquid developed-market currencies.
- Gamma — The rate at which an option's delta changes with spot; long-gamma dealers dampen volatility while short-gamma dealers amplify moves through their hedging.
- GENIUS Act — US legislation requiring USD stablecoins to be fully backed by liquid, short-term USD reserves (such as T-bills) to maintain a 1:1 peg with the dollar.
- Give-up — The process by which a trade executed by a client is transferred ('given up') to its prime broker, becoming two matching trades.
- Global financial cycle — Rey's concept in which the US dollar and global risk appetite drive worldwide leverage, capital flows and financial

conditions.

H

- Hedge accounting — Accounting treatment that defers a hedge's mark-to-market so it does not affect net income until the underlying transaction is recorded in earnings.
- Hedge ratio — The proportion of an exposure that is hedged; typically sticky and adjusted in cycles, and an increasingly first-order driver of FX.
- Herstatt risk — See Settlement risk.
- High-frequency trading (HFT) — Very high-speed algorithmic trading, often conducted by principal trading firms.

I

- Implied volatility — The market's expected volatility embedded in option prices; the price of protection against exchange-rate movements.
- In-the-money (ITM) — An option that has intrinsic value because its strike is favourable relative to spot.
- Inter-dealer market — The 'inner tier' in which dealers trade among themselves to manage inventory imbalances; no longer exclusive to banks.
- Intermediary asset pricing — A framework in which exchange-rates are set at the margin by constrained financial intermediaries with limited, time-varying risk-bearing capacity.
- Internalisation — Dealers matching offsetting client flow on their own books rather than hedging externally, which has reduced their share of inter-dealer trading.
- International Monetary Market (IMM) — The CME division whose weekly CFTC futures positioning data are used to proxy speculative positioning in FX.
- Intervention — Central-bank buying or selling of a currency to influence the exchange-rate; most effective when large, public, communicated and aligned with fundamentals.

K

- Knock-in / knock-out option — Barrier options that are activated (knock-in) or cancelled (knock-out) when spot reaches a predetermined level.

L

- Last look — A brief window (typically milliseconds) allowing a liquidity provider to accept or reject a trade request after receiving it; prohibited on primary venues.
- Leverage ratio — A non-risk-weighted Basel III capital constraint capturing both on- and off-balance-sheet exposures and derivative positions.
- Liquidity consumer (seeker) — A participant that takes liquidity to meet an underlying need, such as corporates, real-money investors and retail.
- Liquidity firmness — The certainty of execution at a quoted price; reduced by the use of last look on secondary venues, MDPs and SDPs.

- Liquidity mirage — The appearance of depth created by aggregated but potentially inaccessible liquidity spread across fragmented venues.
- Liquidity provider (LP) — A participant that supplies two-way prices to the market, principally banks and some PTFs.

M

- Market maker / market making — Providing two-way prices and capturing the bid-offer spread; a liquidity-providing service rather than a source of the 'excess returns' from proprietary trading.
- Market microstructure — The study of how the trading process itself conveys information and moves prices.
- Meese-Rogoff puzzle — The 1983 finding that structural exchange-rate models cannot out-forecast a naive random walk out of sample, even given realised fundamentals.
- Momentum — A factor and technical signal based on the tendency of price trends to persist.
- Monetary model — An exchange-rate model based on relative money supplies and interest rates; in modern form it uses the monetary policy reaction function and Taylor-rule fundamentals.
- Multi-dealer platform (MDP) — A venue that lets clients request quotes (RFQs) from multiple dealers at once (e.g. FX Connect, FXall, 360T); it bears no inventory risk.

N

- Net Foreign Assets / Net International Investment Position (NFA / NIIP) — A country's external assets minus its liabilities; a long-run anchor for the real exchange-rate.
- Nominal exchange-rate — The quoted price of one currency in terms of another.
- Non-deliverable forward (NDF) — A cash-settled forward for a restricted currency, settled (usually in USD) against a fixing rather than by physical delivery.
- Non-deliverable option (NDO) — A cash-settled option on a restricted, non-deliverable currency pair.
- Non-reporting banks — Smaller banks that do not report turnover directly to the BIS and largely intermediate for institutional and non-financial customers.

O

- Official sector — Central banks, sovereign wealth funds, international financial institutions (IMF, BIS), development banks and government agencies.
- One-touch / no-touch option — Options that pay a fixed amount depending on whether spot does (one-touch) or does not (no-touch) reach a predetermined level during the option's life.
- Order flow — Signed transaction volume (buyer-initiated minus seller-initiated trades); the proximate driver that aggregates dispersed

information into price.

- Out-of-the-money (OTM) — An option with no intrinsic value (0–25 delta); the band favoured by hedgers.
- Over-the-counter (OTC) — Bilateral trading conducted away from an exchange; the dominant structure of the FX market.

P

- Payment-versus-Payment (PvP) — The simultaneous exchange of both currency legs of a trade, which eliminates settlement risk (e.g. through CLS).
- Positioning — The stock of existing long and short currency exposures held by investors; most useful as a contrarian signal at extremes and a measure of market vulnerability.
- Pre-trade anonymity — The degree to which counterparties remain unknown to each other before execution; highest on exchanges and lowest on fully disclosed MDPs/SDPs.
- Prime brokerage (PB) — An arrangement in which a bank lends its credit standing and trading relationships to a client, separating execution from credit provision.
- Principal Trading Firm (PTF) — A lightly capitalised non-bank firm (e.g. XTX Markets, Citadel Securities) that trades for its own account and also makes markets, often via HFT.
- Programmability — The embedding of automated rules and conditions into money, usually via smart contracts, enabling conditional and synchronised settlement.
- Proprietary trading — Profit-seeking trading with a firm's own capital; heavily constrained for banks by the Volcker Rule.
- Purchasing Power Parity (PPP) — The theory that exchange-rates adjust so that identical baskets of goods cost the same across countries; a powerful long-run anchor but a poor short-run timing tool.
- Put option — The right, but not the obligation, to sell one currency against another at a predetermined strike price.

R

- Random walk — A price process with no predictable drift; the benchmark that FX forecasting models struggle to beat at short horizons.
- Real effective exchange-rate (REER) — A trade-weighted measure of a currency's real value against a basket of trading partners.
- Real exchange-rate — The nominal exchange-rate adjusted for relative price levels; it mean-reverts under PPP in the long run.
- Real interest-rate differential — The nominal interest differential adjusted for expected or realized inflation; the more relevant concept for medium-run valuation (Frankel's real-interest-differential model).
- Real money investors — Mutual funds, pension funds, insurance companies and endowments that buy currencies to satisfy end-user demand for underlying assets.
- Reporting dealers — The major banks that report turnover to the BIS; their

share of spot trading has fallen as internalisation has grown.

- Request for Quote (RFQ) — A request from a liquidity consumer to a liquidity provider for the price to buy or sell a specified currency pair, amount and settlement date.
- Request for Stream (RFS) — A continuous flow of quotes for a specified size and duration (such as a trading day) — in effect a continuous RFQ providing real-time price updates.
- Restricted markets — Currencies subject to capital controls and/or current-account restrictions, often traded offshore through NDFs.
- Risk reversal — The implied-volatility difference between an out-of-the-money currency call and an equivalent-delta put; a gauge of skew and directional demand.

S

- Settlement risk (Herstatt risk) — The risk of paying out one currency in an FX trade without receiving the other, arising from settlement across time zones; mitigated by PvP/CLS.
- Sharpe ratio — A measure of risk-adjusted return, defined as excess return per unit of volatility.
- Single-dealer platform (SDP) — A bank's own electronic platform for direct client trading (e.g. Deutsche Bank Autobahn), integrated within the bank's risk-management function.
- Skew Volume Indicators (SVA / SVO) — Deutsche Bank indicators built from DTCC data that separate investor flows (at-the-money, SVA) from hedger flows (out-of-the-money, SVO).
- Slippage — The difference between the expected price of a trade and the price at which it is actually executed.
- Smart contract — Self-executing code that performs a transaction automatically when predefined conditions are met.
- Spot / FX spot — The purchase or sale of one currency against another for near-immediate delivery, typically settling two business days later (T+2).
- Stablecoin — A digital token that maintains a peg to a fiat currency (about 99% are USD-linked), backed by reserves; a potential conduit for dollar demand.
- Standardised Approach for Counterparty Credit Risk (SA-CCR) — A regulatory approach for measuring counterparty credit risk on derivatives, relevant when comparing cleared and bilateral margining.
- Sterilised intervention — Central-bank intervention offset by a domestic money-market operation so that the money supply is left unchanged.
- Straight-through processing (STP) — The automated, end-to-end matching, confirmation and settlement of trades.
- Strike price — The predetermined exchange-rate at which an option can be exercised.
- Swap Data Repository (SDR) — A CFTC-registered repository to which in-scope OTC FX derivatives must be reported under Dodd-Frank.
- Systematic investor — A manager who translates observable data into positions through a repeatable, rules-based portfolio-construction and risk-management process.

T

- T+1 / T+2 — Settlement one or two business days after the trade date; spot normally settles T+2, while USD/CAD settles T+1.
- Taylor rule — A monetary-policy reaction function relating the interest rate to inflation and the output gap; central to modern monetary models of the exchange-rate.
- Technical analysis — Forecasting future price moves from patterns in past market data rather than fundamentals (e.g. moving averages, momentum, support and resistance).
- Terms of trade — The price of a country's exports relative to its imports; a key real-exchange-rate driver for commodity-exporting economies.
- Tokenization — The process of creating transferable digital tokens that represent claims on underlying money or assets on a shared infrastructure.
- Top-of-book spread — The best (tightest) bid-offer spread available on a venue; used to track FX market quality over time.
- Total Portfolio Approach (TPA) — Managing currency exposure as an integral part of the whole portfolio rather than as a standalone overlay or transaction cost.
- Treasury basis — The yield gap between US government bonds and currency-hedged foreign government bonds; a measure of the dollar's convenience yield.
- Treasury International Capital data (TIC) — US data on gross cross-border purchases and sales of securities, used to split investor turnover between fixed income and equities.

U

- Uncovered Interest Parity (UIP) — The condition that the expected change in the exchange-rate equals the interest-rate differential; strongly rejected in the data, which underpins the carry trade.

V

- Value (valuation factor) — A strategy that goes long undervalued and short overvalued currencies relative to an equilibrium anchor such as PPP or DBeer.
- Value-at-Risk (VaR) — A statistical estimate of the potential loss on a position or portfolio over a horizon at a given confidence level; used in portfolio hedging.
- Vanilla option — A standard call or put option with no exotic features.
- Voice trading — Trade negotiation by telephone or messaging rather than electronic systems; classified as voice-direct or voice-indirect (via a broker).
- Volatility smile — The pattern of implied volatility across option strikes, conveying the market's perceived balance of tail risks.
- Volcker Rule — A Dodd-Frank provision (implemented 2014–15) banning banks from most proprietary trading, including many foreign-exchange instruments.

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Artificial Intelligence has been used to generate and/or enhance certain diagrams and parts of the text in this handbook, including the cover image.

Appendix 1

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The end of the safety net:
Life without the macroeconomic stabilisers

28 September 2026

For several decades, financial markets and the wider economy benefited from an implicit safety net. If a slowdown or a downturn arrived, there was a monetary put from central banks, and a fiscal put from governments. The monetary put would see interest rates slashed and liquidity injected via QE. The fiscal put would see governments run larger deficits to cushion any downturn. These policy stabilisers smoothed out the business cycle, and supported asset valuations too.

These policy interventions were increasingly normalised from the late-1980s, and became more aggressive with each passing cycle. This reached its peak during the Covid-19 pandemic, when governments ran fiscal deficits unprecedented in peacetime, whilst many central banks expanded their balance sheets with huge asset purchases.

But after an era of ever-bigger interventions, these old stabilisers now face major obstacles. On the fiscal side, policy is constrained today by sovereign yields at multi-year highs, alongside the highest debt-to-GDP ratios in several decades. Then on the monetary side, policy is constrained by persistent inflation that's increasingly supply-driven.

This represents a big shift from the last few decades, but it does echo previous regimes where the macroeconomic stabilisers were either blunted or left unused. A good example is the 1970s experience, when stagflation and supply shocks posed a dilemma to traditional policy tools. Even further back, the pre-WWI environment of the classical gold standard and balanced budget orthodoxy saw governments play a very different and more limited role in macroeconomic stabilisation.

Looking forward, this environment means policymakers will have a much more difficult time stabilising the business cycle in the coming years. So absent a sharp fall in yields or a collapse in inflation, higher macro volatility will be here to stay.

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1. Background: How the stabilisers used to work.

Economic policymakers have long used interventions to stabilise output. This became increasingly fashionable after WWII, when Keynesian economics established a more active use for fiscal policy to stimulate the economy in a downturn. This meant budget deficits became a more regular feature of life, rather than used solely for wartime emergencies.

It was from the late 1980s, however, that the extent of policy interventions became increasingly sizeable. Indeed, it was here that the origins of a central bank "put" for financial markets began, in the crisis of Black Monday in 1987. That was a huge stock market crash, where the S&P 500 fell over 20% in a single day, marking the biggest daily decline in its entire history. The size of the falls even went beyond the Great Depression, raising fears of wider contagion, particularly if it were to spark a liquidity crisis that caused financial institutions to fail. However, the Federal Reserve responded by cutting rates and providing liquidity, offering a macroeconomic stabiliser against the prospect of wider economic stress. This offered support against a wider downturn, and the US economic expansion continued until 1990.

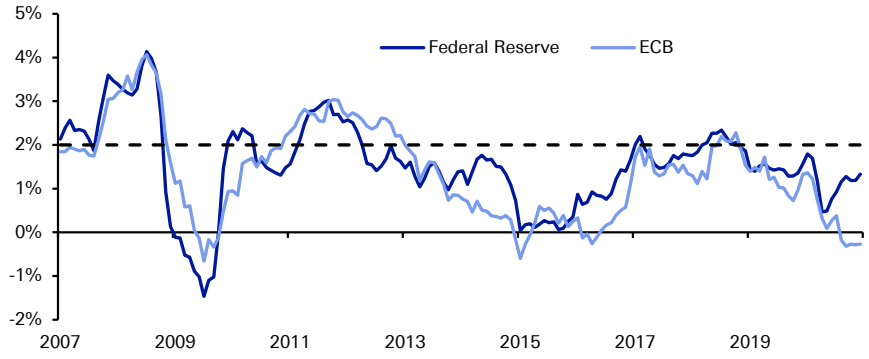
Whilst the genesis of a central bank "put" came in 1987, the idea gained further traction in the late-1990s, with the Asian financial crisis in 1997, followed by the Russian financial crisis in 1998. Fears of an economic contraction mounted, particularly after issues at Long-Term Capital Management (LTCM) raised concerns that the US wasn't immune from these global pressures. So once again, the Fed delivered three 25bp rate cuts pre-emptively, seeking to support the economy.

A few years later, the bursting of the dot com bubble in early 2000 set the stage for yet further stimulus. As equity markets declined, the Fed began to cut rates from early 2001, ultimately delivering 475bps of cuts that year that took rates from 6.5% to just 1.75%. That was followed up by further rate cuts in 2002 and 2003, leaving rates at 1% by mid-2003 into 2004. Moreover, the Bush tax cuts were also delivered in 2001 and 2003, acting as a fiscal stimulus as the wider US economy entered recession.

In hindsight, these moves seem modest compared to what then happened in 2008, when the Global Financial Crisis led to even bigger policy interventions. Suddenly, the world faced the biggest economic contraction since the Great Depression, with genuine fears that a repeat might happen. To avoid this, policymakers launched into the biggest economic stimulus ever seen in a peacetime environment. That included massive fiscal stimulus, with the US federal deficit at 9.8% of GDP in 2008, the highest since 1945. Moreover, central banks slashed rates to the zero lower bound, and many like the Federal Reserve launched into Quantitative Easing programmes, purchasing assets to ease financial conditions even more. Over the coming years, many began pledging to keep rates low well into the future, pioneering the concept of forward guidance.

These levels of policy intervention weren't confined to the immediate crisis however. In the subsequent years, active attempts were made to prolong and extend the business cycle, particularly as the 2010s became an era of low inflation and low interest rates. In this environment, borrowing was historically cheap, and low inflation meant central banks could cut rates without seeing a big overshoot of their targets. Indeed, both the Fed and the ECB undershot their 2% inflation target on average in the 2010s, and the ECB took their own deposit rate into negative territory in 2014.

Figure 1: The Federal Reserve and the ECB frequently missed their 2% inflation targets on the downside in the 2010s (PCE inflation for the US, HICP inflation for the Euro Area)

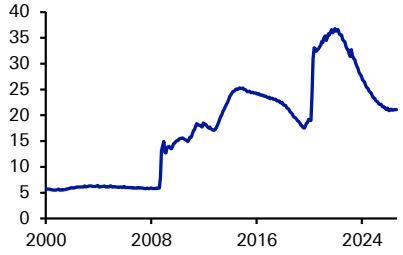


Source : Haver Analytics, Deutsche Bank

By 2019, the year before the pandemic, the intellectual climate had continued to move in the direction of more policy intervention. Government bond yields hit new lows, with Germany's 10-year bund yield at -0.7% in summer 2019. This environment of negative yields meant borrowing was basically a "free lunch", and that summer also saw the Fed begin a 75bp easing cycle, as inflation was below their target and they wanted to sustain the expansion.

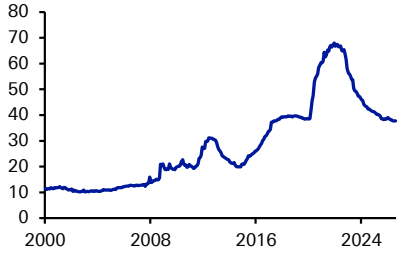
The peak of this policy interventionism was seen with the Covid-19 pandemic in 2020. Just as in 2008, the world was facing a huge and sudden economic contraction. Moreover, the consensus among policymakers was that they could have done more to speed up the recovery after 2008, potentially avoiding years of very high unemployment. As such, 2020 saw the most aggressive economic stimulus packages ever seen in a peacetime environment, well above those in 2008. Central banks immediately slashed rates back to the zero lower bound, pledged to keep rates low for years to come, and launched into major QE programmes again. Then on the fiscal side, record peacetime deficits were run. Indeed, the US federal deficit of 14.5% of GDP in 2020 was the highest since 1945.

Figure 2: Federal Reserve Balance Sheet (% of GDP)



Source : Bloomberg Finance LP, Deutsche Bank

Figure 3: ECB Balance Sheet (% of GDP)



Source : Bloomberg Finance LP, Deutsche Bank

During the pandemic period, many initially expected these interventions to remain for years to come. After all, in mid-2020, financial markets continued to price a continuation of the low inflation and low interest rates of the 2010s. This belief was clear on several fronts, and the 10-year Treasury yield closed at just 0.51% in August 2020. Meanwhile, the Federal Reserve even adjusted its own framework to allow for average inflation targeting, i.e. permitting an overshoot of the inflation target after periods of an undershoot. This reflected the consensus view that inflation was unlikely to be a serious threat.

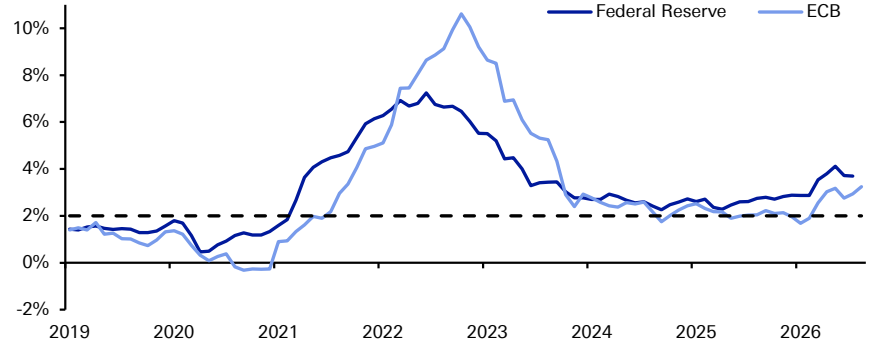
But as we now know, inflation returned in a serious way after the pandemic, forcing central banks into rate hikes. Meanwhile, public debt-to-GDP ratios rose significantly thanks to the pandemic stimulus, helping to push up sovereign bond yields too. In other words, many of the usual policy tools of the previous few decades were either blunted or no longer available.

2. The Monetary Constraint: High inflation means the central bank put isn't available

In the pre-2020 world described above, central banks regularly grappled with how to generate enough inflation. In part, this was thanks to several structural forces that put downward pressure on global prices. For instance, deepening globalisation acted as a positive supply shock that kept costs down. Meanwhile, the expansion of the global labour force in these years also boosted the supply side. In other words, inflation was low and central banks generally had the space to cut interest rates when they needed to.

But since the pandemic, we have entered a very different environment of above-target inflation across many of the world's biggest economies. Initially, that was caused by a huge wave of pent-up demand after the pandemic, alongside disruption to supply chains and labour markets. But the subsequent years also brought further shocks. Those included Russia's invasion of Ukraine in early 2022, driving a big spike in European energy prices, and pushing Euro Area inflation above 10% briefly. In 2026, the closure of the Strait of Hormuz then acted as a fresh energy shock, pushing inflation further away from target and contributing to a fresh tightening cycle from the major central banks.

Figure 4: In the 2020s, inflation has generally been above central bank targets (PCE inflation for the US, HICP inflation for the Euro Area)



Source : Haver Analytics, Deutsche Bank

Alongside those shocks, the structural forces keeping inflation down have gone into reverse. For instance, demographic trends have seen working-age populations shrink in many places. Meanwhile, the trend towards deeper globalisation has stalled, and the focus has turned to near-shoring and supply-chain resilience, rather than locating production in the cheapest place. Simultaneously, other trends like the push for higher defence spending are also adding to demand-side pressures.

Because of all this, the 2020s so far have brought persistently above-target inflation, and central bankers have found it much more difficult to ease policy in response to growth wobbles or financial stability fears.

This initially became apparent in 2022, amidst the biggest inflation wave for several decades. Central banks had to keep hiking rates, even in the face of slowing growth and financial market stress. Indeed, despite the S&P 500 being in bear market territory by mid-2022, the Federal Reserve still launched into a series of 75bp hikes, the biggest hikes since the 1990s. Then in 2023, despite the collapse of Silicon Valley Bank and fears of wider financial contagion, the Fed and ECB still continued with their rate hikes to combat inflation.

That tightening cycle in 2022-23 was a significant moment, because high inflation meant monetary policy could not simply act as a stabiliser against growth or financial stress. After all, policymakers realised that if they didn't act to tackle high inflation, they risked a repeat of the 1970s and the potential for even harsher monetary tightening to get inflation under control again.

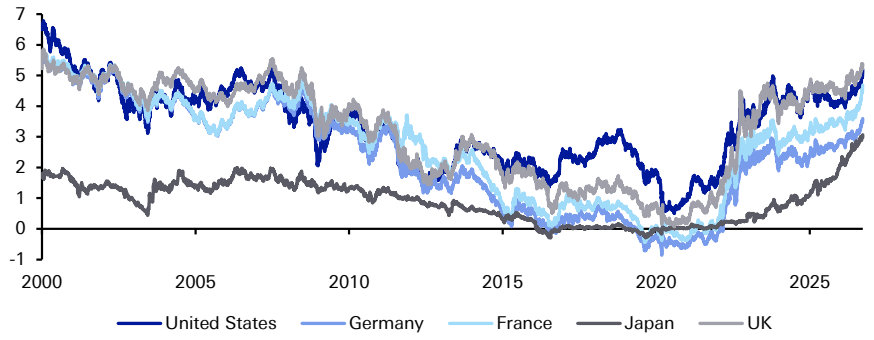
Today in 2026, this is a familiar problem again. We are in a globally synchronised hiking cycle, with the Fed, ECB and Bank of Japan all having delivered rate hikes. So far at least, global growth has been robust and financial conditions remain accommodative by historical standards. But because inflation remains clearly above target, central banks don't have the space they did in the pre-pandemic world to simply cut rates at the first sign of difficulty. There is still an inflation mandate to be achieved, and given we have now seen multiple years of elevated inflation, there is a much higher risk of inflation expectations becoming unanchored in that context, particularly if another shock occurred.

3. The Fiscal Constraint: High yields and high public debt mean fiscal stimulus isn't easily available either

Just as monetary policy is constrained by high inflation, fiscal policy today is also constrained by higher yields and higher public debt. This is a big transition from the world of 1980-2020, when yields were generally falling throughout. Indeed, despite all the fiscal stimulus around the GFC, the continuous decline in yields over the 2010s meant that debt dynamics remained manageable, because lower yields helped to offset the impact of higher debt.

Since the pandemic however, we've been in a very different world, as the dual forces of higher debt and higher yields have made debt dynamics far more unfavourable around the world. And this isn't just a nominal story, as real yields have also risen to multi-year highs as well.

Figure 5: 10yr Government Bond Yields (%) have risen to multi-year highs across several countries

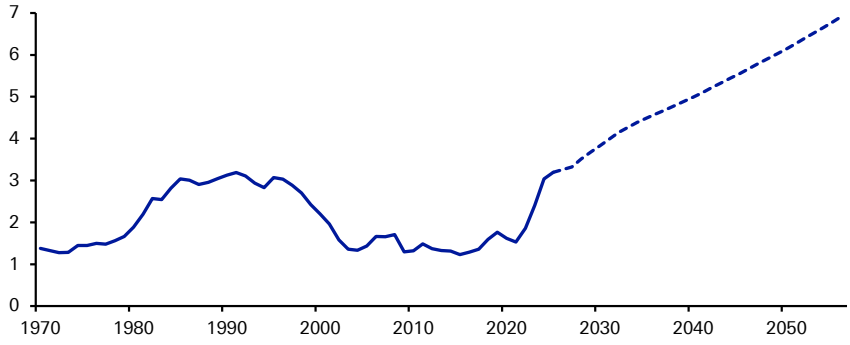


Source : Bloomberg Finance LP, Deutsche Bank

In essence, there are 4 new constraints around fiscal stimulus.

First, is the significant rise in global yields. By definition, the cost of incurring debt has grown, in both nominal and real terms. Relative to the pre-pandemic baseline in early 2020, the 10-year US Treasury yield has risen from 1.9% to 5.2%. Similarly in Germany, the 10-year bund yield has risen from -0.2% to 3.6%. Moreover, with yields already elevated, any further borrowing risks crowding out the private sector, and further dampening growth. And for governments, interest spending has ramped up significantly. For instance, US net interest spending as a share of GDP exceeded its recent peak from the early 1990s in the most recent fiscal year, and the CBO forecast this is set to continue.

Figure 6: US Net Interest Costs (% of GDP) with CBO Forecasts



Source : CBO, Deutsche Bank

The second constraint is that public debt itself is now at a higher level. This interacts with the higher yields mentioned above, because even though yields have only returned to their levels from the mid-2000s, that is much more impactful because they are being paid on a much larger debt pile. Using IMF numbers for consistency, net public debt across the advanced economies has risen from 52% of GDP in 2000, to 80% in 2025, with further increases projected to 2030. This means the fiscal space available to governments is more limited than it used to be.

The third constraint on fiscal stimulus is the high starting level of budget deficits. So to generate a meaningful fiscal stimulus, it would require deficits to be pushed up to even higher levels than they are today. But that in turn would add to the risks of bond market disruption that cascades through the economy, leading to higher borrowing costs for corporates and consumers as well.

The fourth constraint is higher inflation. When inflation is high, any fiscal stimulus that boosts demand risks pushing up prices rather than boosting output, and in turn makes monetary tightening more likely that could offset the impact of the stimulus. This trade-off was apparent in 2022, when central banks were hiking rates aggressively, and even further back in the inflationary environment of the 1970s. In a world of high inflation and supply constraints, fiscal stimulus is more likely to overheat the economy.

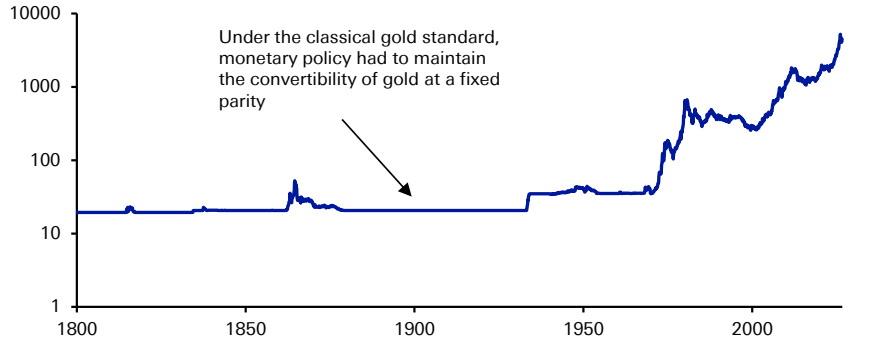
4. Are there any other eras of history where the stabilisers weren't available?

Whilst history never quite repeats exactly, there have been other eras in which the traditional policy stabilisers were either unavailable or severely blunted, which hold some lessons for our own time.

The pre-WWI classical gold standard

Before WWI, the international monetary system was known as the gold standard, in which currencies were convertible into a fixed quantity of gold. In practice however, this acted as a huge constraint on macroeconomic stabilisation. After all, monetary policy had to maintain the convertibility of gold at a fixed parity. This meant contemporary options like QE weren't available in a crisis. Meanwhile on the fiscal side, governments generally adhered to balanced budget orthodoxy. This was before Keynes wrote the General Theory, so the modern understanding of countercyclical deficit spending was not in place. In the US, there wasn't even a central bank until the Federal Reserve's establishment in 1913.

Figure 7: Gold Prices (US Dollars per ounce, log scale)



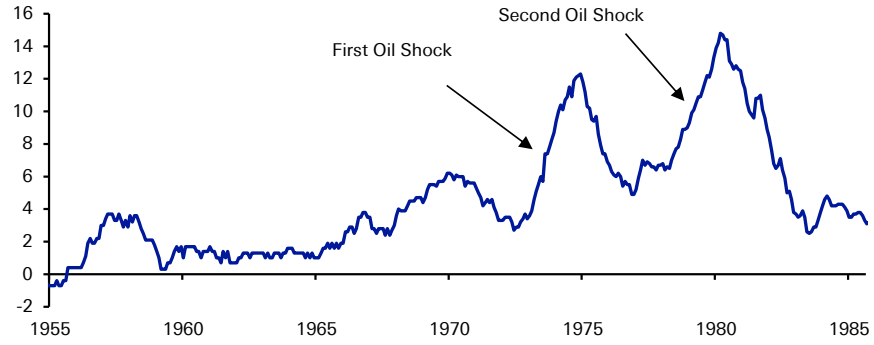
Source : Finaeon, Deutsche Bank

Without modern policy stabilisers however, this was also an era of huge macroeconomic volatility. There was a major financial panic across Europe and the US in 1873, which marked the start of the so-called "Long Depression". In the US, that contraction lasted until 1879, which in length was even longer than the Great Depression contraction of 1929-33. And this was followed by other major panics, including the 1893 panic that led to another US contraction, along with the 1907 panic that saw the Dow Jones almost halve relative to its 1906 peak.

The stagflationary 1970s

The more recent example comes from the 1970s, another era of multiple supply shocks. The biggest were the two oil shocks in 1973 and 1979, which both represented negative supply shocks that pushed up inflation and hurt growth in the US and Europe. However, this meant policymakers faced a dilemma. If central banks cut rates, that would add to the inflationary pressures and risk worsening the wage-price spirals already underway. Meanwhile, if fiscal stimulus were pursued, that would also push up inflation.

Figure 8: US CPI (year-on-year % change)



Source : Haver Analytics, Deutsche Bank

This context meant that policy couldn't smooth out the business cycle as successfully as it had in other decades. Indeed, in the decade from the first oil shock in 1973, the US experienced 3 separate recessions. That is significantly more turbulent than the last 40 years, when there have been just 4 recessions in total, or once a decade on average. For financial markets too, the 1970s were a terrible period for asset performance, with both equities and bonds losing ground in real terms across multiple countries.

5. Implications

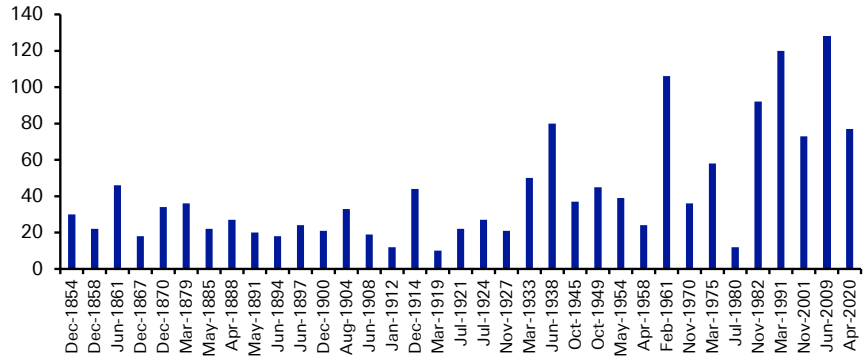
If economic policymakers can't use traditional monetary and fiscal policy tools to stabilise output and smooth the business cycle, then this will result in higher macro volatility.

If we look back at the last 40 years, it is notable that the 5 most recent US expansions all rank in the top 7 longest expansions since business-cycle dating began. This has several causes, and some of it reflects a move away from an agriculture-based economy where output was dependent on factors like the weather. But a key role has also been played by pre-emptive policy interventions, such as cutting rates to stop a slowdown turning into a recession. Indeed, recent examples include the a dovish shifts in monetary policy after the 1987 stock market crash, the 1997-98

financial crises, and the mid-cycle slowdown of 2015-16.

Beyond the actual policy interventions, these moves also had a confidence effect that supported financial markets. The fact that policymakers stood ready to take action helped create a virtuous circle, as optimism on the economic outlook boosted financial asset values, supported wealth effects, and avoided a tightening in financial conditions that would itself cause growth to slow.

Figure 9: Length of US economic expansions (in months) from the starting point



Source : NBER, Deutsche Bank

Clearly, policy is not as constrained today as in some periods of history. There is no longer a gold standard where the convertibility needs to be maintained. And unlike the 1970s, inflation is not in double-digits. Nevertheless, this is still a more constrained world than we have been used to in the last 4 decades. Higher inflation and more frequent supply shocks are constraining monetary policy. Meanwhile, higher yields and higher public debt levels are constraining fiscal policy. We have already seen the impacts since the pandemic, most notably in 2022, when central banks hiked rates to combat the inflation surge, and there was a simultaneous selloff across bonds and equities of a kind unseen in modern times.

Looking forward, this environment will offer new challenges of the sort we haven't been used to in recent decades. The policy trade-offs are now more acute, and if pre-emptive stabilisation is no longer available, then higher macro volatility lies ahead.

Appendix 1

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